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Measurement of the $R(D^*)$ ratio and P_τ in hadronic 1-prong τ decays with the hadronic FEI

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Abstract

This note presents a measurement of the $R(D^*)$ ratio defined as $\frac{B \rightarrow D^* \tau \nu}{B \rightarrow D^* \ell \nu}$ and the polarization of the τ lepton (P_τ) in hadronic one-prong τ decays using the hadronic Full Event Interpretation. The analysis is based on the full Long Shutdown 1 dataset, corresponding to an integrated luminosity of $365.29 \pm 1.70 \text{ fb}^{-1}$ collected on the $\Upsilon(4S)$ resonance. This represents the first simultaneous measurement of $R(D^*)$ and P_τ at Belle II. The analysis aims to shed light on the longstanding $R(D^*)$ anomaly — the discrepancy between experimental measurements and Standard Model predictions observed over the past 15 years — and to explore potential contributions from New Physics, to which P_τ is particularly sensitive. We report $R(D^*) = xxx \pm xxx(stat) \pm xxx(syst.)$ and $P_\tau = xxx \pm xxx(stat) \pm xxx(syst.)$ with a correlation coefficient of $\rho = xxx$ as determined from the simultaneous fit.

Todo list

Contents

0	Changelog	5
1	Introduction	7
2	Physics motivation	9
2.1	Theoretical motivations	9
2.2	Experimental landscape	10
3	Datasets and software	13
3.1	Software	13
3.1.1	basf2	13
3.1.2	rdstar1prong	13
3.1.3	pyhf	14
3.1.4	SysVar	16
3.1.5	cabinetry	17
3.1.6	HAMMER	17
3.2	Simulation Datasets	18
3.2.1	Full Event Interpretation (FEI) skims	18
3.2.2	Generic Monte Carlo (MC)	18
3.2.3	Gap modes	19
3.3	Experimental data	21
3.3.1	Experimental data $\Upsilon(4S)$	22
3.3.2	Off-resonance data	23
4	Online reconstruction	24
4.1	Event signature	24
4.2	Global tags	25
4.3	Event-wise selection	25
4.4	Tag side reconstruction	26
4.5	Final state particles	27
4.5.1	Photons	27
4.5.2	Leptons	27
4.5.3	Charged hadrons	27
4.6	Composite particles	28
4.6.1	π^0	28
4.6.2	K_S^0	28
4.6.3	ρ	28
4.6.4	Charmed mesons	28
4.6.5	B meson candidates	29
4.7	$\Upsilon(4S)$ and Rest of Event	29

5	Offline processing	32
5.1	Truth-matching	32
5.1.1	Btag and Bsig	32
5.1.2	Bsig decay category	33
5.1.3	D^*	33
5.1.4	τ	33
5.1.5	Hadronic decays	35
5.1.6	Final truth matched categories	35
5.2	Helicity angle calculation	35
5.3	Definition of Signal Region	37
5.4	Best Candidate Selection	39
5.5	Corrections from the Performance group	42
5.5.1	Luminosity scaling	42
5.5.2	Tracking efficiency	42
5.5.3	Tracking momentum scale	42
5.5.4	Photon energy bias	42
5.5.5	Photon efficiency	43
5.5.6	π^0 efficiency	43
5.5.7	π_s efficiency	43
5.5.8	PID (with the systematics framework)	44
5.5.9	PID (custom corrections)	47
5.6	Branching Fraction corrections	48
5.6.1	Charmed mesons	48
5.6.2	τ lepton	48
5.6.3	Double charm background	48
5.6.4	Prompt hadronic background	49
5.7	Gap replacement	49
5.8	Form Factor (FF) reweighing	51
5.9	D^0 mass calibration	53
5.10	Continuum calibration	54
5.11	FEI calibration	55
5.12	Photon multiplicity reweighting	58
6	Signal extraction	62
6.1	$R(D^*)$ parametrization	64
6.2	P_τ parametrization	68
6.3	Sideband and Control regions	69
6.4	Sensitivity on Asimov data	74
6.5	Statistical validation of $R(D^*)$ and P_τ fit	78
7	Systematics	84
7.1	Systematic budget for the $R(D^*)$ and P_τ fit	85

8	Agreement of simulated and experimental data	87
8.1	Off-resonance data	87
8.2	Control Region fits	88
8.3	Correctly reconstructed events with constrained normalization	91
8.4	Correctly reconstructed events with free normalization	94
8.5	Misreconstructed events with constrained normalization	95
8.6	Misreconstructed events with free normalization	97
9	Results	100
9.1	Box-opening strategy	100
9.2	Box-closed results	100
9.3	Box opened results	101
A	SysVar	102
B	PDG codes	109
C	Multinomial errors	110
D	D^0 mass calibration	111
E	Form Factor (FF) model parameters	114
F	FEI calibration fit validation	123
G	Statistical tests	125
H	Helicity angle window optimization	128
I	Selection of D^0 modes	129
J	Background composition checks	131
K	FEI calibration postfit plots	143
L	Free parameters of the fit	146
	References	200

0 Changelog

1. v1.0
 - (a) Initial commit.
 - (b) Setting b2template and document structure.
2. v1.0 $\rightarrow$ v2.0
 - (a) Added b2note number
3. v2.0 $\rightarrow$ v3.0
 - (a) Added proper changelog
 - (b) Fixed failing references
 - (c) Corrected spelling typos across the document using `ispell`
4. v3.0 $\rightarrow$ v3.1
 - (a) Added dedicated section [L](#) for working group review answers.
 - (b) Answered to Michele Mantovano's questions
 - (c) Removed some text and the Table with the MC15ri luminosities [3.2.2](#) as part of WG-review answer
 - (d) Updated Table [10](#) with some global tags that been included but the note was not up-to-date.
 - (e) Added a comment in Section [5.1.4](#) in line 622 about the problems in the τ decay mode truth matching.
 - (f) Removed misinformation sentence from Section [5.5.5](#) about the application of photon efficiency corrections to the ROE.
 - (g) Removed all information about the background reweighting from Section [6.3](#) as a result of Michele's question [20](#).
 - (h) Updated Figure [56](#) using the MINOS error for P_τ as a result of Michele's question [24](#).
 - (i) Modified the text in Section to avoid confusion about event level cuts. Brought up in Koga-san's question [34](#).
 - (j) Modified the text in sections [8.5](#) and [8.6](#) to make it more clear that the photon multiplicity reweighting has been used for these Data/MC checks.
 - (k) Added Appendix [I](#) as part of answer to Question [41](#). The $D^0 \rightarrow K^- K^+$ and $D^0 \rightarrow \pi^- \pi^+$ are now excluded from the analysis. I updated Sections [4.6.4](#) and [5.9](#).
 - (l) Updated Table [6](#) as part of Question [32](#).
5. v3.1 $\rightarrow$ v3.2

- (a) Updated online cuts for D^0 and ρ mesons in Section 4
- (b) Updated final selection and SR definition in Section 5.3
- (c) Added a linearity study for P_τ in Section 6.5.
- (d) Added new Section 5.10 for continuum calibration based on off-resonance data.
- (e) Updated FEI calibration in Section 5.11. We now switched to a mode-by-mode calibration contrary to the previous flat calibration factors.
- (f) Updated all plots in Section 6 with the new coarse binning for the leptonic reconstruction and new $E_{\text{ECL}}^{\text{extra}}$ range for the hadronic reconstruction.
- (g) Rearranged Section 6 for better flow and clarity since based on the WG-reviewer comments some parts read of a bit misleading. Added Section 5.10 that outlines our new continuum calibration based on off-resonance data.
- (h) Updated all plots in Section 8.

6. v3.2 $\rightarrow$ v4.0

- (a) Move table of contents to the very top of the document.
- (b) Add and answer comments and questions from FSR.
- (c) Complete answers from WGR.
- (d) Added a new template for the τ crossfeed semitauonic events as described in Question 93.

7. v4.0 $\rightarrow$ v4.1

- (a) Updated Table 44 with systematic budget.
- (b) Moved questions and answers from WGR to the end of the document.
- (c) Moved List of Figures, Tables and Acronyms after the Appendices

8. v4.1 $\rightarrow$ v4.2

- (a) Updated text in 4.5.1 to match Table 14. Mismatch pointed out by the RC.
- (b) Renamed Section 5.5.8 and added Section 5.5.9 for the customly derived efficiency corrections for muons faking pions.
- (c) Added explanation about electron momentum used for momentum correction bin with respect to Brems recovery in Line 796.
- (d) Updated Table 44 with systematic budget, including fake rates where a muon is faking a pion.

1 Introduction

The Standard Model of Particle Physics (SM) provides a solid description of the Universe at the elementary level but fails to account for several well-established phenomena documented in the scientific literature. Therefore, a primary goal of the physics community is to develop a more comprehensive framework that incorporates the SM while addressing these unexplained observations [1]. Lepton Flavour Universality (LFU) is a fundamental postulate of the SM that can be tested experimentally. Any observed discrepancies from its predictions could serve as a gateway to new physics beyond the SM [2].

Over the past decade, several experimental results have hinted at the existence of Lepton Flavor Violating processes [3]. In this analysis, we aim to measure the $R(D^*)$ ratio which is defined as

$$\frac{\mathcal{B}(B \rightarrow D^* \tau \nu)}{\mathcal{B}(B \rightarrow D^* \ell \nu)} \quad (1)$$

using 1-prong hadronic τ decays. The $R(D^*)$ ratio is an excellent test of LFU since it has a relatively clear theoretical prediction with several factorizable experimental and theoretical uncertainties canceling in the ratio. Additionally, since the τ lepton decays into two particles (a charged hadron and a neutrino), we can also determine the polarization of the τ lepton P_τ . Such a measurement becomes impractical for leptonic τ decays where two undetectable neutrinos are present.

This will be the first measurement of $R(D^*)$ in hadronic τ decays using Belle II data. Furthermore, it will mark only the second-ever measurement of P_τ in B-semileptonic decays—the first being a Belle result from 2017 [4]. The theoretical and experimental motivation for carrying out this measurement are outlined in 2.

We analyze the Long Shutdown 1 (LS1) dataset which accounts to collisions of an integrated luminosity of $365.29 \pm 1.70 \text{ fb}^{-1}$ on the $\Upsilon(4S)$ production energy. The datasets and the software used to analyze the data are described in section 3.

For the reconstruction of events we first use the Full Event Interpretation (FEI) algorithm [5] to reconstruct the second B meson of the event in fully hadronic modes and kinematically constrain the signal B meson of interest. We then combine D^* candidates with a charged π meson or a charged ρ meson for the signal B meson reconstruction. For the denominator of $R(D^*)$ in equation 1 i.e. the normalization mode, we combine the D^* candidates with an electron or a muon. We ignore any neutrinos from the reconstruction as these will not be captured by the Belle II detector. We then combine the above with the candidate from the FEI to build $\Upsilon(4S)$ candidates. We select only events where no extra charged tracks can be found. Naturally correctly reconstructed $B \rightarrow D^* \tau \nu$ and $B \rightarrow D^* \ell \nu$ decays will peak at small values of the extra energy in the calorimeter, since all particles have been assigned either to the signal or to the tag side B meson, with background decays being assigned a higher value for the aforementioned quantity. Our reconstruction is described step by step in section 4.

In order to account for any mismodelling effects in the simulation we correct our simulated dataset by considering three different kinds of corrections. Firstly the ones recommended by the Belle II Performance group [6]. Branching Fraction (BF) corrections where we correct the values of the `DECAY.dec` file according to the latest values of the

Particle Data Group (PDG) [7]. Lastly calibrations of the hadronic FEI algorithm and background reweighing to account for composition differences in the background decays in the signal enriched region and other regions used for calibrations. All the above are described in section 5.

In order to extract our signal we perform a binned maximum likelihood fit [8]. We implement a simultaneous two dimensional fit using pyhf [9], [10] in bins of the missing mass squared and the extra energy in the calorimeter in a kinematic region where we expect to be signal enriched. The main challenge of measuring $R(D^*)$ in hadronic τ decays compared to the leptonic modes is the significantly higher contamination of the retained sample from background events, mainly from fully hadronic decays and continuum events. Many of the former ones have never been measured experimentally, so the assumed BF is often purely a guess. Continuum events often also suffer from mismodelling effects given that the PYTHIA [11] parameters are not finely tuned in the Belle II simulation. Several Control Region (CR)s are added in order to normalize the background in the signal enriched region. The signal extraction method, the definition of the CR and the systematic uncertainties are discussed in sections 6 and ??.

We perform several tests to ensure that the modeling in our corrected simulated data is good. All these tests are described in section 8. Lastly we present our results in section 9.

2 Physics motivation

In the SM semileptonic B decays involving a D^* meson proceed via tree level Feynman diagrams with couplings to the $W^\pm$ boson. Figure 1 shows the Feynman diagram at the quark level. The relatively large value of $|V_{cb}|$ makes such a decay process experimentally favorable, as it provides sufficient data compared to other rare processes [7]. The coupling of the W boson to both light leptons (e and μ) and to the heavier τ lepton should be identical according to the LFU [12] except for kinematic phase-space differences which are calculated trivially. This enables the conduction of LFU tests as the couplings of the W boson to the leptons are precisely predicted within the SM.

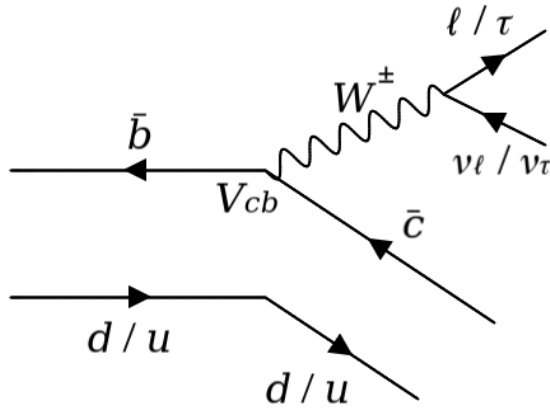


Figure 1: Feynman diagram of a $b \rightarrow c$ transition at the quark level.

2.1 Theoretical motivations

The $b \rightarrow c$ transition at the quark level is not directly accessible in experiments, as quarks are confined within hadrons [13]. Instead, one studies the transition at the hadronic level, where the B meson decays into a D^* meson along with a lepton-neutrino pair. This process is governed by the weak interaction, but due to the presence of strong interactions, the hadronic effects must be carefully modeled.

To describe the hadronic transition between the B and D^* mesons, form factors (Form Factor (FF)s) are introduced to parameterize the non-perturbative QCD effects governing the decay. These FFs encapsulate the dynamics of the heavy-to-heavy quark transition and depend on the squared four-momentum transfer, q^2 , between the initial and final hadronic states. The FFs are essential in determining the decay rate and kinematic distributions of $B \rightarrow D^* \tau \nu$ decays [14].

While the $R(D^*)$ ratio directly probes the decay rate of $B \rightarrow D^* \tau \nu$ decays, an additional key observable is the polarization of the τ lepton, denoted as P_τ . P_τ is determined by the spin structure of the underlying interaction and depends on the helicity amplitudes of the $B \rightarrow D^*$ transition. These amplitudes are directly influenced by the FFs, which govern the relative strength of different helicity contributions. Thus, a measurement of

P_τ provides valuable insight into the balance between longitudinal and transverse helicity states of the D^* , complementing differential decay rate measurements [15].

Furthermore in the SM, the τ lepton is expected to be highly polarized due to the purely left-handed chiral structure of the weak interaction. The polarization is sensitive to different helicity structures of the $b \rightarrow c$ transition and can be affected by new physics contributions. For instance, if tensor or scalar interactions exist, they would introduce right-handed or mixed-chirality contributions. Scalar interactions could be explained by the existence of a charged Higgs boson described in Two-Higgs Doublet Models [16]. Tensor interactions could be mediated by hypothetical particles called leptoquarks, that couple to both leptons and quarks [17], while a heavy W' boson could alter the chirality of the $b \rightarrow c$ transition by introducing right handed currents [18]. Precise measurements of P_τ , in combination with $R(D^*)$ would provide a robust way to differentiate between these new physics scenarios and test the validity of SM predictions.

2.2 Experimental landscape

The consistent study of B-hadrons started with B-factories such as the KEKB collider in Japan [19] and the PEP-II collider in the United-States [20], even though the bottom quark has been observed already in 1977 [21]. The reason for that is the challenging experimental environment, especially in semileptonic decays due to the presence of missing, undetected neutrinos. Several measurements of $R(D^*)$ have been carried out since the beginning of the B-factory era. The first measurement was conducted by *BABAR* in 2012 [22]. It used hadronic tagging to and leptonic τ decays and reported a tension of 2.7σ from the SM expectation. To this date this is measurement with the largest discrepancy. Several measurement followed from Belle that used hadronic or SemiLeptonic (SL) tagging and leptonic or hadronic τ decays demonstrated similar tensions, without however being able to rule out or confirm the discrepancy reported by *BABAR*. In recent years new results from LHCb and Belle II have reported newest results but still haven't give a final answer to the discrepancy due to the lack of statistical precision. The results mentioned above are summarized in Table 1 with the first uncertainty being statistical and the second systematic.

The experimental average of all those measurements is determined by the Heavy Flavor Averaging Group (HFLAV) [3]. The average is found to be 0.287 ± 0.012 with a total tension of 2.54σ . All the above results are summarized in Figure 2

In contrast to $R(D^*)$, the P_τ has been measured only once by the Belle collaboration [4]. The measured value of P_τ at Belle is -0.38 ± 0.51 (stat) $^{+0.21}_{-0.16}$ (syst), which is in agreement with the Standard Model, excluding polarization values greater than 0.5 at the 90% confidence level. The measurement of P_τ is not feasible in leptonic τ decays due to the presence of two missing neutrinos, while resolution effects make it particularly challenging in three-prong hadronic decays. At present, Belle II is the only experiment capable of measuring P_τ with the current experimental techniques used by both Belle II and LHCb. An improved measurement of the polarization with reduced uncertainty would further constrain form factor models and provide valuable insights into potential new physics contributions, as previously discussed.

Table 1: Overview of experimental measurements of $R(D^*)$ [23]

Experiment	$R(D^*)$	Tension	Year	Tagging	τ Decays	Reference
<i>BABAR</i>	$0.332 \pm 0.024 \pm 0.018$	2.7σ	2012	Hadronic	$\tau \rightarrow \ell \nu_\tau \nu_\ell$	[22]
Belle	$0.293 \pm 0.038 \pm 0.015$	1.8σ	2015	Hadronic	$\tau \rightarrow \ell \nu_\tau \nu_\ell$	[24]
Belle	$0.270 \pm 0.035 \pm_{-0.025}^{+0.028}$	0.4σ	2017	Hadronic	$\tau \rightarrow \pi \nu_\tau$ $\tau \rightarrow \rho \nu_\tau$	[4]
Belle	$0.283 \pm 0.018 \pm 0.014$	1.1σ	2020	SL	$\tau \rightarrow \ell \nu_\tau \nu_\ell$	[25]
LHCb	$0.281 \pm 0.018 \pm 0.024$	0.9σ	2023	-	$\tau \rightarrow \mu \nu_\tau \nu_\mu$	[26]
LHCb	$0.267 \pm 0.012 \pm 0.019$	0.7σ	2023	-	$\tau \rightarrow \pi \pi \pi \nu_\tau$	[27]
Belle II	$0.262 \pm_{-0.039}^{+0.041} \pm_{-0.032}^{+0.035}$	0.2σ	2024	Hadronic	$\tau \rightarrow \ell \nu_\tau \nu_\ell$	[28]
^a LHCb	$0.402 \pm 0.081 \pm 0.085$	1.3σ	2024	-	$\tau \rightarrow \ell \nu_\tau \nu_\ell$ $\tau \rightarrow \pi \pi \pi \nu_\tau$	[29]
Belle II	$0.306 \pm 0.034 \pm 0.018$	1.3σ	2025	acsl	$\tau \rightarrow \ell \nu_\tau \nu_\ell$	[30]

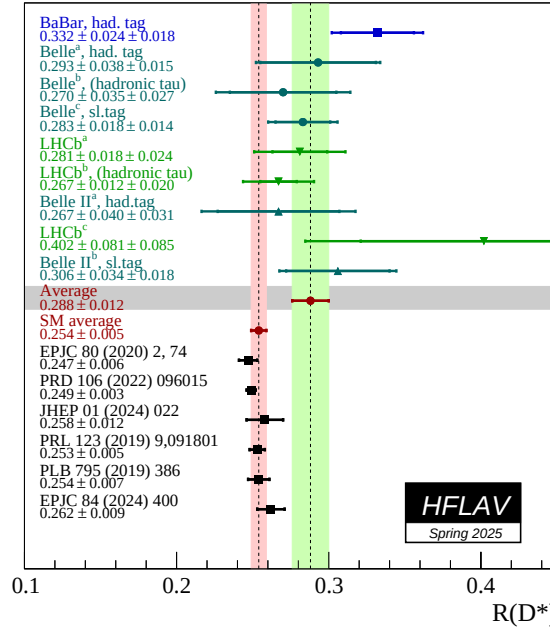


Figure 2: Recent experimental results and theoretical predictions from HFLAV [23]

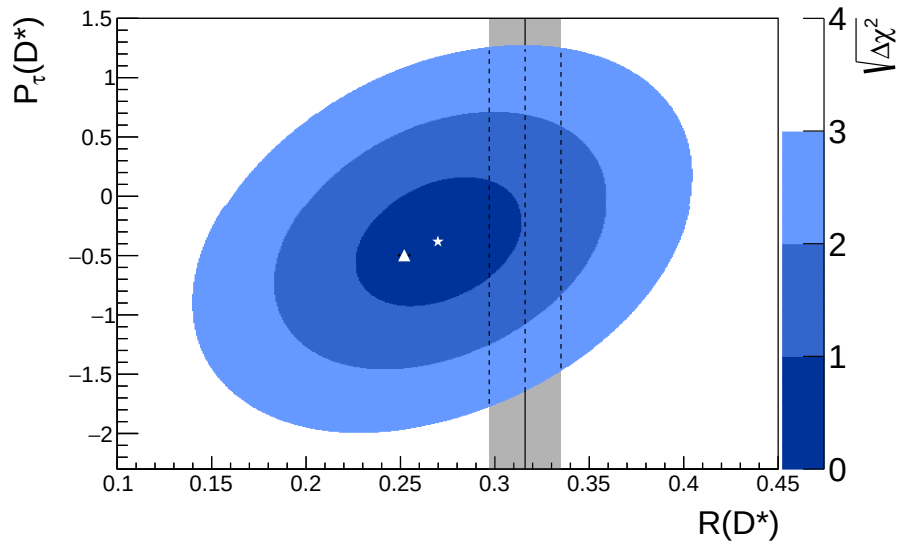


Figure 3: Unique experimental result of the P_τ in B semitauonic decays by Belle. The result is presented in contour with the measured value of $R(D^*)$ in the same event topology and with the same dataset. [4]

3 Datasets and software

The following section outlines the software and the simulated and experimental data that are used for the analysis. Extracting the $R(D^*)$ ratio and the P_τ requires a series of operations on the data, namely, reconstructing the decays of interest, correcting the simulated data with respect to the experimental data, developing a statistical inference model to extract information from the experimental data, and understanding the complicated correlations at play in order to assign consistent systematic uncertainties. Therefore a brief overview of the main software utilized throughout the different steps of the analysis is presented below. Furthermore a description of the official simulated and experimental data is provided.

3.1 Software

In what follow we describe the main software that is used for the analysis.

3.1.1 basf2

The Belle II Analysis Software Framework (**basf2**) is a software developed by the Belle II collaboration to carry out several tasks across the data processing chain [31]. For the analysis described in this document, only the high-level features of the software are required. These high-level features allow the user to build Final State Particle (FSP) candidates from tracks and clusters in the detector. These FSP candidates are then combined into composite particles, e.g., B mesons. A set of different algorithms performs various tasks such as building the Rest of Event (ROE), i.e. collecting all the tracks and clusters that do not belong to the particle of interest like the $\Upsilon(4S)$ resonance, vertex fitting, B-meson tagging, suppressing light quark processes, which we term as continuum, etc. Throughout the analysis, the light release `light-2403-persian` of **basf2** is used. [32]

3.1.2 rdstar1prong

For the purposes of this analysis we have developed a python library based on **basf2**. The library firstly wraps the reconstruction code around **basf2**. Secondly it implements all the functionalities necessary to perform a post-**basf2** reconstruction which we term *offline processing*. This includes the calculation of several quantities outside of **basf2**. It also enables the extraction of generator level information of the decay termed *truth matching*, useful for the identification of different decay processes in simulated data. It also applies all the necessary corrections on simulated data, and selects sub samples of the dataset that **basf2** outputs in order to define of kinematic regions enriched in signal or other background processes. Lastly it implements the plotting code necessary to visualize our findings. All the figures illustrating information extracted from simulated and experimental data are produced with the **rdstar1prong** library unless stated otherwise. Most importantly the **rdstar1prong** library implements a task pipeline around the **b2luigi** software [33] that automates the **basf2** reconstruction and the *offline processing*. The programming language that we use in the **rdstar1prong** library is **python** and the code lives in a **gitlab** repository [34].

3.1.3 pyhf

When presented with a dataset comprising of a mixture of different physical processes, inferring the number of events coming from a particular one requires the design of a statistical model that compares the measured data to the theoretical expectations. In cases that the underlying observable distributions cannot be expressed analytically, it is a common practice to build non-parametric functions from simulated data. Such an approach is termed as a Monte Carlo (MC)-template fit [35]. In this analysis no parametric functions that describe the signal extraction observables can be easily found, therefore, we also adopt such a strategy.

In order to perform a binned maximum likelihood MC-template fit we use the `pyhf` library [9], [10]. `pyhf` provides a `python` implementation of the HistFactory method [36] which has been well established in Large Hadron Collider (LHC) experiments. The HistFactory method provides a set of tools to build complex statistical models, in the form of a likelihood function. Such models are built from non parametric probability density functions, which have been derived by histogramming simulated data. It also allows the user to introduce systematic uncertainties and correlate those across different reconstruction channels and templates. A more detailed explanation of the HistFactory method can be found in [36].

We choose to use `pyhf` as our fitting framework as it is currently the state-of-the-art fitting software in the `python` ecosystem of the High Energy Physics (HEP) community. The software is well tested and documented and the increasing community spans across different experiments. Furthermore publishing the full likelihood model that is used to derive an experimental result is technically trivial in `pyhf`. This is something that the theory and the statistics communities have been persistently advocating for [37], as it simplifies the theoretical reinterpretation of a result, enables the combination of different measurements etc. [38].

In the following we introduce some nomenclature that stems from the HistFactory method and is established across the `pyhf` community. We only present the minimum information necessary to follow the statistical inference part described in 6 and the systematic uncertainties in ???. For a more detailed discussion the reader is referred to [9], [10], [36]. `pyhf` uses a JSON data format to define the statistical model in a declarative way. The contents of such a JSON file constitute a `pyhf workspace`. `pyhf` allows to perform simultaneous fits across different reconstruction *channels*. Usage of the `iminuit` minimizer [39], the `scipy` minimizer [40] or a combination of both is possible. `pyhf` is implementing nuisance parameters to model systematic uncertainties in a source-wise fashion. That is, every template, or *sample*, undergoes modifications through *modifiers* which implement different constraint terms and different correlations across channels or samples. For a full list of the available modifications in `pyhf` we point the reader to [9], [10], [36]. In Table 2 we present the modifiers that are used throughout the analysis under consideration.

`pyhf` allows to fully correlate systematic uncertainties across different samples, meaning that the same nuisance parameter can be used across different templates with the only thing changing being the error amplitude in the respective template. As it is clear from Table 2 `pyhf` also allows to fully correlate systematics across all bins. However in certain cases, arbitrary correlations with coefficients between zero and one might need to

Table 2: List of **pyhf** modifier used in this analysis. All the above modifier are controlled by a single nuisance parameter. The subscripts stand for: s for sample, c for channel, b for bin. f and g are functions determined by interpolating the nominal template between $+1$ and -1 sigma. a is an auxiliary measurement and delta is the relative uncertainty of a template in a particular bin with respect to the total uncertainty.

Modifier	Modification	Constraint term	Input	Short description
normfactor	-	-	-	This is a free unconstrained parameter. It is primarily used to determine the yields or strengths of different template components.
statererror	$\kappa_{scb}(\gamma_b) = \gamma_b$	$\prod_b \text{Gauss}(a_{\gamma_b} = 1 \gamma_b, \delta_b)$	$\delta_b^2 = \sum_s \delta_{sb}^2$	Implements the lightweight Barlow-Beeston method [41] for MC-stat errors. One nuisance parameter. per bin is assigned. The error amplitude. associated to each template depends on the relative contribution to every particular bin
normsys	$\kappa_{scb}(\alpha) = g_p(\alpha \kappa_{scb,\alpha=-1}, \kappa_{scb,\alpha=1})$	$\text{Gauss}(a = 0 \alpha, \sigma = 1)$	$\kappa_{scb,\alpha=\pm 1}$	Normalization factor. This accounts for a flat up or down shift of the whole template in all bins.
histosys	$\Delta_{scb}(\alpha) = f_p(\alpha \Delta_{scb,\alpha=-1}, \Delta_{scb,\alpha=1})$	$\text{Gauss}'s(a = 0 \alpha, \sigma = 1)$	$\Delta_{scb,\alpha=\pm 1}$	Correlated shape. Implements a shape uncertainty, fully correlated across all bins. The error amplitudes depend per bin on the template variations with respect to the nominal one .

be deployed. Currently there is no implementation in **pyhf** that can take into account an arbitrary correlation matrix between the different bins of the signal extraction fitted variable. However we can achieve the same effect by performing a Principal Component Analysis (PCA) in a space defined by the product of the number of reconstruction channels, templates and bins [42].

PCA is a statistical technique used for dimensionality reduction, while preserving as much of the original information as possible. The main idea is that a set of correlated variables are transformed into a smaller number of uncorrelated variables which are called *principal components*. The first principal component describes the direction with the largest variability of the data. The second principal component describes the direction with the second largest variability of the data orthogonal to the first and so on and so forth. Describing each principal component with a **histosys** modifier is effectively equivalent to implementing nuisance parameters with an arbitrary correlation across different bins. In the following we describe in more detail how we carry out the PCA for the needs of this analysis.

3.1.4 SysVar

We have developed the **SysVar** software library [43] in order to treat the systematic uncertainties of the analysis in a consistent fashion. Performing a PCA in the context of a template-fit requires:

1. computing a covariance matrix across all reconstruction channels, templates and bins.
2. performing the eigendecomposition of this matrix
3. determining the necessary number of principal components to describe the original correlation

An excellent explanation of such a procedure in a Belle II analysis that uses **pyhf** has already been given in [44] (Belle II internal) In the following we try to reiterate the most important points.

The covariance between the bins of the signal extraction variable, associated with a particular source of systematic uncertainty can be expressed as

$$C_{syst}^{ij} = \sum_{v=1}^{N_v} \sum_{c=1}^{N_c} \sum_{s=1}^{N_s} \Gamma_{vcsi} \Gamma_{vcsj} \quad (2)$$

where Γ_{vcsi} describes the variation v of bin i in template s in reconstruction channel c , with respect to the bin content of the nominal histogram, due to a particular systematic source. The various variations v are typically drawn from distributions that are normally distributed around the nominal correction event weights. Building the full covariance matrix becomes trivial by extending Eq. 2 to all bins. Then the covariance matrix can be decomposed into its principal components as

$$C_{syst} = VUV^T = (V\sqrt{U})(V\sqrt{U})^T = \Gamma'\Gamma'^T \quad (3)$$

where V is a matrix describing the eigenvectors and U is the diagonalized matrix with the eigenvalues. Now the eigenvariation is defined as

$$\Gamma' = V\sqrt{U} \quad (4)$$

A covariance matrix C'_{syst} can be computed using the eigenvariations from Eq. 4 plugged in Eq. 2. If all the eigenvariations are considered then the original covariance matrix C_{syst} and the new one from the eigendecomposition C'_{syst} are completely identical. Thankfully it is observed empirically that only a handful of orthogonal principal components are necessary to describe the original covariance matrix to a, subjectively defined, acceptable precision. This significantly reduces the number of nuisance parameters that are needed to be implemented in the signal extraction fit e.g. in **pyhf**.

With all the above in mind we have developed **SysVar** which carries out the following tasks:

1. Applying corrections to a MC dataframe

2. Generating variations of the Data/MC corrections
3. Histogramming simulated and experimental data to create the nominal templates for the fit.
4. Producing eigenvariations in the space of reconstruction channels, templates, bins.
5. Determining the number of necessary eigendirections to be implemented in the fit.
6. Saving the nominal templates and the eigenvariations of it in a format that the can be used to build a statistical model in `pyhf`

For a more detailed discussion the reader should refer to Appendix A, [45] (technical talk at the analysis tools meeting), [46] (recommendations talk for end users at the (S)L working group meeting). Furthermore [47] outlines new gateways that are opening for measurement combinations with `SysVar`.

3.1.5 `cabinetry`

We use the `cabinetry python` library to define `pyhf` workspaces in a very intuitive and declarative way [48]. One needs to define the Parameter Of Interest (POI)s, the input data or histograms, the reconstruction channels, the templates and the systematics that have to be considered in a YAML format. `cabinetry` then used the YAML files to read the data and create the `pyhf` workspace. It also provides a visualization module that helps the user to debug the statistical model during development and enables high level visualization of the results. For all the plots that we have generated through the `cabinetry` API we add a respective note in the caption in the rest of the document.

3.1.6 `HAMMER`

The simulated data produced by the Belle II experiment assumes certain FF parametrizations for hadronic decays. Even if the state-of-the-art FF models are used at the time of the generation of the simulated data, often enough these turn out to be outdated when a physics analysis is performed. As described in 2 the FF model and its parametrization can alter the decay rates as well as the kinematic and angular distributions of physical processes. Therefore the choice of the FF parametrization in the generated simulated data may bias the measured result, since non-parametric signal extraction fits make use of the MC template method. Reproducing MC samples with a different FF parametrizations in order to study such biases is computationally unsustainable for the large samples that are required for B-semileptonic analyses.

We use the Helicity Amplitude Module for Matrix Element Reweighting (`HAMMER`) software [49] to get a handle on the FF parametrization that is used throughout the analysis. The `HAMMER` software library provides an interface that allows the reweighting of a dataset from an assumed FF parametrization to a target one, implemented as event weights circumventing the need to reproduce large MC samples. The event weights can then be included when building the template histograms. Such an approach accounts for the alteration of efficiencies or acceptances that may arise from the usage of different FF

parametrizations. For a more detailed description of the HAMMER software we refer the reader to [50]

3.2 Simulation Datasets

3.2.1 FEI skims

In order to avoid using unnecessary computational resources we analyze simulated and experimental data skimmed with the hadronic FEI algorithm [5]. The FEI is B-tagging algorithm implemented in `basf2`. The FEI utilizes an array of Boosted Decision Tree (BDT)s to reconstruct B-meson candidates in a hierarchical manner starting from tracks and clusters. Subsequently in FSPs and composite particles are reconstructed in six distinct steps. These can be seen in Figure 4.

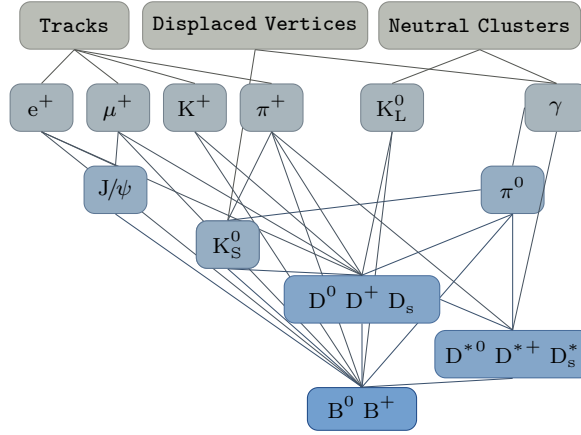


Figure 4: Overview of the FEI hierarchical reconstruction approach. At first step clusters and tracks are used to form final state particle candidates. These are given as input to subsequent BDTs in order to build composite particle candidates until a valid B-meson candidate is formed.

The FEI skim selects applies event-wise selections to ensure that B-meson candidates in the tag side can be reconstructed [51]. It later reconstructs B-meson candidates and performs a Best Candidate Selection (BCS) retaining the first twenty best candidates based on the signal probability. The FEI skim selections are outlined in Table 3

Unless stated otherwise all the simulated and experimental data mentioned in the document have been centrally FEI skimmed by the Belle II Data Production (DP) group [52].

3.2.2 Generic MC

For the nominal version of the analysis we use the official generic MC15rd generated by the DP group [52].

The amount of MC15rd available is less by roughly a factor of two compared to MC15ri which however comes with the advantage of better modeling of the beam conditions,

Table 3: Pre-selection cuts of the FEI skim.

Category	Cuts
CleanedTrackCuts	$\text{abs}(z0) < 2.0$ and $\text{abs}(d0) < 0.5$ and $\text{pt} > 0.1$
CleanedClusterCuts	$E > 0.1$ and $0.296706 < \text{theta} < 2.61799$
EventCuts	$\text{nCleanedTracks} \geq 3$ and $\text{nCleanedECLClusters} \geq 3$ and $\text{visibleEnergyOfEventCMS} > 4$
B-meson Cuts	$M_{bc} > 5.2 \text{ GeV}$ $ \Delta E < 0.3 \text{ GeV}$ signal probability > 0.001 (omitted for decay mode 23)
BCS	20 candidates with highest signal probability

therefore better Data/MC agreement in certain quantities. The luminosity used per production and per sample for the generic MC15rd campaign can be seen in Table 4.

Table 4: Luminosity for different samples in the Generic MC for MC15rd

Data Processing	Sample	Luminosity (fb^{-1})
proc13 exp 7-18	mixed	744.764
	charged	744.764
	ccbar	744.764
	ssbar	744.764
	uubar	744.764
	ddbar	744.764
prompt exp 20-26	mixed	699.235
	charged	699.235
	ccbar	699.235
	ssbar	699.235
	uubar	699.235
	ddbar	699.235

3.2.3 Gap modes

The experimentally measured BF of inclusive $B \rightarrow X_c \ell \nu$ decays does not agree with the sum of the BFs measured in exclusive modes [53]. This mismatch is commonly referred to as the *gap*. To address this gap several non-resonant decay modes that might contribute, but have never been experimentally measured, have been added to the Belle II generic MC. The non-resonant decays include $B \rightarrow D^{(*)} \pi \pi \nu$ and $B \rightarrow D^{(*)} \eta \nu$ final states. However such non-resonant multi-body decays result in a much softer momentum spectrum for the lepton, which as observed in previous Belle II analyses is not representative of the observed experimental data [54]. To resolve this issue it has become standard practice within

Belle II to replace these final states with a dedicated signal MC production containing modes produced through one of the excited D^{**} resonances. A detailed discussion on the matter can be found in [55].

Since the DP group provides us only the number of generated events, we have to calculate the luminosity ourselves. This is because the assumed BF's for the gap modes are only educated guesses, and their estimation depends on new experimental and theoretical results. The usual luminosity equation $\mathcal{L} = \frac{N_{events}}{\sigma}$ holds here, which transforms into Equation 5.

$$\mathcal{L} = \frac{N_{events}}{2 \times N_{B\bar{B}} \times N_{leptons} \times f \times BF} \quad (5)$$

where the factor 2 comes from the fact that we can have two B mesons in the event, both of which can decay into the final state of interest. $N_{B\bar{B}}$ is the number of $B\bar{B}$ pairs assumed per 100 fb^{-1} . $N_{leptons}$ accounts for the fact that the gap samples are not generated separately for the two light lepton generations. An extra factor f needs to be added as multiple resonances can contribute to the same final state. The assumed BF's for the gap modes are taken from the latest HFLAV averages [3].

All the above information for the light lepton gap modes are provided in Table 5. $N_{B\bar{B}}$ is assumed to be 5.1×10^4 per fb^{-1} for B^0 and 5.4×10^4 per fb^{-1} for B^+ as per the official `DECAY.dec` file of the Belle II simulated data.

Table 5: Event types and corresponding decays with number of events and luminosity for gap modes including light leptons. $N_{leptons}$ has been set 2 in Eq 5 to account for the fact that the light lepton gap modes are not produced separately. The calculated luminosity is used to scale the events that survive the full set of event selections to the luminosity of experimental data.

Decay	Event Type	Prod.	$N_{events}(10^6)$	f	BF (10^{-2})	$\mathcal{L} (\text{fb}^{-1})$
$B^0 \rightarrow D_1'(D\pi\pi)\ell^-\bar{\nu}_\ell$	1196700001	26218	8	0.5	0.07 ± 0.08	11,204.48
$B^0 \rightarrow D_1'(D^*\pi\pi)\ell^-\bar{\nu}_\ell$	1196700002	26222	8	0.5	0.20 ± 0.10	3,921.57
$B^0 \rightarrow D^{*0}(D\pi\pi)\ell^-\bar{\nu}_\ell$	1196700003	26224	8	0.5	0.07 ± 0.08	11,204.48
$B^0 \rightarrow D^{*0}(D^*\pi\pi)\ell^-\bar{\nu}_\ell$	1196700004	26228	8	0.5	0.20 ± 0.10	3,921.57
$B^0 \rightarrow D_1'(D^*\eta)\ell^-\bar{\nu}_\ell$	1196708000	26220	8	1.0	0.86 ± 0.19	456.0
$B^0 \rightarrow D^{*0}(D\eta)\ell^-\bar{\nu}_\ell$	1196708001	26226	8	1.0	0.86 ± 0.19	456.0
$B^+ \rightarrow D_1'(D\pi\pi)\ell^-\bar{\nu}_\ell$	1296700001	26235	8	0.5	0.07 ± 0.09	10,582.01
$B^+ \rightarrow D_1'(D^*\pi\pi)\ell^-\bar{\nu}_\ell$	1296700002	26239	8	0.5	0.22 ± 0.10	3,367.0
$B^+ \rightarrow D^{*0}(D\pi\pi)\ell^-\bar{\nu}_\ell$	1296700003	26241	8	0.5	0.07 ± 0.09	10,582.01
$B^+ \rightarrow D^{*0}(D^*\pi\pi)\ell^-\bar{\nu}_\ell$	1296700004	26245	8	0.5	0.22 ± 0.10	3,367.0
$B^+ \rightarrow D_1'(D^*\eta)\ell^-\bar{\nu}_\ell$	1296708000	26237	8	1.0	0.90 ± 0.20	411.52
$B^+ \rightarrow D^{*0}(D\eta)\ell^-\bar{\nu}_\ell$	1296708001	26243	8	1.0	0.90 ± 0.20	411.52

We follow a similar procedure for the gap modes that include a τ lepton. However, in this case, we must first determine the predicted BF for the semitauonic gap modes, as they are not provided in the latest averages by HFLAV [3]. To achieve this, we start with the BF's of the gap modes involving a light lepton and the predicted averages for $R(D^{(*)})$ from [3]. We then incorporate the predicted $R(H_c)$ ratios from [56] to calculate the BF's

of the corresponding semitauonic decays, where H_c represents the excited D mesons. For the non-resonant modes, since these are in the end produced via an excited resonance in the MC, we apply the same ratio used for the experimentally measured modes involving excited D mesons. All the above information is summarize in Table 6.

Table 6: Latest values of BF's for the light lepton $B \rightarrow H_c \ell \nu$ decays and $R(H_c)$ ratios. The BF of the semitauonic modes is calculated using $R(H_c) = \frac{B \rightarrow H_c \tau \nu}{B \rightarrow H_c \ell \nu}$

hadronic system	$\mathcal{B}(B \rightarrow H_c \ell \nu)$ (%) (measured)		$R(H_c)$ (prediction) $B^0/B^\pm$	$\mathcal{B}(B \rightarrow H_c \tau \nu)$ (%) (prediction)	
	B^0	$B^\pm$		B^0	$B^\pm$
D	2.11 ± 0.05 [3]	2.27 ± 0.06 [3]	0.296 ± 0.06 [3]	0.62 ± 0.02	0.67 ± 0.02
D^*	4.90 ± 0.11 [3]	5.27 ± 0.12 [3]	0.254 ± 0.005 [3]	1.24 ± 0.04	1.34 ± 0.04
D_0^*	0.12 ± 0.26 [3]	0.13 ± 0.19 [3]	0.08 ± 0.03 [56]	0.0096 ± 0.0211	0.0104 ± 0.0235
D_1^*	0.26 ± 0.04 [3]	0.28 ± 0.04 [3]	0.05 ± 0.02 [56]	0.013 ± 0.0056	0.014 ± 0.0059
D_1^*	0.59 ± 0.10 [3]	0.64 ± 0.10 [3]	0.10 ± 0.02 [56]	0.059 ± 0.015	0.064 ± 0.016
D_2^*	0.299 ± 0.027 [3]	0.321 ± 0.029 [3]	0.07 ± 0.01 [56]	0.021 ± 0.0035	0.022 ± 0.0038
$\sum D^{(*,**)}$	8.23 ± 0.17	8.86 ± 0.18	-	1.968 ± 0.044	2.117 ± 0.048
X_c	10.27 ± 0.15 [3]	10.99 ± 0.16 [3] ¹	0.223 ± 0.005 [56]	2.290 ± 0.061	2.451 ± 0.066
$D \pi \pi$ (via D_1')	0.07 ± 0.08 [3]	0.07 ± 0.09 [3]	-	0.3221 ± 0.0756	0.3335 ± 0.0816
$D^* \pi \pi$ (via D_0^*)	0.20 ± 0.10 [3]	0.22 ± 0.10 [3]	-		
$D^* \eta$ (via D_1')	0.86 ± 0.19 [3]	0.90 ± 0.20 [3]	-		
$D \eta$ (via D_0^*)	0.86 ± 0.19 [3]	0.90 ± 0.20 [3]	-		

To calculate the luminosity corresponding to the semitauonic gap MC samples, we use Equation 5 once again. The value of $N_{B\bar{B}}$ remains unchanged from the case of gap modes involving light leptons, as it corresponds to generic MC. For $N_{leptons}$, we set it to 1, since the τ lepton samples are generated separately. Additionally, the extra factor f is assigned a value of 1. This is because, in the production of semitauonic gap modes, which followed the one for the light lepton modes, the $B \rightarrow D^{(*)} \pi \pi \nu$ decays are merged into a single resonance. The decision to merge them stems from the realization that the 50-50 split assumed for the light modes was arbitrary and introduced unnecessary technical complexity (for more details, see the corresponding PR [57]). The calculated BF's and the corresponding luminosity are presented in Table 7.

3.3 Experimental data

We use the LS1 dataset acquired by the Belle II detector between 2019-2023. In the following we outline the different collision energies, data processing, and luminosity of the experimental data.

¹We have excluded $D_s^+ K$ and $D_s^+ K^*$ from the inclusive calculation

Table 7: Event types and corresponding decays with number of events and luminosity for gap semitauonic modes. The calculated luminosity is used to scale the events that survive the full set of event selections to the luminosity of experimental data.

Decay	Event Type	Prod.	$N_{events}(10^6)$	f	BF (10^{-2})	$\mathcal{L}$ (fb^{-1})
$B^0 \rightarrow D_1'(D\pi\pi)\tau^-\bar{\nu}_\tau$	1196300001	28067	3	1/8	0.3221 ± 0.0756	24,350.01
$B^0 \rightarrow D^{*0}(D^*\pi\pi)\tau^-\bar{\nu}_\tau$	1196300004	28068	3	3/8		73,050.02
$B^0 \rightarrow D_1'(D^*\eta)\tau^-\bar{\nu}_\tau$	1196308000	28069	3	3/8		73,050.02
$B^0 \rightarrow D^{*0}(D\eta)\tau^-\bar{\nu}_\tau$	1196308001	28070	3	1/8		24,350.01
$B^+ \rightarrow D_1'(D\pi\pi)\tau^-\bar{\nu}_\tau$	1296300001	28072	3	1/8	0.3335 ± 0.0816	23,517.65
$B^+ \rightarrow D^{*0}(D^*\pi\pi)\tau^-\bar{\nu}_\tau$	1296300004	28073	3	3/8		70,552.96
$B^+ \rightarrow D_1'(D^*\eta)\tau^-\bar{\nu}_\tau$	1296308000	28074	3	3/8		70,552.96
$B^+ \rightarrow D^{*0}(D\eta)\tau^-\bar{\nu}_\tau$	1296308001	28075	3	1/8		23,517.65

3.3.1 Experimental data $\Upsilon(4S)$

We use the on resonance data to reconstruct the signal processes of interest. The total luminosity that correspond to the on resonance LS1 dataset is $365.29 \pm 1.70 \text{ fb}^{-1}$ as per the calculation of the Data Production group [52]. This dataset contains the officially reprocessed data (proc) and data that a revision of the processing strategy might occur in the future (buckets). An overview of the different experiments and the corresponding luminosity for on resonance data can be found in Table 8.

Table 8: Experiment Offline Luminosity for 4S

Experiment	Processing	Offline Luminosity (fb ⁻¹)
26	bucket35-bucket36	54.795 ± 0.009 ± 0.351
24		85.642 ± 0.011 ± 0.549
22		32.060 ± 0.007 ± 0.206
20		3.788 ± 0.003 ± 0.025
18		89.900 ± 0.011 ± 0.576
17		10.715 ± 0.004 ± 0.069
16		10.294 ± 0.004 ± 0.066
14	proc13	16.500 ± 0.006 ± 0.106
12		54.373 ± 0.005 ± 0.350
10		3.655 ± 0.002 ± 0.024
8		1.663 ± 0.003 ± 0.011
7		0.506 ± 0.002 ± 0.004
Total		
all	proc13 & prompt	365.29 ± 1.70 fb ⁻¹

3.3.2 Off-resonance data

Off-resonance data is acquired with the Belle II detector at energies less than the Center-Of-Mass (CoM) energies necessary to produce the $\Upsilon(4S)$ resonance. Therefore this data sample is rich in $q\bar{q}$ processes, since the collision energy is not sufficient enough to produce a $B\bar{B}$ pair. The off-resonance sample is a great choice for checking the Data/MC modeling of the continuum processes. An overview of the different experiments and the corresponding luminosity for off-resonance data can be found in Table 9.

Table 9: Experiment Offline Luminosity for 4S_offres

Experiment	Processing	Offline Luminosity (fb^{-1})
25	bucket35	$24.574 \pm 0.006 \pm 0.158$
18		$8.482 \pm 0.004 \pm 0.055$
12	proc13	$8.691 \pm 0.002 \pm 0.056$
8		$0.808 \pm 0.001 \pm 0.006$
Total		
all	proc13 & prompt	$42.556 \pm 0.007 \pm 0.273$

4 Online reconstruction

In the following section we're outlining the online reconstruction strategy and the offline selections we are applying in our analysis.

4.1 Event signature

The overarching goal of this analysis is to measure $R(D^*)$ in hadronic 1-prong τ decays. Therefore a fully hadronic Final State (FS) is reconstructed on the signal side. Figure 5 illustrates the reconstruction on the signal and tag side.

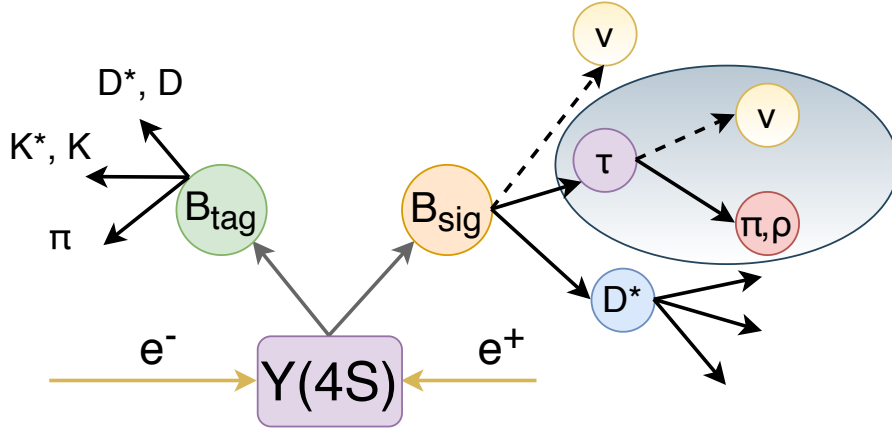


Figure 5: Illustration of event reconstruction for signal.

We're using the FEI to reconstruct fully hadronic states on the tag side. On the signal side we're building D^* candidates by combining D^0 candidates with charged or neutral slow pions. We then combine the D^* candidates with one charged pion or one charged pion and one neutral pion to reconstruct signal B-meson candidates. The latter one facilitates the reconstruction of $\tau \rightarrow \rho \nu_\tau$ candidates, since the ρ^+ meson decays almost exclusively in this mode. Since the τ lepton decays into a charged particle and one undetectable neutrino in the decays modes of interest, it is not necessary to reconstruct the τ lepton explicitly. The two τ decay modes considered here account for $\simeq 36.31\%$ of the total BF of the τ lepton, as illustrated in Figure 6. The relatively large BF is however counterbalanced by experimental challenges. Such a fully hadronic FS is susceptible to a large number of different hadronic backgrounds that can imitate the signal. This is contrary to the leptonic τ decays where only a handful of background processes such as $B \rightarrow D^{**} \ell \nu$ may imitate the intended mode of reconstruction.

For the normalization mode which appears in the denominator of $R(D^*)$ we combine the D^* candidates with a light lepton i.e. an electron or a muon.

In the rest of this section we describe in detail the reconstruction steps and all the selections applied in `basf2` online reconstruction.

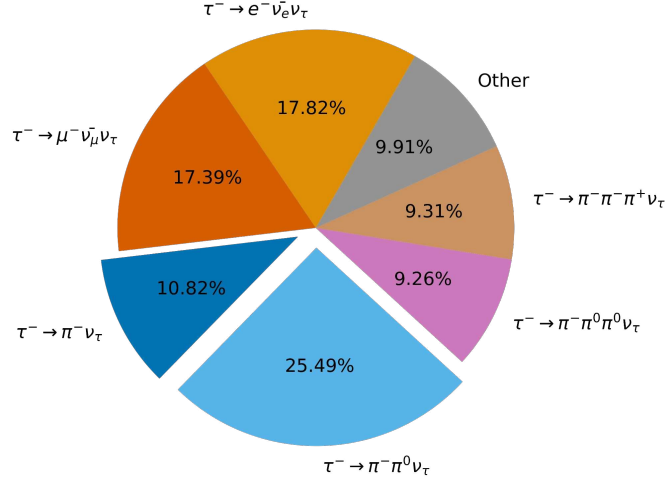


Figure 6: Overview of τ lepton BFs

4.2 Global tags

At the beginning of our steering file we use the following global tags to align with the recommendations of the Performance group.

Table 10: Global tags used in the steering file. The names of the global tags are used as input to the `basf2` helper function `b2.conditions.prepend_globaltag`

<code>ma.getAnalysisGlobaltag()</code>
<code>analysis_tools_light-2305-korat</code>
<code>analysis_tools_light-2403-persian</code>
<code>data_beam_conditions_proc13prompt</code>
<code>tracking_data_Moriond23_v1</code>
<code>neutrals_2024</code>
<code>Legacy_CollisionAxisCMS</code> (only for gap samples)
<code>leptonid_official_rel6_mc15rd</code>
<code>chargedpidmva_rel6_v5</code>

4.3 Event-wise selection

As mentioned already in Section 3.2.1 we use the hadronic FEI skims. A pre-selection of events is carried out by the skim to ensure that the reconstruction of hadronic B^{tag} candidates is possible. These selections have been outlined already in Table 3. These selections exclude events where there are insufficient particle candidates to reconstruct at least one hadronic B meson decay. [51] We do not apply any further event-level filtering of the events.

4.4 Tag side reconstruction

The first step after the event preselection is to select suitable B^{tag} candidates. We load the `B0:feiHadronic` and `B+:feiHadronic` ParticleLists provided by the FEI skim. We first suppress continuum events. To that end we reconstruct an ROE against the B^{tag} candidates from the skim. We will refer to this ROE as B_{ROE}^{tag} . The tracks and clusters that are considered for the construction of the B_{ROE}^{tag} are summarized in Table 11

Table 11: Selections for tracks and cluster considered in the B_{ROE}^{tag}

tracks	$ dr < 2 \text{ cm}$ $ dz < 4 \text{ cm}$ $p_T > 0.2 \text{ GeV}$ $\text{thetaInCDCAcceptance} == 1$
clusters	$\text{clusterNHits} > 1.5$ $((\text{clusterReg}==1 \text{ and } E > 0.080 \text{ GeV}) \text{ or } (\text{clusterReg}==2 \text{ and } E > 0.030 \text{ GeV}) \text{ or } (\text{clusterReg}==3 \text{ and } E > 0.060 \text{ GeV}))$ $0.2967 < \text{clusterTheta} < 2.6180$ $ \text{clusterTiming} < 200 \text{ ns}$

Once the B_{ROE}^{tag} is build we retain only candidates whose `cosTBT0` is smaller than 0.9. `cosTBT0` is the cosine of the angle between the thrust axis of the B^{tag} and the thrust axis of B_{ROE}^{tag} . Such a variable provides discrimination power between $B\bar{B}$ and $q\bar{q}$ events as the latter have a more jet like event shape. That is contrary to the former where the momentum is distributed more spherically because of the small available kinematic phase-space, between the mass of the $\Upsilon(4S)$ and the sum of the masses of the two B mesons.

We then keep only B^{tag} candidates whose M_{bc} is larger than 5.272 GeV and ΔE between -0.150 and 0.100 GeV. These last two selections are commonly used for B^{tag} candidates in Belle II analyses. For a definition of M_{bc} and ΔE the reader is referred to [2].

Lastly we keep only one B^{tag} candidate based on the `SignalProbability` i.e. the output BDT score of the FEI algorithm. We keep only one B^{tag} candidate before reconstructing the B^{sig} meson in an attempt to keep the B^{tag} and B^{sig} reconstruction efficiencies as factorizable as possible. This effect has been very nicely illustrated in this Belle II Physics week talk [58]. The *factorizability* of these efficiencies becomes crucial at a later stage of the analysis. When measuring ratios like $R(D^*)$, the assumption is that most efficiencies not dependent on the B^{sig} will cancel out. By selecting the B^{tag} candidate completely independently of the B^{sig} , we are following the safest approach to avoid mingling the two efficiencies.

If no B^{tag} candidate survives the process described above, we skip the event in order to avoid wasting CPU time.

4.5 Final state particles

After having selected a single B^{tag} candidate we proceed with reconstructing FSPs that will be used for B^{sig} .

4.5.1 Photons

All the photon candidates we select are later used to reconstruct π^0 candidates. Since the π^0 candidates are divided in two categories i.e. normal π^0 with $p > 0.2$ GeV and soft π^0 with $p < 0.2$, we collect photons in two ParticleLists. The first ParticleList is used for the normal π^0 . In this case the cluster selection criteria that we apply for the photon candidates are identical to the ones prescribed from the π^0 **eff40** list recommendations from the Neutrals Group [59]. For the soft photons we use the π^0 **eff50** recommendations [59] but we additionally require $|\text{clusterTiming}| < 200$ ns and an extra selection on a combination of **minC2Dist** and **clusterZernikeMVA** for the three different ECL regions as outlined in Table 14. The extra cuts are aligned with the ones used for the determination of Data/MC ratios for soft π^0 specifically [60].

4.5.2 Leptons

For electrons and muons we follow the recommendations of the Lepton IDentification (LID) group [61]. We first require that the lepton candidates satisfy $|\text{dr}| < 2$ cm and $|\text{dz}| < 4$ cm. We then profit by the predefined **std.Charged** lists in **basf2** [32] and the options we are using are summarized in Table 12

Table 12: LID selection criteria

	e	μ
working point	FixedThresh09	FixedThresh09
method	bdt	likelihood
classification	global	global
lid weights gt	leptonid_official_rel6_mc15rd	leptonid_official_rel6_mc15rd
release	6	6
brems_angleThreshold	0.2	-

4.5.3 Charged hadrons

For charged hadrons i.e. pions and kaons we again require that the track candidates satisfy $|\text{dr}| < 2$ cm and $|\text{dz}| < 4$ cm. We then discard track candidates with less than 20 hits in the CDC to ensure high performance on Hadron IDentification (HID) [62]. Lastly we use the global PID variables of **basf2** based on the likelihood method and apply **pionID>0.1** for pions and **kaonID>0.1** for kaons.

4.6 Composite particles

After having selected the FSP we then combine those to form composite particle candidates.

4.6.1 π^0

As already mentioned in ?? we reconstruct π^0 candidates using the π^0 `eff40` recommendations. As its name indicates this set of selections target an overall reconstruction efficiency of 40% for π^0 candidates. For normal π^0 we apply a mass window on the invariant mass of the candidate between 0.120 and 0.145 GeV, as prescribed by the recommendations of the Neutrals Group. For soft π^0 this invariant mass range is relaxed to a range of 0.105 and 0.150 GeV. For both lists we accept only candidates that have a non-failing vertex fit using the `kFit` algorithm with an additional mass constrain on the mass of the π^0 .

4.6.2 K_s^0

We reconstruct K_s^0 candidates using the predefined helper function `stdKshorts` from the `stdV0s` module in `basf2`. `stdKshorts` combines two oppositely charged pions into a K_s^0 candidate by. Only candidates that survive a vertex fit and with an invariant mass between 0.450 and 0.550 GeV are kept [63]. We additionally apply a set of selections adopted from a previous $R(D^*)$ analysis at Belle II [64]. These selections require that the `flightDistance` and the logarithm of the `significanceOfDistance` are positive. Additionally the cosine of the angle between the momentum and the vertex vector has to be larger than 0.8.

4.6.3 ρ

We combine a charged and a neutral pion into ρ candidates. These are only used for the τ decay of the signal. We require that the invariant mass of the pair of pions is between 0.560 and 1.060 GeV.

4.6.4 Charmed mesons

We present the reconstruction modes of D^* and D^0 in Table 13 along with their BFs taken from [7]. The charge conjugate reconstruction is implied in the rest of the document unless stated otherwise. We avoid reconstructing decay modes with two π^0 as empirically we have observed that these are very noisy. A longer discussion on our final choice of D^0 reconstruction modes is given in Appendix I.

For the D^0 meson candidates we require a loose mass window between 1.78 and 1.90 GeV. After combining the D^0 candidates with a soft pion we require that the mass difference between the D^* and the D^0 candidates is between 0.130 and 0.170 GeV for $D^{*\pm}$ and between 0.100 and 0.170 for D^{*0} . Since the reconstruction of the latter includes a soft pion we expect a worse resolution, hence the looser mass difference window. The difference between the two invariant masses is preferred over the invariant mass of the D^* alone, as systematic effects present in both reconstructions cancel out in the subtraction, making the peak in $\Delta M(D^*)$ more prominent. We also require that the momentum of

Table 13: Charmed meson reconstruction modes

Decay mode	BF (%)
$D^{*+} \rightarrow D^0 \pi^+$	67.7 ± 0.5
$D^{*0} \rightarrow D^0 \pi^0$	64.7 ± 0.9
$D^0 \rightarrow K^- \pi^+$	3.95 ± 0.03
$D^0 \rightarrow K_S^0 \pi^0$	1.24 ± 0.02
$D^0 \rightarrow K^- \pi^+ \pi^0$	14.40 ± 0.60
$D^0 \rightarrow K_S^0 \pi^+ \pi^-$	2.80 ± 0.18
$D^0 \rightarrow K_S^0 K^+ K^-$	0.44 ± 0.03
$D^0 \rightarrow K^- \pi^+ \pi^- \pi^+$	8.22 ± 0.14

all D^* candidates is less than 2.5 GeV as this suppresses continuum events, candidates originating from $c\bar{c}$ events in particular.

4.6.5 B meson candidates

We combine the D^* candidates with either a charged pion, a charged rho or a lepton to reconstruct the two tauonic signal modes, or the normalization mode respectively. We also add an extra reconstruction channel by combining D^* candidates with a light lepton and a pion, which defines a CR, enriched in $B \rightarrow D^{**} \ell \nu$ decays. The only requirement on the B^{sig} candidates whose vertex fit did not fail are kept.

4.7 $\Upsilon(4S)$ and Rest of Event

We lastly combine the two B meson into an $\Upsilon(4S)$ candidate. We take into account the charge allowed combinations i.e. $B^0 \bar{B}^0$ and $B^+ B^-$. For the neutral B mesons we also considered charge combinations which are not allowed i.e. $B^0 B^0$ and its charge conjugate. We assume that any flavor inference inconsistencies will not affect the measurement of $R(D^*)$, but they will increase the available statistics. We define the ROE with respect to the $\Upsilon(4S)$ candidate. Only tracks and clusters that satisfy the selections of Table 15 are taken into account.

We keep only $\Upsilon(4S)$ candidates whose ROE contains one or none extra charged tracks. The latter enables us to kinematically constraint the B^{sig} decays of interest with missing neutrinos, given that the B^{tag} is reconstructed fully in hadronic modes. The sample with one extra charged tracks is saved for later sideband checks. All the reconstruction channels are summarized in Table 16.

Table 14: Overview of online cuts for building B meson signal candidates

good track	
$dr < 2 \text{ cm}$ and $ dz < 4 \text{ cm}$ and $nCDCHits > 20$	
unflavored hadrons	
π^+	[good track] and $\text{pionID} > 0.1$
π_{slow}^+	$p > 0.05 \text{ GeV}$
K^+	[good track] and $\text{kaonID} > 0.1$
π^0	pi0:eff40_May2020
	pi0:eff50_May2020 and $ \text{clusterTiming} < 200 \text{ ns}$ and $0.1183 < M_{\gamma\gamma} < 0.147 \text{ GeV}$ and $p > 0.05 \text{ GeV}$
π_{soft}^0	$(\text{clusterReg} == 1 \text{ and } (\frac{\text{minC2Dist}}{40})^2 + (\frac{\text{clusterZernikeMVA}}{0.7})^2 > 1$ or $\text{clusterReg} == 2 \text{ and } (\frac{\text{minC2Dist}}{40})^2 + (\frac{\text{clusterZernikeMVA}}{0.4})^2 > 1$ or $\text{clusterReg} == 3 \text{ and } (\frac{\text{minC2Dist}}{85})^2 + (\frac{\text{clusterZernikeMVA}}{0.2})^2 > 1)$
ρ^+	$0.560 < M_{\pi^+\pi^0} < 1.060 \text{ GeV}$
leptons	
e	$\text{pidChargedBDTScore} > 0.9$ and $\text{brems_angleThreshold:0.2}$
μ	$\text{muonID_noSVD} > 0.9$
flavored hadrons	
K_s^0	$0.450 < M_{\pi^+\pi^+} < 0.550 \text{ GeV}$ and $\text{flightDistance} > 0$ and and $\log_{10}(\text{significanceOfDistance}) > 0$ and $\text{cosAngleBetweenMomentumAndVertexVector} > 0.8$
D^0	$1.78 < M_{D^0} < 1.90$
D^{*+}	$0.130 < M_{D^{*+}} - M_{D^0} < 0.170 \text{ GeV}$ and $p < 2.5 \text{ GeV}$
D^{*0}	$0.100 < M_{D^{*0}} - M_{D^0} < 0.170 \text{ GeV}$ and $p < 2.5 \text{ GeV}$
B^{sig}	non-failed vertex fit

Table 15: Selections for tracks and cluster considered in the ROE against the $\Upsilon(4S)$ candidate

tracks	$ dr < 2 \text{ cm}$ $ dz < 4 \text{ cm}$ $\text{thetaInCDCAcceptance} == 1$
clusters	$\text{clusterNHits} > 1.5$ $((\text{clusterReg}==1 \text{ and } E > 0.100 \text{ GeV}) \text{ or } (\text{clusterReg}==2 \text{ and } E > 0.060 \text{ GeV}) \text{ or } (\text{clusterReg}==3 \text{ and } E > 0.150 \text{ GeV}))$ $\text{clusterTheta} > 0.2967$ and $\text{clusterTheta} < 2.6180$ $\text{minC2TDist} > 50 \text{ cm}$

Table 16: $\mathcal{R}(4S)$ reconstructions modes and decays of interest

Reconstruction mode	Decay of interest
$B^{\text{tag}} D^{*-} \pi^+$	$B^0 \rightarrow D^{*-} \tau^+ (\pi^+ \nu_\tau) \nu_\tau$
$B^{\text{tag}} D^{*-} \pi^+ \pi^0$	$B^0 \rightarrow D^{*-} \tau^+ (\rho^+ \nu_\tau) \nu_\tau$
$B^{\text{tag}} D^{*-} e^+$	$B^0 \rightarrow D^{*-} e^+ \nu_e$
$B^{\text{tag}} D^{*-} \mu^+$	$B^0 \rightarrow D^{*-} \mu^+ \nu_\mu$
$B^{\text{tag}} D^{*0} \pi^+$	$B^+ \rightarrow D^{*0} \tau^+ (\pi^+ \nu_\tau) \nu_\tau$
$B^{\text{tag}} D^{*0} \pi^+ \pi^0$	$B^+ \rightarrow D^{*0} \tau^+ (\rho^+ \nu_\tau) \nu_\tau$
$B^{\text{tag}} D^{*0} e^+$	$B^+ \rightarrow D^{*0} e^+ \nu_e$
$B^{\text{tag}} D^{*0} \mu^+$	$B^+ \rightarrow D^{*0} \mu^+ \nu_\mu$

5 Offline processing

A series of operations are applied after the online reconstruction. These include truth matching of decays in simulated data, kinematic selections that define the signal and normalization-enhanced regions, as well as sideband regions used for calibrations and Data/MC checks. Additionally, best candidate selection is performed, and various corrections are applied to simulated data. In the following section, we describe each of these steps in detail.

5.1 Truth-matching

In the following section we describe the truth-matching strategy. We based our truth matching on information that we save from the `basf2` software and we later analyze offline. The truth matching is based on different modules which we describe below.

5.1.1 Btag and Bsig

We first try to identify the B^{tag} meson. In order to do that we employ the `mostcommonBTagIndex` variable of `basf2` [32]. This variable matches an `arrayIndex`, to a B^{tag} FEI candidate. The `arrayIndex` is an internal enumerator for the particles that are generated from the MC generator. For B mesons originating from the $\Upsilon(4S)$ resonance, the `arrayIndex` must be either 1 or 2. The `mostcommonBTagIndex` determines the `arrayIndex` by traversing down the tree of the B^{tag} FEI candidate. For every reconstructed object in the tree, `basf2` tries to match this reconstructed object with a generator level particle. Then `basf2` finds the `arrayIndex` of the B meson that the MC particle descends from. All these `arrayIndexes` are collected and stored. If no MC particle could be found, or if the MC particle does not descend from a B meson -1 is assigned. Out of this collection of `arrayIndexes` `basf2` determines the most common one and assigns it to the `mostcommonBTagIndex` variable. For the rest of the truth matching purposes we consider that this `arrayIndex` correspond to the true particle in the generator level. Knowing the `arrayIndex` of the matched MC particle to the FEI B^{tag} candidate enables us to determine the flavor of the B meson, through other `basf2` functionalities. Clearly for all continuum events the entry of the `mostcommonBTagIndex` will always be -1. However we do not base the identification of these events on this variable. On the contrary we trace back the type of input file that a event originates from. The categories of files are defined in Table 23. If the event originates from a `cckbar`, `ssbar`, `uubar` or `ddbar` input file, then we consider this event a continuum event without carrying out any further checks.

We base the rest of our truth matching on the lepton candidate or τ daughter candidate that originates from the signal B meson candidate. We traverse the decay tree of the reconstructed signal B meson, starting from the lepton candidate or τ daughter candidate and up to five generations until we find a B meson on the generator level. If the `arrayIndex` of this true B meson is the same as the one we determined from the `mostcommonBTagIndex`, or if no true B meson could be found we consider this signal B meson candidate as *combinatorial background*. No further checks are carried out for combinatorial background candidates. If the `arrayIndex` of the true B meson ancestor of the

the lepton candidate or τ daughter candidate is different than the `mostcommonBTagIndex`, then we spawn a second layer of truth matching, that tries to identify the true generated decay of the B signal candidate. This is described in the following.

5.1.2 Bsig decay category

We traverse the B signal candidate tree again until we find the B meson from the same starting point as before. We then save the `mcPDG` of the first six daughters of the true B meson. Based on the `mcPDG` codes of those daughters we make a first rough categorization of the decay. The categories are summarized in Table 17.

Table 17: Main decay truth matching categories. For D^* both charged and neutral mesons are considered. For D^* and D , τ , ℓ , ν , charge conjugation is taken into account. The definitions of the PDG codes that are used for D^{**} , H_c , H_c^{gap} , $X_{u/s}^{gap}$, X_s , and X_u are given in Appendix B.

ID	True decay	true B daughters					
		0	1	2	3	4	5
11	$B \rightarrow D^* \tau \nu$	D^*	τ	ν_τ	γ/NaN	γ/NaN	γ/NaN
12	$B \rightarrow D \tau \nu$	D	τ	ν_τ	γ/NaN	γ/NaN	γ/NaN
21	$B \rightarrow D^* \ell \nu$	D^*	ℓ	ν_ℓ	γ/NaN	γ/NaN	γ/NaN
22	$B \rightarrow D \ell \nu$	D	ℓ	ν_ℓ	γ/NaN	γ/NaN	γ/NaN
31	$B \rightarrow D^{**} \ell \nu$	D^{**}	ℓ/τ	ν_ℓ/ν_τ	γ/NaN	γ/NaN	-
32	$B \rightarrow D^{**} \ell \nu$	H_c^{gap}	ℓ/τ	ν_ℓ/ν_τ	γ/NaN	γ/NaN	-
33	$B \rightarrow D^{**} \ell \nu$	H_c^{gap}	$X_{u/s}^{gap}$	$X_{u/s}^{gap}$	ℓ/τ	ν_ℓ/ν_τ	γ/NaN
44	$B \rightarrow H_c H_c$	H_c	H_c	NaN	NaN	NaN	-
443	$B \rightarrow H_c H_c X_s$	H_c	H_c	X_s	NaN	NaN	-
500	$B \rightarrow D^* + n \cdot h_u$	D^*	$X_u/\gamma/\text{NaN}$	$X_u/\gamma/\text{NaN}$	$X_u/\gamma/\text{NaN}$	$X_u/\gamma/\text{NaN}$	$X_u/\gamma/\text{NaN}$

5.1.3 D^*

We do not truth match the decay of the D^* , if this is found in the previous step of the truth matching. The reason is that we don't really care about the reconstruction of the D^* , as long as the D^* and the D candidates can be matched to a MC true particle and they both fall into the mass windows that we define in the reconstruction. Our reasoning here is that it is very likely that in the D^* reconstruction the slow pion is misreconstructed, or in the D meson one of the daughters was missed, or a wrong particles was added. Cross-feeds between the different 8 D modes that we reconstruct are of course possible. For the reasons above we still accept such events as signal, without requiring that the D^* or the D mesons are correctly reconstructed.

5.1.4 τ

If we find a τ particle as a daughter of the B meson i.e. if the truth matched signal B meson falls into categories 11 or 12 as defined in Table 17, we then need to truth match

the τ decay. We have observed an issue with the order of the generator level particles in τ decays. Event if the decay grammar instructs that the MC particles should be generated in mass descending order, a lot of times we truth match τ leptons whose daughters follow a random ordering. Another issue is that the decay $\tau \rightarrow \rho\nu_\tau$ is never truth matched. A ρ^+ meson is never found as a descendant of the τ lepton, but the resonance is skipped in favor of the daughters of the ρ^+ meson. As example we would expect to truth match a $\tau \rightarrow \rho\nu_\tau$ but on the contrary we find ourselves truth matching $\tau \rightarrow \pi^+\pi^0\nu_\tau$. Furthermore we have found events where the truth matched τ decay is $\tau \rightarrow \nu_\tau\pi^+\pi^0$ or any other possible permutation of the τ daughters. The last element that adds an extra complexity to the τ truth matching is the presence of radiative photons. Strangely enough the order of those appears to be random in many cases in the Belle II MC. For example a $\tau \rightarrow \nu_\tau\pi^+\gamma\pi^0$ could also be truth matched. We presume that such pathological cases arise from bugs of the TAUOLA MC generator. In order to face these unpredictable complications we have developed a dedicated truth matching algorithm for τ decays that considers all possible permutations of the τ daughters.

During the WG review we found that such problems were more prominent in previous MC campaigns and are not relevant for MC15rd anymore. These problems might have been solved by changes in the truth matching modules of **basf2** or modification of the Nevertheless we still use the same algorithm that we had developed since it's robust against correctly ordered and misordered τ daughters.

We always consider the first 5 daughters of the τ decay in 15 distinct modes. There are presented in Table 18. In order to account for photons coming from PHOTOS we pad the target queries for the τ daughters up to 5 daughters, with NaNs and photons. All possible combinations of NaNs and photons are considered as well as all possible permutations in conjunction with the true daughters of every decay mode.

Table 18: τ decay truth matching categories. The order of the truth matched daughters is not important as all possible permutation are considered.

ID	True decay	Daughters
1	$\tau \rightarrow \pi^+\nu_\tau$	$[\pi^+, \nu_\tau]$
2	$\tau \rightarrow \pi^+\pi^0\nu_\tau$	$[\pi^+, \pi^0, \nu_\tau]$
3	$\tau \rightarrow \pi^+\pi^0\pi^0\nu_\tau$	$[\pi^+, \pi^0, \pi^0, \nu_\tau]$
4	$\tau \rightarrow \pi^+\pi^+\pi^-\nu_\tau$	$[\pi^+, \pi^+, \pi^-, \nu_\tau]$
5	$\tau \rightarrow \mu\nu_\tau\nu_\mu$	$[\mu, \nu_\tau, \nu_\mu]$
6	$\tau \rightarrow e\nu_\tau\nu_e$	$[e, \nu_\tau, \nu_e]$
7	$\tau \rightarrow K^+\nu_\tau$	$[K^+, \nu_\tau]$
8	$\tau \rightarrow \pi^+K^0K^0\nu_\tau$	$[\pi^+, K^0, K^0, \nu_\tau]$
9	$\tau \rightarrow \pi^+\pi^0\pi^0\pi^0\nu_\tau$	$[\pi^+, \pi^0, \pi^0, \pi^0, \nu_\tau]$
10	$\tau \rightarrow \pi^+K^{*0}\nu_\tau$	$[\pi^+, K^{*0}, \nu_\tau]$
11	$\tau \rightarrow \pi^+\pi^+\pi^-\pi^0\nu_\tau$	$[\pi^+, \pi^+, \pi^-, \pi^0, \nu_\tau]$
12	$\tau \rightarrow K^+\pi^+\pi^-\nu_\tau$	$[K^+, \pi^+, \pi^-, \nu_\tau]$
13	$\tau \rightarrow \pi^+K^0\nu_\tau$	$[\pi^+, K^0, \nu_\tau]$
14	$\tau \rightarrow K^{*+}\nu_\tau$	$[K^{*+}, \nu_\tau]$
15	$\tau \rightarrow \pi^+K^0\pi^0\nu_\tau$	$[\pi^+, K^0, \pi^0, \nu_\tau]$

5.1.5 Hadronic decays

In case a hadronic decay is truth matched in the first step i.e. categories 44, 443 or 500 in Table 17 then we need to further identify the decay. This is important as some of the hadronic decays need to be calibrated or their BF need to be corrected. Since there is large number of possible hadronic backgrounds it becomes impractical to hard code these decays in the same way we do for the τ decays. To tackle this problem we adopt a different approach. We truth match all the daughters of the B meson and we create string with the absolute values of the PDG code of the B meson itself and the ones of its daughters. Before we create the string, we sort the absolute values of the PDG codes of the daughters in descending order. This makes our method robust against misorderings like the ones that we described for the τ truth matching. It is also true for fully hadronic decays that we often truth match the same decay in different orderings of the B meson daughters. Empirically we report that is true for decays that include modes that decay through complicated resonances. This approach allows us to distinguish all the possible hadronic decays that mimic our signal.

5.1.6 Final truth matched categories

We combine information from all the above truth matching queries to create final MC categories. Table 19 summarizes the final truth matching categories that are used for visualization purposes throughout the analysis. All the stacked histogram plots of simulated and experimental data in this document use the categories as defined in Table 19 unless stated otherwise.

Table 19: Final truth matching categories. The entries of the column B^{sig} mcID are described in Table 17. The entries of the column τ mcID are described in Table 18.

MC category	B^{sig} mcID	τ mcID	wrong B meson	input file
$B \rightarrow D^* \tau \nu$ ($\tau \rightarrow \pi \nu_\tau$)	11	1	False	mixed/charged/gap
$B \rightarrow D^* \tau \nu$ ($\tau \rightarrow \rho \nu_\tau$)	11	2	False	mixed/charged/gap
$B \rightarrow D^* \ell \nu$	21	-	False	mixed/charged/gap
$B \rightarrow D^{**} \ell \nu$	31, 32, 33	-	False	mixed/charged/gap
$B \rightarrow H_c H_c X_s$	44, 443	-	False	mixed/charged/gap
$B \rightarrow D^* + n \cdot h_u$	500	-	False	mixed/charged/gap
Other $B\bar{B}$	-1, 12, 22	-	False	mixed/charged/gap
τ missID	11	not 1 and not 2	False	mixed/charged/gap
Combinatorial	-	-	True	mixed/charged/gap
Continuum	-	-	-	ccbar/ssbar/uubar/ddbar

5.2 Helicity angle calculation

As mentioned in 1 one of the main objectives of this analysis is to measure P_τ along $R(D^*)$. P_τ can be defined through the differential decay rate as

$$\frac{1}{\Gamma} \frac{d\Gamma}{d(\cos \theta_{hel})} = \frac{1}{2} [1 + \alpha P_\tau \cos \theta_{hel}] \quad (6)$$

where $\cos \theta_{hel}$ is the helicity angle of the τ lepton and α is a coefficient expressing the sensitivity of the different τ decays to P_τ .

We aim at measuring the polarization of τ lepton as an asymmetry between the forward and backward produced τ leptons in the helicity plane of the τ rest frame. Even though due to the missing neutrino in the event the helicity angle cannot be fully determined, equivalent kinematic information can be retrieved by using the $\tau\nu_\tau$ rest frame. The method for determining this kinematic information has been developed as part of the first Belle analysis [4]. however in this section we reiterate the method for completion.

We begin by boosting the laboratory frame along with the three momentum component of the momentum transfer vector.

$$q = p_{e^+e^-} - p_{tag} - p_{D^*} \quad (7)$$

where p denotes the four-momentum of the e^+e^- beam, the B^{tag} and the D^* respectively. Now in this frame one can extract the momentum and energy of the τ lepton as

$$E_\tau = \frac{q^2 + m_\tau^2}{2\sqrt{q^2}} \quad (8)$$

$$p_\tau = \frac{q^2 - m_\tau^2}{2\sqrt{q^2}} \quad (9)$$

where m_τ is the mass of the τ lepton. $\cos \theta_d$ is the angle between the the τ lepton and the charged hadron daughter of the τ in its two body decay and can be computed as

$$\cos \theta_d = \frac{2E_\tau E_d - m_\tau^2 - m_d^2}{2|\vec{p}_\tau||\vec{p}_d|} \quad (10)$$

The true momentum of the charged hadron in the τ rest frame can be calculated as $p_d = \frac{m_\tau^2 - m_d^2}{2m_\tau}$.

Finally by boosting this frame to the $\tau\nu_\tau$ rest frame we obtain

$$|\vec{p}_d^*| \cos \theta_{hel} = -\gamma|\vec{\beta}|E_d + \gamma|\vec{p}_d| \cos \theta_d \quad (11)$$

with $\gamma = \frac{E_\tau}{m_\tau}$ and $\beta = \frac{|\vec{p}_\tau|}{E_\tau}$

Now by integrating 6 in $[0,1]$ and $[-1, 0]$, we define P_τ as the asymmetry of forward-backward events in the helicity plane.

$$P_\tau = \frac{2}{\alpha} \frac{N_F - N_B}{N_F + N_B} \quad (12)$$

where N_F and N_B are events with positive and negative helicity angle respectively. For $\tau \rightarrow \pi\nu_\tau$ the coefficient α is 1 while for $\tau \rightarrow \rho\nu_\tau$ is 0.45 [65].

5.3 Definition of Signal Region

After truth matching the events, we apply a set of tighter selections on the events that survived the `basf2` reconstruction. Such an approach enables the definitions of sidebands and the conduction of optimization tests without rerunning the `basf2` reconstruction which is computationally intensive. In Table 20 we present the set of tight selections that we apply. For every selection, we save a sample with the respective anti-cut as well in a different file. That way we ensure that orthogonal samples can be used for sideband checks. This set of selections defines what we call Signal enhanced Region (SR) where we expect most of $B \rightarrow D^* \tau \nu$ events and Normalization enhanced Region (NR) where we expect most of $B \rightarrow D^* \ell \nu$ events in the rest of the document.

Table 20: List of final tight cuts applied in the sample.

cut	variable definition	motivation	region
<code>nROE.Charged < 1</code>	charged tracks present in the ROE of $\Upsilon(4S)$	Apply the completeness constraint	SR, NR
<code>roeEextra < 1.25 GeV</code>	total energy in the ECL attributed to the ROE of $\Upsilon(4S)$	Signal peaking close to zero as no missing particles are expected	SR, NR
$< 0.1413 \Delta M(D^{*\pm}) < 0.154 \text{ GeV}$ $< 0.135 \Delta M(D^{*0}) < 0.149 \text{ GeV}$	$M(D^{*\pm}) - M(D^0)$ $M(D^{*0}) - M(D^0)$	Tighten D^* mass window	SR, NR
$q^2 > 4 \text{ GeV}^2$	$q^2 = (\mathbf{p}_{beam} - \mathbf{p}_{Btag} - \mathbf{p}_{D^*})^2$	kinematic constraint from τ mass	SR, NR
<code>Pleading FSP > 0.5 GeV</code>	lepton momentum or charged hadron momentum	suppress fakes	SR, NR
$-2 < \cos \theta_{hel}^\tau < 2.2$ in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ $-1.2 < \cos \theta_{hel}^\tau < 3$ in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ $-2 < \cos \theta_{hel}^\tau < 2.2$ in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ $-1.2 < \cos \theta_{hel}^\tau < 3$ in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	$\frac{-\gamma \beta E_d + \gamma p_d \cos \theta_{\tau d}}{p_d^{rest}}$	cosine of τ helicity angle that minimizes the uncertainty of P_τ . Optimization procedure outlined in Appendix H	SR
$-1 < m_{miss}^2 < 7 \text{ GeV}^2$ in $B \rightarrow D^*(D^0 \pi) h^+$	Squared recoil mass of the signal side	identify signal events that peak in higher values	SR
$-1 < m_{miss}^2 < 2 \text{ GeV}^2$ in $B \rightarrow D^*(D^0 \pi) \ell^+$	Squared recoil mass of the signal side	identify normalization events that peak around zero	NR
$1.84 < M_{D^0} < 1.89 \text{ GeV}$	$M(D^0)$	Tighten D^0 mass window	SR, NR
$0.590 < M_{\rho^+} < 1.050 \text{ GeV}$	$M(\rho^+)$	Tighten ρ^+ mass window	SR

Figures 7, 8 and 9 present the impact of every offline selection in the $B^0 \rightarrow D^* \tau \nu$, $B^+ \rightarrow D^* \tau \nu$, $B^0 \rightarrow D^* \ell \nu$ and $B^+ \rightarrow D^* \ell \nu$ templates in the signal reconstruction channels in terms of number of candidates that survive, sample efficiency and sample purity. It clear that all selections have the desired impact since the purity is always increasing especially for the templates of interest if one ignores cross-feeds between the channels.

Figures 10, 11 and 12 present the impact of every offline selection in the $B^0 \rightarrow D^* \tau \nu$, $B^+ \rightarrow D^* \tau \nu$, $B^0 \rightarrow D^* \ell \nu$ and $B^+ \rightarrow D^* \ell \nu$ templates in the normalization reconstruction channels in terms of number of candidates that survive, sample efficiency and sample purity. It clear that all selections have the desired impact since the purity is always increasing especially for the templates of interest if one ignores cross-feeds between the channels.

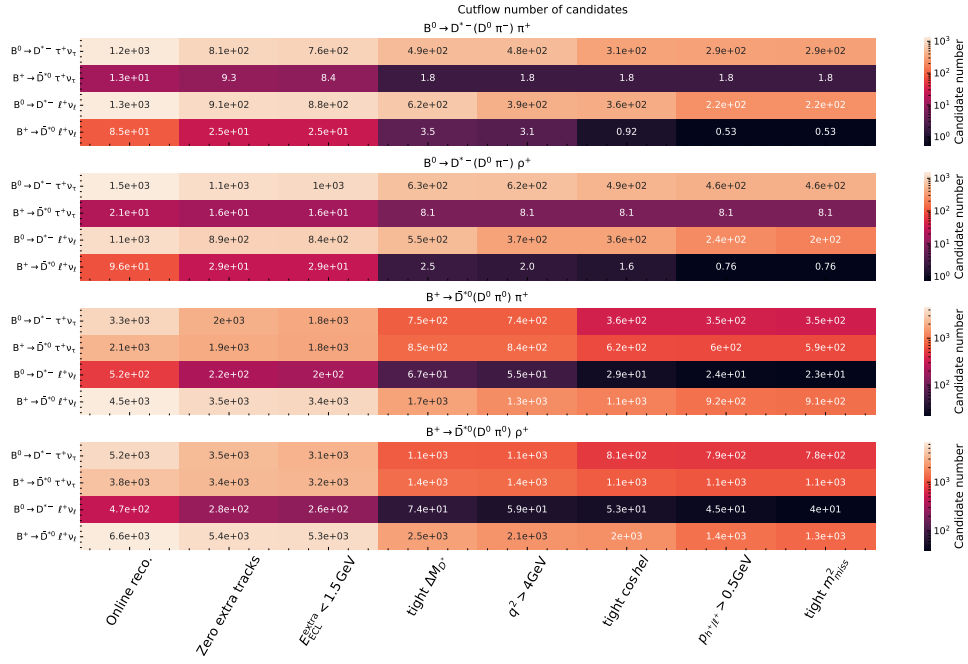


Figure 7: Total number of events that survive after every offline selection for the most important templates in the signal reconstruction channels.



Figure 8: Efficiency after every offline selection for the most important templates in the signal reconstruction channels.

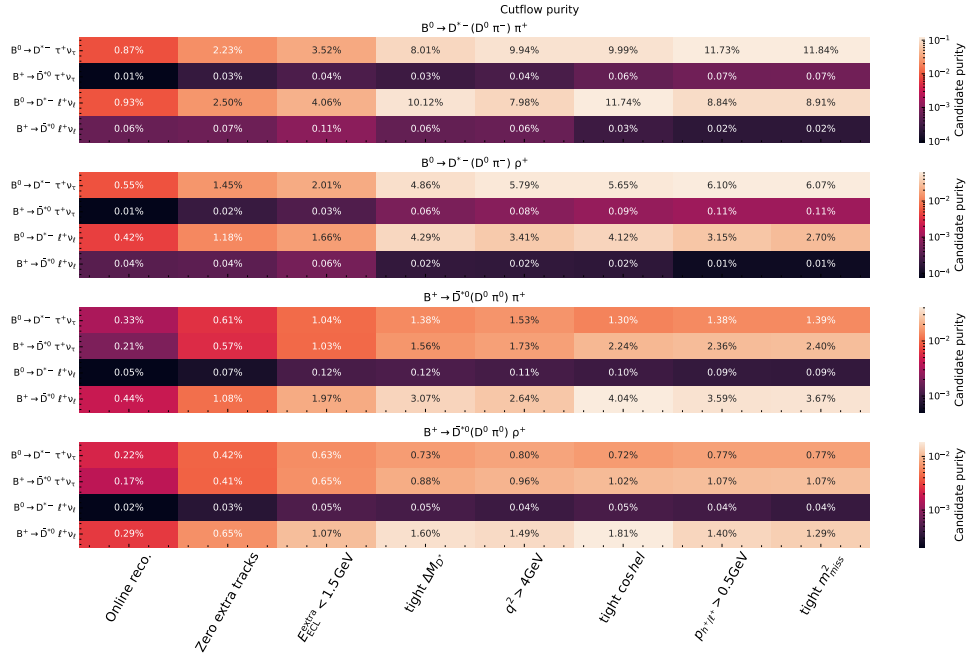


Figure 9: Purity after every offline selection for the most important templates in the signal reconstruction channels.

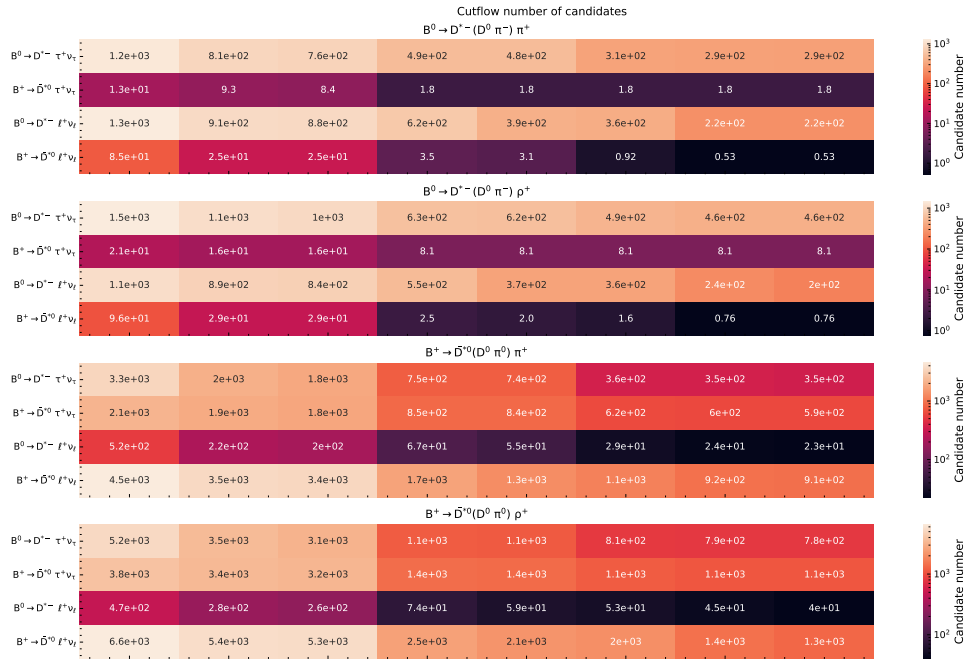


Figure 10: Total number of events that survive after every offline selection for the most important templates in the normalization reconstruction channels.

5.4 Best Candidate Selection

The sample we're left with after all selections still contains multiple $\mathcal{T}(4S)$ candidates. We have selected only one B^{tag} candidate in the online reconstruction already as mentioned

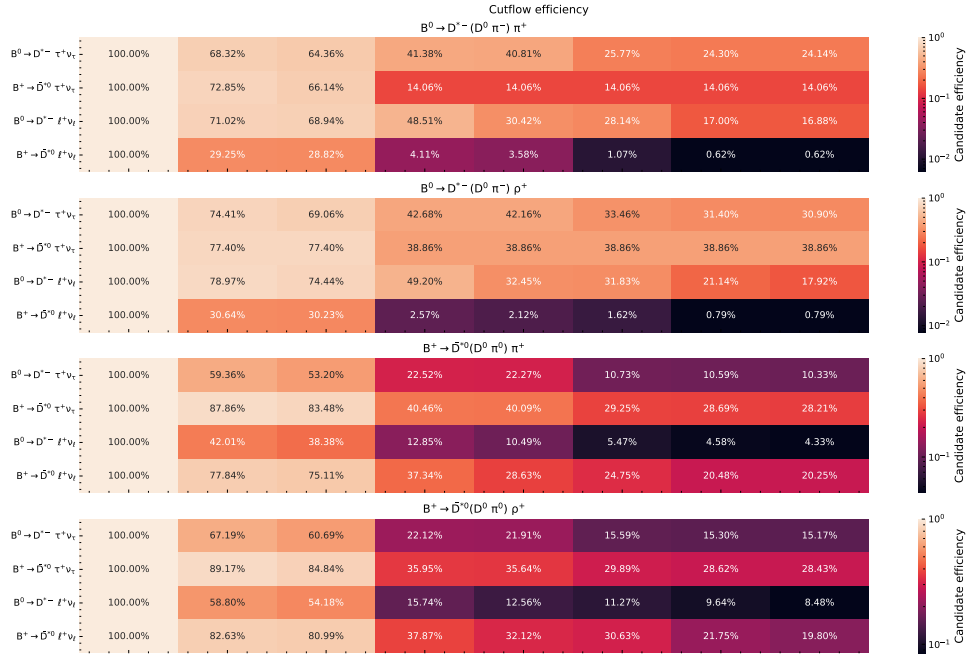


Figure 11: Efficiency after every offline selection for the most important templates in the normalization reconstruction channels.

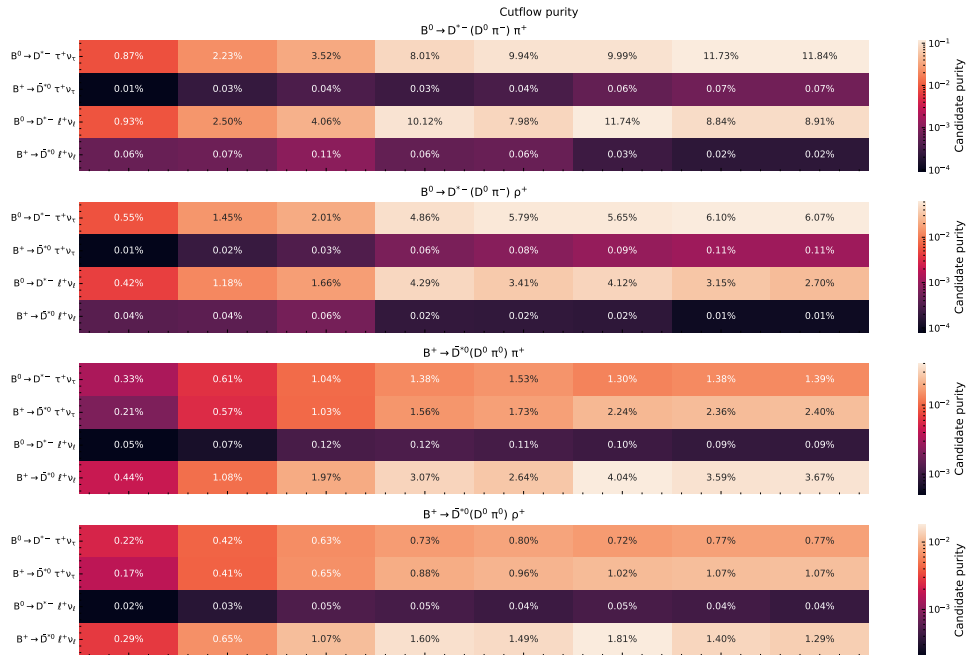


Figure 12: Purity after every offline selection for the most important templates in the normalization reconstruction channels.

in 4.4. Therefore the multiple candidates are arising from duplicate D^0 candidates, π_s candidates or multiple ℓ , π^+ and ρ^+ candidates. We have developed a multi-step BCS that is tackling this problem.

In order to identify duplicate $\Upsilon(4S)$ candidates we create a unique eventID based on `--experiment--`, `--run--`, `--event--` and `--production--` for simulated data. For experimental data the `--production--` is redundant. The reasoning behind defining the uniqueness of an event is described very nicely in this Belle II internal forum [66]. We proceed with theBCS as follows:

1. Drop all duplicate D^0 candidates by keeping the one whose reconstructed invariant mass is closer to the theoretical one, namely $1,864.84 \pm 0.05$ GeV.
2. Drop all duplicate π_s candidates
 - For $\pi_s^\pm$ we keep the candidate with the highest momentum
 - For π_s^0 we keep the candidate whose reconstructed invariant mass is closer to the theoretical one, namely 0.1349768 GeV.
3. For the reconstruction of $\tau \rightarrow \rho\nu_\tau$ modes, if multiple ρ^+ candidates are reconstructed, we select the candidate whose π^0 daughter satisfies the same criteria as those applied in the case of π_s^0 .
4. If duplicate B meson candidates are found between two signal reconstruction modes, we keep the one that has been reconstructed in the $\tau \rightarrow \rho\nu_\tau$ mode instead of the $\tau \rightarrow \pi\nu_\tau$ mode as the former one has almost 2.5 times larger BF than the latter. This is also illustrated in Figure 6.
5. For the reconstruction modes with the extra pion for the calibration of $B \rightarrow D^{**}\ell\nu$ decays we drop duplicate pion candidates using the same criteria we listed for π_s candidates.
6. If there are still multiple candidates in the sample, we pick one at random

Tables 21 and 22 shows the effect of the series of BCSs in terms of number of candidates and signal and normalization efficiencies for charged and neutral B mesons for the different reconstruction channels. In the end if we find multiple candidates having been reconstructed in two types of reconstruction channels e.g. signal and normalization, then we keep one candidate at random.

Table 21: BCS, number of candidates and event efficiencies for the signal reconstruction channels. Results are not scaled to data luminosity.

BCS criterion	$B^0 \rightarrow D^{*-}(D^0 \pi^-) h^+$			$B^+ \rightarrow D^{*0}(D^0 \pi^0) h^+$		
	# cand.	multipl.	$B \rightarrow D^* \tau \nu$ eff.	# cand.	multipl.	$B \rightarrow D^* \tau \nu$ eff.
After offline selections	17,563	2.855	100% (781)	155,990	4.885	100% (1502)
Best D^0 meson	13,967	2.27	100%	109,621	3.433	99.9%
Best π_s	13,861	2.253	100%	78,449	2.457	99.9%
Best π^0 from ρ^+	8,526	1.386	100%	44,835	1.404	99.9%
Favor $\tau \rightarrow \rho\nu_\tau$ over $\tau \rightarrow \pi\nu_\tau$	6,178	1.004	100%	32,160	1.007	99.9%
Random candidate	6,152	1.0	100%	31,933	1.0	99.9%

Table 22: BCS, number of candidates and event efficiencies the normalization reconstruction channels. Results are not scaled to data luminosity.

BCS criterion	$B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$			$B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$		
	# cand.	multipl.	$B \rightarrow D^* \ell \nu$ eff.	# cand.	multipl.	$B \rightarrow D^* \ell \nu$ eff.
After offline selections	18,192	1.074	100% (14,996)	62633	1.576	33390
Best D^0 meson	16,970	1.002	100%	53,862	1.355	100%
Best π_s	16,952	1.0	100%	39,816	1.002	100%
Random candidate	16,944	1.0	100%	39,745	1.0	100%

5.5 Corrections from the Performance group

Here we describe the corrections that we use based on the recommendations of the Performance group.

5.5.1 Luminosity scaling

Due to the different luminosity assumed in the MC production we apply the scaling factors mentioned in Table 23 to the the respective samples.

5.5.2 Tracking efficiency

For the medium to high momentum tracks i.e. > 0.2 GeV, we assign a flat systematic uncertainty on the track finding efficiency as per the Tracking group's recommendation [67]. No correction is applied on simulated data as the Data/MC ratio is expected to be one. We calculate the number of tracks on the signal side and assign a 0.27% uncertainty per track. Therefore the total uncertainty that is assigned per event is defined by Equation

$$\sigma_{track\,eff.} = 1.0027^{N_{tracks}^{B^{sig}}} - 1 \quad (13)$$

We do not take into account the number of tracks on the tag side, as these will be taken into account in the FEI calibration.

5.5.3 Tracking momentum scale

We scale the momentum of track on experimental data using the `ma.scaleTrackMomenta` helper function of `basf2`. The arguments used here are `tracking_scale_factor_payload: tracking_momentumScaleFactor_global` and `tracking_scale_factor_name: sf_global_central` as per the Performance group recommendation [68].

5.5.4 Photon energy bias

We account for the energy bias in photons on experimental data using the `ma.correctEnergyBias` helper function of `basf2`. We use the `PhotonEnergyBiasCorrection_MC15rd_June2023` option for the photon energy bias table [68].

Table 23: Luminosity and scaling factors for different samples simulated data. The assumed $\mathcal{L}$ for experimental data for the LS1 dataset is $365.29 \pm 1.70 \text{ fb}^{-1}$

Sample	$\mathcal{L} \text{ (fb}^{-1}\text{) in MC}$	scaling factor
mixed		
charged		
ccbar	1,443.999	0.25045
ssbar		
uubar		
ddbar		
$B^0 \rightarrow D'_1(D\pi\pi)\ell^-\bar{\nu}_\ell$	11,204.48	0.0326
$B^0 \rightarrow D'_1(D^*\pi\pi)\ell^-\bar{\nu}_\ell$	3,921.57	0.0931
$B^0 \rightarrow D^{*0}(D\pi\pi)\ell^-\bar{\nu}_\ell$	11,204.48	0.0326
$B^0 \rightarrow D^{*0}(D^*\pi\pi)\ell^-\bar{\nu}_\ell$	3,921.57	0.0931
$B^0 \rightarrow D'_1(D^*\eta)\ell^-\bar{\nu}_\ell$	456.0	0.8011
$B^0 \rightarrow D^{*0}(D\eta)\ell^-\bar{\nu}_\ell$	456.0	0.8011
$B^+ \rightarrow D'_1(D\pi\pi)\ell^-\bar{\nu}_\ell$	10,582.01	0.0345
$B^+ \rightarrow D'_1(D^*\pi\pi)\ell^-\bar{\nu}_\ell$	3,367.0	0.1085
$B^+ \rightarrow D^{*0}(D\pi\pi)\ell^-\bar{\nu}_\ell$	10,582.01	0.0345
$B^+ \rightarrow D^{*0}(D^*\pi\pi)\ell^-\bar{\nu}_\ell$	3,367.0	0.1085
$B^+ \rightarrow D'_1(D^*\eta)\ell^-\bar{\nu}_\ell$	411.52	0.8877
$B^+ \rightarrow D^{*0}(D\eta)\ell^-\bar{\nu}_\ell$	411.52	0.8877
$B^0 \rightarrow D'_1(D\pi\pi)\tau^-\bar{\nu}_\tau$	7,305.0	0.05
$B^0 \rightarrow D^{*0}(D^*\pi\pi)\tau^-\bar{\nu}_\tau$	2,435.0	0.15
$B^0 \rightarrow D'_1(D^*\eta)\tau^-\bar{\nu}_\tau$	2,435.0	0.15
$B^0 \rightarrow D^{*0}(D\eta)\tau^-\bar{\nu}_\tau$	7,305.0	0.05
$B^+ \rightarrow D'_1(D\pi\pi)\tau^-\bar{\nu}_\tau$	6,663.33	0.0548
$B^+ \rightarrow D^{*0}(D^*\pi\pi)\tau^-\bar{\nu}_\tau$	2,221.11	0.1645
$B^+ \rightarrow D'_1(D^*\eta)\tau^-\bar{\nu}_\tau$	2,221.11	0.1645
$B^+ \rightarrow D^{*0}(D\eta)\tau^-\bar{\nu}_\tau$	6,663.33	0.0548

5.5.5 Photon efficiency

We do not correct the photon finding efficiency on simulated data for photons that are used to reconstruct π^0 candidates. Discussions among members of the the S(L) working group lead to the conclusion that these are accounted in the π^0 efficiency correction.

5.5.6 π^0 efficiency

We follow the Neutrals group recommendations and we apply a momentum and $\cos\theta$ dependent efficiency corrections for π^0 . The corrections and their uncertainties are provided in [69]

5.5.7 π_s efficiency

We follow the Tracking and Neutrals groups recommendations and we apply a momentum dependent efficiency correction for π_s . Table 24 shows the correction central values and the corresponding uncertainties as determined by the Tracking group [68].

Table 24: Correction tables for $\pi_s^\pm$ efficiency corrections derived by the Tracking group.

$\pi_s^\pm$ p (GeV)	value	stat. unc. correlated	stat. unc. uncorrelated	syst. unc.
0.05-0.12	0.947	0.011	0.016	0.0027
0.12-0.16	0.985	0.011	0.013	0.0028
0.16-0.20	0.983	0.011	0.015	0.0028

For π_s^0 we apply corrections as describe in Table 25. Following private communication with the expert who derived the corrections we multiply by 0.818 in order to take the absolute corrections, as the ones mentioned on [68] are relative to the bin 0.2 - 0.4 GeV for the cancellation of systematics.

Table 25: Correction tables for π_s^0 efficiency corrections derived by the Neutrals group.

π_s^0 p (GeV)	value	stat. unc.	syst. unc.
0.05-0.10	0.991	0.064	0.032
0.10-0.15	0.877	0.036	0.018
0.15-0.20	0.871	0.038	0.038

5.5.8 PID (with the systematics framework)

We correct the lepton ID efficiency and fakes based on the recommendations of the LID group. For the efficiency corrections we use the centrally produced tables by the LID group under the version `v0_coarse` (which is just a symbolic link to `v1` on `kekcc`) [70]. These corrections are summarized in Figures 13 and 14. For these and all figures that follow empty bins imply that control samples were not available in that particular kinematic region or that the statistics of the control samples were that low that the derived correction weight was negative, therefore no correction is applied.

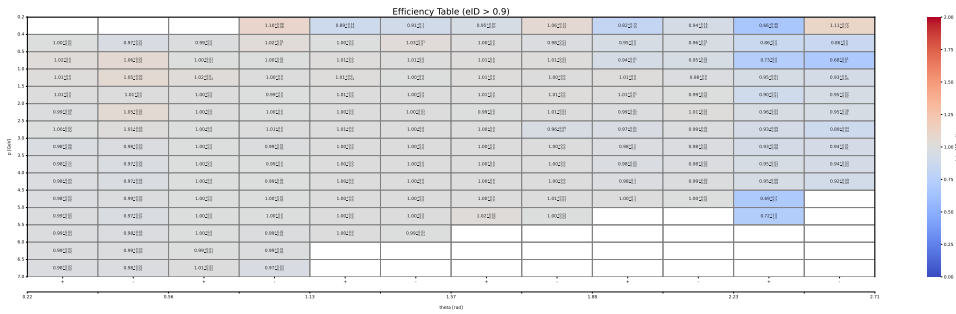


Figure 13: eID efficiency tables.

For lepton fakes and hadron efficiencies and fake we generate corrections tables for using the Systematics Framework for PID [71]. We align the selections with for leptons and charged hadrons to the ones that are outlined in 4. We derive different corrections

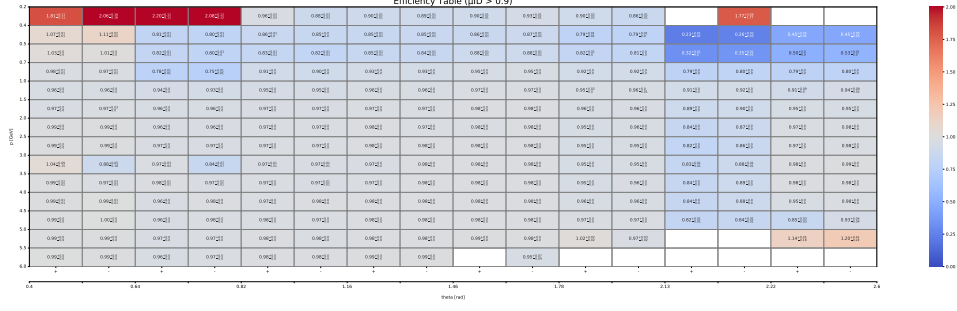


Figure 14: μ ID efficiency tables.

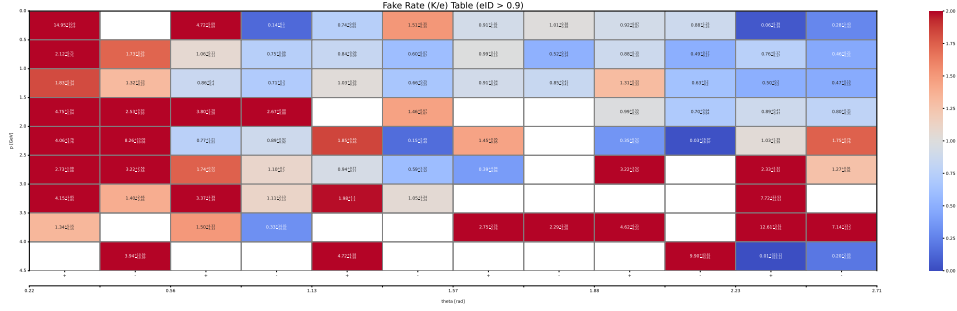


Figure 15: eID fake rate tables where a kaon is faking an electron.

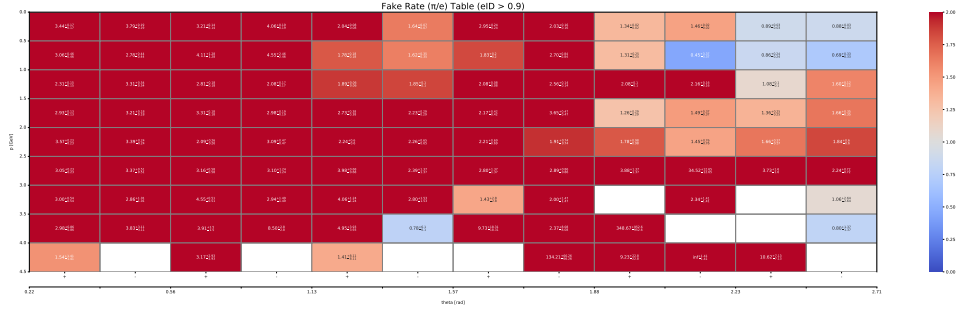


Figure 16: eID fake rate tables where a pion is faking an electron.

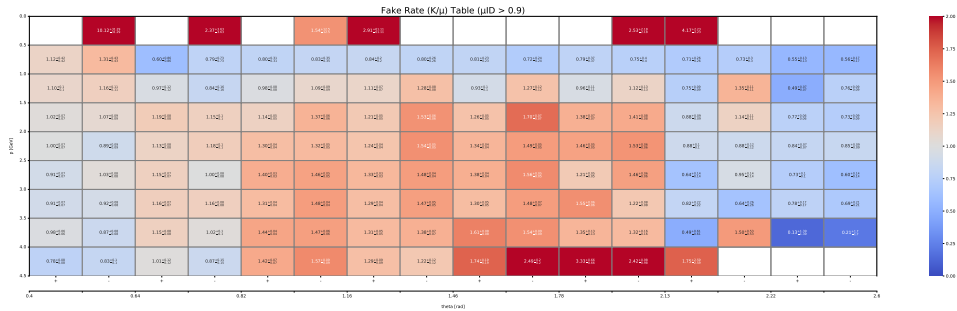


Figure 17: μ ID fake rate tables where a kaon is faking a muon.

791 tables for kaons and for pions faking the reconstructed lepton. We present the correction
 792 for lepton fakes in Figures 15, 16, 17 and 18,

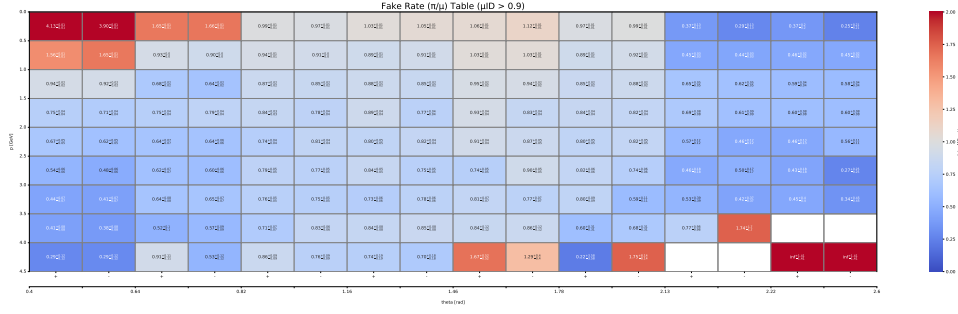


Figure 18: μ ID fake rate tables where a pion is faking a muon.

Figures 19, 20, 21 and 22 illustrate the HID Data/MC ratios that we derive using the Systematics Framework in bins of momentum and $\cos\theta$ of the track for correctly reconstructed and misreconstructed hadrons.

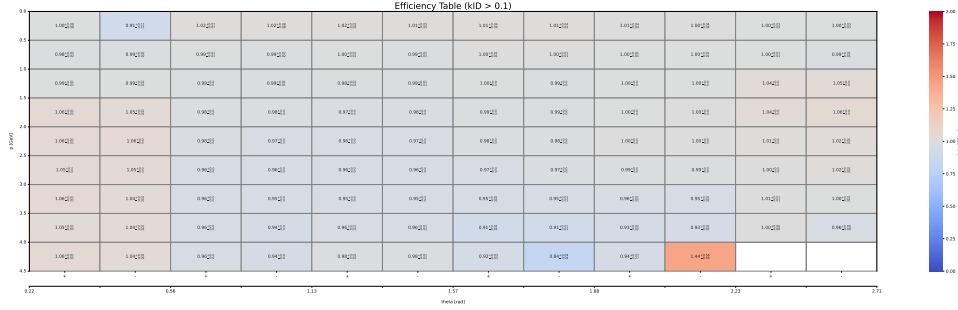


Figure 19: kaonID efficiency tables

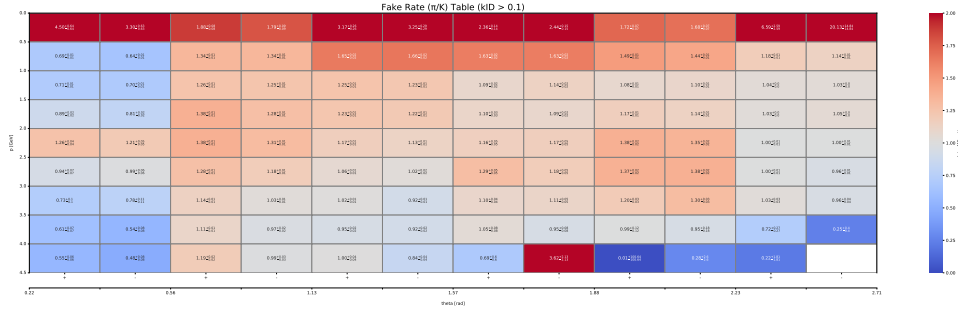


Figure 20: kaonID fake rate tables where a pion is faking a kaon

A subtle but important point is that, since we recover the bremsstrahlung (Brems) momentum for electrons as described in Section 4.5.2, we use the electron momentum before Brems recovery to determine the kinematic bin in which an event falls. This follows the LID recommendations [72], as all electron efficiency corrections in the Systematics Framework are derived without applying any Brems recovery. Accessing the pre-Brems momentum in **basf2** is straightforward. When Brems recovery is performed using the dedicated **basf2** modules, the electron object before Brems recovery is stored as a daughter

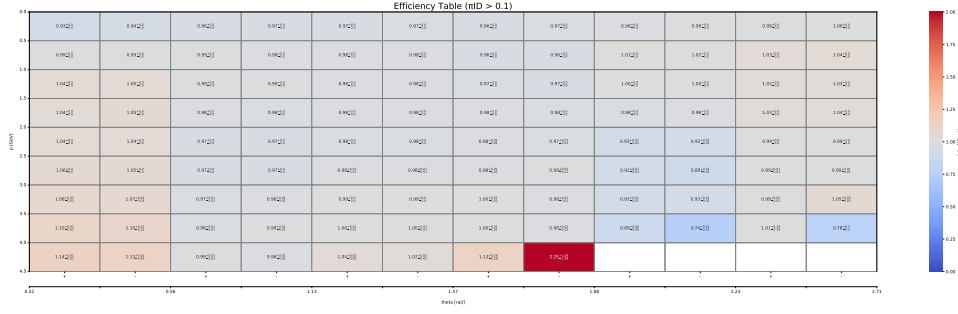


Figure 21: π ID efficiency tables

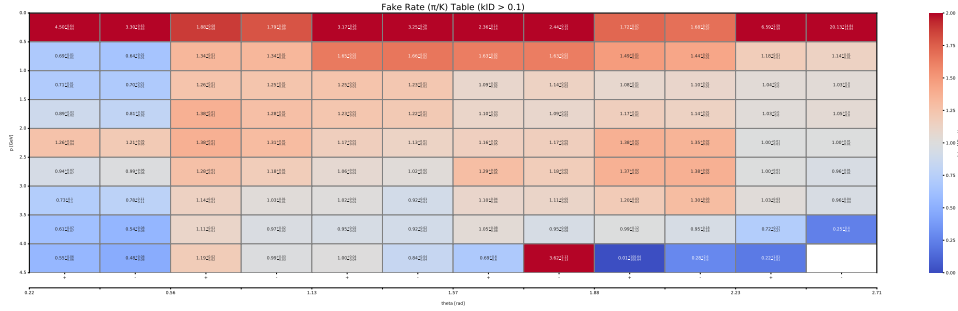


Figure 22: π ID fake rate tables where a kaon is faking a pion.

of the post-Brems electron. This behaviour is also explained clearly in the corresponding Belle II discussion [73]. The considerations above apply to both the eID efficiency corrections and the fake-rate corrections.

5.5.9 PID (custom corrections)

In this analysis, a significant fraction of muons are observed to be misidentified as pions. This component plays a crucial role in our signal extraction, particularly for $R(D^*)$, since the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ and $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ are especially contaminated by $B \rightarrow D^* \tau \nu$ events where the τ lepton decays $\tau \rightarrow \mu \nu_\tau \nu_\mu$. Such events have kinematic properties very similar to $B \rightarrow D^* \tau \nu$ events where the τ lepton decays as $\tau \rightarrow \pi \nu_\tau$, with the only differences being that the muon is misidentified as a pion and an additional neutrino is present in the event. Consequently, these events can be easily mistaken for the signal in the extraction fit, as their shapes closely resemble the correctly reconstructed signal component.

Given their significant impact on the $R(D^*)$ measurement, a reliable calibration of the muon-to-pion fake rate efficiency is essential. Unfortunately, the current Systematics Framework for PID [71] does not support corrections for muons faking pions in MC15rd. After consultation with PID experts, dedicated efficiency correction tables were derived specifically for this analysis. The samples used to derive these corrections are described in [74]. A clean sample of muons is obtained by reconstructing the process $e^+e^- \rightarrow e^+e^-\mu^+\mu^-$. The muons are reconstructed using a tag-and-probe method. The probe muon selection is chosen to match our pion reconstruction criteria, as outlined in Section

4.5.3. Although [74] details the reconstruction and efficiency correction derivation for MC13a and proc11, the methods and code used for and `proc13 + prompt` are identical; the only difference is that the code is run on different samples. The code used for these corrections is available in the GitLab repository [75].

The resulting corrections are shown in Figure 23. The specified momentum and angular ranges cover the majority of the relevant sample. Only 1.43 out of the 37.98 expected $B \rightarrow D^* \tau \nu$ events where the τ decays as $\tau \rightarrow \mu \nu_\tau \nu_\mu$ remain uncorrected, as they fall within the range $0 < \theta < 0.4$, for which corrections are not available from the $e^+e^- \rightarrow e^+e^-\mu^+\mu^-$ sample. This fraction is considered negligible, and no further corrections are applied to these events.

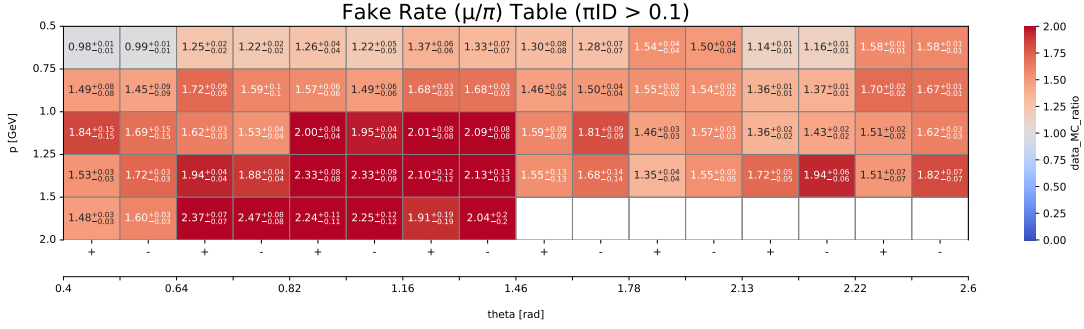


Figure 23: πID fake rate tables where a muon is faking a pion.

5.6 Branching Fraction corrections

We update the BF's of several reconstructed decays to align with the latest values from PDGLive. Each decay is identified using the truth matching strategy outlined in Section 5.1 All the assumed values in the Belle II MC are taken from the `DECAY.dec` file of release 6.

5.6.1 Charmed mesons

We update the BF of D^0 , D^{*0} and $D^{*\pm}$ meson decays with the ratios described in Table 26

5.6.2 τ lepton

We update the BF of the τ lepton decays with the ratios described in Table 27

5.6.3 Double charm background

We update the BF of a handful of double charm decays with the ratios described in Table 28. These decays include decays where the B meson decays into two charmed mesons, excited or not and optionally a meson including an s quark. For decays where no prediction from PDG [7] or their final state is generated via PYTHIA [11] in the Belle II MC we attribute a correction weight of one and assign a 100% uncertainty on the BF.

Table 26: BF corrections for charmed meson decays

DECAY.dec			pdgLive		
decay	value	line	value	entry	ratio
$D^{*+} \rightarrow D^0 \pi^+$	0.677	7913	0.677 ± 0.0005	$\Gamma 1$	1.0000 ± 0.0007
$D^{*+} \rightarrow D^+ \pi^0$	0.307	7914	0.307 ± 0.0005	$\Gamma 2$	1.0000 ± 0.0016
$D^{*+} \rightarrow D^+ \gamma$	0.016	7915	0.016 ± 0.0004	$\Gamma 3$	1.0000 ± 0.0250
$D^{*0} \rightarrow D^0 \pi^0$	0.647	7940	0.647 ± 0.0009	$\Gamma 1$	1.0000 ± 0.0014
$D^{*0} \rightarrow D^0 \gamma$	0.353	7941	0.353 ± 0.0009	$\Gamma 2$	1.0000 ± 0.0025
$D^0 \rightarrow K^- \pi^+$	0.0395	8512	0.03947 ± 0.0030	$\Gamma 38$	0.9992 ± 0.0759
$D^0 \rightarrow K^+ K^-$	0.00408	8659	0.00408 ± 0.00006	$\Gamma 211$	1.0000 ± 0.0147
$D^0 \rightarrow \pi^+ \pi^-$	0.001455	8624	0.001454 ± 0.000024	$\Gamma 141$	0.9993 ± 0.0165
$D^0 \rightarrow K_S^0 \pi^0$	0.0124	8514	0.0124 ± 0.00022	$\Gamma 38$	1.0000 ± 0.0177
$D^0 \rightarrow K^- \pi^+ \pi^0$	0.144	8539	0.144 ± 0.006	$\Gamma 59$	1.0000 ± 0.0417
$D^0 \rightarrow K_S^0 \pi^+ \pi^-$	0.028	8546	0.0280 ± 0.0018	$\Gamma 44$	1.0000 ± 0.0643
$D^0 \rightarrow K_S^0 K^+ K^-$	0.00442	8606	0.00442 ± 0.00032	$\Gamma 126$	1.0000 ± 0.0724
$D^0 \rightarrow \pi^+ \pi^- \pi^0$	0.0149	8629	0.0149 ± 0.0007	$\Gamma 141$	1.0000 ± 0.0470

Table 27: BF corrections for τ lepton decays

DECAY.dec			pdgLive		
decay	value	line	value	entry	ratio
$\tau \rightarrow \pi \nu$	0.1091	7855	0.1082 ± 0.0005	$\Gamma 9$	0.9917 ± 0.0046
$\tau \rightarrow \rho \nu$	0.2551	7857	0.2549 ± 0.0009	$\Gamma 14$	0.9992 ± 0.0035
$\tau \rightarrow \pi \pi^0 \pi^0 \nu$	0.079301562	7859	0.0932 ± 0.0010	$\Gamma 19$	1.1753 ± 0.0126
$\tau \rightarrow \pi \pi \pi \nu$	0.0932	7874	0.0931 ± 0.0005	$\Gamma 67$	0.9989 ± 0.0054
$\tau \rightarrow \mu \nu \nu$	0.170	7854	0.1739 ± 0.0004	$\Gamma 3$	1.0229 ± 0.0024
$\tau \rightarrow e \nu \nu$	0.154002925	7853	0.1782 ± 0.0004	$\Gamma 5$	1.1571 ± 0.0026
$\tau \rightarrow K \nu$	0.00696	7856	0.00696 ± 0.0001	$\Gamma 10$	1.0000 ± 0.0144
$\tau \rightarrow \pi K^0 K^0 \nu$	0.0017	7869	0.00155 ± 0.0024	$\Gamma 47$	0.9118 ± 1.4118
$\tau \rightarrow \pi \pi^0 \pi^0 \pi^0 \nu$	0.008508640	7861	0.0104 ± 0.0007	$\Gamma 27$	1.2222 ± 0.0822
$\tau \rightarrow K^* \pi \nu$	0.0022	7892	0.0038 ± 0.0017	$\Gamma 140$	1.7273 ± 0.7732
$\tau \rightarrow \pi \pi \pi \pi^0 \nu$	0.0461	7875	0.0274 ± 0.0007	$\Gamma 76$	0.5944 ± 0.0152
$\tau \rightarrow K \pi \pi \nu$	0.00342	7878	0.00345 ± 0.0007	$\Gamma 95$	1.0088 ± 0.2041
$\tau \rightarrow \pi K^0 \nu$	0.000319939	7876	0.000838 ± 0.00014	$\Gamma 36$	2.61857 ± 0.4483
$\tau \rightarrow K^* \nu$	0.012	7888	0.012 ± 0.0007	$\Gamma 136$	1.0000 ± 0.0583
$\tau \rightarrow \pi K^0 \pi^0 \nu$	0.004	7871	0.00382 ± 0.00013	$\Gamma 41$	0.9550 ± 0.0325

5.6.4 Prompt hadronic background

We update the BF of a handful of prompt hadronic decays with the ratios described in Table 29. These decays include decays where the B meson decays into a D^* meson and a few light unflavored mesons. For decays where no prediction from PDG [7] or their final state is generated via PYTHIA [11] in the Belle II MC we attribute a correction weight of one and assign a 100% uncertainty on the BF.

5.7 Gap replacement

In Section 3.2.3, we outlined the way we calculate the luminosity for the dedicated samples that replace the gap modes. The gap modes include $B \rightarrow D^{(*)} \pi \pi \nu$ and $B \rightarrow D^{(*)} \eta \nu$ decays. Now, we replace all the non-resonant decays that have survived the reconstruction with their resonant counterparts. We expect that such a replacement will lead to an increase in the expected yields of $B \rightarrow D^{**} \ell \nu$ decays since the lepton momentum of the original four- and five-body decays present in the generic MC tends to be smaller. Such

Table 28: BF corrections for double charm decays

DECAY.dec			pdgLive		
decay	value	line	value	entry	ratio
$B^0 \rightarrow D_{s0}^* D^*$	0.017687170	1.734	0.0015 ± 0.006	Γ 96	0.0848 ± 0.339
$B^0 \rightarrow D_{s1} D^*$	0		0 ± 0	Γ 96	1.0 ± 1.0
$B^0 \rightarrow D^* D K^0$	0.001710000	1.620	0.0064 ± 0.0005	Γ 179	3.743 ± 0.293
$B^0 \rightarrow D^* D K^{*0}$	0.002500000	1.637	0 ± 0		1.0 ± 1.0
$B^0 \rightarrow D^* D^* K^0$	0.008109240	1.622	0.0081 ± 0.0007	Γ 180	0.999 ± 0.086
$B^0 \rightarrow D^* D^* K^{*0}$	0.005000000	1.639	0 ± 0		1.0 ± 1.0
$B^0 \rightarrow D^{*0} D^* K$	0.010600000	1.616	0.0106 ± 0.00009	Γ 177	1.000 ± 0.009
$B^0 \rightarrow D_s D^*$	0.008044570	1.587	0.008 ± 0.0011	Γ 84	0.995 ± 0.137
$B^0 \rightarrow D_s^* D^*$	0.017700000	1.587	0.0177 ± 0.0014	Γ 86	1.000 ± 0.079
$B^0 \rightarrow D_s^* D_2^*$	0.004000000	1.595	0 ± 0	Γ 86	1.0 ± 1.0
$B^0 \rightarrow D^{*0} \bar{D}^* K^0$	0.004000000	1.595	0 ± 0	Γ 86	1.0 ± 1.0
$B^+ \rightarrow D_{s0}^* D^0$	0.007617290	1.4184	0.0008 ± 0.000145	Γ 173	0.105 ± 0.019
$B^+ \rightarrow D_{s0}^* D^{*0}$	0.017063010	1.4185	0.0009 ± 0.0007	Γ 175	0.053 ± 0.041
$B^+ \rightarrow D_{s1} D^{*0}$	0.011960470	1.4342	0 ± 0	Γ 175	1.0 ± 1.0
$B^+ \rightarrow D^* D^0 K^0$	0.003810000	1.4096	0.0038 ± 0.0004	Γ 203	0.997 ± 0.105
$B^+ \rightarrow D^{*0} D K^0$	0.002060000	1.4097	0.0021 ± 0.0005	Γ 202	1.019 ± 0.242
$B^+ \rightarrow D^{*0} D K^{*0}$	0.002500000	1.4113	0 ± 0		1.0 ± 1.0
$B^+ \rightarrow D^{*0} D^* K^0$	0.009170000	1.4098	0.0092 ± 0.0012	Γ 204	1.003 ± 0.131
$B^+ \rightarrow D^{*0} D^* K^{*0}$	0.005000000	1.4115	0 ± 0		1.0 ± 1.0
$B^+ \rightarrow D^{*0} D^0 K$	0.008580000	1.4102-4103	0.00856 ± 0.00055	Γ 206-207	0.998 ± 0.064
$B^+ \rightarrow D^{*0} D^0 K^*$	0.005000000	1.4120	0 ± 0		1.0 ± 1.0
$B^+ \rightarrow D_s D^0$	0.009009770	1.4069	0.009 ± 0.0009	Γ 172	0.999 ± 0.100
$B^+ \rightarrow D_s D^{*0}$	0.008218450	1.4070	0.0082 ± 0.0017	Γ 193	0.998 ± 0.207
$B^+ \rightarrow D_s D_2^*$	0.004200000	1.4077	0 ± 0		1.0 ± 1.0
$B^+ \rightarrow D^* D^0$	0.007600000	1.4071	0.0076 ± 0.0016	Γ 192	1.000 ± 0.211
$B^+ \rightarrow D_s^* D^{*0}$	0.017100000	1.4072	0.0171 ± 0.0024	Γ 194	1.000 ± 0.140
$B^+ \rightarrow D_s^* D_2^*$	0.004000000	1.4078	0 ± 0		1.0 ± 1.0

Table 29: BF corrections for prompt hadronic decays

DECAY.dec			pdgLive		
decay	value	line	value	entry	ratio
$B^0 \rightarrow D^* \pi$	0.002742680	1.649	0.00274 ± 0.00013	Γ 45	1.00098 ± 0.0474
$B^0 \rightarrow D^* \pi \pi^0$	0.01515771	1.656-657	0.015 ± 0.005	Γ 51	1.01052 ± 0.33674
$B^0 \rightarrow D^* \pi \pi^0 \pi^0$	0.002	1.678-679	0 ± 0		1.0 ± 1.0
$B^0 \rightarrow D^* \pi \pi^0 \pi^0 \pi^0$	0		0 ± 0		1.0 ± 1.0
$B^0 \rightarrow D^* \pi \pi^0 \pi^0 \eta$	0		0 ± 0		1.0 ± 1.0
$B^0 \rightarrow D^* \pi \pi^0 \eta$	0		0 ± 0		1.0 ± 1.0
$B^0 \rightarrow D^* \pi \pi \pi$	0.0187204	1.675	0.013 ± 0.0027	Γ 61	1.43926 ± 0.32385
$B^0 \rightarrow D^* \pi \pi \pi \pi^0$	0.02006	1.769+745	0.02006 ± 0.002706	Γ 64 + Γ 66	1.000 ± 0.1349
$B^0 \rightarrow D^* \pi \pi \pi \pi^0 \pi^0$	0.0047	1.770	0.0047 ± 0.0009	Γ 65	1.000 ± 0.191
$B^0 \rightarrow D^* \pi \pi \pi \pi^0 \pi^0 \pi^0$	0		0 ± 0		1.0 ± 1.0
$B^0 \rightarrow D^* \pi \eta$	0.000256000	1.733	0 ± 0		1.0 ± 1.0
$B^0 \rightarrow D^* \pi \pi \pi \pi^0 \pi^0 \pi^0$	0		0 ± 0		1.0 ± 1.0
$B^+ \rightarrow D^* \pi \pi^0 \pi^0$	0.01715771	1.4161-4162	0.015 ± 0.007	Γ 145	1.14385 ± 0.5332
$B^+ \rightarrow D^{*0} \pi$	0.004900860	1.4137	0.00517 ± 0.00015	Γ 127	0.94777 ± 0.02752
$B^+ \rightarrow D^{*0} \pi \pi^0$	0.01031176	1.4143-4144	0.0098 ± 0.0017	Γ 131	1.05234 ± 0.18178
$B^+ \rightarrow D^{*0} \pi \pi^0 \eta$	0		0 ± 0		1.0 ± 1.0
$B^+ \rightarrow D^{*0} \pi \pi \pi$	0.0138	1.4150-4152	0.019 ± 0.005	Γ 140	0.72632 ± 0.19142
$B^+ \rightarrow D^{*0} \pi \pi \pi \pi^0$	0.018000000	1.4292	0.018 ± 0.004	Γ 141	1.000 ± 0.2222
$B^+ \rightarrow D^{*0} \pi \pi \pi \pi^0 \pi^0$	0		0 ± 0		1.0 ± 1.0
$B^+ \rightarrow D^{*0} \pi \pi \pi \pi \pi$	0.0057	1.4198	0.0057 ± 0.0012	Γ 142	1.000 ± 0.21053
$B^+ \rightarrow D^{*0} \pi \pi \pi \pi \pi^0 \pi^0$	0		0 ± 0		1.0 ± 1.0

863 events are discarded due to the lower bound of the lepton momentum, which is set as
864 described in Table 20.

To test this assumption, we plot the momentum distribution of the charged hadron and lepton. We compare the the total MC expectation before and after the replacement. Figure 24 shows the momentum distribution for the three different reconstruction modes of B^0 mesons. We observe an overall increase in the total yield expectation, which is more prominent in the lower momentum regions. 30 presents the expected percentage increase in the yield of different decay processes. The increase observed in processes other than $B \rightarrow D^{**}\ell\nu$ is due to the misreconstruction of the signal side. In the gap samples, the second B meson of the event decays generically, and the increase in the $B\bar{B}$ category is attributed to events where the second B meson was reconstructed as a signal B meson instead of the resonant gap one.

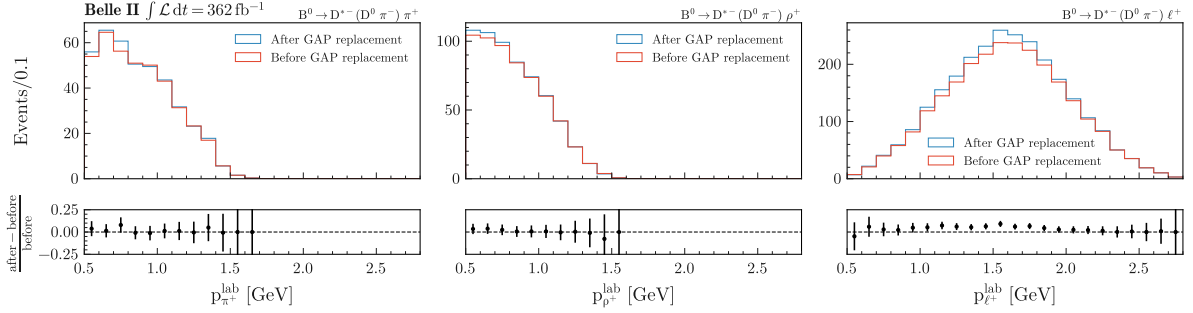


Figure 24: Change of total MC expectation for B^0 . The largest differences are observed in the low momentum region of the charged hadron or lepton as expected.

Table 30: Percentage increase in MC expectation for different templates in the B^0 reconstruction

Decay process	Percentage increase in MC expectation (%)		
	$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$	$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	$B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$
$B \rightarrow D^* \tau \nu$	+ 0.15	+ 0.00	+ 0.00
$B \rightarrow D^* \ell \nu$	+ 0.00	+ 0.00	+ 0.01
$B \rightarrow D^{**} \ell \nu$	+ 28.15	+ 43.78	+ 76.69
$B\bar{B}$	+ 1.06	+ 1.38	+ 1.97
$q\bar{q}$	+ 0.00	+ 0.00	+ 0.00

We present the same information for B^+ mesons in Figure 25 and Table 31. A similar situation is observed with an total increase in the expected yield prominent in the low momentum region. The main contribution in this increase comes from $B \rightarrow D^{**}\ell\nu$ decays, with and additional small contribution from other $B\bar{B}$ decays, due to reconstruction of the second B meson in the gap samples.

5.8 FF reweighting

As mentioned in 3.1.6 we use the HAMMER software to reweight the FF models and parameters of the B semileptonic decays that survive our full selection. In Table we

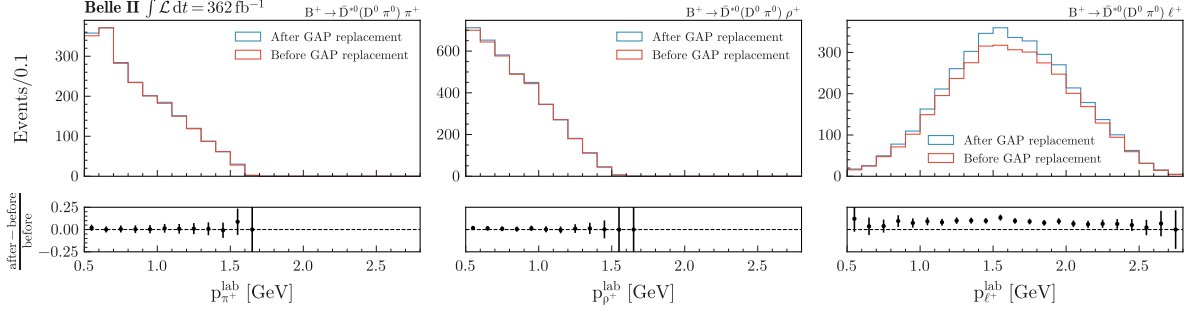


Figure 25: Change of total MC expectation for B^+ . The largest differences are observed in the low momentum region of the charged hadron or lepton as expected.

Table 31: Percentage increase in MC expectation for different templates in the B^+ reconstruction

Decay process	Percentage increase in MC expectation (%)		
	$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$	$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	$B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$
$B \rightarrow D^* \tau \nu$	+ 0.00	+ 0.00	+ 0.00
$B \rightarrow D^* \ell \nu$	+ 0.00	+ 0.00	+ 0.00
$B \rightarrow D^{**} \ell \nu$	+ 16.71	+ 17.99	+ 119.27
$B\bar{B}$	+ 1.38	+ 1.57	+ 1.21
$q\bar{q}$	+ 0.00	+ 0.00	+ 0.00

883 provide and overview of the FF models and references of the values of the parameters
884 that were assumed in the Belle II simulation as well as the target models and parameters
885 that the events are reweighted to

Table 32: FF models and parameters used for the HAMMER reweighting

Decay	FF in <code>DECAY.dec</code>	Parameters	Target FF model	Parameters		
$B \rightarrow D \ell \nu$	BGL	[76]	BLPRXP	[77]		
$B \rightarrow D \tau \nu$	CLN	[78]				
$B \rightarrow D^* \ell \nu$	BGL	[79]				
$B \rightarrow D^* \tau \nu$	CLN	[78]				
$B \rightarrow D_1^{**} \ell \nu$						
$B \rightarrow D_1^{**} \tau \nu$			BLR	[81]		
$B \rightarrow D_0^{**} \ell \nu$	BLR	[80]				
$B \rightarrow D_0^{**} \tau \nu$						
$B \rightarrow D_{1'}^{**} \ell \nu$						
$B \rightarrow D_{1'}^{**} \tau \nu$						
$B \rightarrow D_2^{**} \ell \nu$						
$B \rightarrow D_2^{**} \tau \nu$						

886 Below we provide the set of parameters that have been used for the production of

887 signal decays with the CLN model in the Belle II MC and the set of BLPRXP parameters
 888 that we use as a target as an example. The rest set of source and target parameters are
 889 provided in t Appendix E.

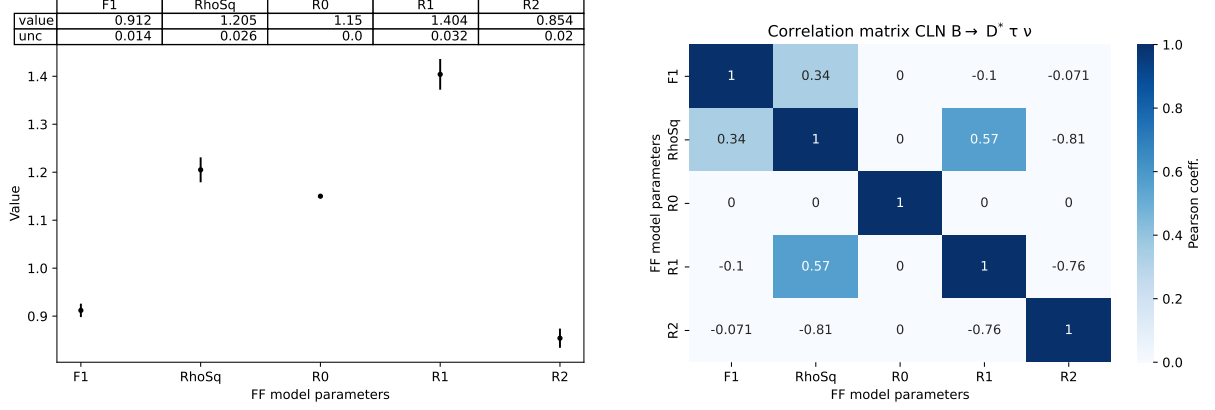


Figure 26: Parameters used in the CLN parametrization for the generation of signal decays in the Belle II MC

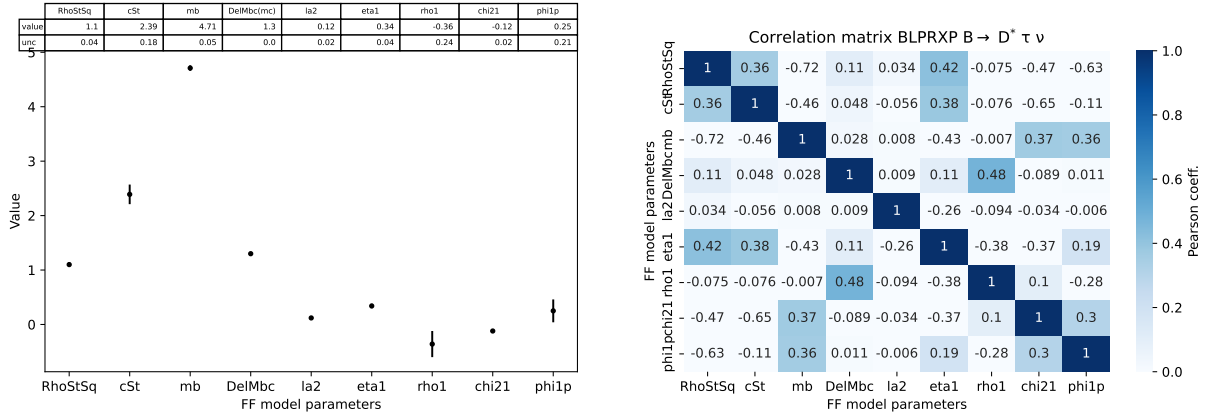


Figure 27: Target parameters used in the BLPRX parametrization for the reweighting with HAMMER

890 5.9 D^0 mass calibration

891 The mass resolution of the D^0 meson is known to be overestimated in Belle II simulated
 892 data compared to experimental data. This mismatch likely arises from a combination
 893 of factors, including mismodelling of tracking parameters, the Belle II magnetic field,
 894 charged hadron identification performance, vertex fitting algorithm efficiency, and poten-
 895 tial inaccuracies in the simulation of the D^0 decay resonance structure and associated fake
 896 rates for different reconstructed modes. To correct this effect, we use a control channel to
 897 calibrate the D^0 mass resolution. We determine calibration factors, which are then ap-
 898 plied to smear the reconstructed invariant mass distribution of D^0 mesons across different

reconstruction modes. Our results are summarized in Table 33, and the method used to determine these factors is detailed in Appendix D.

Table 33: Smearing factors determined for every different D^0 decay mode reconstruction.

Reconstructed decay mode	Smearing factor
$D^0 \rightarrow K^- \pi^+$	0.00198
$D^0 \rightarrow K_s^0 \pi^0$	0.00236
$D^0 \rightarrow K^- \pi^+ \pi^0$	0.00103
$D^0 \rightarrow K_s^0 \pi^+ \pi^-$	0.00193
$D^0 \rightarrow K_s^0 K^+ K^-$	
$D^0 \rightarrow K^- \pi^+ \pi^- \pi^+$	

5.10 Continuum calibration

Since the quality of the modelling of $q\bar{q}$ processes is a priori unknown in Belle II we use off resonance data to constrain the overall normalization of continuum. As described in Section 3.3.2 Belle II has collected $42.556 \pm 0.007 \pm 0.273 \text{ fb}^{-1}$ of off-resonance data at center of mass energies below the $\Upsilon(4S)$ production energy. We reconstruct these experimental events using exactly the same reconstruction code and selections as for on-resonance experimental data and simulated data. The only difference is that we have to shift the cuts on B_{Mbc}^{tag} and $B_{\Delta E}^{tag}$ since the center of mass energy is different. The way that we shift the values of the observables is outlined in Appendix J.

In order to extract scaling factors from the off-resonance data we have to compare the observed yield to an expected one based on simulated data. Since off-resonance MC is not available at Belle II, we still use the on-resonance one, but ideally the kinematic region that we will use has to be statistically independent from the one that is used for all other calibrations and inference. To that end we choose the $M(D^0) \in [1.78, 1.84] \cup [1.89, 1.90] \text{ GeV}$ sideband. We also include the double sidebands defined by the selections $1.5 < E_{ECL} < 2 \text{ GeV}$, 1 extra charged track and $0.154 < \Delta M_{D^{*+}} < 0.4 \text{ GeV}$ for B^0 and $0.149 < \Delta M_{D^{*0}} < 0.170 \text{ GeV}$ for B^+ , in order to increase statistics. These regions and the available statistics are presented in Figure 28.

Deriving calibration factors corresponds to simply dividing the observed yield on experimental off-resonance data and the expected yield on the continuum MC. For reasons that will be discussed further in Section 6.3 we extract calibration factors on a per charge, per reconstruction mode and per sign of helicity angle basis, without linking the categories between different charges or different reconstruction modes. For $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ and $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ we also calibrate the backgrounds in the two different regions of m_{miss}^2 . Table 34 provides the calibration factors and their respective uncertainties. The uncertainties are purely statistical. For the rest of the document these factors are applied to all continuum processes on the hadronic reconstruction channels.

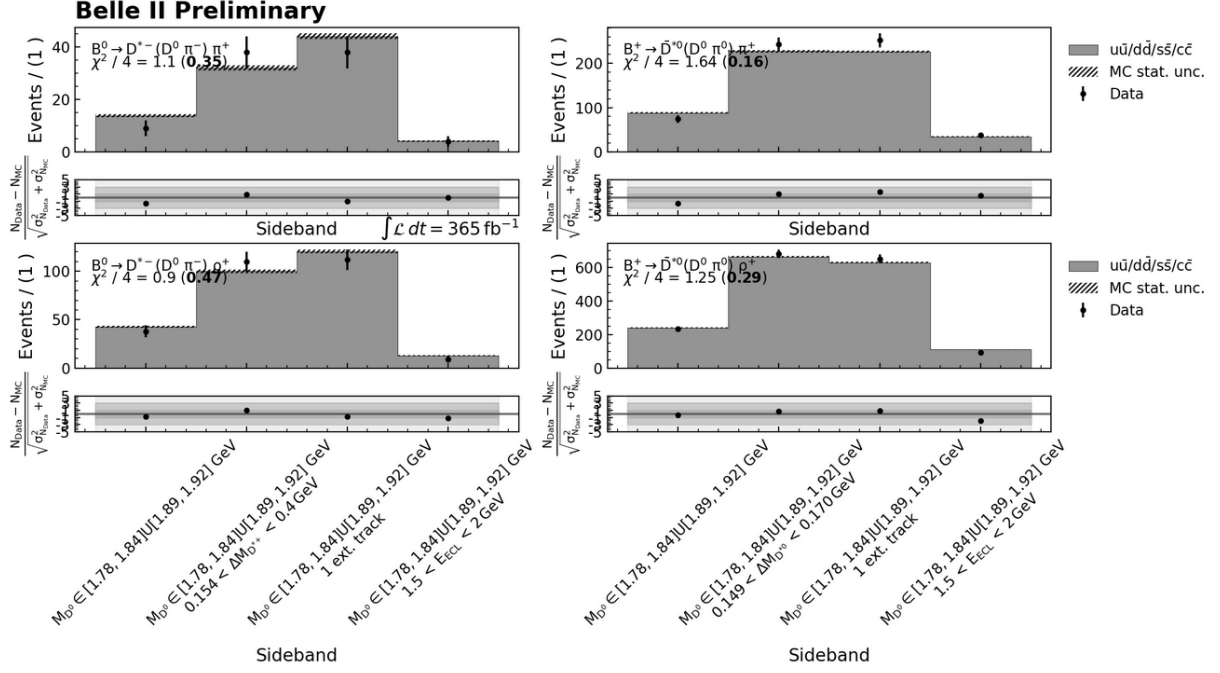


Figure 28: Available statistics used for the off-resonance calibration of the continuum.

Table 34: Continuum calibration factors as derived by off-resonance data.

channel and selection	$q\bar{q}$ cal. factor
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ and $\cos \theta_{\text{hel}}^\tau < 0$	1.118 ± 0.078
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ and $\cos \theta_{\text{hel}}^\tau > 0$	0.922 ± 0.098
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ and $\cos \theta_{\text{hel}}^\tau < 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	0.858 ± 0.221
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ and $\cos \theta_{\text{hel}}^\tau < 0$ and $m_{\text{miss}}^2 > 1 \text{ GeV}^2$	1.076 ± 0.051
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ and $\cos \theta_{\text{hel}}^\tau > 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	0.916 ± 0.107
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ and $\cos \theta_{\text{hel}}^\tau > 0$ and $m_{\text{miss}}^2 > 1 \text{ GeV}^2$	0.95 ± 0.056
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ and $\cos \theta_{\text{hel}}^\tau < 0$	0.957 ± 0.162
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ and $\cos \theta_{\text{hel}}^\tau > 0$	0.913 ± 0.276
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ and $\cos \theta_{\text{hel}}^\tau < 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	1.215 ± 0.64
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ and $\cos \theta_{\text{hel}}^\tau < 0$ and $m_{\text{miss}}^2 > 1 \text{ GeV}^2$	0.98 ± 0.114
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ and $\cos \theta_{\text{hel}}^\tau > 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	1.077 ± 0.293
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ and $\cos \theta_{\text{hel}}^\tau > 0$ and $m_{\text{miss}}^2 > 1 \text{ GeV}^2$	0.931 ± 0.139

5.11 FEI calibration

The performance of the FEI algorithm varies across simulated and experimental data, making it a common practice to calibrate these discrepancies using a control mode [82]. We have chosen to not utilize the official FEI calibration factors. Currently, the S(L) group is discussing the potential impact of the signal side on the calibration of the tag side. As this issue remains unresolved, we have decided to be cautious and seek a relatively pure sideband region enriched in our normalization mode which we will use as a control mode.

Given that $B \rightarrow D^* \ell \nu$ decays have been extensively studied in various experiments [83] and the BF's have been precisely measured [7], we do not anticipate any new physics-related effects in this mode. Although the choice of the FF parametrization might influence the shape of the $B \rightarrow D^* \ell \nu$ spectrum for any given observable, the overall normalization should remain unaffected.

The calibration factors we intend to derive will only be applied to $B \rightarrow D^* \tau \nu$ and $B \rightarrow D^* \ell \nu$ decays, i.e. to correctly reconstructed decays. For background processes, we will employ a different calibration strategy. This approach ensures that our signal side remains consistent for the normalization modes. The signal modes differ only by the presence of a charged pion (and an additional neutral pion in the case of $\tau \rightarrow \rho \nu_\tau$ modes) compared to $B \rightarrow D^* \ell \nu$ decays, giving us confidence that the calibration factors will be applicable in the SR and NR.

We choose m_{miss}^2 as the fitting variable for the extraction of the calibration factors as it provides very good discrimination power between $B \rightarrow D^* \ell \nu$ decays and other processes. Since only an undetectable neutrino is present in correctly reconstructed $B \rightarrow D^* \ell \nu$ decays we expect these events to be distributed around zero. However since the q^2 sideband is very clean we are going to perform the fit with a very coarse binning, as the extraction of the calibration factors is almost a counting experiment. Figure 29 shows the prefit distributions for B^0 and $B^\pm$ respectively.

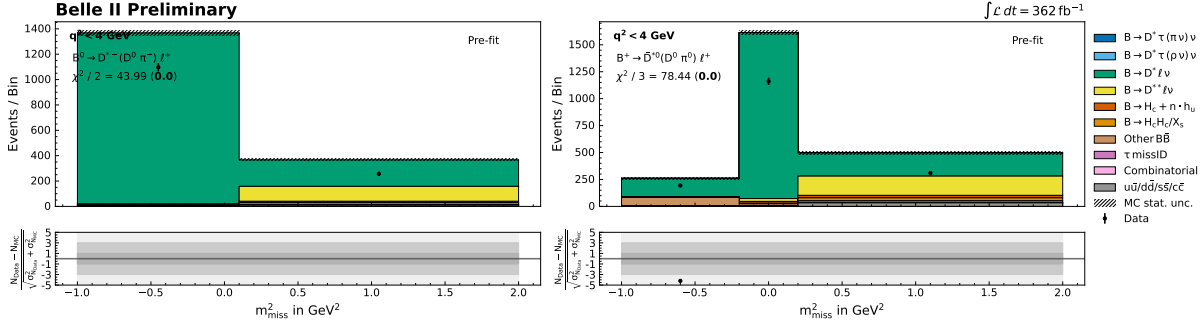


Figure 29: Prefit distributions for the FEI calibration in the q^2 sideband for B^0 (left) and $B^\pm$ (right).

The main source of background in this sideband is $B \rightarrow D^{**} \ell \nu$ decays in the high m_{miss}^2 region. For $B^\pm$ we also find some background events in the negative m_{miss}^2 region, mainly coming from $B \rightarrow D \ell \nu$ decays. This is also what motivates the binning of the fit. For B^0 we have one pure bin of $B \rightarrow D^* \ell \nu$ decays and second where both signal and background processes can be found. For $B^\pm$ we opt for the choice of three bins. The central bin is very pure in $B \rightarrow D^* \ell \nu$ decays, the first bin mainly contains $B \rightarrow D \ell \nu$ decays while the last is enriched in $B \rightarrow D^{**} \ell \nu$ decays. Before we perform the fit we additionally check the modeling of the m_{miss}^2 observable. We normalize MC to the total number of events observed in experimental data and we plot the m_{miss}^2 distribution with the same binning that will be used in the fit. The distributions are presented in Figure 30. The derived p-values from the χ^2 test imply that there is no mismodelling of the shape of the variable at the 5% confidence level, making it a valid choice for extracting the yield of $B \rightarrow D^* \ell \nu$ decays.

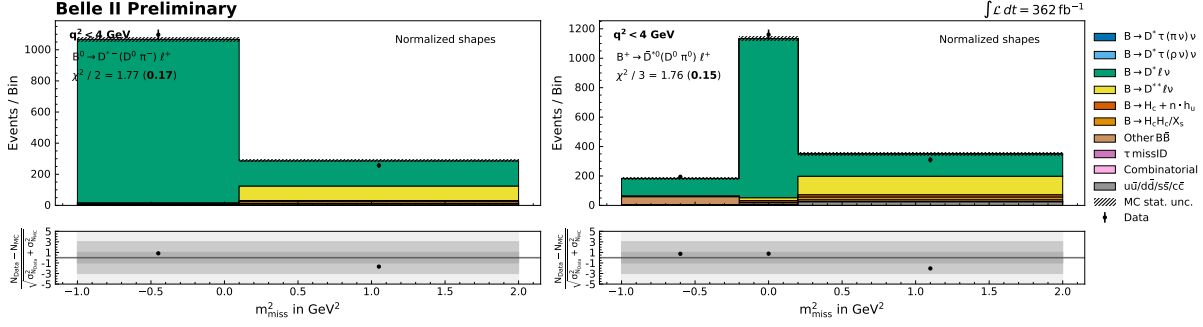


Figure 30: Shape comparison between simulated data and experimental data for the FEI calibration in the q^2 sideband.

We will derive the calibration factors using the same aggregation scheme that is being used for the official FEI calibration. That is, we calibrate 10 distinct modes for in the neutral B meson reconstruction and 11 distinct modes in the charged B meson reconstruction. We then group all the rest modes that are reconstructed by the FEI in a common category. We fit these 23 FEI reconstructions modes simultaneously. For every FEI reconstruction mode we extract a calibration factor for the $B \rightarrow D^* \ell \nu$ decays while we are merging all the background processes into on common templates as we're interested in the strength of $B \rightarrow D^* \ell \nu$ decays which will be later used as a calibration factor in the SR and NR. These free parameters are not shared between any of the FEI modes, meaning that the calibration factors are statistically independant across all FEI reconstruction modes. However the nuisance parameteres for systematic uncertainties are being shared between all reconstruction channels. Those have been derived with the SysVar software using the method that has been described in Section 3.1.4 and the eigenvariations have been computed simultaneously for all reconstruction channels.

We first proceed by performing the fit on an Asimov dataset. In Tables 35 and 36 we provide the expected uncertainties on the Asimov dataset. The first uncertainty is the statistical one, while the second one the systematic. For the systematic uncertainty we have considered mid to high momentum track finding efficiency, charged and neutral slow pion efficiency, mid to high energy neutral pion efficiency, $D^{(*)}BFs$, $B \rightarrow D^{**} \ell \nu$ BFs, lepton ID efficiency and pion and kaon fakes, hadron ID efficiency for pions and kaons and fakes, FF uncertainties and $B \rightarrow D \ell \nu$ BF uncertainties.

We further conduct statistical tests to confirm the validity of the estimator, namely investigate potential biases on the calibration factors and test the linear response of the estimator. These tests are presented in Appendix F. We conclude that our simultaneous fit is a good estimator of the calibration factors that are needed to tackle the Data/MC discrepancy of the FEI efficiency.

Lastly we're performing the fit on experimental data. Table 37 shows the extracted calibration factors on B^0 and 38 on B^+ . We provide all the post fit distributions in Appendix K. We get a Goodness of Fit (GoF) p-value of 10.19%. In the rest of this document we use the values from Table 37 and 38 to scale $B \rightarrow D^* \ell \nu$ and $B \rightarrow D^* \tau \nu$ events in the SR and NR.

Table 35: Signal strengths for $B \rightarrow D^* \ell \nu$ which are interpreted as the FEI calibration factors fit on an Asimov dataset in the B^0 channels. The first uncertainty is statistical while the second is the total systematic uncertainty.

FEI mode	calibration factor
$D^- \pi^+$	$1.0 \pm 0.122 \pm 0.069$
$D^- \pi^+ \pi^0$	$1.0 \pm 0.101 \pm 0.057$
$D^- \pi^+ \pi^+ \pi^-$	$1.0 \pm 0.078 \pm 0.047$
$D^- \pi^+ \pi^+ \pi^- \pi^0$	$1.0 \pm 0.085 \pm 0.051$
$\bar{D}^0 \pi^+ \pi^-$	$1.0 \pm 0.128 \pm 0.068$
$D^{*-} \pi^+$	$1.0 \pm 0.131 \pm 0.071$
$D^{*-} \pi^+ \pi^0$	$1.0 \pm 0.077 \pm 0.047$
$D^{*-} \pi^+ \pi^+ \pi^-$	$1.0 \pm 0.077 \pm 0.048$
$D^{*-} \pi^+ \pi^+ \pi^- \pi^0$	$1.0 \pm 0.084 \pm 0.05$
$\Lambda_c^- p \pi^+ \pi^-$	$1.0 \pm 0.216 \pm 0.113$
Rest	$1.0 \pm 0.073 \pm 0.045$

Table 36: Signal strengths for $B \rightarrow D^* \ell \nu$ which are interpreted as the FEI calibration factors fit on an Asimov dataset in the B^+ channels. The first uncertainty is statistical while the second is the total systematic uncertainty.

FEI mode	calibration factor
$\bar{D}^0 \pi^+$	$1.0 \pm 0.087 \pm 0.057$
$\bar{D}^0 \pi^+ \pi^0$	$1.0 \pm 0.075 \pm 0.053$
$\bar{D}^0 \pi^+ \pi^+ \pi^-$	$1.0 \pm 0.064 \pm 0.051$
$\bar{D}^0 \pi^+ \pi^+ \pi^- \pi^0$	$1.0 \pm 0.069 \pm 0.051$
$\bar{D}^{0*} \pi^+$	$1.0 \pm 0.12 \pm 0.072$
$\bar{D}^{0*} \pi^+ \pi^0$	$1.0 \pm 0.135 \pm 0.08$
$\bar{D}^{0*} \pi^+ \pi^+ \pi^-$	$1.0 \pm 0.107 \pm 0.065$
$\bar{D}^{0*} \pi^+ \pi^+ \pi^- \pi^0$	$1.0 \pm 0.128 \pm 0.074$
$D^- \pi^+ \pi^+$	$1.0 \pm 0.2 \pm 0.107$
$D^- \pi^+ \pi^+ \pi^0$	$1.0 \pm 0.152 \pm 0.085$
$\Lambda_c^- p \pi^+ \pi^- \pi^+$	$1.0 \pm 0.156 \pm 0.086$
Rest	$1.0 \pm 0.068 \pm 0.051$

5.12 Photon multiplicity reweighting

Since several Belle II analysis have observed a mismodelling of the shape of $E_{\text{ECL}}^{\text{extra}}$ we are performing a reweighting of the number of photons that are found in the ROE build against our $\Upsilon(4S)$ candidates. To that end we choose the $\Delta M(D^*)$ sideband in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$ and $B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$ reconstruction channels as these have a significant fraction of correctly reconstructed events.

Table 37: Signal strengths for $B \rightarrow D^* \ell \nu$ and all other background processes in the FEI calibration fit on experimental data in the B^0 channels. The first uncertainty is statistical while the second is the total systematic uncertainty. We get a GoF p-value of 10.19%.

FEI mode	calibration factor
$D^- \pi^+$	$1.019 \pm 0.124 \pm 0.069$
$D^- \pi^+ \pi^0$	$1.124 \pm 0.106 \pm 0.064$
$D^- \pi^+ \pi^+ \pi^-$	$0.876 \pm 0.072 \pm 0.041$
$D^- \pi^+ \pi^+ \pi^- \pi^0$	$0.74 \pm 0.073 \pm 0.037$
$\bar{D}^0 \pi^+ \pi^-$	$0.854 \pm 0.117 \pm 0.061$
$D^{*-} \pi^+$	$1.252 \pm 0.146 \pm 0.087$
$D^{*-} \pi^+ \pi^0$	$0.499 \pm 0.054 \pm 0.024$
$D^{*-} \pi^+ \pi^+ \pi^-$	$0.583 \pm 0.059 \pm 0.027$
$D^{*-} \pi^+ \pi^+ \pi^- \pi^0$	$0.673 \pm 0.069 \pm 0.034$
$\Lambda_c^- p \pi^+ \pi^-$	$0.74 \pm 0.186 \pm 0.084$
Rest	$1.091 \pm 0.066 \pm 0.045$

Table 38: Signal strengths for $B \rightarrow D^* \ell \nu$ and all other background processes in the FEI calibration fit on experimental data in the B^+ channels. The first uncertainty is statistical while the second is the total systematic uncertainty. We get a GoF p-value of 10.19%.

FEI mode	calibration factor
$D^- \pi^+$	$1.019 \pm 0.124 \pm 0.069$
$D^- \pi^+ \pi^0$	$1.124 \pm 0.106 \pm 0.064$
$D^- \pi^+ \pi^+ \pi^-$	$0.876 \pm 0.072 \pm 0.041$
$D^- \pi^+ \pi^+ \pi^- \pi^0$	$0.74 \pm 0.073 \pm 0.037$
$\bar{D}^0 \pi^+ \pi^-$	$0.854 \pm 0.117 \pm 0.061$
$D^{*-} \pi^+$	$1.252 \pm 0.146 \pm 0.087$
$D^{*-} \pi^+ \pi^0$	$0.499 \pm 0.054 \pm 0.024$
$D^{*-} \pi^+ \pi^+ \pi^-$	$0.583 \pm 0.059 \pm 0.027$
$D^{*-} \pi^+ \pi^+ \pi^- \pi^0$	$0.673 \pm 0.069 \pm 0.034$
$\Lambda_c^- p \pi^+ \pi^-$	$0.74 \pm 0.186 \pm 0.084$
Rest	$1.091 \pm 0.066 \pm 0.045$

In order to derive the photon multiplicity weights we first normalize the MC expectation to the one observed in experimental data. The reason we are not applying the FEI calibration factors that we derived in the low q^2 sideband, is that in such case we would have to confirm that the derived calibration factors are compatible with the $\Delta M(D^*)$ sideband. However, since we're not interested in deriving weights that would affect the normalization of the $E_{\text{ECL}}^{\text{extra}}$ distribution but only the shape, we argue that deriving scaling factors on the normalized samples is a sufficiently good approximation.

We then split the sideband into two regions for every B meson charge, one which is enriched in correctly reconstructed candidates and one enhanced in misreconstructed

1013 ones. For B^0 the selected range is $m_{\text{miss}}^2 \in [-1, 0.7] \text{ GeV}^2$ for correctly reconstructed
 1014 events and $m_{\text{miss}}^2 \in [0.7, 2] \text{ GeV}^2$ for misreconstructed ones. For B^+ the selected range
 1015 is $m_{\text{miss}}^2 \in [-1, 1] \text{ GeV}^2$ for correctly reconstructed events and $m_{\text{miss}}^2 \in [1, 2] \text{ GeV}^2$ for
 1016 misreconstructed ones. Figure 31 shows the distributions based on which we do this split.

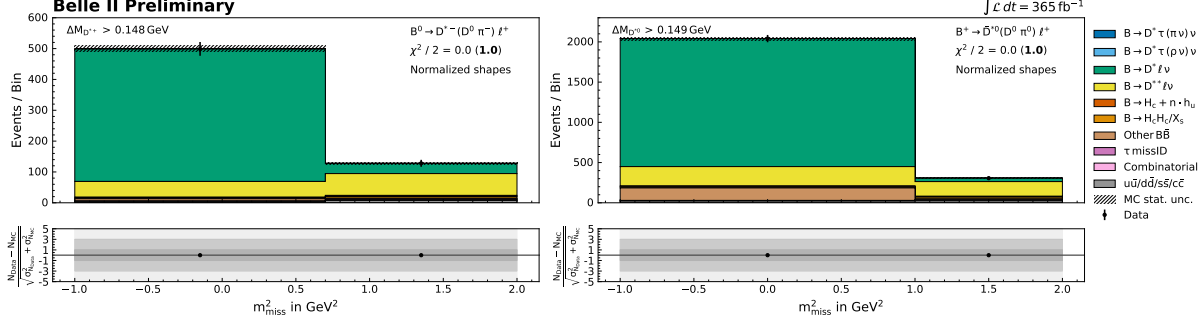


Figure 31: Normalized shape distribution of m_{miss}^2 which defines the correctly reconstructed and misreconstructed enhanced regions where the photon multiplicity reweighting is derived for B^0 (left) and $B^\pm$ (right).

1017 The weights derived in the correctly reconstructed enhanced samples will be applied
 1018 to signal $B \rightarrow D^* \tau \nu$ events and normalization $B \rightarrow D^* \ell \nu$ events. Our Data/MC studies
 1019 which are outlined later in this document (Section 8) indicate that misreconstructed events
 1020 do not contribute to the mismodelling of the $E_{\text{ECL}}^{\text{extra}}$ observable, so it is not necessary to
 1021 apply correction weights to those events. The weights are derived by dividing the yield
 1022 in experimental data over the normalized expected one on simulated data in different
 1023 bins of the photon multiplicity in the ROE. Table 39 shows the derived weights and their
 1024 respective uncertainties. Figure 32 shows the actual distribution of the ROE photons.
 1025 The MC has been normalized to Data to isolate shape differences in this observable while
 1026 disregarding any overall normalization discrepancies that may occur in this sideband.

Table 39: Photon multiplicity weights for B^0 and B^+ for correctly and misreconstructed events.

B^0				B^+		
N_γ	$m_{\text{miss}}^2 \in [-1, 0.7] \text{ GeV}^2$	$m_{\text{miss}}^2 \in [0.7, 2] \text{ GeV}^2$		N_γ	$m_{\text{miss}}^2 \in [0, 1] \text{ GeV}^2$	$m_{\text{miss}}^2 \in [1, 2] \text{ GeV}^2$
0	0.92 ± 0.20	2.3 ± 1.5		0.0	1.30 ± 0.10	2.7 ± 1.0
1	1.10 ± 0.23	0.49 ± 0.30		1.0	1.06 ± 0.06	1.37 ± 0.29
2	1.09 ± 0.26	1.1 ± 0.5		2.0	1.06 ± 0.07	1.08 ± 0.19
3	1.3 ± 0.4	1.5 ± 0.6		3.0	0.86 ± 0.07	0.85 ± 0.16
4-10	0.57 ± 0.22	0.65 ± 0.25		4.0	0.73 ± 0.09	0.82 ± 0.16
				5.0	0.49 ± 0.10	0.98 ± 0.24
				6.0-10	0.47 ± 0.11	0.52 ± 0.20

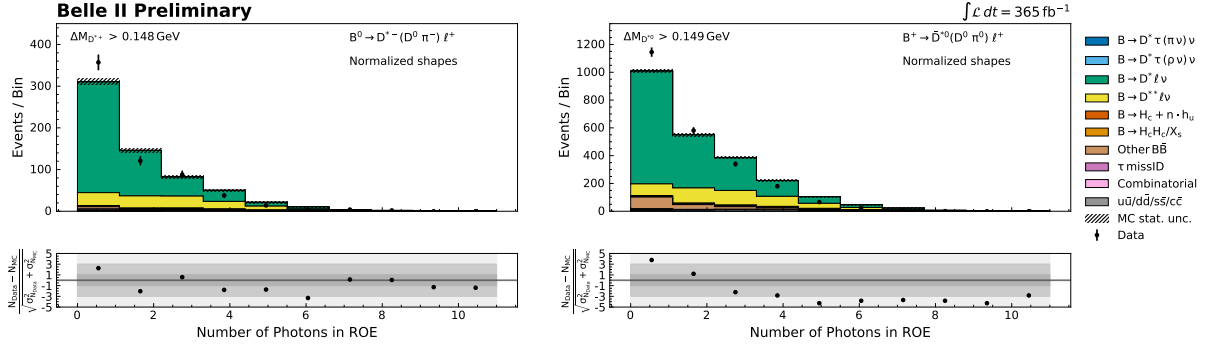


Figure 32: Normalized shape distribution of the ROE photon multiplicity in the high $\Delta M(D^*)$ where the corrections of Table 39 are derived from for B^0 (left) and $B^\pm$ (right)

6 Signal extraction

In Section 5, we have described the set of corrections applied to the simulated data. All these corrections are implemented as correction weights. Table 16 provides an overview of the SRs and NRs used to extract the numerator and denominator of $R(D^*)$. For the NR, we merge the electron and muon reconstruction modes, as we are not interested in distinguishing between them.

In hadronically tagged analyses, extracting the yield of $B \rightarrow D^* \ell \nu$ decays is typically achieved by fitting the m_{miss}^2 distribution. This variable offers strong discriminating power, as only one missing neutrino is expected in the event. Therefore, for $B \rightarrow D^* \ell \nu$ decays, the m_{miss}^2 distribution is expected to peak near zero, in contrast to other misreconstructed decays, which will show larger values of m_{miss}^2 . This trend is confirmed in Figure 33. We also observe that the main source of background events in the NR are $B \rightarrow D^{**} \ell \nu$ decays; however, these can be easily distinguished from $B \rightarrow D^* \ell \nu$ decays using the m_{miss}^2 distribution.

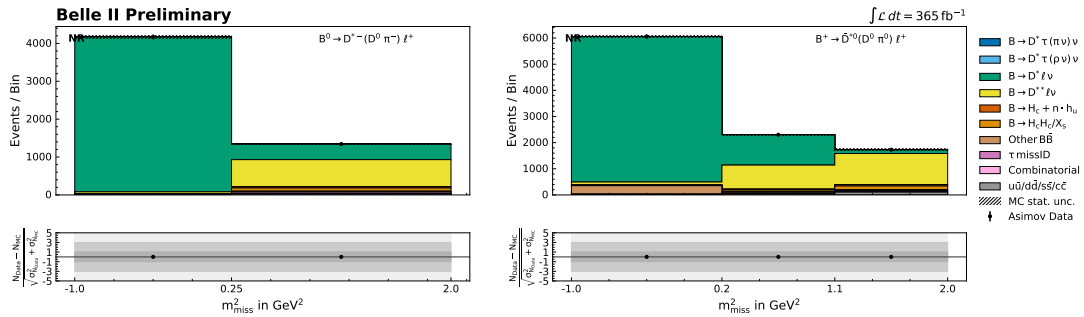


Figure 33: m_{miss}^2 distribution in the NR. $B \rightarrow D^* \ell \nu$ decays peak near zero while all other contributions are spread in the tails of the distribution.

Even if most of $B \rightarrow D^* \ell \nu$ decays are around the peak at 0 GeV^2 we still keep the region up to 2 GeV^2 in order to extrapolate the overall normalization of the yield for background processes (mainly the $B \rightarrow D^{**} \ell \nu$) component to the handful of events that can be found under the m_{miss}^2 peak. This becomes even more evident if we plot m_{miss}^2 using a log scale for the y-axis in Figure 34.

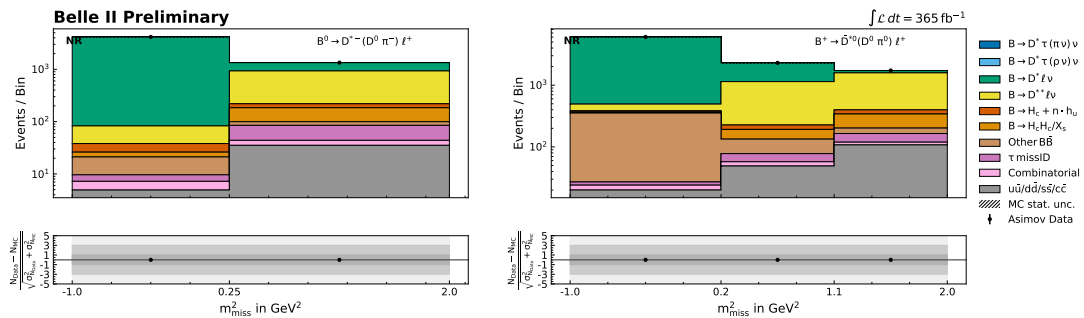


Figure 34: m_{miss}^2 distribution in the NR using a log scale on the y axis for better visibility of the background distributions.

In $B \rightarrow D^* \tau \nu$ decays, where the τ lepton decays hadronically, there are two missing neutrinos in the event. As a result, these decays do not peak at zero in the m_{miss}^2 squared distribution but instead exhibit a continuous spectrum with positive m_{miss}^2 values. Another effective method for distinguishing such decays is by analyzing the extra energy detected in the calorimeter ($E_{\text{ECL}}^{\text{extra}}$). Since the undetectable neutrinos leave no signal in the detector, correctly reconstructed decays are expected to peak at values close to zero, while background events typically result in larger values of $E_{\text{ECL}}^{\text{extra}}$. This holds true for nearly all background processes, except for cases where a lepton is mistakenly identified as the charged hadron reconstructed as the daughter of the τ lepton. This is mainly true for the reconstruction modes where the τ lepton decays into a charged pion and a neutrino. For the modes where the τ lepton decays into a ρ meson and a neutrino, this effect is less prominent due to the presence of an additional neutral pion. This is illustrated in Figure 35.

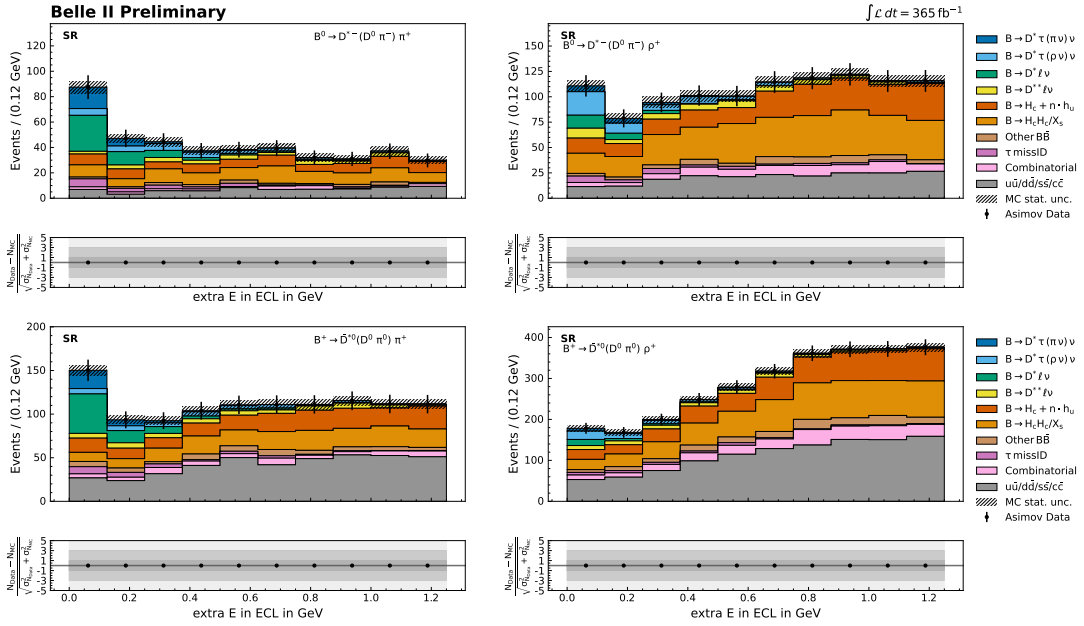


Figure 35: $E_{\text{ECL}}^{\text{extra}}$ distribution in the SR. $B \rightarrow D^* \tau \nu$ decays peak near zero. All other background sources peak toward larger values of $E_{\text{ECL}}^{\text{extra}}$ except $B \rightarrow D^* \ell \nu$ decays, making it hard to distinguish them from signal processes.

These fake hadrons correspond to $B \rightarrow D^* \ell \nu$ events, whose distribution of unassigned energy in the calorimeter closely resembles that of $B \rightarrow D^* \tau \nu$ decays, making it extremely challenging to distinguish between the two. Misreconstructed decays like these will peak around zero in m_{miss}^2 since they involve only one undetected neutrino. To improve separation, we choose to fit the $E_{\text{ECL}}^{\text{extra}}$ distribution in two bins of m_{miss}^2 : one for values around 0, i.e., $[-1, 1]$ GeV, and another for higher m_{miss}^2 values. This approach enhances the discrimination power between signal and normalization events. The effectiveness of this strategy is evident in Figure 36, where we present the final 2D projections for the SR and the m_{miss}^2 distributions for the NR.

In Figure 36 a finer categorization of the events has been chosen for illustration pur-

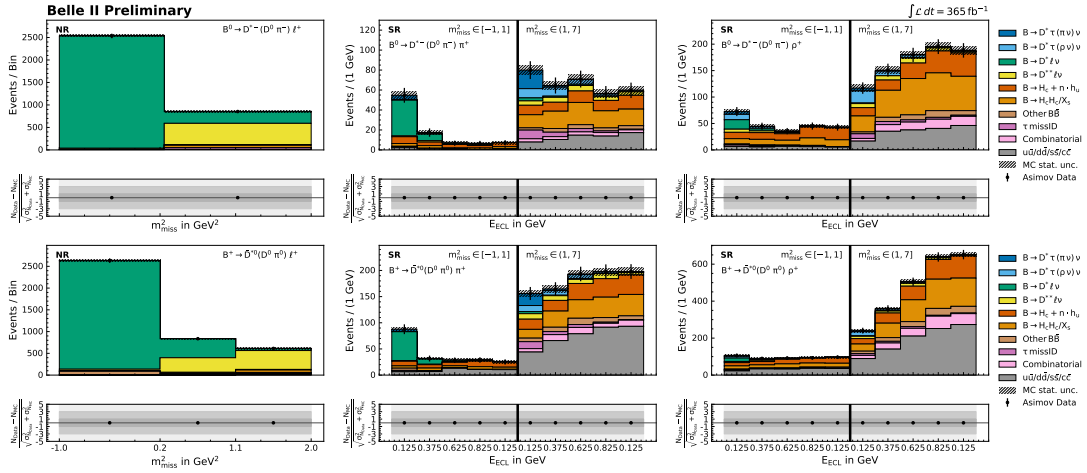


Figure 36: Final fit distributions used for the $R(D^*)$ fit.

poses. We will perform the final fit for the extraction of $R(D^*)$ in five different categories as this gives us the highest sensitivity. The final categories for the fit are presented in Table 40.

Table 40: Final templates for the fit

$B^0 \rightarrow D^* \tau \nu$	$B^0 \rightarrow D^* \tau (\pi \nu) \nu, B^0 \rightarrow D^* \tau (\rho \nu) \nu$
$B^+ \rightarrow D^* \tau \nu$	$B^+ \rightarrow D^* \tau (\pi \nu) \nu, B^+ \rightarrow D^* \tau (\rho \nu) \nu$
$B^0 \rightarrow D^* \ell \nu$	$B^0 \rightarrow D^* \ell \nu$
$B^+ \rightarrow D^* \ell \nu$	$B^+ \rightarrow D^* \ell \nu$
$B \rightarrow D^{**} \ell \nu$	$B \rightarrow D^{**} \ell \nu$
tau_misID	$B \rightarrow D^* \tau \nu$ (τ crossfeed)
$B\bar{B}$	$B \rightarrow D^* + n \cdot h_u, B \rightarrow H_c H_c X_s, B \rightarrow D \tau \nu, B \rightarrow D \ell \nu, \text{other } B\bar{B}$
$q\bar{q}$	uubar, ddbar, ssbar, ccbar

6.1 $R(D^*)$ parametrization

The theoretical definition of $R(D^*)$ is given by Equation 14, where ℓ stands for the two light leptons, electrons and muons.

$$R(D^*) = \frac{\mathcal{B}(B \rightarrow D^* \tau \nu)}{\mathcal{B}(B \rightarrow D^* \ell \nu)} \quad (14)$$

The $R(D^*)$ ratio is an excellent test of LFU and is commonly preferred over a direct determination of the $B \rightarrow D^* \tau \nu$ branching fraction. This preference is due to the assumption that several systematic uncertainties, which are common to both the numerator and denominator, will cancel out in the ratio. Experimentally, the ratio is typically expressed as:

$$R(D^*) = 2 \cdot \frac{\nu_{D^*0\tau^+} \cdot \epsilon_{D^*0\tau^+}^{-1} + \nu_{D^*-\tau^+} \cdot \epsilon_{D^*-\tau^+}^{-1}}{\nu_{D^*0\ell^+} \cdot \epsilon_{D^*0\ell^+}^{-1} + \nu_{D^*-\ell^+} \cdot \epsilon_{D^*-\ell^+}^{-1}} \quad (15)$$

where, ν represents the determined yield of the fitted template, while ϵ accounts for any reconstruction efficiency that needs to be considered when performing the fit. For the assumption to hold when fitting neutral and charged B mesons simultaneously, the reconstruction efficiencies for both charges must be identical. However, it is well known within Belle II that certain reconstruction efficiencies differ between charged and neutral B mesons. A key example, especially relevant for semileptonic analyses like this one, is the discrepancy in the FEI reconstruction efficiency for the two B meson charges. This effect becomes apparent when examining the official FEI calibration factors [82].

To address this issue, instead of directly fitting the yields of $B \rightarrow D^* \tau \nu$ and $B \rightarrow D^* \ell \nu$, we employ a parametrization based on isospin symmetry, allowing us to fit the $R(D^*)$ ratio directly. This approach ensures that most reconstruction efficiencies appear as ratios in the parametrization, leading to the cancellation of common effects, while those that do not cancel are accounted for appropriately. The first time that this parametrization was used was in the hadronically tag $R(D^*)$ measurement by Belle [84]. Here we derive the parametrization again to adapt it to the terminology that is required for a fit implemented with `pyhf`.

We begin considering only one charge for the B mesons. $R(D^*)$ is expressed as

$$R(D^*) = \frac{1}{2} \cdot \frac{N_s}{N_n} \quad (16)$$

where N_s and N_n are the true signal and normalization events, while the factor of 2 arises from the fact that two generation of light leptons are used in the denominator.

The true events are connected to the experimental yields via

$$\nu_{s/n} = N_{s/n} \cdot \epsilon_{s/n} \cdot \prod_{i=k}^K \zeta_{s/n}^i \quad (17)$$

where ν are the experimental yields, ϵ the reconstruction efficiency on MC and ζ^i the efficiency associated a to normalization effect i arising from corrections applied to simulated data, e.g. charged slow pion efficiency correction or LID efficiency correction etc. ζ^i controls solely the impact of every systematic effect in the absolute yield of every templates and doesn't affect the shape of the template. The latter effect is treated separately.

Isospin symmetry connect the decay rates or BF's of for neutral and charged B mesons as

$$\tau_{+0}^{MC} = \frac{1}{\tau_{0+}^{MC}} = \frac{\tau_{B^\pm}}{\tau_{B^0}} = \frac{\Gamma(B^0)}{\Gamma(B^\pm)} = \frac{\mathcal{B}(B^0)}{\mathcal{B}(B^\pm)} \quad (18)$$

where τ_B is the lifetime of B mesons and τ_{+0} the ratio of lifetimes for charged and neutral B mesons.

By combining Equations 16, 17 and 18 we arrive at the base of our parametrization for the experimental yields

$$\nu_{cb}^{s^0} = \frac{1}{2} R(D^*) \cdot \nu_{cb}^{n^\pm} \cdot \frac{\epsilon_{s^0}}{\epsilon_{n^\pm}} \cdot \tau_{+0}^{MC} \cdot \prod_{i=k}^K \frac{\zeta_{s^0}^i}{\zeta_{n^\pm}^i} \quad (19)$$

$$\nu_{cb}^{s^\pm} = \frac{1}{2} R(D^*) \cdot \nu_{cb}^{n^\pm} \cdot \frac{\epsilon_{s^\pm}}{\epsilon_{n^\pm}} \cdot \prod_{i=k}^K \frac{\zeta_{s^\pm}^i}{\zeta_{n^\pm}^i} \quad (20)$$

$$\nu_{cb}^{n^0} = \nu_{cb}^{n^\pm} \cdot \frac{\epsilon_{n^0}}{\epsilon_{n^\pm}} \cdot \tau_{+0}^{MC} \cdot \prod_{i=k}^K \frac{\zeta_{n^0}^i}{\zeta_{n^\pm}^i} \quad (21)$$

$$\nu_{cb}^{n^\pm} = \nu_{cb}^{n^\pm} \quad (22)$$

where the superscript s^0 denotes signal events in B^0 meson decays, n^0 $B^0 \rightarrow D^* \ell \nu$ decays and so on. The cb superscript stands for channel and bin, as these are taken into account separately when the likelihood is being built.

Now all the experimental yields associated with the signal and normalization processes are a function of the experimental yield of charged B meson $B \rightarrow D^* \ell \nu$ processes $\nu_{n_{B^\pm}}$. As described in Table 2 the experimental yields are determined via **normfactor** modifiers in **pyhf**. This translates the experimental yield of the charged B meson normalization modes into:

$$\nu_{n^\pm} = \mu_{n^\pm} \cdot N_{n^\pm}^{reco} \quad (23)$$

where $N_{n_{B^\pm}}^{reco}$ are the reconstructed events of the charged B meson $B \rightarrow D^* \ell \nu$ template in all the reconstruction regions and $\mu_{n_{B^\pm}}$ is the **normfactor** modifier associated to the yield.

Now for the ratios of reconstruction efficiencies ϵ that appear in the parametrization the usual efficiency formula hold $\epsilon = \frac{N_{reco}}{N_{gen}}$ with N_{gen} being the true number of generated events $N_{gen} = N_{B\bar{B}} \cdot \mathcal{B}(\Upsilon(4S) \rightarrow B\bar{B}) \cdot 2 \cdot \mathcal{B}(\text{decay of interest})$ with the factor of 2 appearing since both B mesons of the event can decay into the decay of interest. Therefore the relevant efficiencies transform into

$$\epsilon_{s^0} = \frac{N_{s^0}^{reco}}{2 \cdot N_{B\bar{B}} \cdot \mathcal{B}(\Upsilon(4S) \rightarrow B^0 \bar{B}^0) \cdot \mathcal{B}(B^0 \rightarrow D^* \tau \nu)} \quad (24)$$

$$\epsilon_{s^\pm} = \frac{N_{s^\pm}^{reco}}{2 \cdot N_{B\bar{B}} \cdot \mathcal{B}(\Upsilon(4S) \rightarrow B^+ B^-) \cdot \mathcal{B}(B^+ \rightarrow D^* \tau \nu)} \quad (25)$$

$$\epsilon_{n^0} = \frac{N_{n^0}^{reco}}{2 \cdot N_{B\bar{B}} \cdot \mathcal{B}(\Upsilon(4S) \rightarrow B^0 \bar{B}^0) \cdot 2 \cdot \mathcal{B}(B^0 \rightarrow D^* \ell \nu)} \quad (26)$$

$$\epsilon_{n^\pm} = \frac{N_{n^\pm}^{reco}}{2 \cdot N_{B\bar{B}} \cdot \mathcal{B}(\Upsilon(4S) \rightarrow B^+ B^-) \cdot 2 \cdot \mathcal{B}(B^+ \rightarrow D^* \ell \nu)} \quad (27)$$

with the extra factor of two appearing in the efficiencies of the normalization modes if we assume that the BFs of $B \rightarrow D^* e \nu$ and $B \rightarrow D^* \mu \nu$ are exactly the same. Now if we use Equation 18 to introduce isospin symmetry and consider the ratio of $\Upsilon(4S)$ to two charged B mesons over neutral B mesons that was assumed in the Belle II simulation

$$f_{+0}^{MC} = \frac{f_{+-}^{MC}}{f_{00}^{MC}} = \frac{\mathcal{B}(\Upsilon(4S) \rightarrow B^+ B^-)}{\mathcal{B}(\Upsilon(4S) \rightarrow B^0 \bar{B}^0)} \quad (28)$$

then the efficiencies become

$$\epsilon_{s^0} = \frac{N_{s^0}^{reco}}{2 \cdot N_{B\bar{B}} \cdot \mathcal{B}(\Upsilon(4S) \rightarrow B^0 \bar{B}^0) \cdot \mathcal{B}(B^+ \rightarrow D^* \tau \nu) \cdot \tau_{+0}^{MC}} \quad (29)$$

$$\epsilon_{s^\pm} = \frac{N_{s^\pm}^{reco}}{2 \cdot N_{B\bar{B}} \cdot f_{+0}^{MC} \cdot \mathcal{B}(\Upsilon(4S) \rightarrow B^0 \bar{B}^0) \cdot \mathcal{B}(B^+ \rightarrow D^* \tau \nu)} \quad (30)$$

$$\epsilon_{n^0} = \frac{N_{n^0}^{reco}}{2 \cdot N_{B\bar{B}} \cdot \mathcal{B}(\Upsilon(4S) \rightarrow B^0 \bar{B}^0) \cdot 2 \cdot \mathcal{B}(B^+ \rightarrow D^* \ell \nu) \cdot \tau_{+0}^{MC}} \quad (31)$$

$$\epsilon_{n^\pm} = \frac{N_{n^\pm}^{reco}}{2 \cdot N_{B\bar{B}} \cdot f_{+0}^{MC} \cdot \mathcal{B}(\Upsilon(4S) \rightarrow B^0 \bar{B}^0) \cdot 2 \cdot \mathcal{B}(B^+ \rightarrow D^* \ell \nu)} \quad (32)$$

Now if we also use Equation 23, Equations 19, 20, 21 and 22 become

$$\nu_{s^0} = R(D^*) \cdot \mu_{n^\pm} \cdot N_{s^0}^{reco} \cdot \frac{f_{+0}^{MC}}{\mathcal{R}_{MC}^*} \prod_{i=k}^K \frac{\zeta_{s^0}^i}{\zeta_{n^\pm}^i} \quad (33)$$

$$\nu_{s^\pm} = R(D^*) \cdot \mu_{n^\pm} \cdot N_{s^\pm}^{reco} \cdot \frac{1}{\mathcal{R}_{MC}^*} \cdot \prod_{i=k}^K \frac{\zeta_{s^\pm}^i}{\zeta_{n^\pm}^i} \quad (34)$$

$$\nu_{n^0} = \mu_{n^\pm} \cdot N_{n^0}^{reco} \cdot f_{+0}^{MC} \cdot \prod_{i=k}^K \frac{\zeta_{n^0}^i}{\zeta_{n^\pm}^i} \quad (35)$$

$$\nu_{n^\pm} = \mu_{n^\pm} \cdot N_{n^\pm}^{reco} \quad (36)$$

where $\mathcal{R}_{MC}^*$ is the SM expectation for $R(D^*)$, and f_{+0}^{MC} the ratio of $\Upsilon(4S)$ that are expected to decay into two charged B mesons over neutral B mesons as defined in Equation 28. This value is found to be 1.066 in the `DECAY.dec` file of `relase-06`, while the BF of $\tau \rightarrow \pi \nu_\tau$ is 0.109 and the one of $\tau \rightarrow \rho \nu_\tau$ 0.255. All of these values are the ones used in the Belle II simulation. All other term that are appearing in the ratio are canceling out as demonstrated in the above.

We implement the fit using the aforementioned parametrization as a maximum likelihood fit using `pyhf`. The general form of the specified likelihood is

$$f(\eta, \alpha | \eta, \chi) = \prod_{c \in \text{channels}} \prod_{b \in \text{bins}_c} \text{Pois}(n_{cb} | \nu_{cb}(\eta, \chi)) \prod_{\chi \in \chi} c_\chi(\alpha_\chi | \chi) \quad (37)$$

where the second term constraints the χ terms from some auxiliary measurements α_χ of the parameters χ . The term $\nu_{cb}(\eta, \chi)$ contains the set of unconstrained parameters η and the constrained ones χ . Taking into account all the templates in the fit this is expressed as;

$$\sum_{s \in \text{samples}} \nu_{scb}(\eta, \chi) = \sum_{s \in \text{samples}} \left(\prod \kappa \in \kappa \kappa_{scb}(\eta, \chi) \right) (\nu_{scb}^0(\eta, \chi) + \sum_{\Delta \in \Delta} \Delta_{scb}(\eta, \chi)) \quad (38)$$

where the term $\prod \kappa \in \kappa \kappa_{scb}(\eta, \chi)$ is multiplied to the nominal rate ν_{scb}^0 while the term $\sum_{\Delta \in \Delta} \Delta_{scb}(\eta, \chi)$ is being added to it. These terms are implemented as `pyhf` modifiers in the fit as defined in Table 2.

6.2 P_τ parametrization

An important goal of this analysis is to measure P_τ simultaneously together with $R(D^*)$. As explained in 5.2 we aim at measuring P_τ as an asymmetry of positive and negative τ helicity events. In order to incorporate P_τ into our parametrization we modify Equation 16 to 39 and we introduce Equation 40.

$$R(D^*) = \frac{N_s^{\theta^+} + N_s^{\theta^-}}{N_n} \quad (39)$$

$$P_\tau = \frac{2}{\alpha} \frac{N_s^{\theta^+} - N_s^{\theta^-}}{N_s^{\theta^+} + N_s^{\theta^-}} \quad (40)$$

where $N_s^{\theta^+}$ and $N_s^{\theta^-}$ are events with positive and negative helicity angle respectively and the coefficient α is a coefficient expressing the sensitivity of the different τ decays to P_τ as explained in 5.2.

We are not deriving the full parametrization from scratch as all the steps are identical to 6.1. The final parametrization differs only on an extra free and unconstrained factor $1 \pm \frac{\alpha}{2} P_\tau$ and of course the indices of all efficiencies are modified to their new respective templates.

$$\nu_{s_{\theta^+}^0} = R(D^*) \cdot \left(1 + \frac{\alpha}{2} P_\tau\right) \cdot \mu_{n^\pm} \cdot N_{s_{\theta^+}^0}^{reco} \cdot \frac{f_{+0}^{MC}}{\mathcal{R}_{MC}^* \cdot \left(1 + \frac{\alpha}{2} P_{\tau SM}\right)} \prod_{i=k}^K \frac{\zeta_{s_{\theta^+}^0}^i}{\zeta_{n^\pm}^i} \quad (41)$$

$$\nu_{s_{\theta^-}^0} = R(D^*) \cdot \left(1 - \frac{\alpha}{2} P_\tau\right) \cdot \mu_{n^\pm} \cdot N_{s_{\theta^-}^0}^{reco} \cdot \frac{f_{+0}^{MC}}{\mathcal{R}_{MC}^* \cdot \left(1 - \frac{\alpha}{2} P_{\tau SM}\right)} \prod_{i=k}^K \frac{\zeta_{s_{\theta^-}^0}^i}{\zeta_{n^\pm}^i} \quad (42)$$

$$\nu_{s_{\theta^+}^\pm} = R(D^*) \cdot \left(1 + \frac{\alpha}{2} P_\tau\right) \cdot \mu_{n^\pm} \cdot N_{s_{\theta^+}^\pm}^{reco} \cdot \frac{1}{\mathcal{R}_{MC}^* \cdot \left(1 + \frac{\alpha}{2} P_{\tau SM}\right)} \cdot \prod_{i=k}^K \frac{\zeta_{s_{\theta^+}^\pm}^i}{\zeta_{n^\pm}^i} \quad (43)$$

$$\nu_{s_{\theta^-}^\pm} = R(D^*) \cdot \left(1 - \frac{\alpha}{2} P_\tau\right) \cdot \mu_{n^\pm} \cdot N_{s_{\theta^-}^\pm}^{reco} \cdot \frac{1}{\mathcal{R}_{MC}^* \cdot \left(1 - \frac{\alpha}{2} P_{\tau SM}\right)} \cdot \prod_{i=k}^K \frac{\zeta_{s_{\theta^-}^\pm}^i}{\zeta_{n^\pm}^i} \quad (44)$$

$$\nu_{n^0} = \mu_{n^\pm} \cdot N_{n^0}^{reco} \cdot f_{+0}^{MC} \cdot \prod_{i=k}^K \frac{\zeta_{n^0}^i}{\zeta_{n^\pm}^i} \quad (45)$$

$$\nu_{n^\pm} = \mu_{n^\pm} \cdot N_{n^\pm}^{reco} \quad (46)$$

In order to be able to distinguish between events with positive and negative helicity we now transform the 2D projections shown in Figure 36 into a 3D projection where we include the reconstructed helicity angle. The way that we access this information has been described in 5.2. The final distributions can be seen in Figure 37.

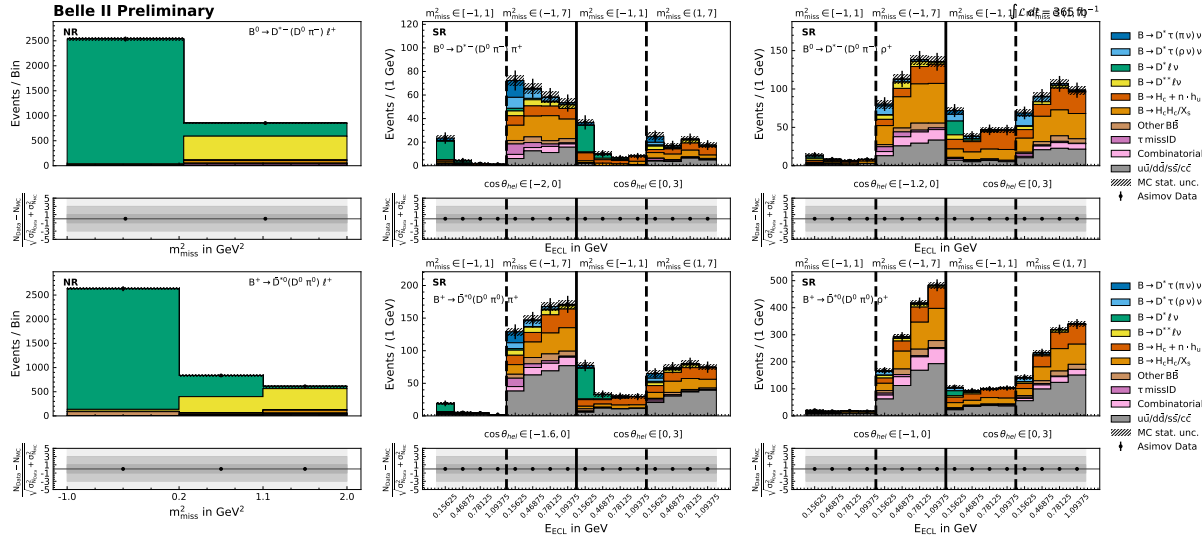


Figure 37: Final fit distributions used for the $R(D^*)$ and P_τ fit.

Of course we are interested in the true helicity angle and not the reconstructed one as resolution effects that could lead to migrations are expected. We calculate the true helicity angle and we provide the confusion matrices in Figure 38. We avoid displaying information on the true helicity in Figure 37 to maintain readability on such an already busy final distribution.

We now split our signal templates even further in order to fit P_τ alongside $R(D^*)$. For the template creation we use the true helicity angle and not the reconstructed one as this makes a later unfolding of our result redundant. The final categories for the $R(D^*)$ and P_τ fit are presented in Table 41. The results of the Asimov fit are presented in 43.

6.3 Sideband and Control regions

Figure 36 shows that a few different backgrounds processes can fake the decays of interest, namely $B \rightarrow D^* \tau \nu$ and $B \rightarrow D^* \ell \nu$, even after the tighter selections. These can be categorized in three main components: $B \rightarrow D^{**} \ell \nu$ decays, other $B\bar{B}$ and continuum processes. Some of these processes such as $B \rightarrow D^{**} \ell \nu$ and the continuum decays are well known to be mismodeled in the Belle II MC. $q\bar{q}$ processes are well constrained by our continuum calibration as this is described in 5.10. For $B\bar{B}$ processes the situation depends heavily on the signal side reconstruction and the level to which these decays are modeled correctly is not a priori known. The overall yield for these processes might differ between simulated and experimental data. In order to ensure that these processes will not affect the extraction of $R(D^*)$ we need to find sideband regions which can be used in order to constraint these background processes in the SR and NR. After extensive

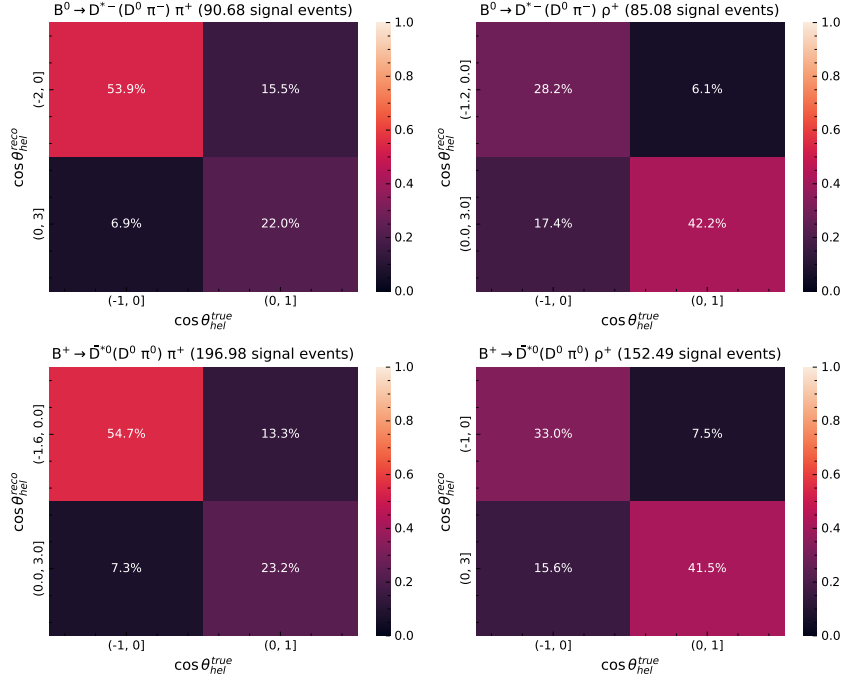


Figure 38: Confusion matrix between the true generated cosine of the helicity angle and the reconstructed one.

Table 41: Final templates for the $R(D^*)$ and P_τ fit

$B^0 \rightarrow D^* \tau (\rightarrow \pi \nu) \nu +$	$B^0 \rightarrow D^* \tau (\pi \nu) \nu$ and $\cos \theta_{\text{hel}}^\tau > 0$
$B^0 \rightarrow D^* \tau (\rightarrow \pi \nu) \nu -$	$B^0 \rightarrow D^* \tau (\pi \nu) \nu$ and $\cos \theta_{\text{hel}}^\tau < 0$
$B^0 \rightarrow D^* \tau (\rightarrow \rho \nu) \nu +$	$B^0 \rightarrow D^* \tau (\rho \nu) \nu$ and $\cos \theta_{\text{hel}}^\tau > 0$
$B^0 \rightarrow D^* \tau (\rightarrow \rho \nu) \nu -$	$B^0 \rightarrow D^* \tau (\rho \nu) \nu$ and $\cos \theta_{\text{hel}}^\tau < 0$
$B^+ \rightarrow D^* \tau (\rightarrow \pi \nu) \nu +$	$B^+ \rightarrow D^* \tau (\pi \nu) \nu$ and $\cos \theta_{\text{hel}}^\tau > 0$
$B^+ \rightarrow D^* \tau (\rightarrow \pi \nu) \nu -$	$B^+ \rightarrow D^* \tau (\pi \nu) \nu$ and $\cos \theta_{\text{hel}}^\tau < 0$
$B^+ \rightarrow D^* \tau (\rightarrow \rho \nu) \nu +$	$B^+ \rightarrow D^* \tau (\rho \nu) \nu$ and $\cos \theta_{\text{hel}}^\tau > 0$
$B^+ \rightarrow D^* \tau (\rightarrow \rho \nu) \nu -$	$B^+ \rightarrow D^* \tau (\rho \nu) \nu$ and $\cos \theta_{\text{hel}}^\tau < 0$
$B^0 \rightarrow D^* \ell \nu$	$B^0 \rightarrow D^* \ell \nu$
$B^+ \rightarrow D^* \ell \nu$	$B^+ \rightarrow D^* \ell \nu$
$B \rightarrow D^{**} \ell \nu$	$B \rightarrow D^{**} \ell \nu$
tau_misID	$B \rightarrow D^* \tau \nu$ (τ crossfeed)
$B\bar{B}$	$B \rightarrow D^* + n \cdot h_u, B \rightarrow H_c H_c X_s, B \rightarrow D \tau \nu, B \rightarrow D \ell \nu, \text{other } B\bar{B}$
$q\bar{q}$	uubar, ddbar, ssbar, ccbbar

studies in different sidebands we managed to find suitable regions that enable us to safely constrain the different background processes in the SR and NR. Our studies tried to ensure that the sidebands that were chosen demonstrate a similar performance of the FEI on those events between the SR and the chosen sidebands. Another important aspect is that the contribution of the different individual decays that make up the aforementioned $B\bar{B}$ subcategories are similar between the SR and the sidebands. Since a lot of these decays have never been measured the assumed BF's in the Belle II `DECAY.dec` file are most of the time purely educated guesses, so we wish to pursue a data driven strategy

of constraining these decays. We constrain the background processes on a per charge, per reconstruction mode and per sign of helicity angle basis, without linking the categories between different charges or different reconstruction modes. For $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ and $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ we also calibrate the backgrounds in the two different regions of m_{miss}^2 as Data/MC studies have revealed that the normalization of the backgrounds in these two kinematics regions is different.

In what follows we demonstrate why these kinematic regions are appropriate choices for constraining certain types of background processes.

Generic $B\bar{B}$ decays is a category of processes that is present in the SR but not well modeled in the Belle II MC. The decay processes that are included in this category include $B \rightarrow H_c H_c X_s$, $B \rightarrow D^* + n \cdot h_u$, Other $B\bar{B}$ and the Combinatorial background. We use the one extra charged track sideband to constrain $B\bar{B}$ decays in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ and $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ channels and the high $E_{\text{ECL}}^{\text{extra}}$ sideband defined as $1.5 < E_{\text{ECL}}^{\text{extra}} < 2$ GeV in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ and $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ to constrain these decays.

We first compare the performance of the FEI on misreconstructed $B\bar{B}$ events between four regions of interest. We include only the FEI modes in which we find at least 20 events, so that the uncertainties and the χ^2 statistic that we calculate are sensible.

Figure 39 and 40 shows the percentages of the different decay modes reconstructed by the FEI for the four regions of interest for the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ channel in the positive and negative helicity bins. Since both the one extra charged track and the high $E_{\text{ECL}}^{\text{extra}}$ regions are compatible with the SR we choose to use the one extra charged track region due to higher statistics and better compatibility of the background composition. In Appendix J we provide several plots that test the compatibility of the background composition as per the request of the WG reviewers.

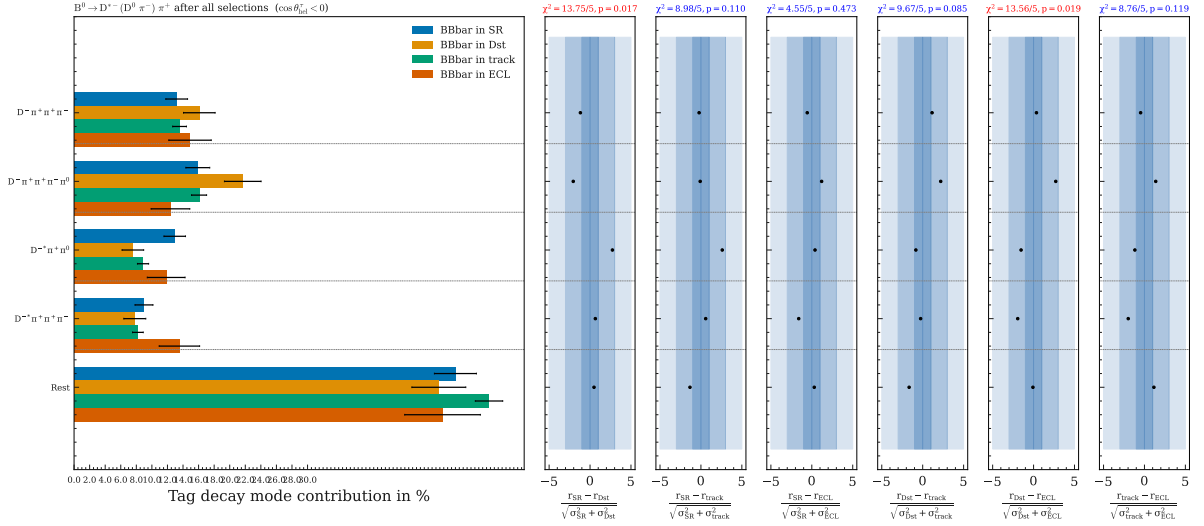


Figure 39: FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ in the negative helicity bin.

Figures 41 - 44 provide the same test for the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$. Based on these findings we choose the high $E_{\text{ECL}}^{\text{extra}}$ sideband.

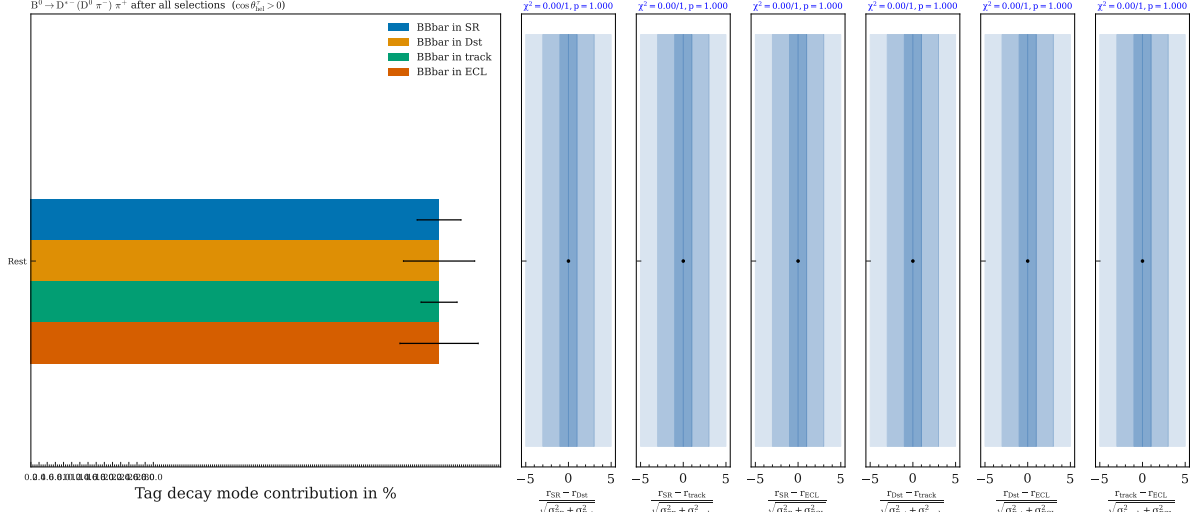


Figure 40: FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ in the positive helicity bin.

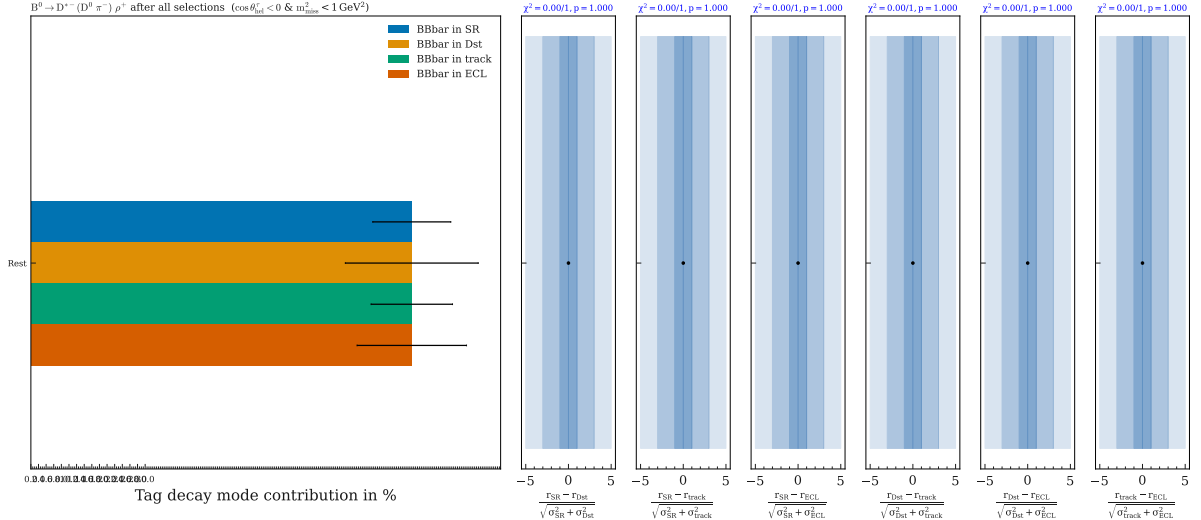


Figure 41: FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ in the negative helicity bin and low m_{miss}^2 .

Figures 45 and 46 provide the same test for the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$. Based on these findings we choose the one extra charged track sideband.

Figures 47 - 50 provide the same test for the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$. Based on these findings we choose the high $E_{\text{ECL}}^{\text{extra}}$ sideband.

For the extraction variable we fine tune 1D or 2D binnings of combinations of the difference of the invariant masses of the $D^{*\pm}$ and D^0 mesons, mass of the D^0 meson or $\cos\text{TBT0}$ as they provide good discrimination power between $B\bar{B}$ events compared to $q\bar{q}$, which are the main backgrounds in these regions.

All the above results together with Appendix J indicate that the chosen sidebands are

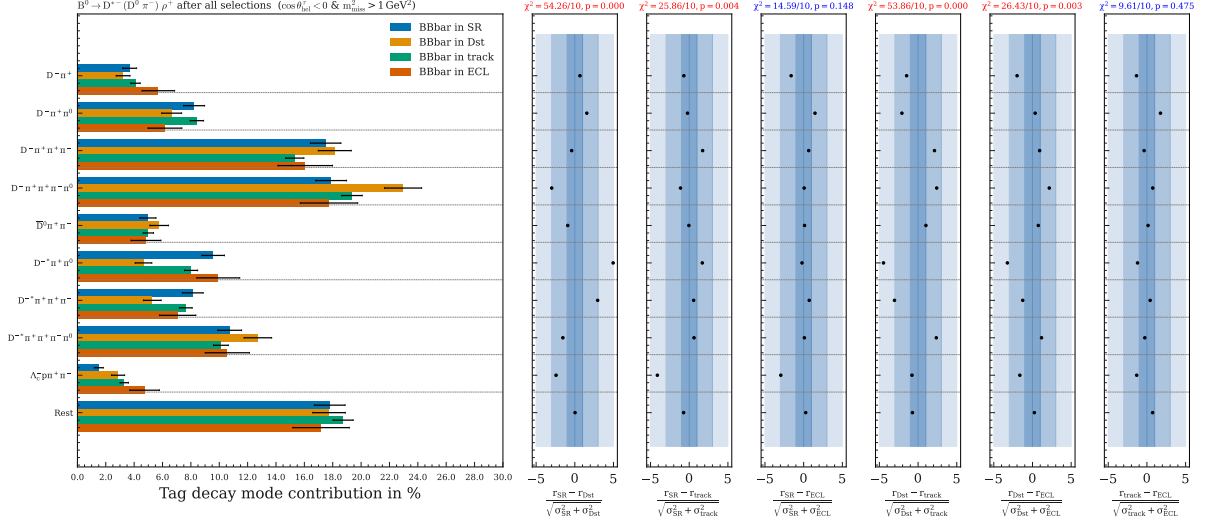


Figure 42: FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ in the negative helicity bin and high m_{miss}^2 .

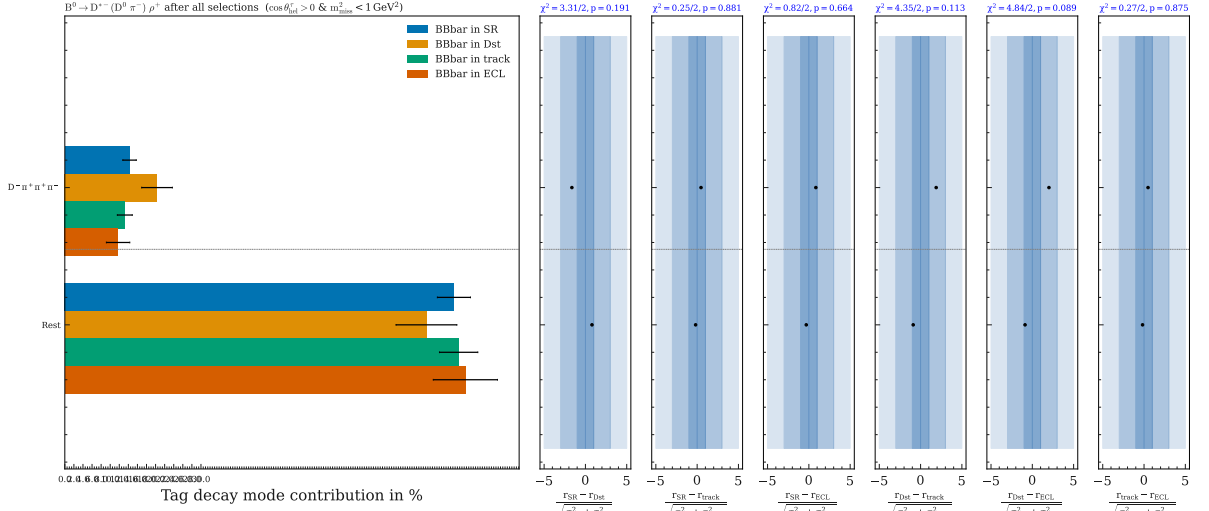


Figure 43: FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ in the positive helicity bin and low m_{miss}^2 .

appropriate regions for constraining the $B\bar{B}$ events in the SR. Combining this with the constraints determined on off-resonance data for $q\bar{q}$ events, which will be applied on both SR and control regions, we are confident that the main two background sources in the SR will be properly constrained.

Rather than using a two-step calibration process —where we would first perform a fit and then apply a calibration factor to the $B\bar{B}$ events in the SR, we will fit these control regions simultaneously with the SR and NR. This approach allows the yields of $B\bar{B}$ template in the SR and NR to be constrained directly from the control regions while consistently addressing systematic uncertainties.

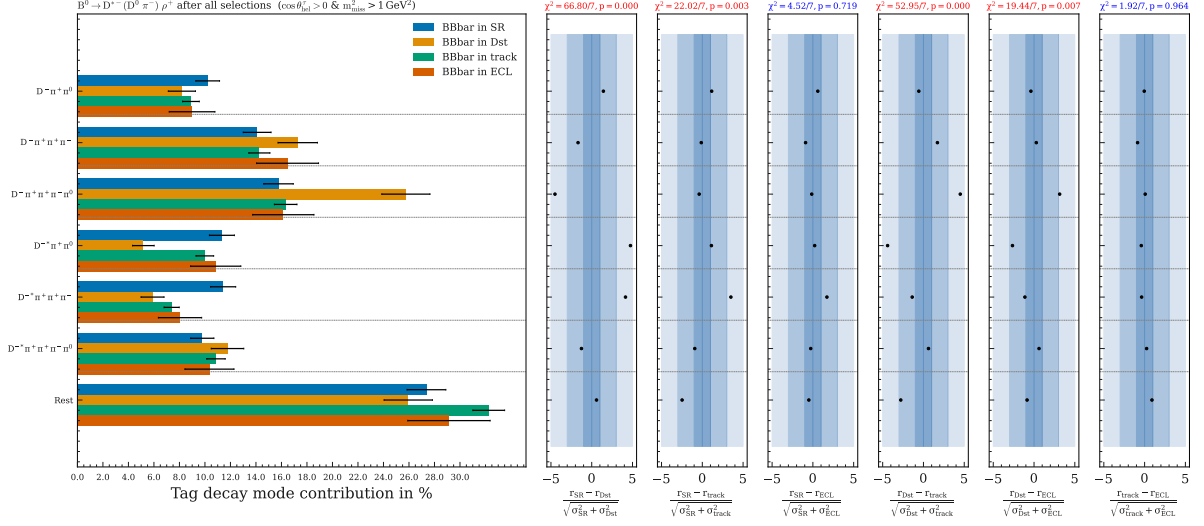


Figure 44: FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ in the positive helicity bin and high m_{miss}^2 .

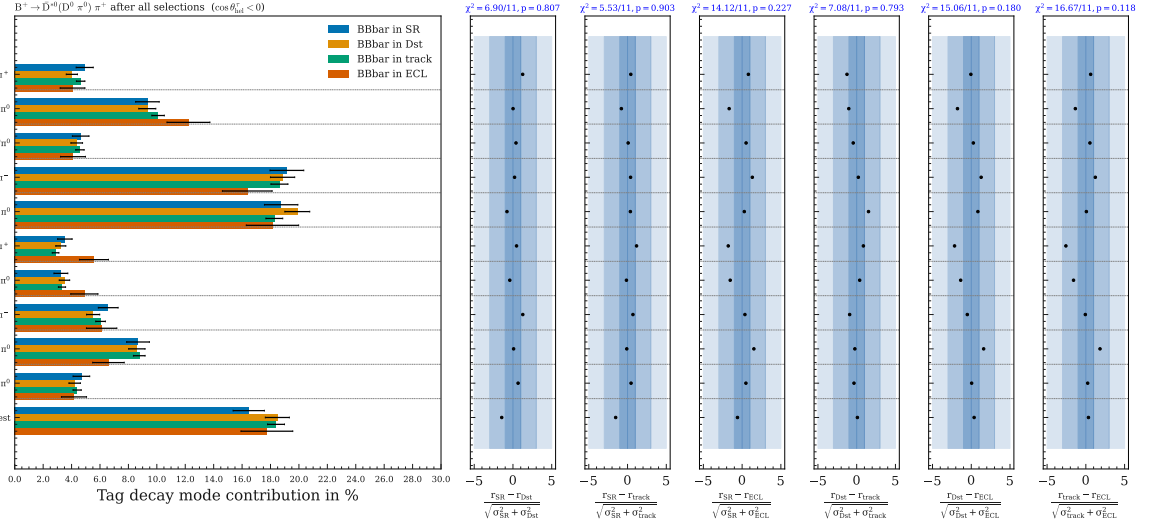


Figure 45: FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ in the negative helicity bin.

In Table 42 we present and overview of the definitions of the regions to be fitted, the motivation, the observable that is fitted and the binning used.

6.4 Sensitivity on Asimov data

To validate that the maximum likelihood fit is performing as expected, we begin by performing the fit on an Asimov dataset. Initially, we conduct the fit without including the control regions that constrain the backgrounds, but instead fixing them to their expected values. This represents an idealized scenario where we fully trust the Belle II MC. How-

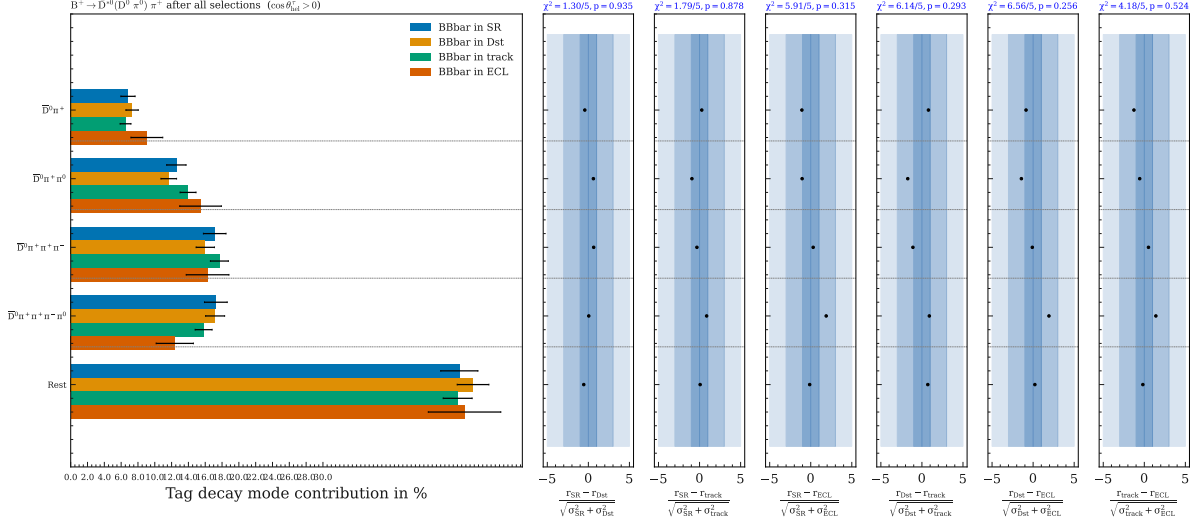


Figure 46: FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ in the positive helicity bin.

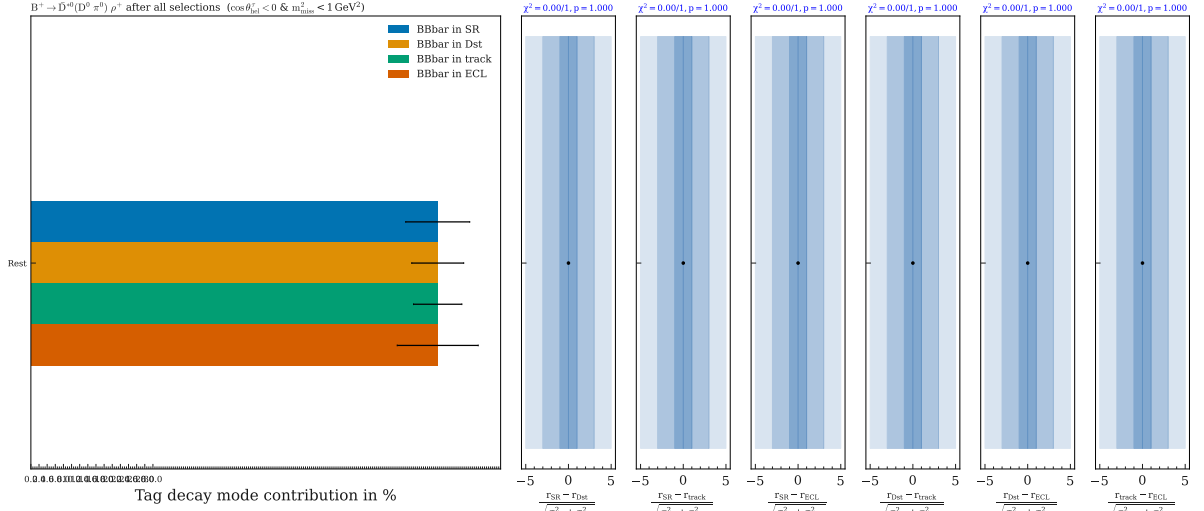


Figure 47: FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ in the negative helicity bin and low m_{miss}^2 .

ever, as discussed in Section 6.3, this assumption is unrealistic, as several $B\bar{B}$ and $q\bar{q}$ processes are known to be mismodeled in the Belle II MC. Next, we explore the opposite extreme, where all backgrounds are allowed to float freely in the fit. This combined with the previous exercise provides the possible uncertainty range for $R(D^*)$ based on the current statistical power, as any background-constraining strategy would yield an uncertainty that falls between these two extremes.

Before performing the full fit with all regions included, we first run the fit using only the control regions. This step confirms that the chosen control regions offer better sensitivity in estimating the background yields compared to the SR and NR. Thus, including

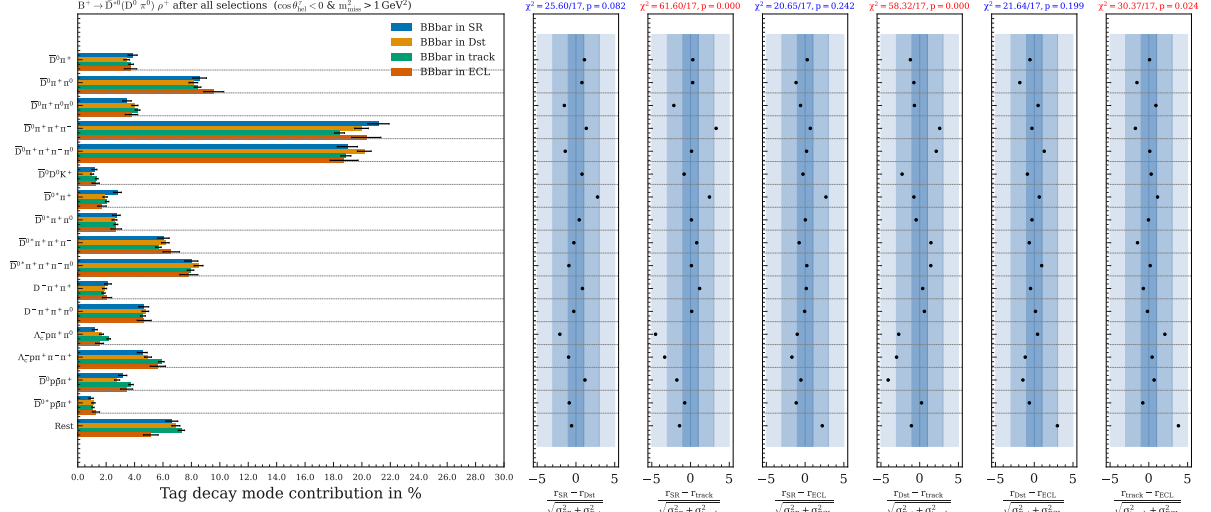


Figure 48: FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ in the negative helicity bin and high m_{miss}^2 .

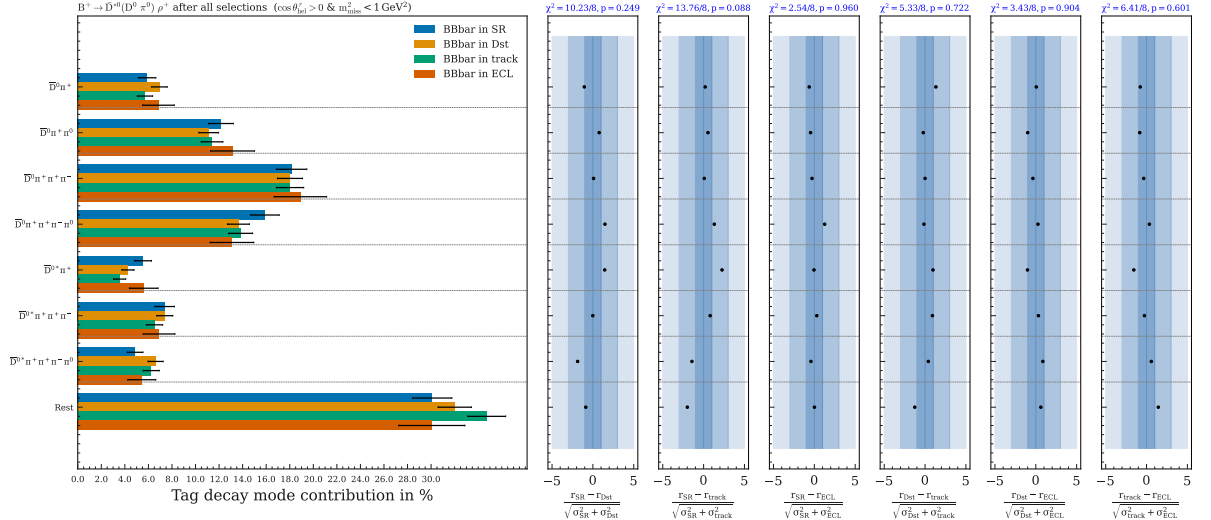


Figure 49: FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ in the positive helicity bin and low m_{miss}^2 .

them in the final fit provides a meaningful improvement. For this fit, we don't allow the SL templates $B \rightarrow D^* \tau \nu$, $B \rightarrow D^* \ell \nu$, and $B \rightarrow D^{**} \ell \nu$ to float freely, but we rather first group $B \rightarrow D^* \tau \nu$ and $B \rightarrow D^* \ell \nu$ together and then let the templates float within 99% of their expected values by implemented a Gaussian constrain on them. This approach offers two key advantages. First, it completely masks the $R(D^*)$ value, allowing us to later run the fit on experimental data to confirm its stability and determine scaling factors without the risk of inadvertently "opening the box." This is particularly useful for studying Data/MC agreement in other sidebands. Second, it enhances the sensitivity of background yield estimations in the control region fits. Given the limited statistics for the

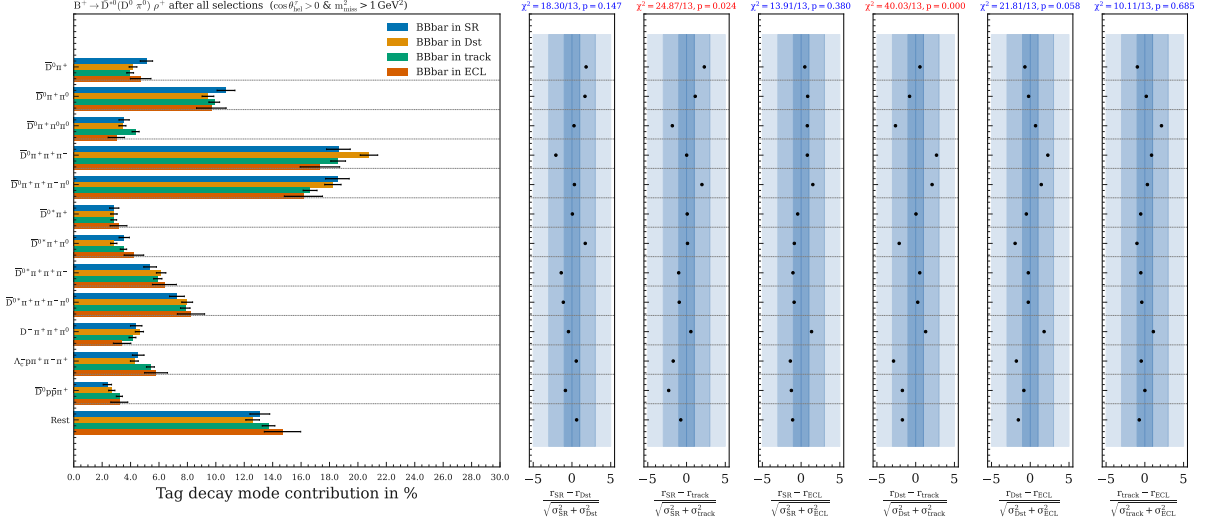


Figure 50: FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ in the positive helicity bin and high m_{miss}^2 .

Table 42: Overview of fit regions

Reconstruction	Target sample	Label	Extra cut (cf. SR)	Variables (bins)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau > 0$	m_{miss}^2 (2), $E_{\text{ECL}}^{\text{extra}}$ (4)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau < 0$	m_{miss}^2 (2), $E_{\text{ECL}}^{\text{extra}}$ (4)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau > 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$E_{\text{ECL}}^{\text{extra}}$ (4)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau < 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$E_{\text{ECL}}^{\text{extra}}$ (4)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau > 0$ and $m_{\text{miss}}^2 > 1 \text{ GeV}^2$	$E_{\text{ECL}}^{\text{extra}}$ (4)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau < 0$ and $m_{\text{miss}}^2 > 1 \text{ GeV}^2$	$E_{\text{ECL}}^{\text{extra}}$ (4)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau > 0$	m_{miss}^2 (2), $E_{\text{ECL}}^{\text{extra}}$ (4)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau < 0$	m_{miss}^2 (2), $E_{\text{ECL}}^{\text{extra}}$ (4)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau > 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$E_{\text{ECL}}^{\text{extra}}$ (4)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau < 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$E_{\text{ECL}}^{\text{extra}}$ (4)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau > 0$ and $m_{\text{miss}}^2 > 1 \text{ GeV}^2$	$E_{\text{ECL}}^{\text{extra}}$ (4)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	$B \rightarrow D^* \tau \nu$	SR	$\cos \theta_{hel}^\tau < 0$ and $m_{\text{miss}}^2 > 1 \text{ GeV}^2$	$E_{\text{ECL}}^{\text{extra}}$ (4)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$	$B \rightarrow D^* \ell \nu$	NR	-	m_{miss}^2 (3)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$	$B \rightarrow D^* \ell \nu$	NR	-	m_{miss}^2 (3)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$	$B\bar{B}$	BR01_pos	1 extra track and $\cos \theta_{hel}^\tau > 0$	$\cos \text{TBT0}$ (2)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$	$B\bar{B}$	BR01_neg	1 extra track and $\cos \theta_{hel}^\tau < 0$	$\cos \text{TBT0}$ (2)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	$B\bar{B}$	BR02_pos.m1	$1.5 < E_{\text{ECL}}^{\text{extra}} < 2 \text{ GeV}$ and $\cos \theta_{hel}^\tau > 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$\Delta M(D^*)$ (5)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	$B\bar{B}$	BR02_neg.m1	$1.5 < E_{\text{ECL}}^{\text{extra}} < 2 \text{ GeV}$ and $\cos \theta_{hel}^\tau < 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$\cos \text{TBT0}$ (2)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	$B\bar{B}$	BR02_pos.m2	$1.5 < E_{\text{ECL}}^{\text{extra}} < 2 \text{ GeV}$ and $\cos \theta_{hel}^\tau > 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$\cos \text{TBT0}$ (2)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	$B\bar{B}$	BR02_neg.m2	$1.5 < E_{\text{ECL}}^{\text{extra}} < 2 \text{ GeV}$ and $\cos \theta_{hel}^\tau < 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$\cos \text{TBT0}$ (2)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$	$B\bar{B}$	BRp1_pos	1 extra track and $\cos \theta_{hel}^\tau > 0$	$\cos \text{TBT0}$ (3) / $M(D^0)$ (5)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$	$B\bar{B}$	BRp1_neg	1 extra track GeV and $\cos \theta_{hel}^\tau < 0$	$\cos \text{TBT0}$ (3) / $M(D^0)$ (5)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	$B\bar{B}$	BRp2_pos.m1	$1.5 < E_{\text{ECL}}^{\text{extra}} < 2 \text{ GeV}$ and $\cos \theta_{hel}^\tau > 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$\cos \text{TBT0}$ (2)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	$B\bar{B}$	BRp2_neg.m1	$1.5 < E_{\text{ECL}}^{\text{extra}} < 2 \text{ GeV}$ and $\cos \theta_{hel}^\tau < 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$\cos \text{TBT0}$ (2) / $M(D^0)$ (5)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	$B\bar{B}$	BRp2_pos.m2	$1.5 < E_{\text{ECL}}^{\text{extra}} < 2 \text{ GeV}$ and $\cos \theta_{hel}^\tau > 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$\cos \text{TBT0}$ (2)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	$B\bar{B}$	BRp2_neg.m2	$1.5 < E_{\text{ECL}}^{\text{extra}} < 2 \text{ GeV}$ and $\cos \theta_{hel}^\tau < 0$ and $m_{\text{miss}}^2 < 1 \text{ GeV}^2$	$\cos \text{TBT0}$ (2) / $M(D^0)$ (5)
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$	$B \rightarrow D^{**} \ell \nu$	Dstst0	1 extra track	m_{miss}^2 (3)
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$	$B \rightarrow D^{**} \ell \nu$	Dststp	1 extra track	m_{miss}^2 (3)

SL processes in these regions, estimating two constrained nuisance parameters per region is more manageable than minimizing three completely free parameters per regions, which would be significantly more challenging.

Finally, we perform the fit with all regions included: SR, NR, BR, CR and BCR. We

confirm that the combined fit enhances sensitivity for all yields compared to individual fits, with the most significant improvement observed for $q\bar{q}$ processes, for which the control regions offer greater sensitivity. The $R(D^*)$ and P_τ uncertainties falls within the expected range, as defined by the fits where all backgrounds are either fixed or allowed to float. The results are summarized in Table 43.

No systematic effects have been included in these fits, so the uncertainties reported are purely statistical. In all fits, closure is achieved, meaning all free parameters converge to their nominal values. In Appendix L we provide a grid modifier plot that presents all the free and fixed parameters of the fit, as this plot is provided by `cabinetry`. As this plot spans across multiple pages we provide it in the Appendix to maintain flow in the main text.

Table 43: Asimov fit uncertainties of all free unconstrained parameters for the simultaneous $R(D^*)$ and P_τ fit. The uncertainties are presented in terms of percentages in case the free parameter's nominal value is not 1. The nominal values of all parameters is presented in parenthesis in the first column. We confirm that we observe full closure in all our Asimov fits, i.e. all free parameters are minimized to their nominal values.

Scale factors/ Regions	SR, NR (BKGs fixed)	SR, NR, (BKGs floating)	CRs	SR, NR, CRs
$R(D^*)$ (0.258)	± 0.0330 (12.79%)	± 0.0443 (16.66%)	-	± 0.0387 (13.91%)
P_τ (-0.497)	± 0.7799	± 0.9281	-	± 0.8223
$B \rightarrow D^* \ell \nu$ (1.0)	± 0.0138	± 0.0142	-	± 0.0141
$B \rightarrow D^{**} \ell \nu$ ($B^0 \bar{B}^0$) (1.0)	-	± 0.0624	± 0.1191	± 0.0552
$B \rightarrow D^{**} \ell \nu$ ($B^+ B^-$) (1.0)	-	± 0.0483	± 0.0842	± 0.0419
$B\bar{B}$ ($B^0 \bar{B}^0$, $\tau \rightarrow \pi \nu_\tau$) $\cos \theta_{hel}^\tau > 0$ (1.0)	-	± 0.1765	± 0.1039	± 0.0894
$B\bar{B}$ ($B^0 \bar{B}^0$, $\tau \rightarrow \pi \nu_\tau$) $\cos \theta_{hel}^\tau < 0$ (1.0)	-	± 0.137	± 0.0567	± 0.0522
$B\bar{B}$ ($B^0 \bar{B}^0$, $\tau \rightarrow \rho \nu_\tau$) $\cos \theta_{hel}^\tau > 0$ $m_{miss}^2 < 1$ (1.0)	-	± 0.308	± 0.3021	± 0.2155
$B\bar{B}$ ($B^0 \bar{B}^0$, $\tau \rightarrow \rho \nu_\tau$) $\cos \theta_{hel}^\tau > 0$ $m_{miss}^2 > 1$ (1.0)	-	± 1.2195	± 0.0754	± 0.0641
$B\bar{B}$ ($B^0 \bar{B}^0$, $\tau \rightarrow \rho \nu_\tau$) $\cos \theta_{hel}^\tau < 0$ $m_{miss}^2 < 1$ (1.0)	-	± 0.1081	± 0.164	± 0.09
$B\bar{B}$ ($B^0 \bar{B}^0$, $\tau \rightarrow \rho \nu_\tau$) $\cos \theta_{hel}^\tau < 0$ $m_{miss}^2 > 1$ (1.0)	-	± 0.073	± 0.1108	± 0.0605
$B\bar{B}$ ($B^+ B^-$, $\tau \rightarrow \pi \nu_\tau$) $\cos \theta_{hel}^\tau > 0$ (1.0)	-	± 0.1108	± 0.0623	± 0.0542
$B\bar{B}$ ($B^+ B^-$, $\tau \rightarrow \pi \nu_\tau$) $\cos \theta_{hel}^\tau < 0$ (1.0)	-	± 0.1021	± 0.0353	± 0.0333
$B\bar{B}$ ($B^+ B^-$, $\tau \rightarrow \rho \nu_\tau$) $\cos \theta_{hel}^\tau > 0$ $m_{miss}^2 < 1$ (1.0)	-	± 0.0928	± 0.1257	± 0.0745
$B\bar{B}$ ($B^+ B^-$, $\tau \rightarrow \rho \nu_\tau$) $\cos \theta_{hel}^\tau > 0$ $m_{miss}^2 > 1$ (1.0)	-	± 0.059	± 0.0854	± 0.0484
$B\bar{B}$ ($B^+ B^-$, $\tau \rightarrow \rho \nu_\tau$) $\cos \theta_{hel}^\tau < 0$ $m_{miss}^2 < 1$ (1.0)	-	± 0.2371	± 0.2323	± 0.1659
$B\bar{B}$ ($B^+ B^-$, $\tau \rightarrow \rho \nu_\tau$) $\cos \theta_{hel}^\tau < 0$ $m_{miss}^2 > 1$ (1.0)	-	± 0.0519	± 0.0582	± 0.0386

In Figure 51 we provide the correlation matrix for all free parameters in the fit. From there one can extract the statistical correlation coefficient between $R(D^*)$ and P_τ which is 0.54. For comparison the reported correlation between the parameters in the Belle analysis [4] was 0.29, however that was determined differently since the Belle analysis did not extract the two parameters simultaneously.

In Figures 52 - 55 we present the post fit distributions on Asimov data. Not pulls are observed in any of the bins.

6.5 Statistical validation of $R(D^*)$ and P_τ fit

To confirm the statistical validity of the $R(D^*)$ estimator, we conduct additional tests using an Asimov dataset. Our primary goal is to ensure that the maximum likelihood fit provides an unbiased estimate of $R(D^*)$ and all other yields. To achieve this, we generate pseudo datasets, referred to as "toys."

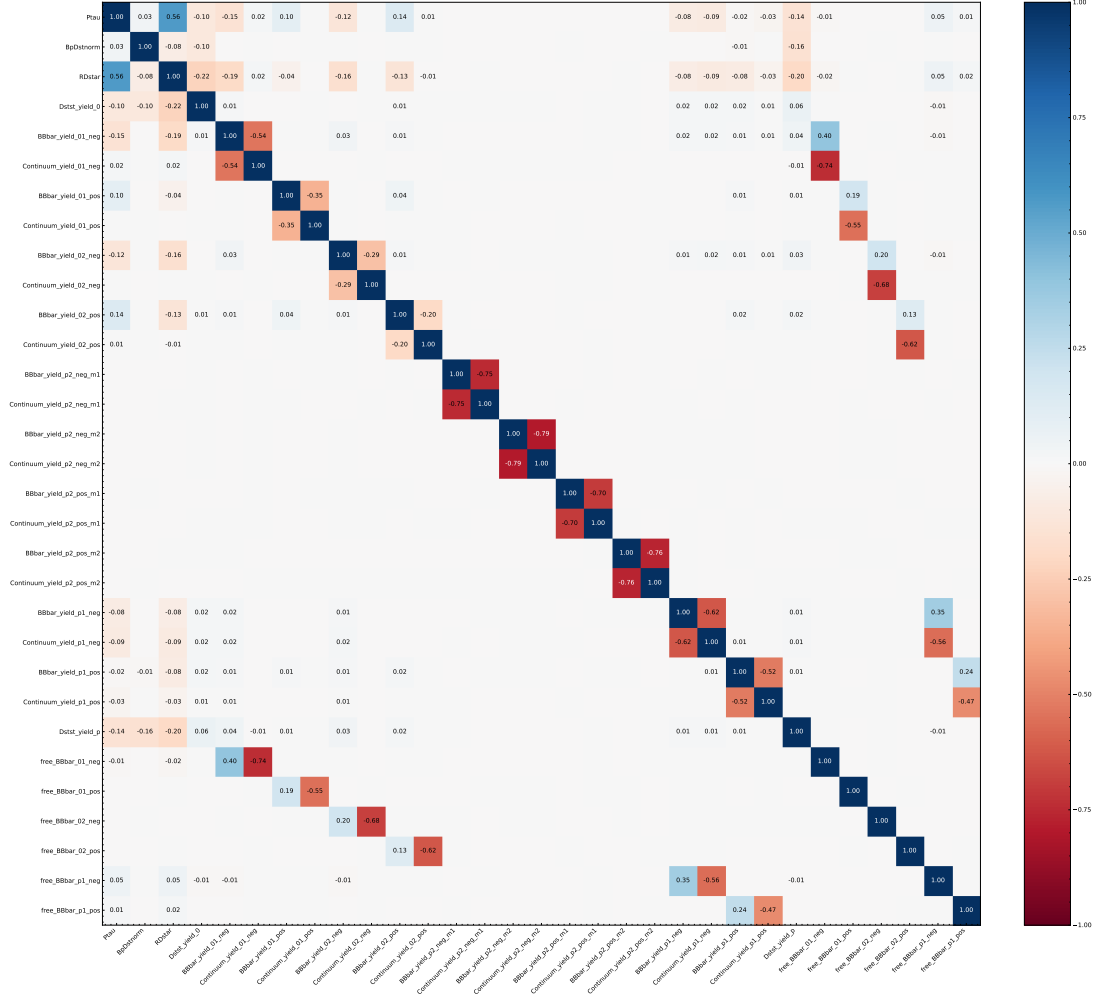


Figure 51: Statistical only correlation matrix of the P_τ and $R(D^*)$ fit.

Each toy dataset is created by taking the expected event counts from the Asimov dataset and re-sampling the counts for each bin from a Poisson distribution. We generate 2,500 toy datasets, where the event counts in each bin are replaced by random draws from these Poisson distributions. We then fit each toy dataset to the nominal templates, determining the value of $R(D^*)$ and the scale factor for the yield of each decay process. For each fitted parameter, we calculate the pull. The pull of an estimated parameter is defined as

$$\text{pull} = \frac{\hat{\theta} - \theta_{\text{true}}}{\sigma_{\hat{\theta}}} \quad (47)$$

We then histogram the pull distribution and fit it to a standard normal distribution. If the maximum likelihood fit provides an unbiased estimate of the true value of the parameter of interest, the pull distribution should be compatible with a standard normal. When the pulls are symmetrically distributed around zero, the numerator of Equation 47 is, on average, zero. This indicates that the estimator does not introduce any bias by favoring larger or smaller values for the parameter.

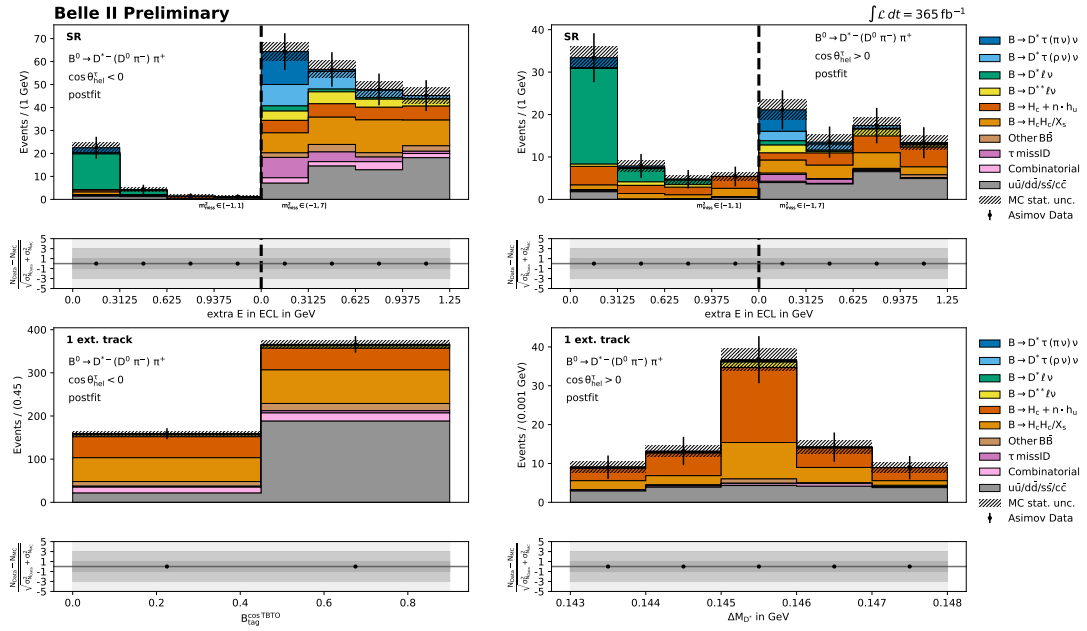


Figure 52: Postfit distributions on Asimov in the SR and control regions for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$

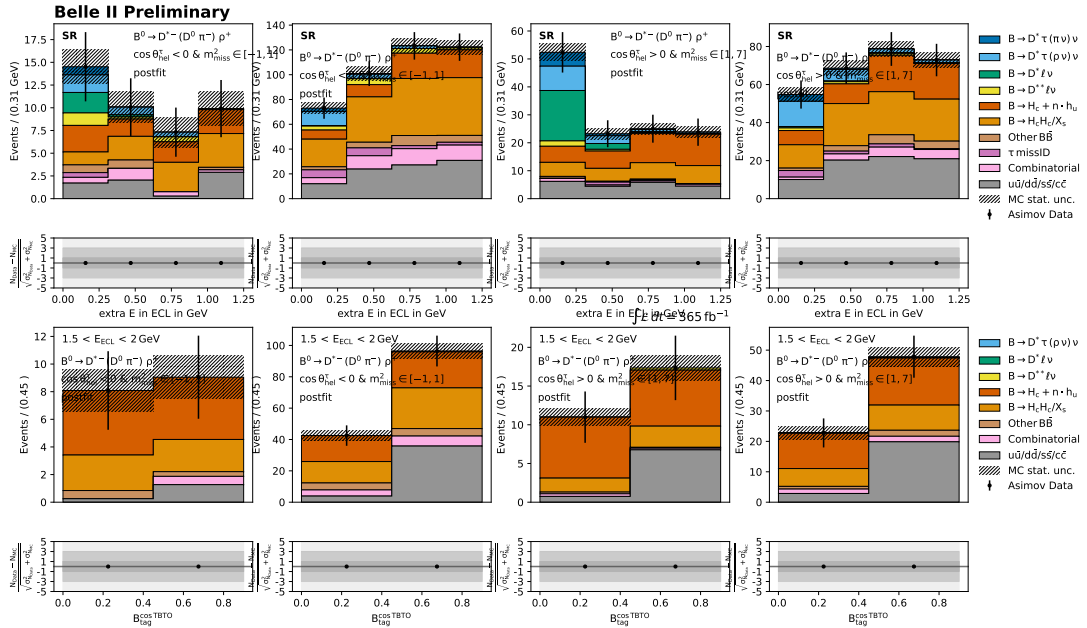


Figure 53: Postfit distributions on Asimov in the SR and control regions for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$

Furthermore, if the standard deviation of the pull distribution is close to one, we can be confident that the estimated uncertainty of the parameter accurately reflects the true variability due to statistical fluctuations. In Figure 56 we show the pull plots for $R(D^*)$, P_τ and the strength of the $B \rightarrow D^* \ell \nu$ template. For $R(D^*)$ and the strength of the

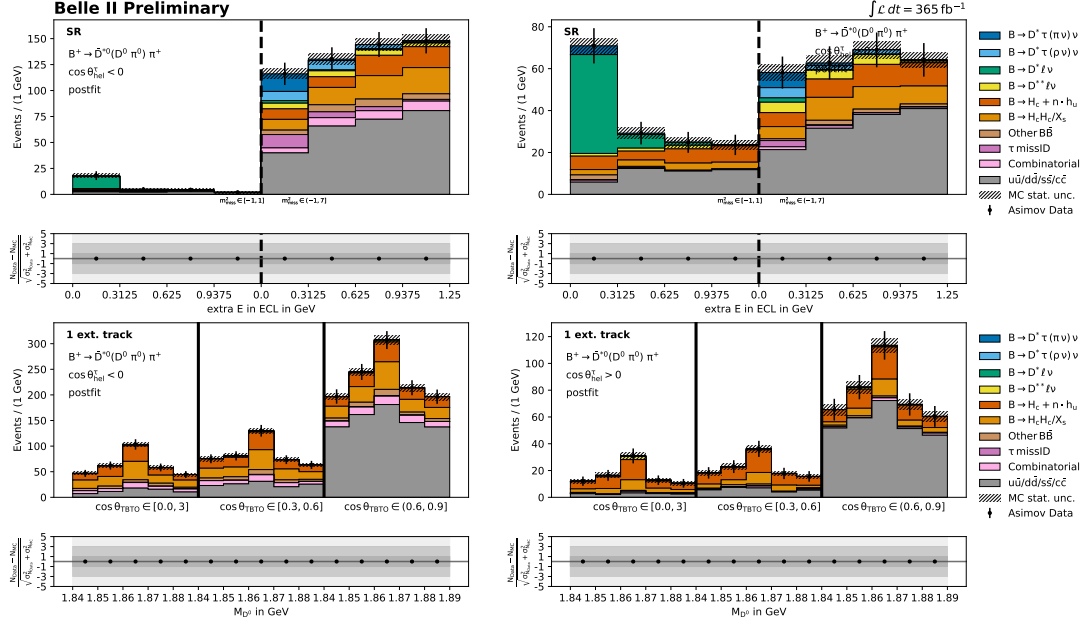


Figure 54: Postfit distributions on Asimov in the SR and control regions for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$

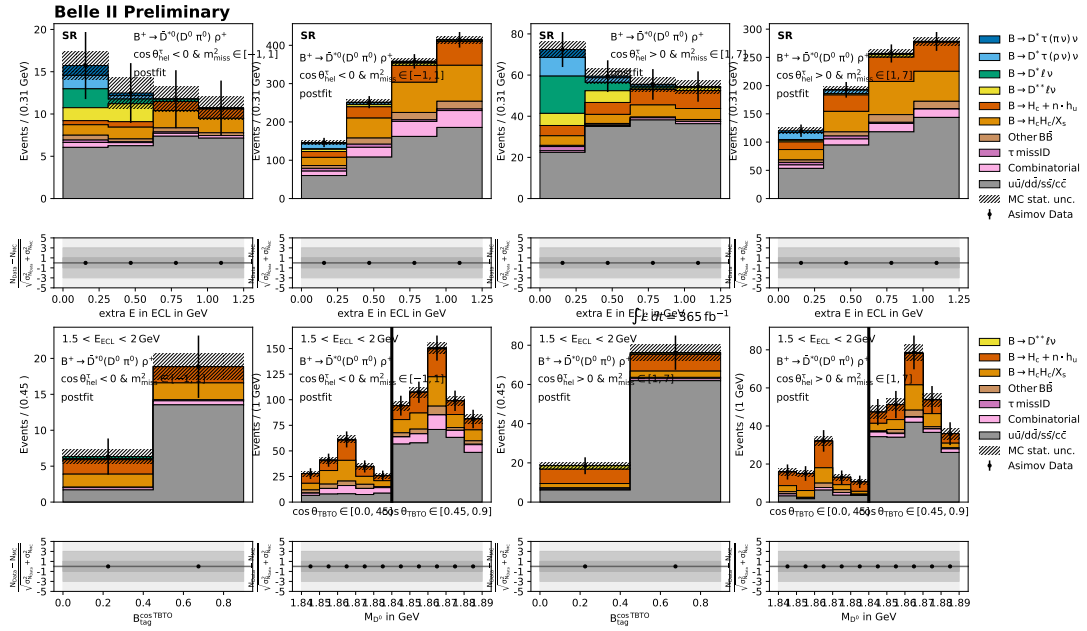


Figure 55: Postfit distributions on Asimov in the SR and control regions for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$

1340 $B \rightarrow D^* \ell \nu$ template we use the HESSE error from the fit. Since P_τ due to it's definition
 1341 and range can have asymmetric errors, especially when throwing toys, we use the MINOS
 1342 error to calculate the pull of every toy. When the nominator of the pull is larger than zero,
 1343 we use the lower MINOS uncertainty. When the nominator of the pull is smaller than zero

we use the upper MINOS uncertainty. In Appendix G we provide the pull distributions for all other free parameters of the fit. We observe that all pull distributions are compatible with a standard normal, confirming that our maximum likelihood fit provides an unbiased estimate of $R(D^*)$ and all other yields, and that the uncertainties accurately represent the true variability in the data.

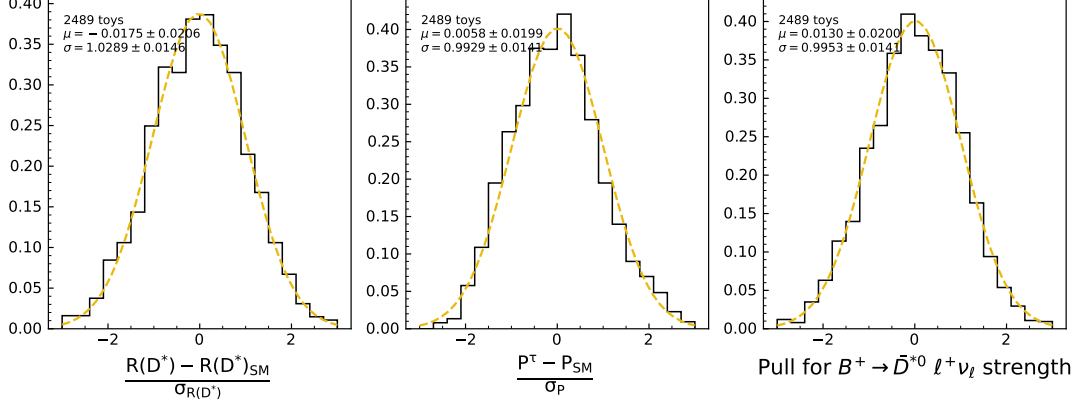


Figure 56: Pull plots for toys for $R(D^*)$, P_τ and the yield of $B \rightarrow D^* \ell \nu$ decays. The MINOS error has been used to calculate the pull of P_τ for every toy. A negligible amount of toys (11/2500) did not converge, therefore we excluded them from the pull distribution. The parameters of the gaussian fit are compatible with a standard normal.

Another important test is evaluating how our estimator responds to changes in the observed data. To test this, we bias the Asimov dataset in favor of a specific decay process. This is done by adjusting the event counts in the Asimov dataset according to the distribution of a particular nominal template.

In Figure 57, we show this procedure, where the Asimov datasets are constructed with contributions from each template varying from 50% to 200%. We fit these biased Asimov datasets to the nominal templates and plot the estimated parameter values. In all cases, we observe that the parameter estimates scale as expected. The estimated value of $R(D^*)$ scales linearly when the contribution from $B \rightarrow D^* \tau \nu$ decays is varied. Similarly, the yield of $B \rightarrow D^* \ell \nu$ scales linearly, while $R(D^*)$ scales as $\frac{1}{x}$ when the $B \rightarrow D^* \ell \nu$ contribution is modified. Finally, all background yields scale linearly with changes in their respective contributions, while both $R(D^*)$ and the yield of $B \rightarrow D^* \ell \nu$ remain constant.

All the above test give us confidence that the maximum likelihood fit that we have implemented is a robust estimator of the $R(D^*)$ and all other related yields.

Lastly we perform a linearity test for the polarization measurement as the linearity of the estimator on $R(D^*)$ the normalization strength and all the background scale factors have been tested previously, but not the one of P_τ . In order to carry out the linearity test on P_τ we evaluate the full likelihood with a certain set of parameters. We keep all parameters to their nominal values except P_τ which is varied from -0.9443 to -0.0497 in 20 steps of -0.0497. By evaluating the likelihood we get a new dataset every time that corresponds to the expected observed data for this set of free parameters. We then fit this expected dataset to our nominal templates. Figure 58 shows the results of this study.

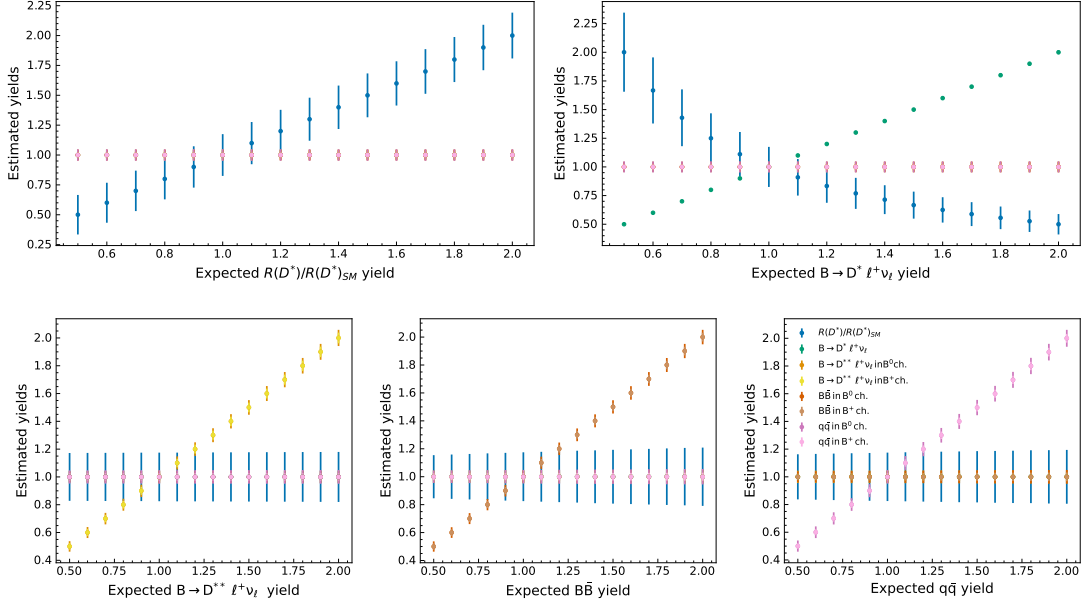


Figure 57: The response of the estimator scales as expected when we bias the Asimov dataset.

1371 P_τ scales as expected while the other POIs are minimized to their nominal values. We
 1372 conclude that the estimator's response scales as expected.

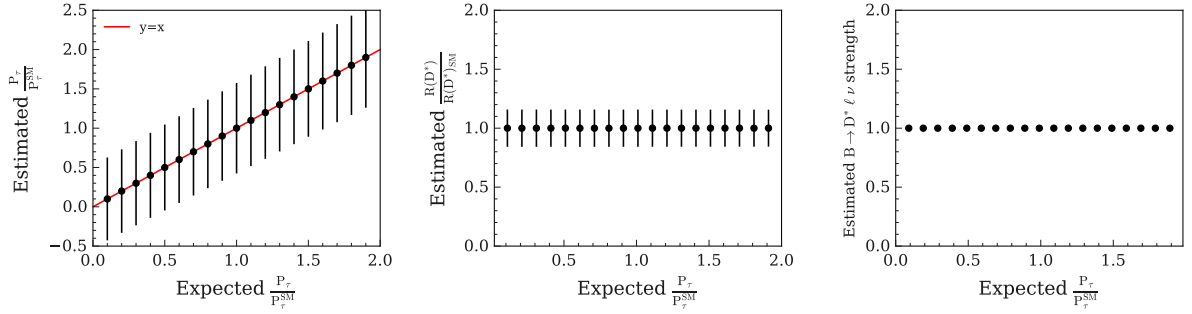


Figure 58: Linearity test for the three POIs in the simultaneous fit. P_τ scales as expected while the other POIs are minimized to their nominal values. We conclude that the estimator's response scales as expected.

7 Systematics

We adopt a unified framework for handling all systematic uncertainties. For each source of systematic uncertainty, we apply the SysVar eigendecomposition procedure, as described in Section 3.1.4. All systematic uncertainties considered in this analysis can be categorized into three distinct classes:

1. Fully correlated correction weights
2. Uncorrelated correction weights
3. Partially correlated correction weights

SysVar has been designed to handle systematics with partial correlations, such as LID. For the first two categories, the eigendecomposition procedure defaults to well-defined limiting cases. In the fully correlated case, a single dominant eigendirection results in **one nuisance parameter in total**. In the uncorrelated case, **each correction weight yields one nuisance parameter**, corresponding to its own dominant eigendirection. While implementing shape systematics for these two limiting cases is relatively straightforward without the full machinery of SysVar, we nonetheless perform the basis transformation to maintain a consistent treatment across all systematic uncertainties. This also serves to validate SysVar’s behavior in these well-understood edge cases.

For each source of systematic uncertainty, the eigendecomposition is performed across all relevant reconstruction channels—including those used in control regions for calibrations, whether included in a simultaneous fit or not—and all applicable templates. For instance, in the case of the tracking efficiency uncertainty (see Section 5.5.2), we include all reconstruction channels and templates, as the impact of this effect is universal. Conversely, for LID fake rates (see Section 5.5.8), we restrict the PCA to reconstruction channels that involve actual lepton reconstruction, and exclude templates that cannot be affected by fake leptons—such as the $B \rightarrow D^* \ell \nu$ template. As a final example, for uncertainties related to BF of the light lepton gap modes (see Section 5.7), we consider all reconstruction channels but include only the relevant template—specifically, the $B \rightarrow D^{**} \ell \nu$ template.

For each systematic uncertainty, we account for both an additive (shape-related) component and a multiplicative (normalization-related) component. The eigenvariations derived from Equation 4 inherently capture both effects within each template. Disentangling the normalization component is straightforward: we compute the sum of bin contents for the up or down eigenvariation and divide it by the sum of the bin contents in the nominal histogram. This ratio gives the relative normalization effect of the systematic, which can be directly implemented as a `normsys` modifier, as defined in Table 2 of the `pyhf` framework. To isolate the shape-only effect, we divide each bin of the eigenvariation by the relative normalization factor. This effectively re-normalizes the variation to match the total yield of the nominal histogram, ensuring that the resulting `histosys` modifier (also defined in Table 2) only alters the shape of the template without affecting its overall normalization. Notably, the absolute up and down variations—whether normalized or not—are already in the exact form required by `pyhf` to implement the `histosys` modifier, making the technical implementation both clean and efficient. Accordingly, we introduce two nuisance parameters per template per eigendirection: one to account for the additive

(shape) component, and one for the multiplicative (normalization) component. These two nuisance parameters may be treated as either fully correlated or fully uncorrelated, depending on the nature of the underlying systematic uncertainty. The choice of correlation is made on a case-by-case basis, guided by physical intuition and the behavior of each specific systematic.

For every source of systematic uncertainty we generate 500 variations of the corrections weights. By building 500 template variations we get the full covariance matrix of the analysis that is used for the eigendecomposition. In order to determine the important number of eigendirections we set the precision to 10^{-8} . That is we implement as many eigendirections as fully correlated NPs as needed to reconstruct the original covariance matrix with differences smaller than 10^{-8} in the orthogonal space.

In the following, we first present the full systematic uncertainty budget for the simultaneous $R(D^*)$ and P_τ fit. We can provide detailed information for each individual systematic uncertainty using SysVar’s visualization API per request during the review. SysVar’s visualization API automatically plots the correction weight variations, the covariance matrix of the corrections, either explicitly defined or build from the different set of uncertainties, the varied templates across all reconstruction channels and the full correlation matrix across all bins, templates, reconstruction channels. We avoid providing this information for all systematics by default in this document since this would result to an extremely large number of plots.

We evaluate the systematic uncertainties in a unified way as follows: We first run the nominal fit on an Asimov dataset without including any systematic uncertainties. The uncertainty on the parameter of interest we acquire from this fit corresponds to the statistical only uncertainty. We then include all NP for a particular source or category of systematic uncertainties. We then determine the total uncertainty that includes both the statistical and the uncertainty associated to that particular systematic effect. In order to isolate the systematic part of the uncertainty we subtract those two in quadrature as described in Equation 48, since the statistical and systematic part should be completely independant from each other.

$$\sigma_{syst} = \sqrt{\sigma_{tot}^2 - \sigma_{stat}^2} \quad (48)$$

In the following we present the systematic uncertainty per source for both $R(D^*)$ and P_τ .

7.1 Systematic budget for the $R(D^*)$ and P_τ fit

Table 44: Summary of Systematic Uncertainties

Source	uncertainties			
	# NP	$R(D^*)$ (0.258) %	P_τ (-0.497)	ρ (shape-norm)
data sample size	-	13.91	0.8223	-
simulation sample size	138	7.011	0.4131	-
Prompt hadronic BFs	20	3.132	0.1257	1
Double Charm BFs	24	1.144	0.0330	1
D** and gap BF	16	0.301	0.0167	1
$\tau/D^{(*)}$ BFs	1+1+1+1	~ 0	~ 0	1
π^0 efficiency	7	1.765	0.0125	0
π_{slow} efficiency	3+3	0.472	0.0201	0
FEI efficiency	23	1.973	0.0283	0
tracking efficiency	1	0.795	0.0050	0
μ ID eff.	4	0.02	0.0008	0
eID eff.	8	0.006	~ 0	0
LID fakes	9+9+9+9	~ 0	0.009	0
kID	9+9	2.818	0.0196	0
π ID	15+15	3.167	0.0206	0
μ fake π	22	2.526	0.0259	0
D ^(*) FF	9	0.143	0.0067	1
D ^(**) FF	4	1.319	0.0487	1
Photon multiplicity reweighting	14	1.283	0.0245	shape only
$\sum \sigma_{\text{syst}}$	-	9.893	0.4404	-
Total systematic unc.	-	10.094	0.4422	-

8 Agreement of simulated and experimental data

Before performing the measurement of $R(D^*)$ and P_τ in the signal region (SR) and normalization region (NR), we must ensure that all corrections and calibrations lead to an accurate modeling of the data. To achieve this, we first run the signal extraction fit exclusively in the control regions to extract scaling factors for the background templates, specifically for the $B\bar{B}$ and $q\bar{q}$ components. Our validation strategy follows a structured approach. First, we verify that the signal extraction variables are well-modeled in regions where the overall event normalization is already constrained, either through FEI calibration for correctly reconstructed events or through control fits for misreconstructed events. Once this is established, we repeat the same tests in regions where the normalization is not explicitly fixed. This allows us to simultaneously verify the shape consistency of the observables of interest and assess the validity of the derived calibration and scaling factors. The regions used in this validation process are summarized in Table 45.

Table 45: Sideband checks summary

Reconstruction	Sideband	Normalization	Motivation	Target Sample
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	off-resonance data off-resonance data off-resonance data off-resonance data	Free	Overall modelling in continuum MC	$q\bar{q}$
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	1 extra track $1.5 < E_{\text{ECL}}^{\text{extra}} < 2 \text{ GeV}$ 1 extra track $1.5 < E_{\text{ECL}}^{\text{extra}} < 2 \text{ GeV}$	Fixed	Observable modelling in misreconstructed events	$B\bar{B}$ and $q\bar{q}$
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$ $B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$	$q^2 < 4 \text{ GeV}^2$	Fixed	Observable modelling in correctly reconstructed events	$B \rightarrow D^* \ell \nu$
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	$1.25 < E_{\text{ECL}}^{\text{extra}} < 1.5 \text{ GeV}$	Free	Overall agreement in misreconstructed events	$B\bar{B}$ and $q\bar{q}$
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$ $B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$	NR	Free	Overall agreement in correctly reconstructed events	$B \rightarrow D^* \ell \nu$

In the rest of this section we present the Data/MC validation in more detail.

8.1 Off-resonance data

Since we calibrate the continuum processes using off resonance data in the sidebands that are described in Section 5.10 we begin by applying the scaling factors to the SR and the regions that are used to constrain $B\bar{B}$ processes. This tests serves as a way of validating the overall normalization and shapes of important variables of continuum events. This will give us confidence that any potential mismodelling we observe in those regions is not coming from $q\bar{q}$ events. Figures 63 - 66 show the Data/MC agreement on off resonance data with the SR selection applied. Due to relatively low statistics we used a coarse binning on $E_{\text{ECL}}^{\text{extra}}$ when needed to have a meaningful χ^2 test. We conclude that with the current statistical power there's now disagreement in continuum events.

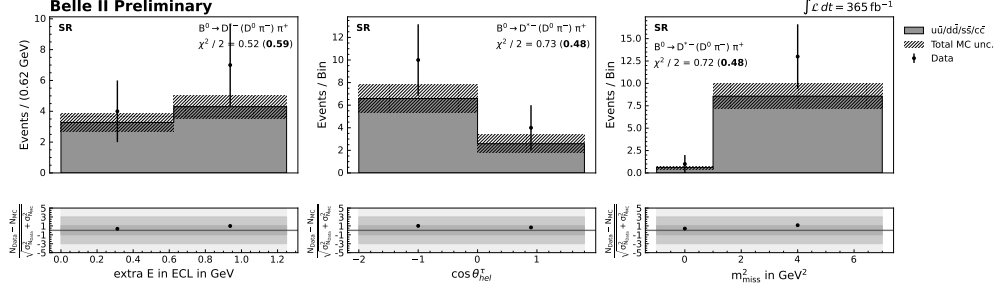


Figure 59: Fitting variables in continuum processes on off-resonance data in SR of $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$

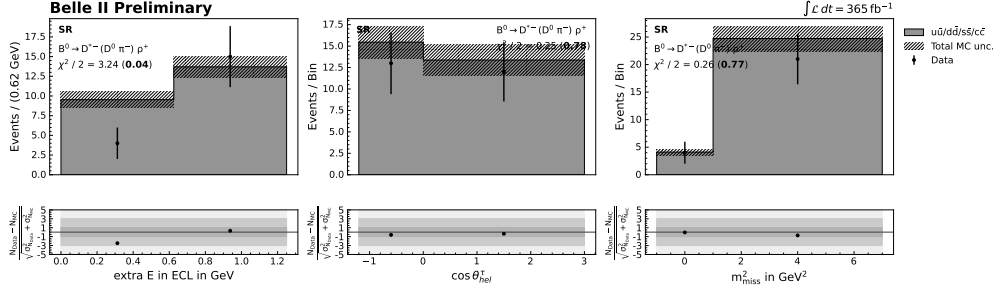


Figure 60: Fitting variables in continuum processes on off-resonance data in SR of $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$

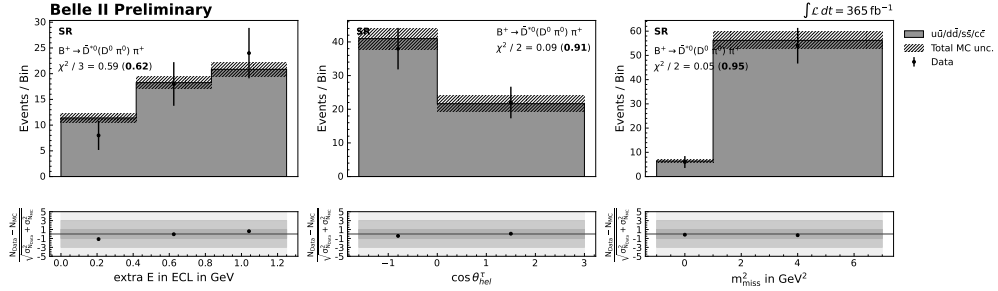


Figure 61: Fitting variables in continuum processes on off-resonance data in SR of $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$

We continue by performing the same test on off resonance data but with the selection that is used in the regions that constrain $B\bar{B}$ events. We see overall good agreement in these regions as well, therefore we conclude that all continuum processes in the regions that are included in the fit are well modelled.

8.2 Control Region fits

We first perform the fit in the control regions defined in Table 42. These regions are initially fitted separately and then simultaneously. Although the event yields are not linked between the different regions, the nuisance parameters for most systematic uncertainties

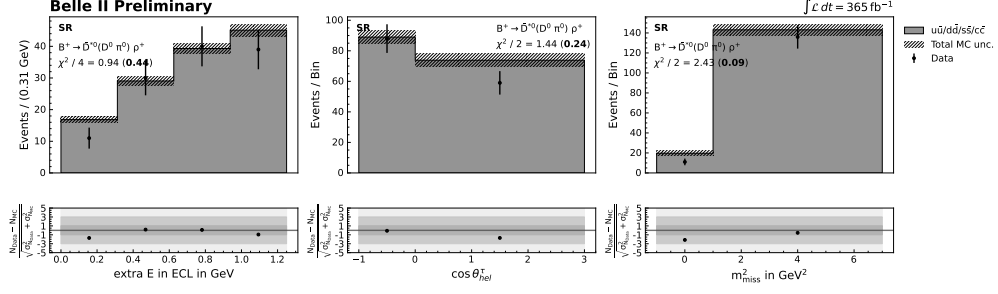


Figure 62: Fitting variables in continuum processes on off-resonance data in SR of $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$

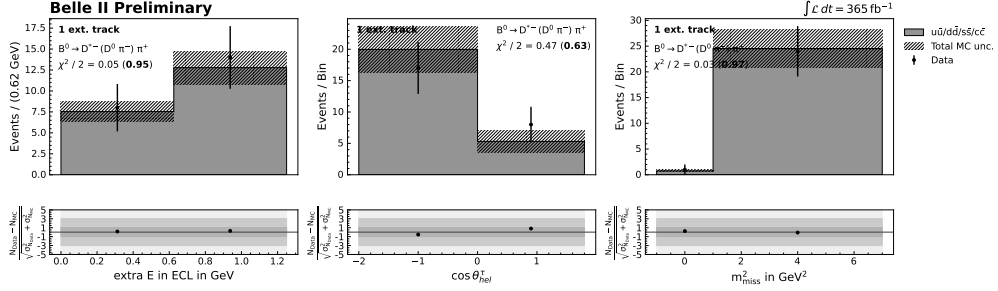


Figure 63: Fitting variables in continuum processes on off-resonance data in SR of $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$

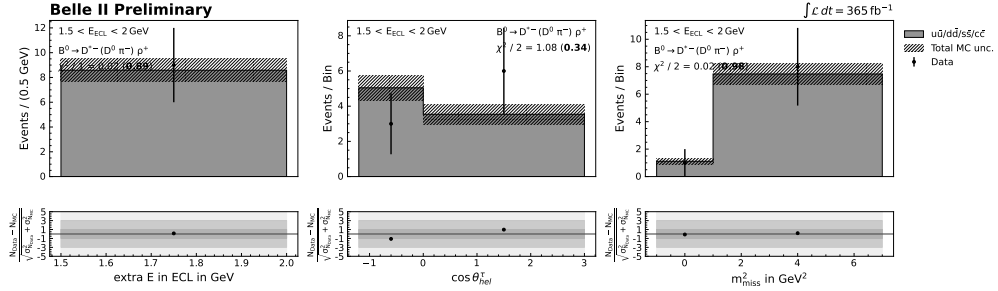


Figure 64: Fitting variables in continuum processes on off-resonance data in SR of $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$

are shared as the eigendecomposition has taken place simultaneously with the SR. As a result, we expect an improvement in the overall systematic uncertainty, as each region benefits from the information contained in the others. The scaling factors derived for the $B\bar{B}$ and $q\bar{q}$ templates, along with the GoF p-values for all fits, are presented in Table 46. In the last part of the table where we perform the fit simultaneously the first uncertainty is the statistical one and the second the total systematic one. The same systematics as in the signal regions are considered. The rest of the templates namely $B \rightarrow D^* \tau \nu$, $B \rightarrow D^* \ell \nu$ and $B \rightarrow D^{**} \ell \nu$ are allowed to float within 99% of their MC expectation controlled by single NP with a Gaussian constraint. For the $B \rightarrow D^{**} \ell \nu$ events we have a separate NP, while $B \rightarrow D^* \tau \nu$ and $B \rightarrow D^* \ell \nu$ events are controlled by a shared one, to

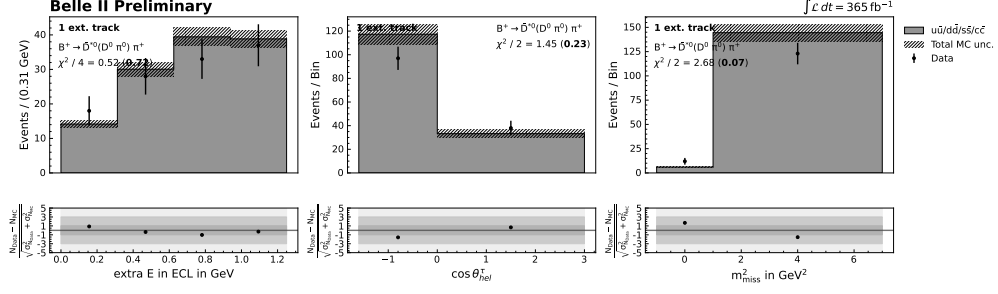


Figure 65: Fitting variables in continuum processes on off-resonance data in SR of $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$

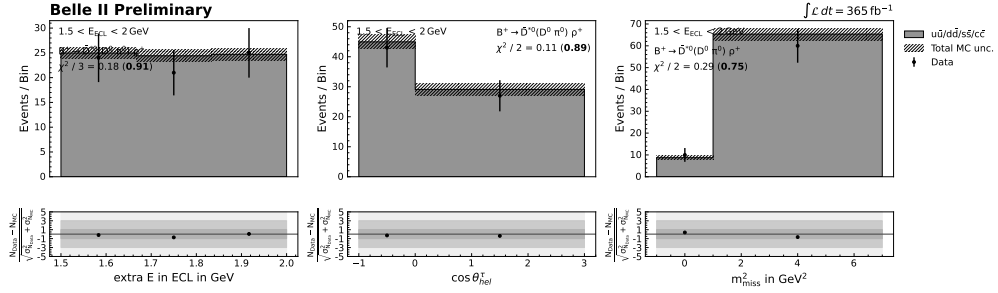


Figure 66: Fitting variables in continuum processes on off-resonance data in SR of $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$

avoid revealing any information about $R(D^*)$ from the signal and normalization events that are present in the sideband regions.

In Figures 67 - 70 we present the postfit distributions of the control fit. We observe a good Data/MC agreement confirming the good GoF p-values that we get from the fit.

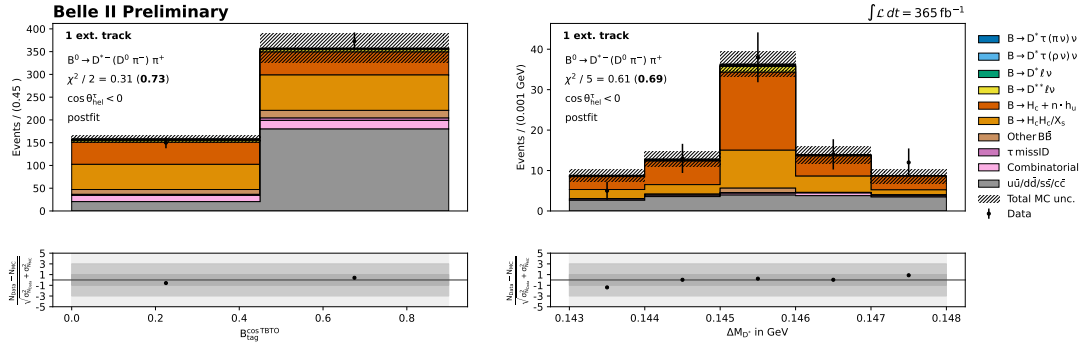
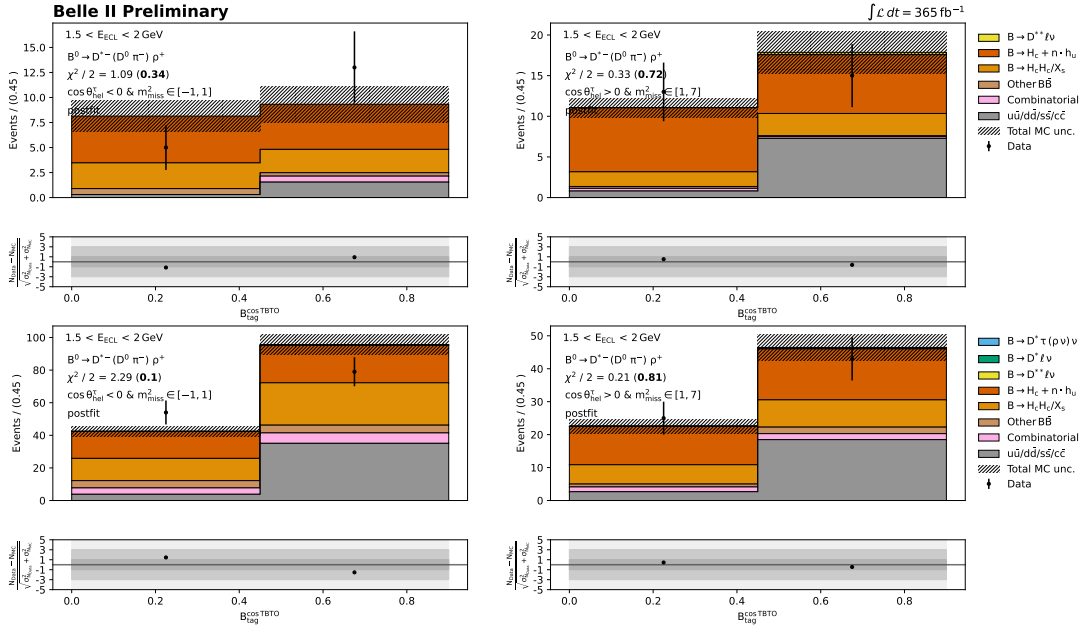


Figure 67: Post fit distributions of the control fit for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ that are constraining $B\bar{B}$ decays.

We apply these these scaling factors as well as the FEI calibration factors derived in 5.11 to all simulated data data for the rest of the Data/MC checks that we perform.

Table 46: Results for control fits

Reco channel	$B\bar{B}$	GoF p-value
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+ \cos \theta_{hel}^\tau > 0$	$0.6521 \pm 0.1271 \pm 0.0470$	
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+ \cos \theta_{hel}^\tau < 0$	$0.6735 \pm 0.0792 \pm 0.0489$	
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+ \cos \theta_{hel}^\tau > 0 (m_{miss}^2 < 1)$	$0.4604 \pm 0.1228 \pm 0.0672$	63.92%
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+ \cos \theta_{hel}^\tau < 0 (m_{miss}^2 < 1)$	$1.2749 \pm 0.3563 \pm 0.0837$	
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+ \cos \theta_{hel}^\tau > 0 (m_{miss}^2 > 1)$	$0.6704 \pm 0.1202 \pm 0.0423$	
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+ \cos \theta_{hel}^\tau < 0 (m_{miss}^2 > 1)$	$0.8999 \pm 0.1079 \pm 0.0129$	
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+ \cos \theta_{hel}^\tau > 0$	$0.6396 \pm 0.074 \pm 0.0423$	
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+ \cos \theta_{hel}^\tau < 0$	$0.7026 \pm 0.0583 \pm 0.0129$	
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+ \cos \theta_{hel}^\tau > 0 (m_{miss}^2 < 1)$	$0.2473 \pm 0.1011 \pm 0.0472$	
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+ \cos \theta_{hel}^\tau < 0 (m_{miss}^2 < 1)$	$0.3731 \pm 0.1887 \pm 0.1089$	
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+ \cos \theta_{hel}^\tau > 0 (m_{miss}^2 > 1)$	$0.7113 \pm 0.0786 \pm 0.0308$	
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+ \cos \theta_{hel}^\tau < 0 (m_{miss}^2 > 1)$	$0.7891 \pm 0.0605 \pm 0.0325$	
$B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$	$0.4379 \pm 0.0631 \pm 0.0325$	
$B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$	$0.4504 \pm 0.042 \pm 0.0225$	

Figure 68: Post fit distributions of the control fit for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ that are constraining $B\bar{B}$ decays.

8.3 Correctly reconstructed events with constrained normalization

Here we present the Data/MC agreement in the q^2 sideband, mainly targeting the modeling of important observables for correctly reconstructed $B \rightarrow D^* \ell \nu$ events, given that the overall normalization in this region is constrained by the FEI calibration.

We begin checking distribution on the four reconstructed object that we combined to build $\Upsilon(4S)$ candidates, namely the B^{tag} meson, the D^0 , π_s and the lepton. Figure 71 show the M_{bc} , $M(D^0)$, lepton momentum and π_s momentum for B^0 candidates in the q^2

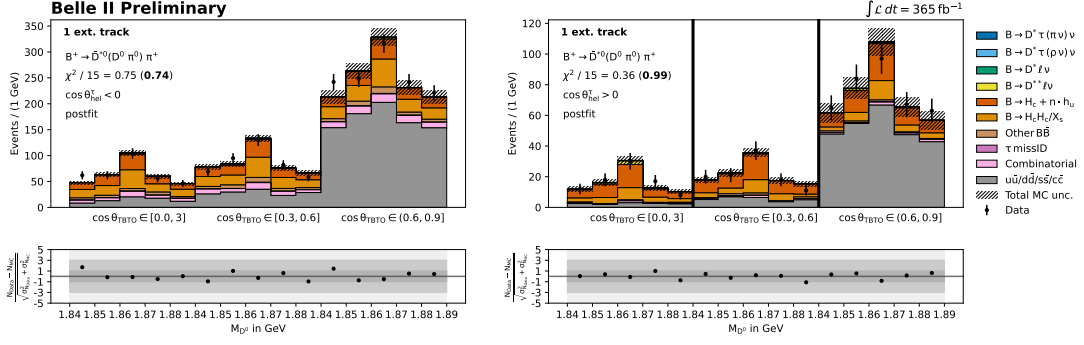


Figure 69: Post fit distributions of the control fit for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ that are constraining $B\bar{B}$ decays.

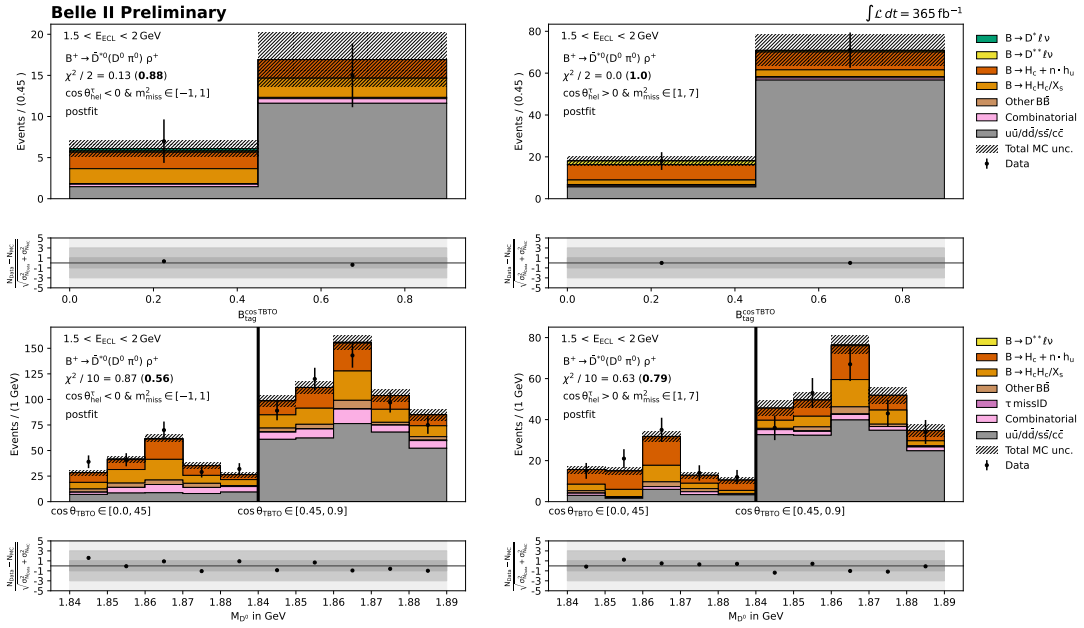


Figure 70: Post fit distributions of the control fit for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ that are constraining $B\bar{B}$ decays.

sidebands after applying the FEI calibration, for the electron and muon channels. All quantities demonstrate good Data/MC agreement.

We continue with $B^\pm$ candidates. Figure 72 shows the M_{bc} , $M(D^0)$, lepton momentum and π_s momentum for $B^\pm$ candidates in the q^2 sidebands after applying the FEI calibration. All quantities demonstrate good Data/MC agreement except for the momentum of the slow neutral pion in the electron channel. We presume that there is something off with the calibration of the neutral slow pions from the Performance group. The corrections that we have applied have been presented in Table 25. Nevertheless we are confident that any overall mismatch in the yield is taken care of by all our calibrations for all the different components, therefore such a mismodelling will not affect our extracted parameters of interest.

In Figure 73 we present the m_{miss}^2 distribution of both B^0 and $B^\pm$. This uses the same

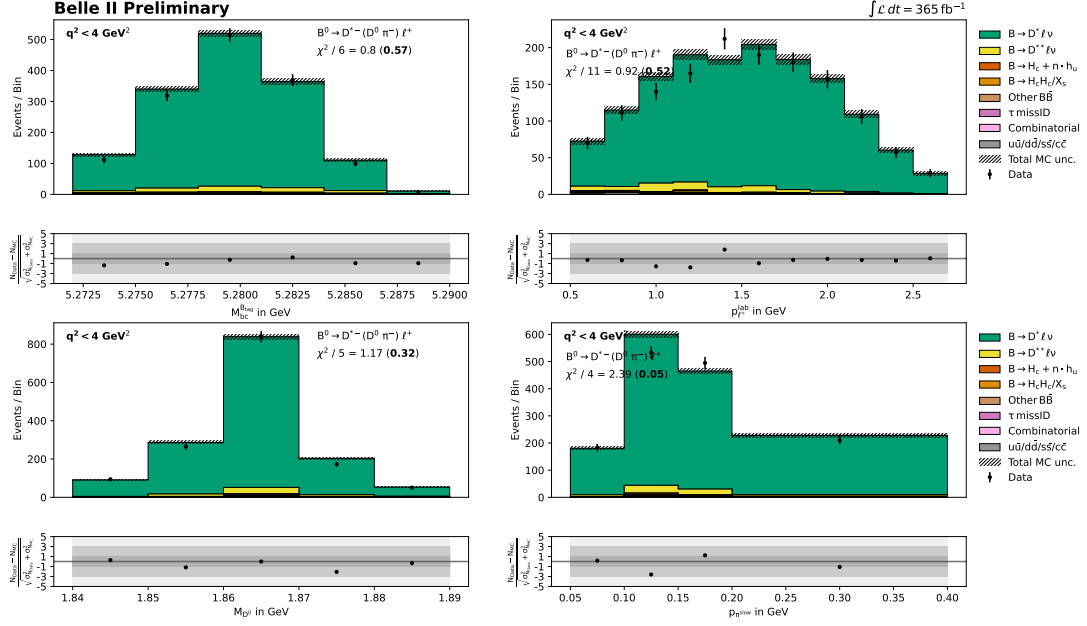


Figure 71: M_{bc} (top left) and $M(D^0)$ (bottom left), lepton momentum (top right) and π_s momentum (bottom right) for the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$ in the q^2 sideband.

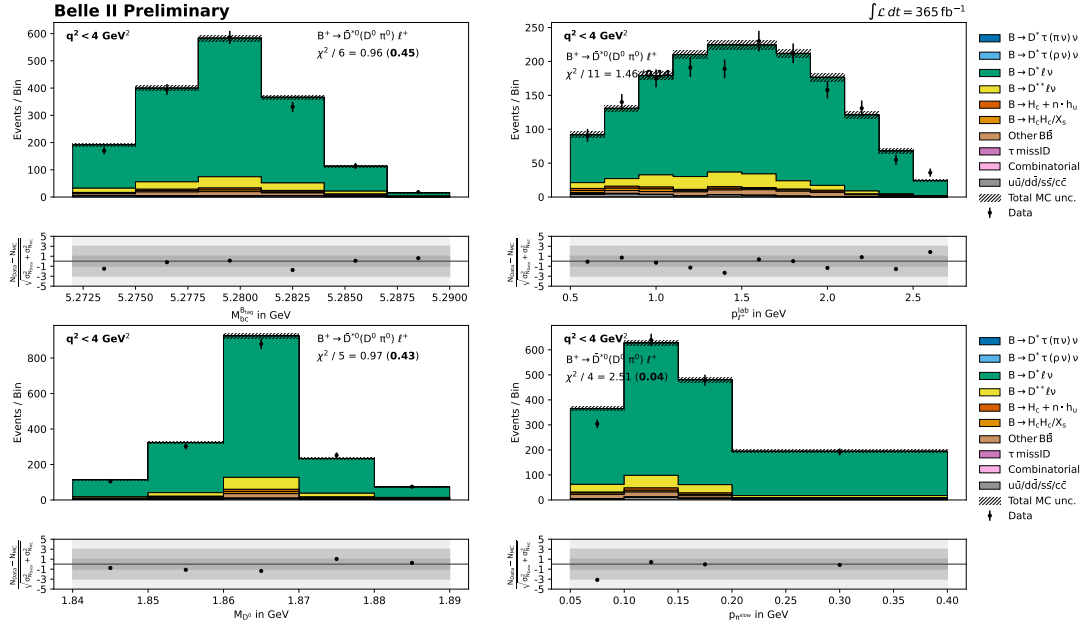


Figure 72: M_{bc} (top left) and $M(D^0)$ (bottom left), lepton momentum (top right) and π_s momentum (bottom right) for the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$ in the q^2 sideband.

1516 binning that will be used for the extraction of $B \rightarrow D^* \ell \nu$ decays in the NR.

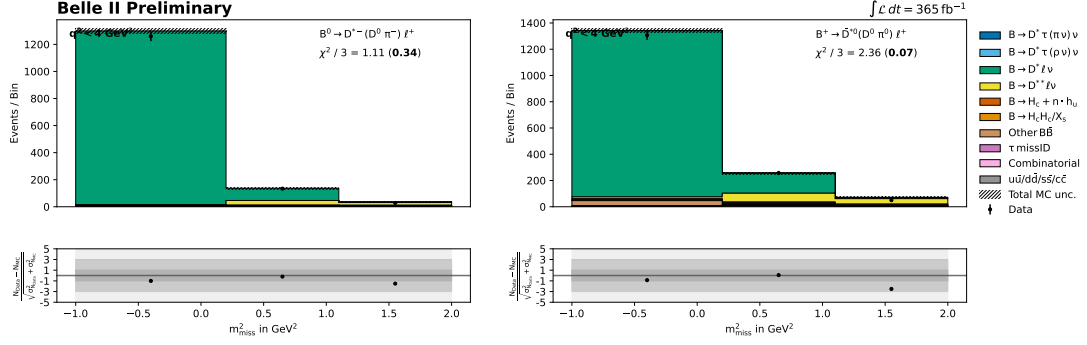


Figure 73: m_{miss}^2 distribution for B^0 (left) and $B^\pm$ (right) mesons in the q^2 sideband

8.4 Correctly reconstructed events with free normalization

Here we present the Data/MC agreement in the NR, mainly targeting the modeling of important observables for correctly reconstructed $B \rightarrow D^* \ell \nu$ events. Even though this region is used in the final fit, we do not expect any new physics impact since $B \rightarrow D^* \ell \nu$ decays have been measured multiple times in the past. We have also received WG premission to look into this region. We first plot the $E_{\text{ECL}}^{\text{extra}}$ distribution first without applying the photon multiplicity correction in Figure 74. We see that there is a severe mismodelling of the $E_{\text{ECL}}^{\text{extra}}$ variable as observed in other analyses as well. We therefore apply the photon multiplicity correction and in Figure 75 we observe that the Data/MC agreement is restored.

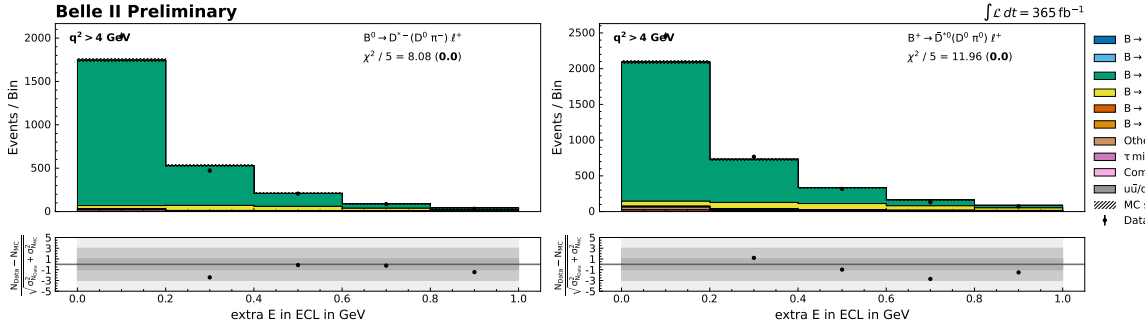


Figure 74: $E_{\text{ECL}}^{\text{extra}}$ distribution for B^0 (left) and $B^\pm$ (right) mesons in the NR sideband

In Figures 76 and 77 we plot the same variables that we plotted in the low q^2 sideband, except m_{miss}^2 which is the fitting variable we're going to use in the final fit.

We see that in all distributions except for the momentum of the slow pion -just like in the case of the q^2 sideband, the photon multiplicity correction is restoring Data/MC agreement. We avoid looking at the m_{miss}^2 distribution as this would essentially open the box for one of our fitting distributions which we want to do at a later stage. We conclude that no mismodelling is observed in the relevant observables and that the photon multiplicity correction is restoring Data/MC agreement in the $E_{\text{ECL}}^{\text{extra}}$ distribution, therefore can be safely applied to correctly reconstructed events, i.e. $B \rightarrow D^* \tau \nu$ events the SR.

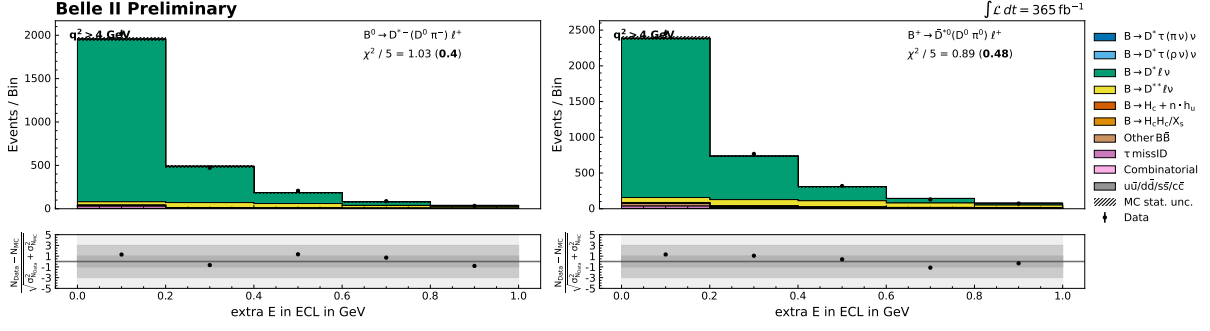


Figure 75: $E_{\text{ECL}}^{\text{extra}}$ distribution for B^0 (left) and $B^\pm$ (right) mesons in the NR sideband after applying the photon multiplicity correction.

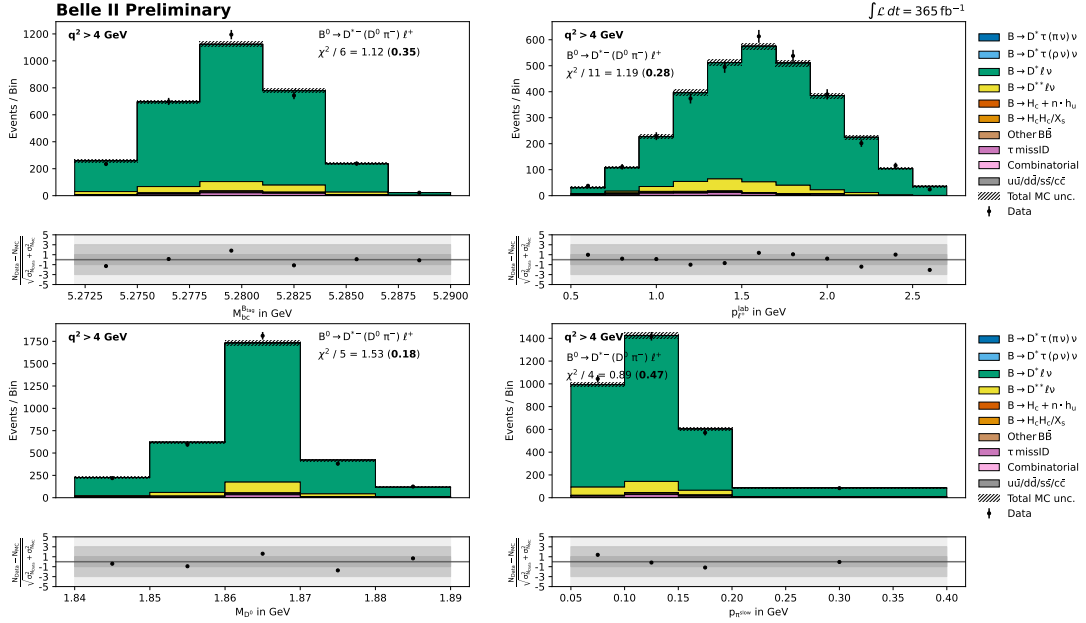


Figure 76: M_{bc} (top left) and $M(D^0)$ (bottom left), lepton momentum (top right) and π_s momentum (bottom right) for the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$ in the NR.

8.5 Misreconstructed events with constrained normalization

Here we present the Data/MC agreement in the regions used in the control fit. Since the normalization here is fixed by the control fit we mainly seek to validate the shapes of the important variables that will be used in the final fit. In particular we're looking at the $E_{\text{ECL}}^{\text{extra}}$ distribution, the reconstructed helicity angle and m_{miss}^2 since these are the observables used in final fit in the SR. On top of the weights derived by the control fits we also apply the photon multiplicity reweighting to ensure its validity in these constrained sidebands. Figure 78 - 81 show the distributions of these variables in all the control regions. Since the normalization of those is fixed by the control fit, we don't expect any significant differences in the overall yield. However the shape of those variables could be mismodeled.

Based on Figures 78 - 81 we conclude the shape of all important variables is well

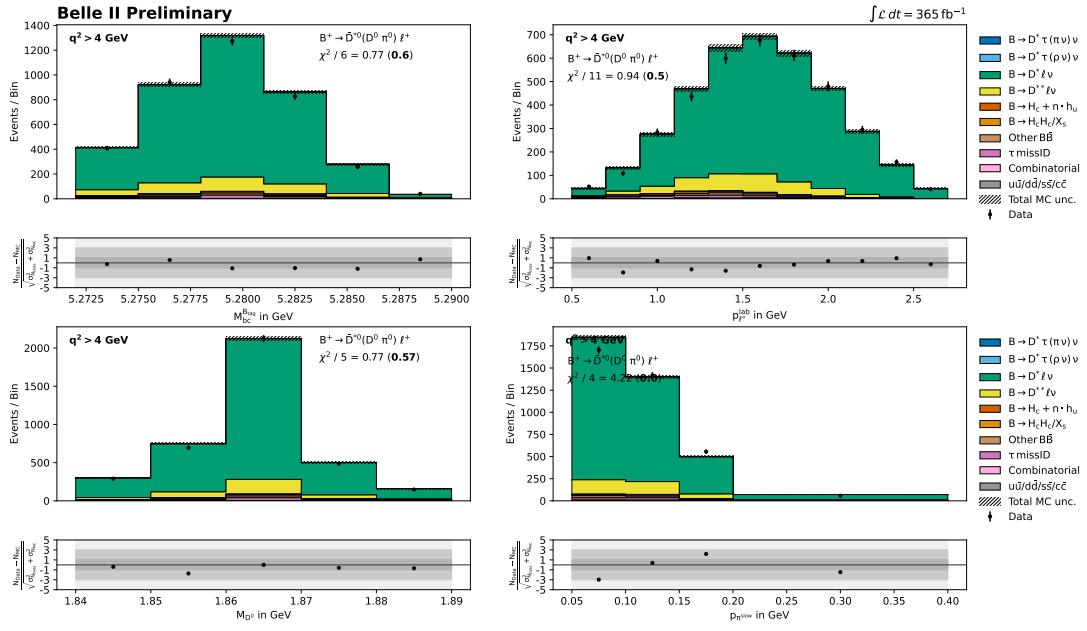


Figure 77: M_{bc} (top left) and $M(D^0)$ (bottom left), lepton momentum (top right) and π_s momentum (bottom right) for the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$ in NR.

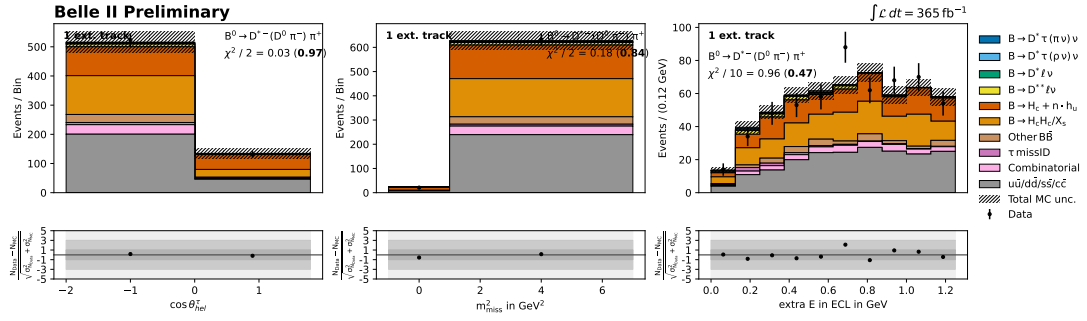


Figure 78: Data/MC agreement of the fitting variables in the control fit region for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$

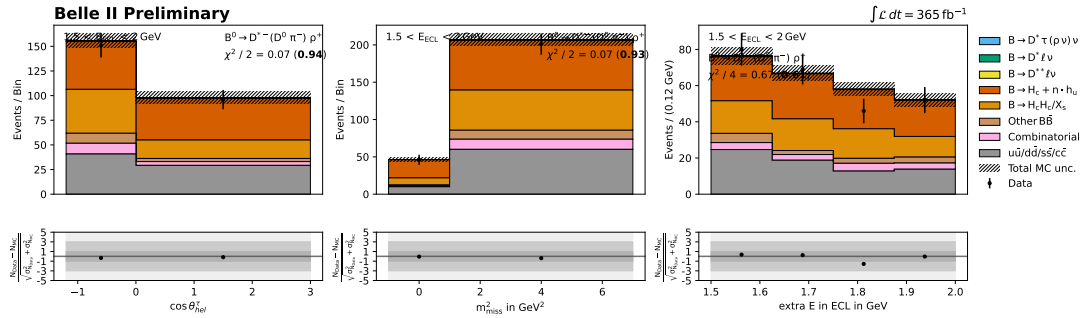


Figure 79: Data/MC agreement of the fitting variables in the control fit region for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$

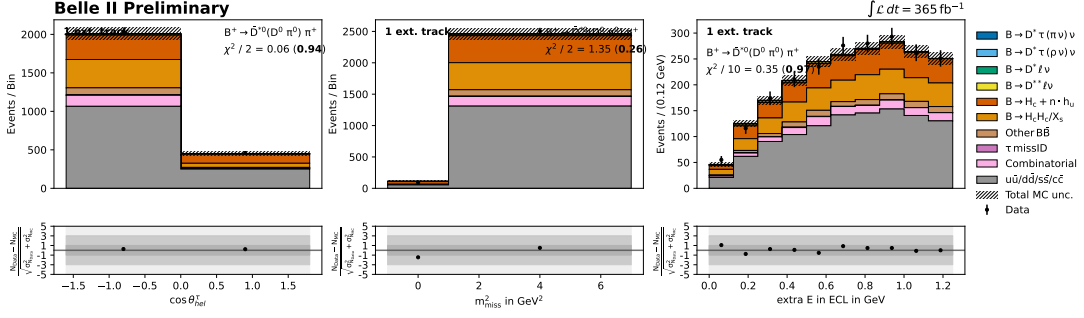


Figure 80: Data/MC agreement of the fitting variables in the control fit region for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$

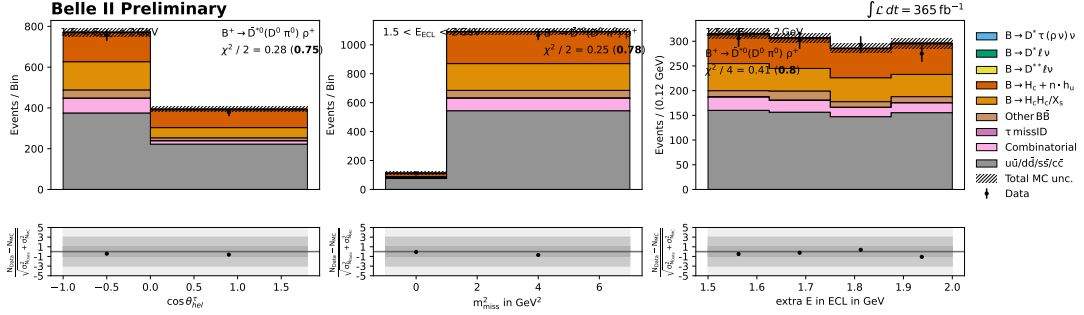


Figure 81: Data/MC agreement of the fitting variables in the control fit region for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$

modeled in the relevant resolution of this analysis.

8.6 Misreconstructed events with free normalization

Here we present the Data/MC agreement in the validation region defined as $1.25 < E_{\text{ECL}}^{\text{extra}} < 1.5 \text{ GeV}$. Here we're mainly targeting to confirm that the modeling of important observables that play a significant role in the final for misreconstructed events are well modeled. This also validates all the calibrations that we're carrying out throughout the analysis i.e. FEI calibration (5.11), photon multiplicity reweighting (5.12) and the control fits (8.2), as all the relevant weights are applied in the following plots. In Figures 82- 85 we present the Data/MC agreement for the four hadronic reconstruction channels in the high $E_{\text{ECL}}^{\text{extra}}$ sideband that we keep for validation. This has been defined in Table 45. The relative good p-values indicate that all our calibrations are sound and can be safely applied to the SR.

We conclude that no mismodelling is observed in the relevant observables.

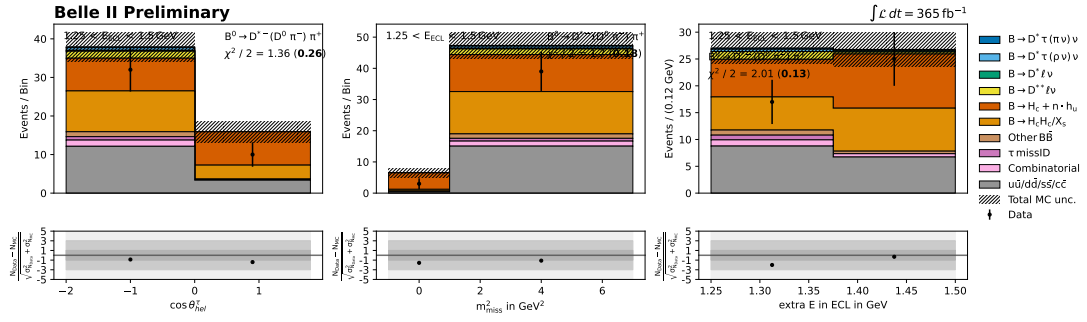


Figure 82: Data/MC agreement of the fitting variables in the validation region for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$

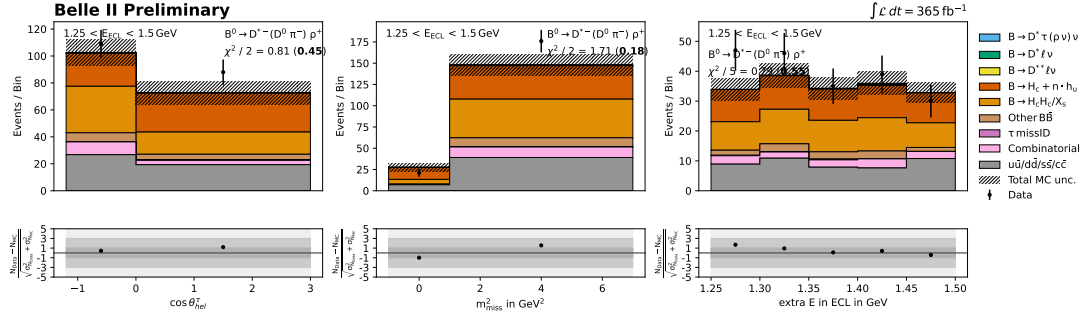


Figure 83: Data/MC agreement of the fitting variables in the validation region for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$

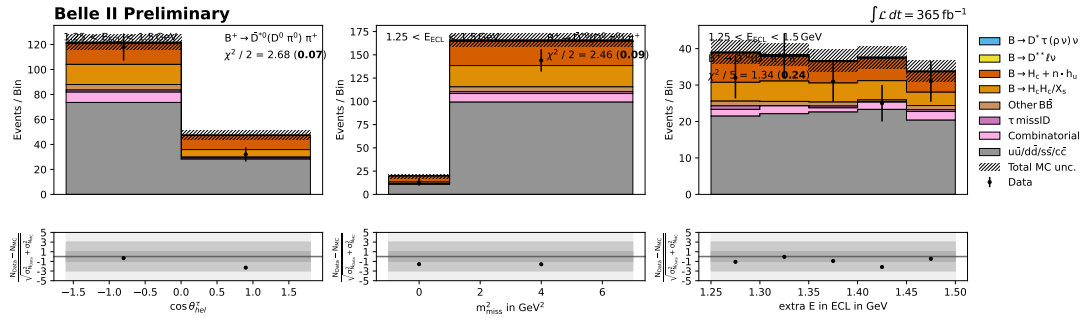


Figure 84: Data/MC agreement of the fitting variables in the validation region for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$

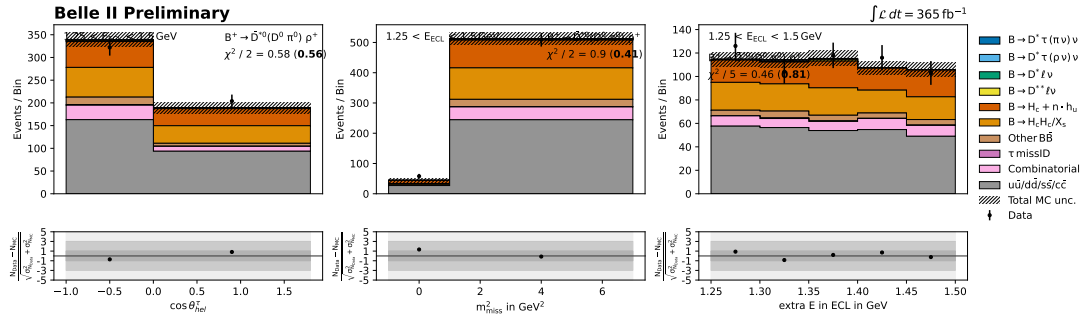


Figure 85: Data/MC agreement of the fitting variables in the validation region for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$

9 Results

In this sections we present the proposed box opening strategy and at a later stage of the review, we will first present the boxed closed, partially box opened and fully box opened results.

9.1 Box-opening strategy

We propose the the following distinct steps towards the full box opening of the $R(D^*)$ and P_τ result. Every step will be carried out once the previous step has been successfully validated.

1. Fit in NR and validation region

- Check Data/MC agreement of m_{miss}^2 in the NR. This is the only variable we haven't looked at in the NR since we're fitting it. If no problems proceed to;
- Run fit including only $B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$ and $B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$ and hadronic validation region of [1.25-1.5]. Check value of $R_{e/\mu}$.
- Check behavior of all nuisance parameters on experimental data. Provide pull plots.

2. Fit in high $E_{\text{ECL}}^{\text{extra}}$ SR and NR region

- Check Data/MC agreement of signal extraction variable in the high $E_{\text{ECL}}^{\text{extra}}$ region [0.75, 1.25] GeV of the SR. If no problems proceed to;
- Run fit including NR and high $E_{\text{ECL}}^{\text{extra}}$ [0.75-1.25]. This region should have small sensitivity on $R(D^*)$ and P_τ .
- Repeat studies for behavior of nuisance parameters.

3. Fit in full $E_{\text{ECL}}^{\text{extra}}$ range of SR and NR region and splits.

- Check Data/MC agreement of signal extraction variables in the full $E_{\text{ECL}}^{\text{extra}}$ range. If no problems proceed to;
- Run fit in SR with the $R(D^*)$ and P_τ central values masked including only all possible combinations of lepton (μ, e) and hadron channels (π, ρ). Include splits based on proc/prompt, positive/negative lepton or hadron and evaluate with 0. Evaluate that the difference of derived results from all the different splits are compatible with 0.
- Repeat studies for the behavior of nuisance parameters.

4. Full unboxing of SR and final result.

9.2 Box-closed results

the title

1594 **9.3 Box opened results**

1595 The title

A SysVar

The software uses class composition and class inheritance as its main design patterns making is easily extendable to any source of systematic uncertainty. An overview of the software modules can be seen in Figure 86

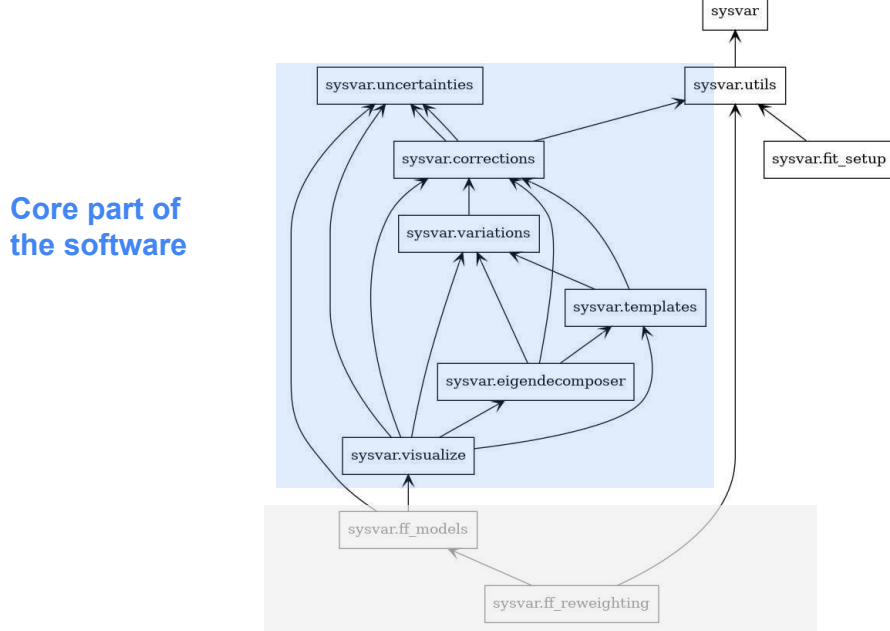


Figure 86: Module overview of the **SysVar** software

SysVar implements Data/MC corrections and their respective uncertainties based on the recommendations of the performance group. A few examples are slow pion reconstruction efficiencies, LID efficiencies or fake rates etc. It also allows for the creation of custom corrections like BF corrections. Currently corrections for the MC15ri campaign and for the Belle MC have been implemented. The **SysVar** package provides a growing list of visualization methods that enable the user to follow all the steps in an educating way and carefully run diagnostic tests. We provide a quick walkthrough of the **SysVar** workflow using a toy example implementing a pseudocorrection.

Imagine a scenario where one wants to perform a simultaneous 2D fit in two reconstruction channels using the pseudovariables displayed in Figure 87.

Now imagine that there is a momentum dependant correction that is applied on simulated data. The momentum dependance of the correction is displayed in Figure 88. The histograms in Figure 88 are binned based on the bins of the pseudocorrection. The correlations between the different correction bins may be well defined, however the correlation effect that those may have on the different bins of the signal extraction variables is a priori not known.

SysVar provides the tools to visualize the Data/MC correction central values, uncertainties and underlying correlations. The central values and the uncertainties can be seen in 89. The correlations between the different corrections bins of course depend on the way that the corrections have been determined. **SysVar** provides the tools to

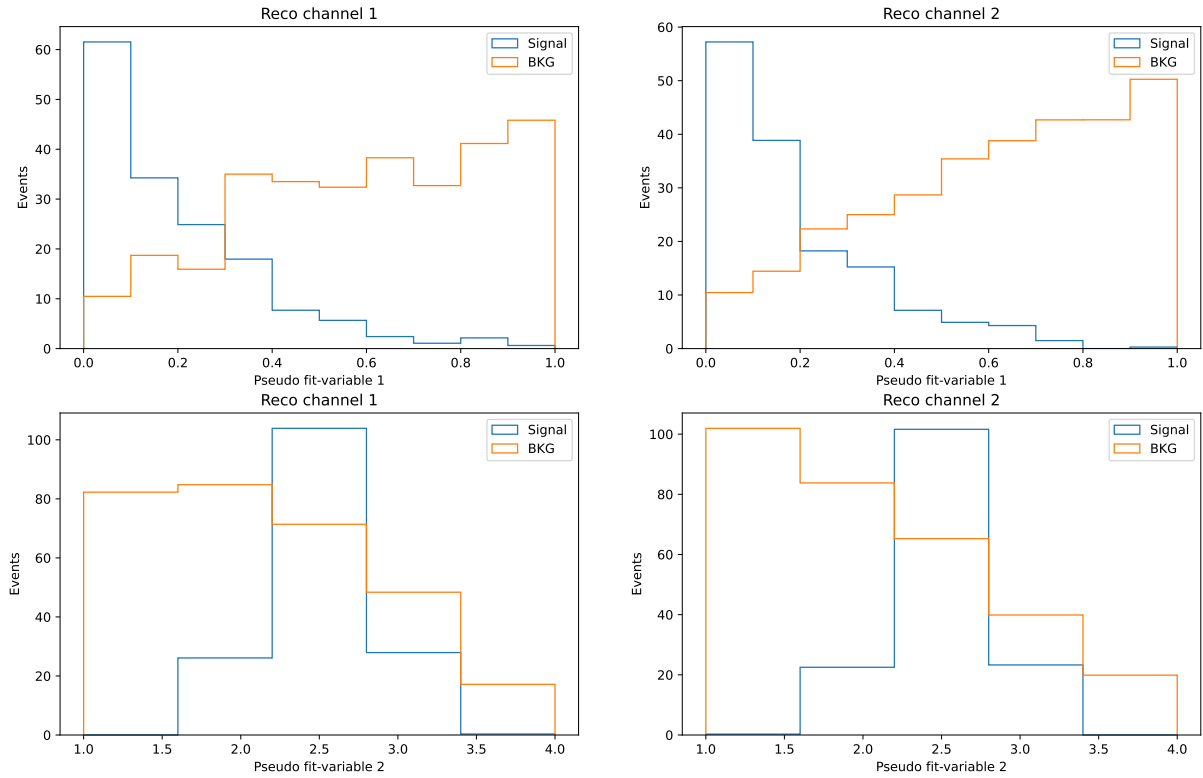


Figure 87: Pseudovariabels for signal extraction

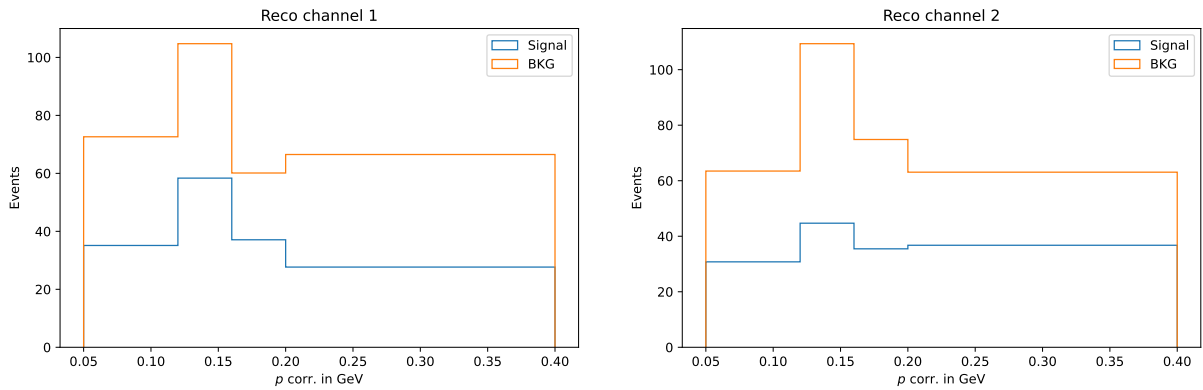


Figure 88: Pseudovariabel for correction bins

visualize these correlation as well as can be seen in Figures 90, 91 and 92.

SysVar subsequently builds a total covariance matrix from the three different sources of uncertainty on the correction central value. Such a covariance matrix is illustrated in Figure 93a. Then **SysVar** samples from multi gaussian distributions to generate variations of the correction central values based on the total covariance matrix. The number of variations has to be declared by the user. The variations can be found in 93b.

SysVar will then histogram the simulated data and will also create varied histograms based on the correction weight variations. In this example 201 histograms will be created. One for the nominal template and 200 with the varied correction weights. A

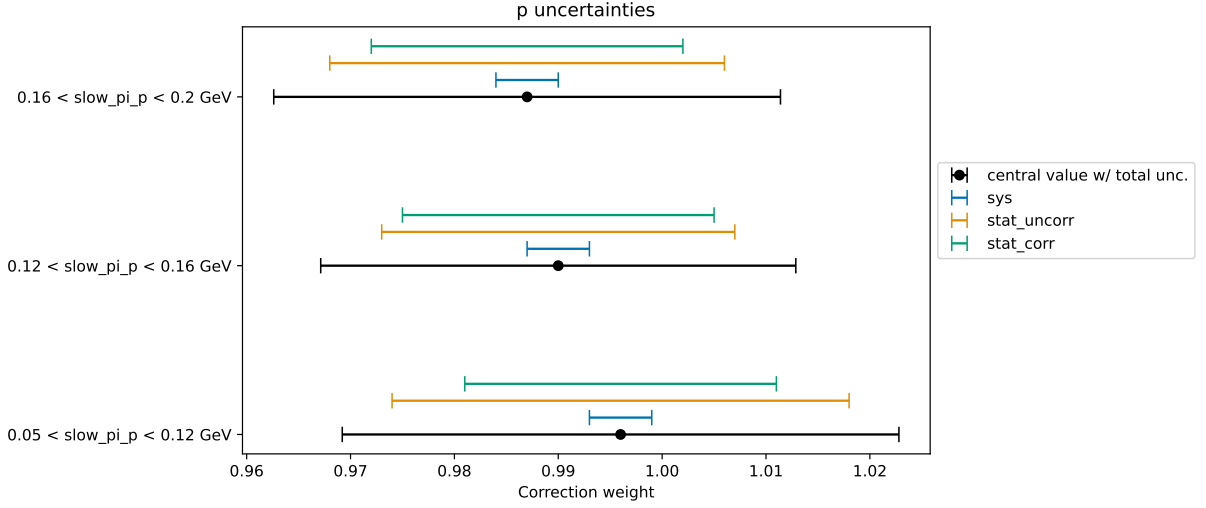


Figure 89: Different corrections and error sources. In this example three sources of uncertainty are considered for the correction. Two statistical in nature and one systematic

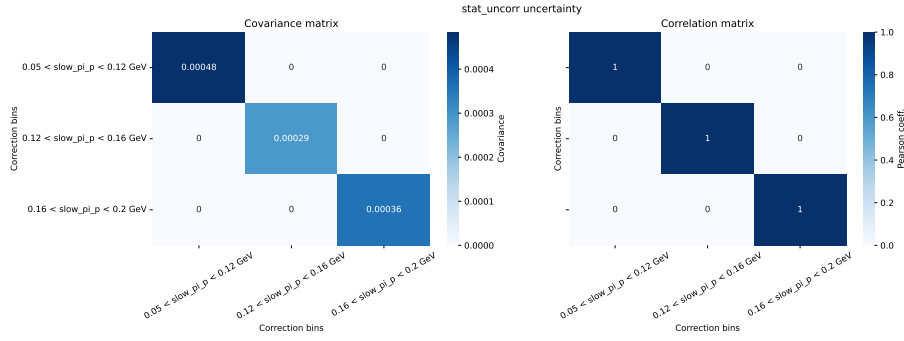


Figure 90

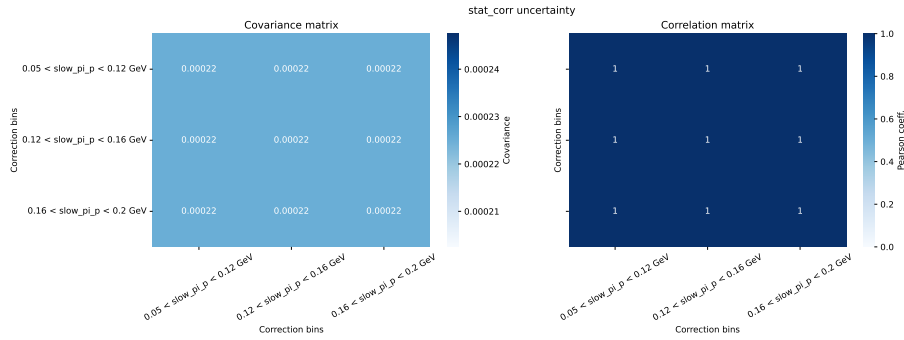


Figure 91

set of overview plots allows the user to examine closely the effect of the variations in each template. In Figure 94 we provide an example of the signal template in the first reconstruction channel.

Finally SysVar performs the eigendecomposition simultaneously in all bins, for all templates and for all reconstruction channels. This enables for consistent treatment of

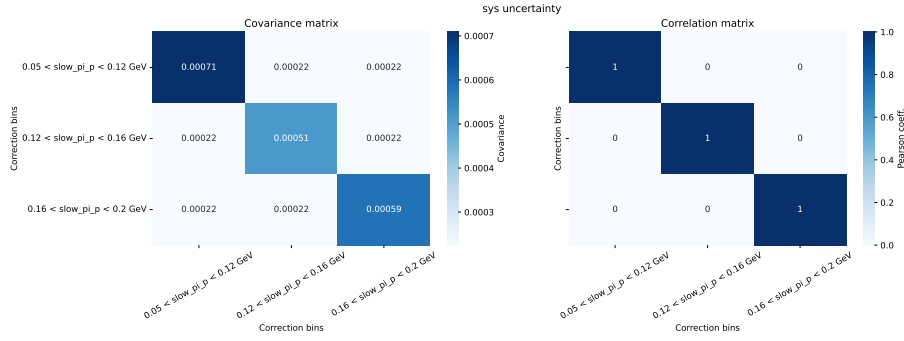
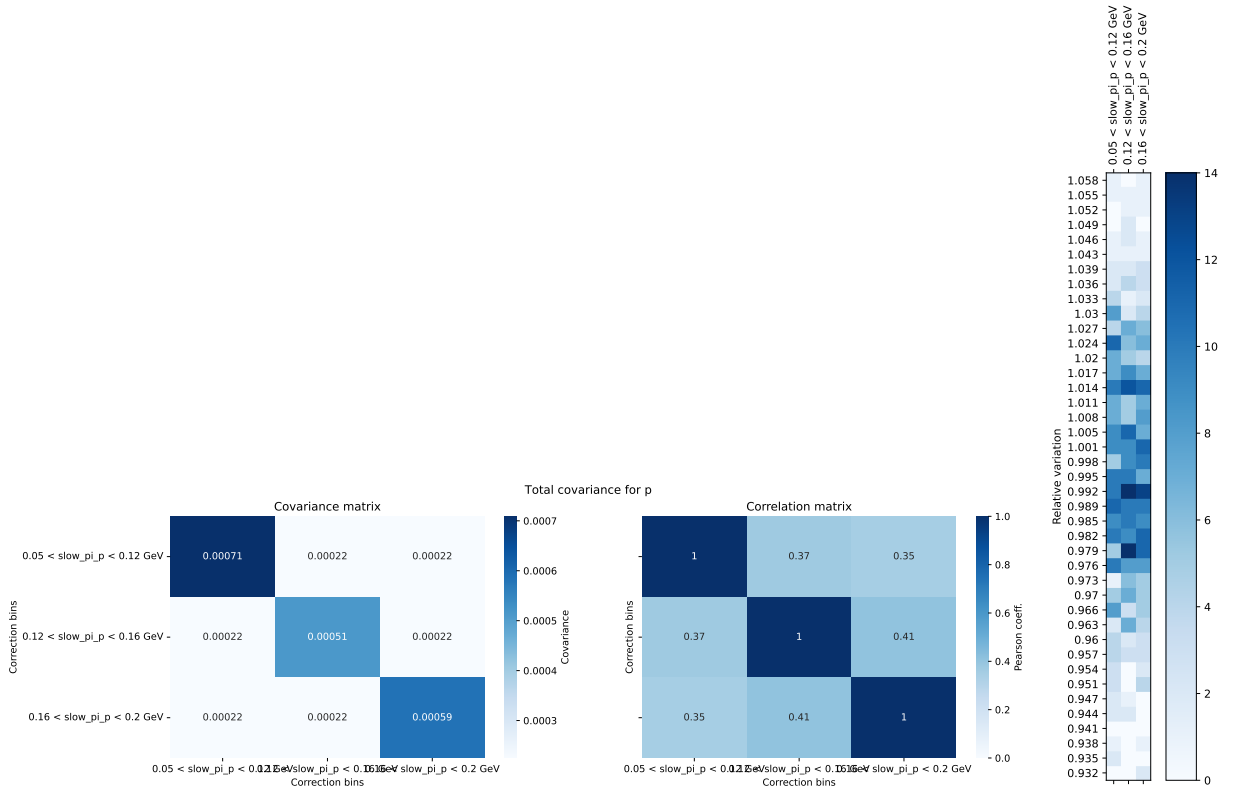


Figure 92



(a) The total covariance matrix from all corrections

(b) The variations of the central values. Here 200 variations have been used.

Figure 93: A summary of the variations of the correction weights.

the correlations across all reconstruction channels and all templates. The correlation matrix obtained can be seen in Figure 95.

SysVar allows to find the number of important eigendirections using two criteria:

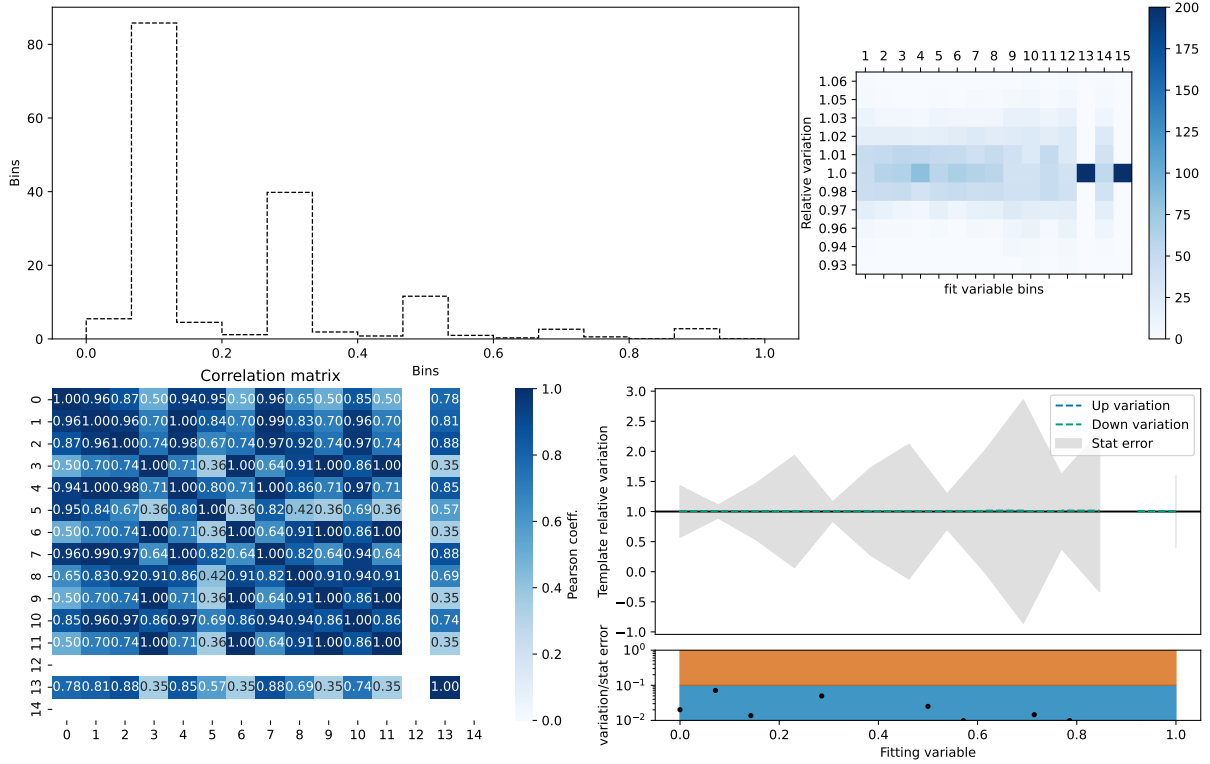


Figure 94

- Taking the normalized differences of the reconstructed covariance matrix and the initial covariance matrix.
- Calculating the trace in the eigenvalue space.

Here we present an example of comparing the reconstructed covariance matrix compared to the original one. If the normalized difference of the two covariance matrices is below 0.5% we conclude that the eigendirections describe the original covariance matrix sufficiently enough. We keep those eigenvariations and we introduce one nuisance parameter per eigendirection in the signal extraction fit. The normalized differences of the two covariance matrices can be seen in Figure 96

For the treatment of the systematics uncertainties we make use of all the aforementioned features of **SysVar**. This approach provides a framework to consistently study and evaluate systematic uncertainties by making all the steps of the PCA abundantly clear.

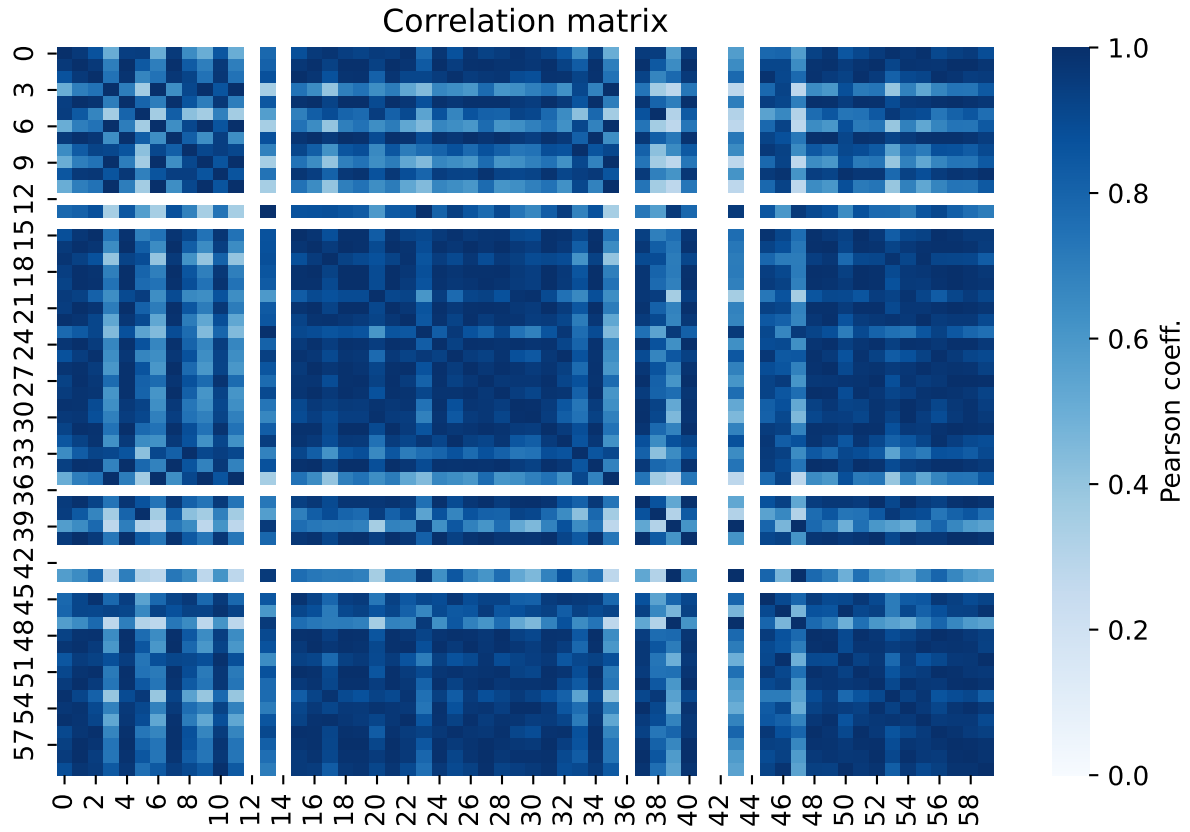


Figure 95: The correlation matrix arising from all the varied templates. It is clear that the correlation structure in this space becomes highly not trivial to model. Therefore the eigendecomposition of it, provides a framework that simplifies the treatment of these correlations in a easier way

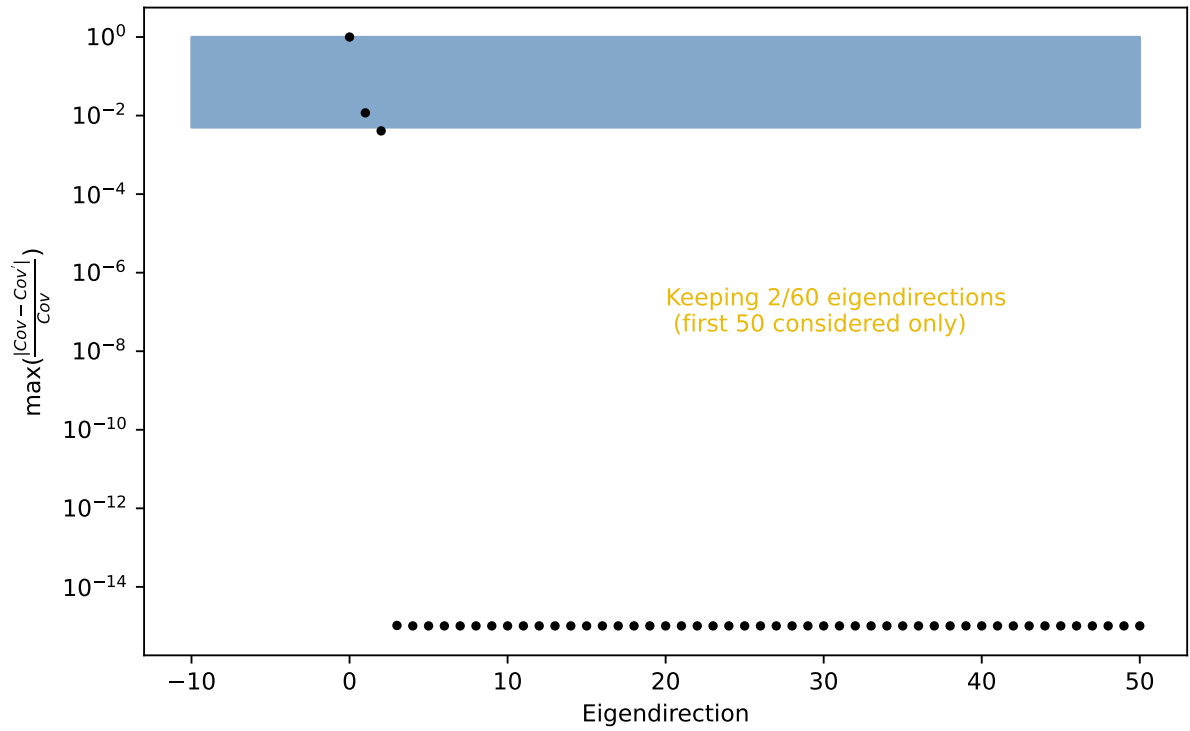


Figure 96: The normalized covariance matrix differences as a function of the eigendirections considered. The target precision at 0.5% is set arbitrarily by the user.

B PDG codes

Table 47: PDG codes of categories for the first layer of truth matching. Charge conjugation is used for all PDG codes.

Category	PDG codes
D	[411,421]
D^*	[413,423]
D^{**}	[415, 425, 10411, 10421, 10413, 10423, 20413, 20423]
H_c^{gap}	[411, 421, 413, 423, 431, 433]
$X_{u/s}^{gap}$	[111, 211, 221, 331, 321]
H_c	[411, 421, 431, 413, 423, 433, 415, 425, 433, 435, 10411, 10413, 10421, 10423, 10431, 10433, 20433, 20423, 20413]
X_s	[130, 310, 311, 313, 321, 323, 333, 10313, 10323, 331]
X_u	[43,44,111,113,221,223,331,211,213, 20113, 20213]

C Multinomial errors

In Figure 97 one can see how the difference of the ratio and the uncertainty of the ratio calculated assuming a multinomial approximation, scales with the total number of occurrences. It is clear that for a small number of total occurrences the multinomial approximation lead to unphysical values for the uncertainties as these lead to negative rates. Therefore we need to be careful that the combination of total occurrences and the rates that are used in the FEI tag side mode composition do not lead to negative rates.

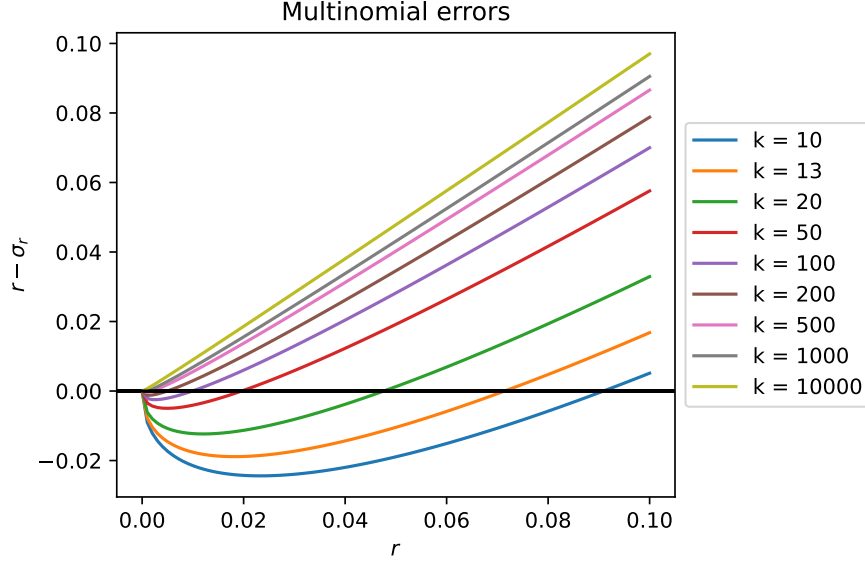


Figure 97: Multinomial error scaling with number of total occurrences and rates

D D^0 mass calibration

We use the q^2 sideband for this calibration, as it provides a clean sample of $B \rightarrow D^* \ell \nu$ decays, meaning we have a large number of well-reconstructed D^0 mesons. However, we do not use true $B \rightarrow D^* \ell \nu$ decays to determine the calibration factors. Instead, we define the signal differently:

- A perfectly reconstructed D^0 meson with `isSignal == 1`, or
- An event where a D^0 meson appears in the descendant tree of a B meson.

Using MC we first determine the shape of the signal by performing an unbinned likelihood fit. Our study finds that a double Gaussian function best describes the distribution. To assess the goodness of fit, we compute a pp-value, excluding the last 5% quantiles since the fit lacks sensitivity in the tails of the distribution. This exclusion is justified as we are primarily interested in the peak region rather than the tails.

The fitted Double Gaussian parameters are summarized in Table 33. Additionally, we present the pre-fit signal shape, the post-fit distribution on both signal and background and the goodness-of-fit test results comparing our parametric function to MC in Figures 98, 99, 100 and 101.

Figure 98: Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K^- \pi^+$, $D^0 \rightarrow \pi^- \pi^+$ and $D^0 \rightarrow K^- K^+$ on MC.

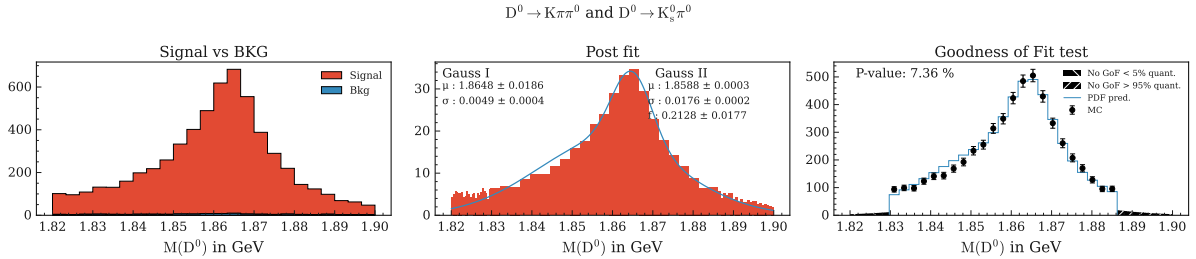


Figure 99: Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K_s^0 \pi^0$ and $D^0 \rightarrow K^- \pi^+ \pi^0$ on MC.

We repeat the same double Gaussian fit on experimental data. The corresponding fit parameters are provided in Table 33. We also provide the postfit distributions and the goodness of fit test for the fits that we run on experimental data in Figures 102, 103, 104 and 105.

Since our Probability Density Function (PDF) consists of multiple Gaussian components, we employ the Central Limit Theorem (CLT) to estimate the overall resolution. We first generate 20,000 toy experiments, sampling from a multivariate normal distribution with the nominal fit parameters as means, and the covariance matrix from the minimization procedure. Using the toy-generated parameters, we construct the Cumulative Distribution Function (CDF) of the double Gaussian acpdf. For each toy experiment

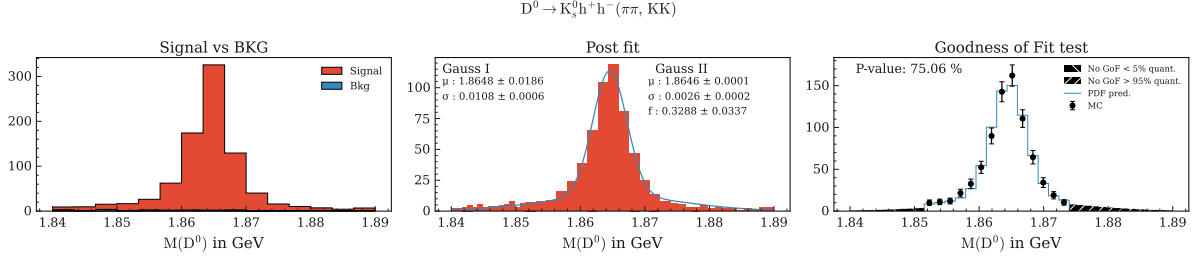


Figure 100: Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K_s^0 \pi^+ \pi^-$ and $D^0 \rightarrow K_s^0 K^+ K^-$ on MC.

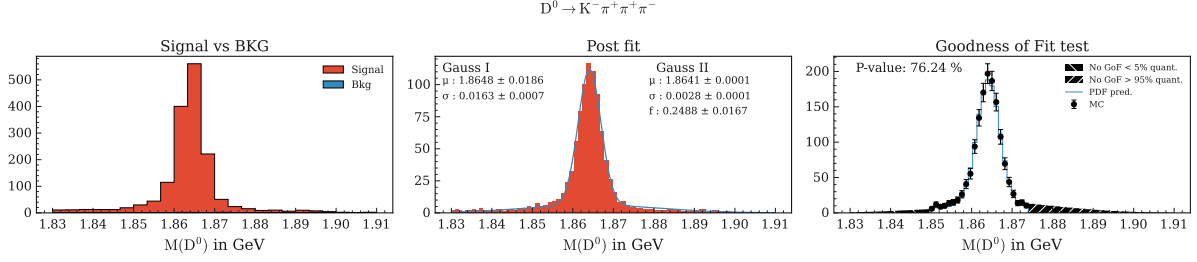


Figure 101: Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K^- \pi^+ \pi^- \pi^+$ on MC.

Figure 102: Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K^- \pi^+$, $D^0 \rightarrow \pi^- \pi^+$ and $D^0 \rightarrow K^- K^+$ on Data.

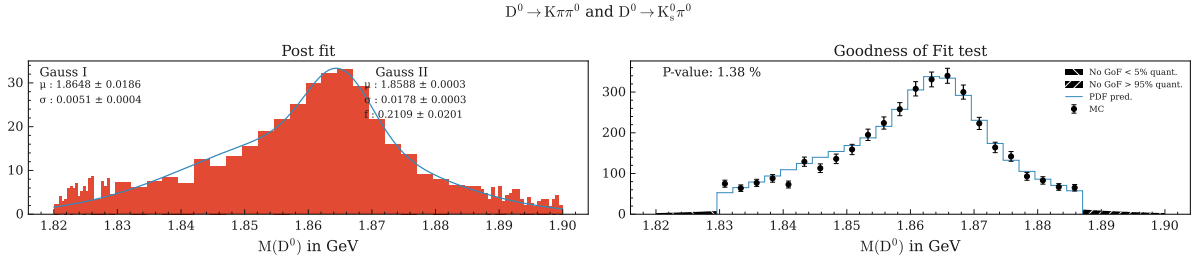


Figure 103: Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K_s^0 \pi^0$ and $D^0 \rightarrow K^- \pi^+ \pi^0$ on Data.

random values between 0 and 1 are drawn. These values are mapped onto the allowed invariant mass range via the CDF. Finally the 68.3% quantile of the resulting distribution is used to estimate the resolution of the fitted PDF.

To account for differences in mass resolution between MC and Data, we compute the smearing factor as

$$r = \sqrt{\mu_{MC}^2 - \mu_{DATA}^2} \quad (49)$$

This factor corrects the resolution mismatch between simulated and real events. The results are presented in Table 33. Once the smearing factor is determined, we correct the

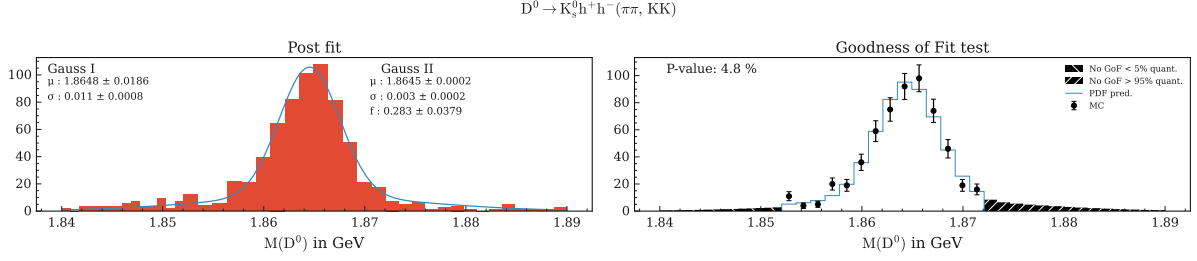


Figure 104: Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K_s^0 \pi^+ \pi^-$ and $D^0 \rightarrow K_s^0 K^+ K^-$ on Data.

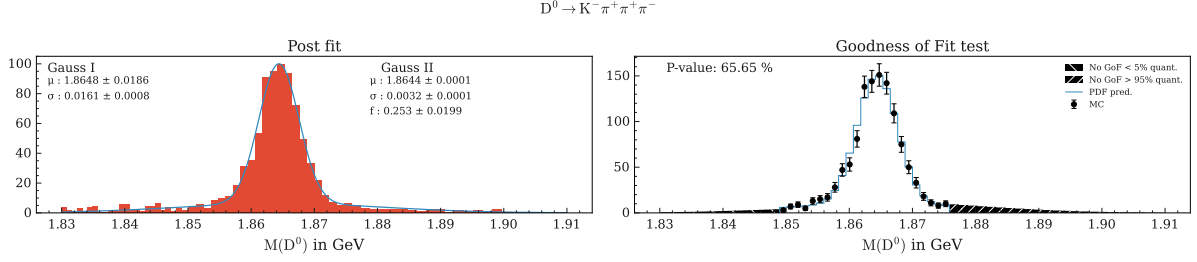


Figure 105: Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K^- \pi^+ \pi^- \pi^+$ on Data.

reconstructed invariant mass for each event. A random shift is drawn from a Gaussian distribution centered at zero with a width equal to the smearing factor. Finally this shift is added to the originally reconstructed invariant mass, formally as

$$M_{D^0} = M_{D^0} + \mathcal{G}(0, r) \quad (50)$$

E FF model parameters

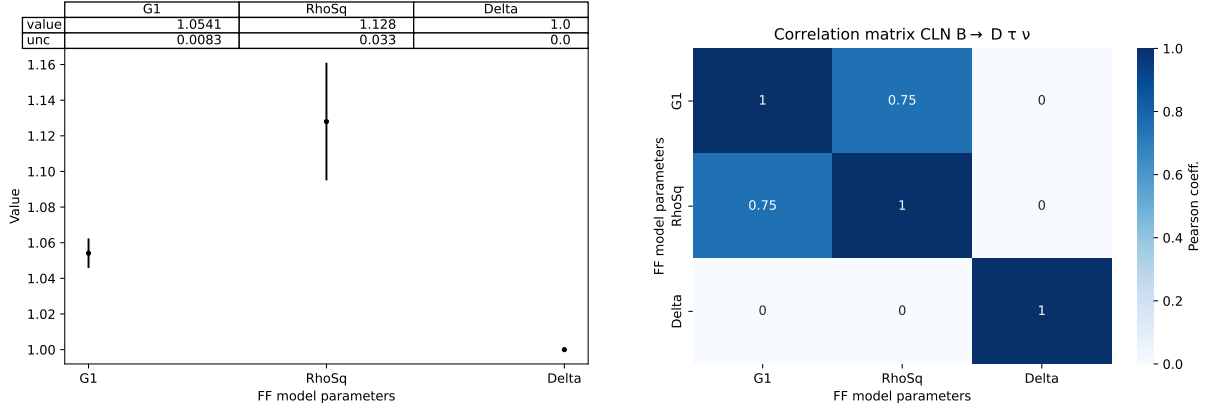


Figure 106: Parameters used in the CLN parameterization for the generation of $B \rightarrow D\tau\nu$ decays in the Belle II MC

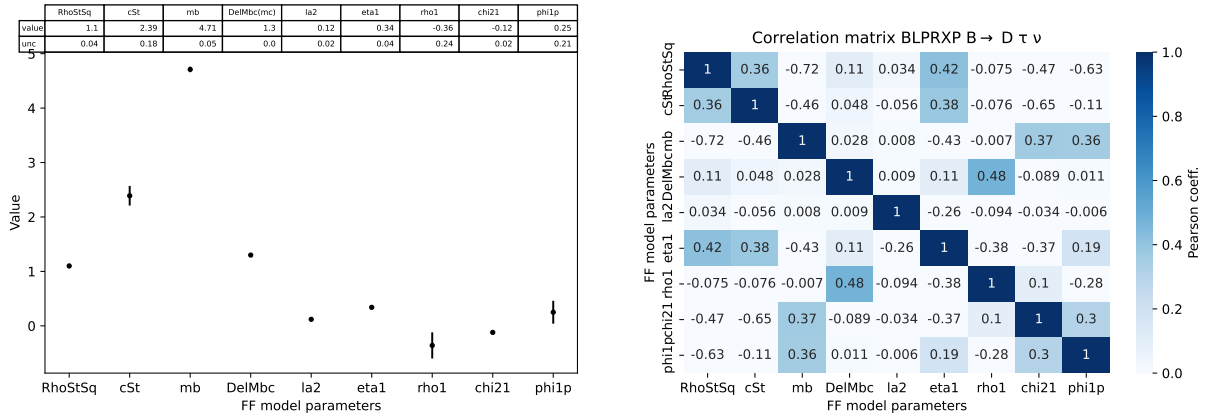


Figure 107: Target parameters used in the BLPRXP parameterization for the reweighting of $B \rightarrow D\tau\nu$ with HAMMER

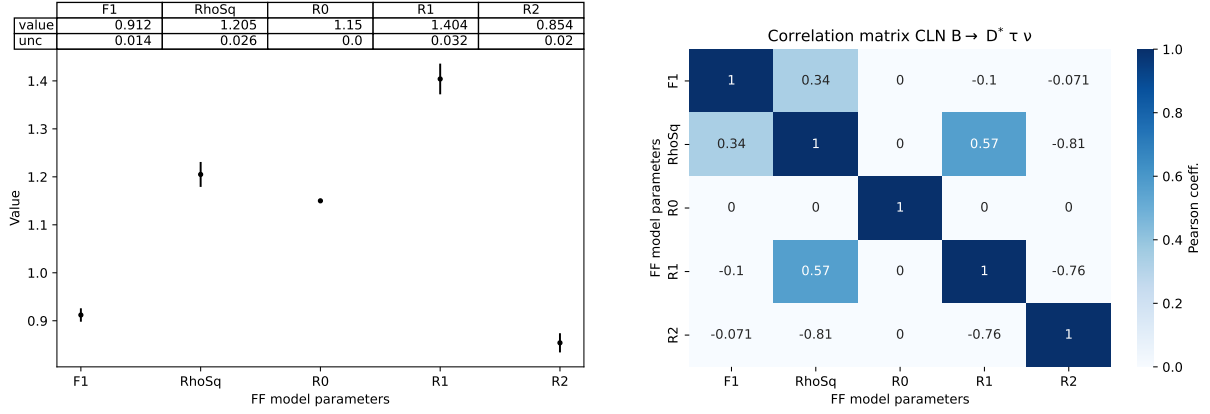


Figure 108: Parameters used in the CLN parameterization for the generation of $B \rightarrow D^* \tau \nu$ decays in the Belle II MC

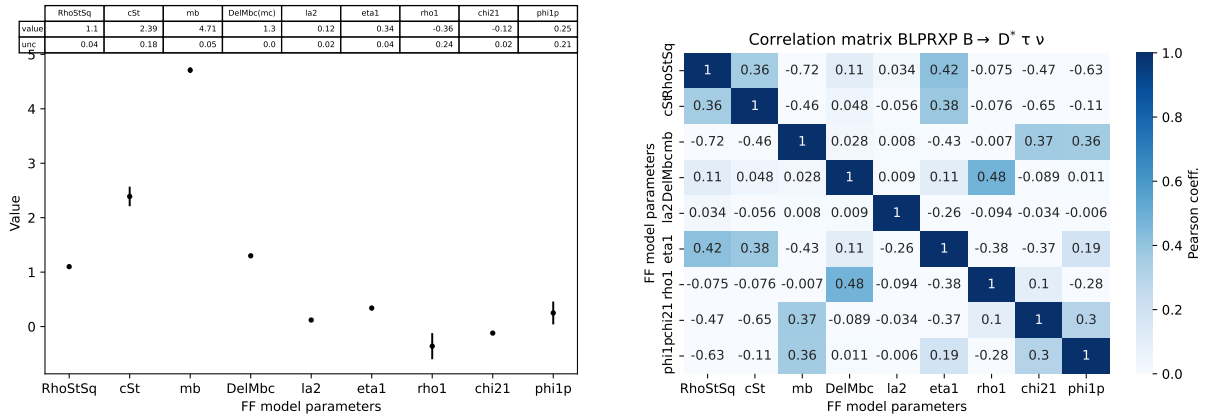


Figure 109: Target parameters used in the BLPRXP parameterization for the reweighting of $B \rightarrow D^* \tau \nu$ with HAMMER

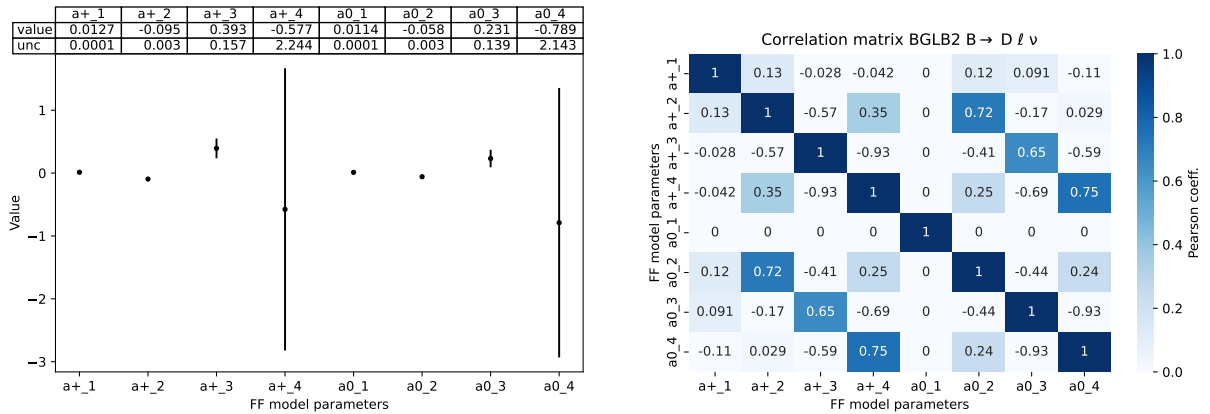


Figure 110: Parameters used in the BGL parameterization for the generation of $B \rightarrow D \ell \nu$ decays in the Belle II MC

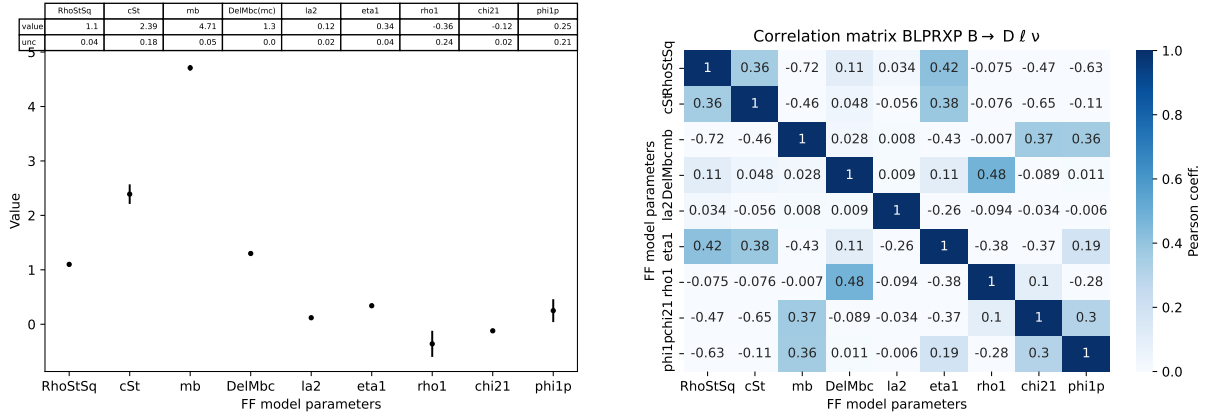


Figure 111: Target parameters used in the BLPRXP parameterization for the reweighting of $B \rightarrow D \ell \nu$ with HAMMER

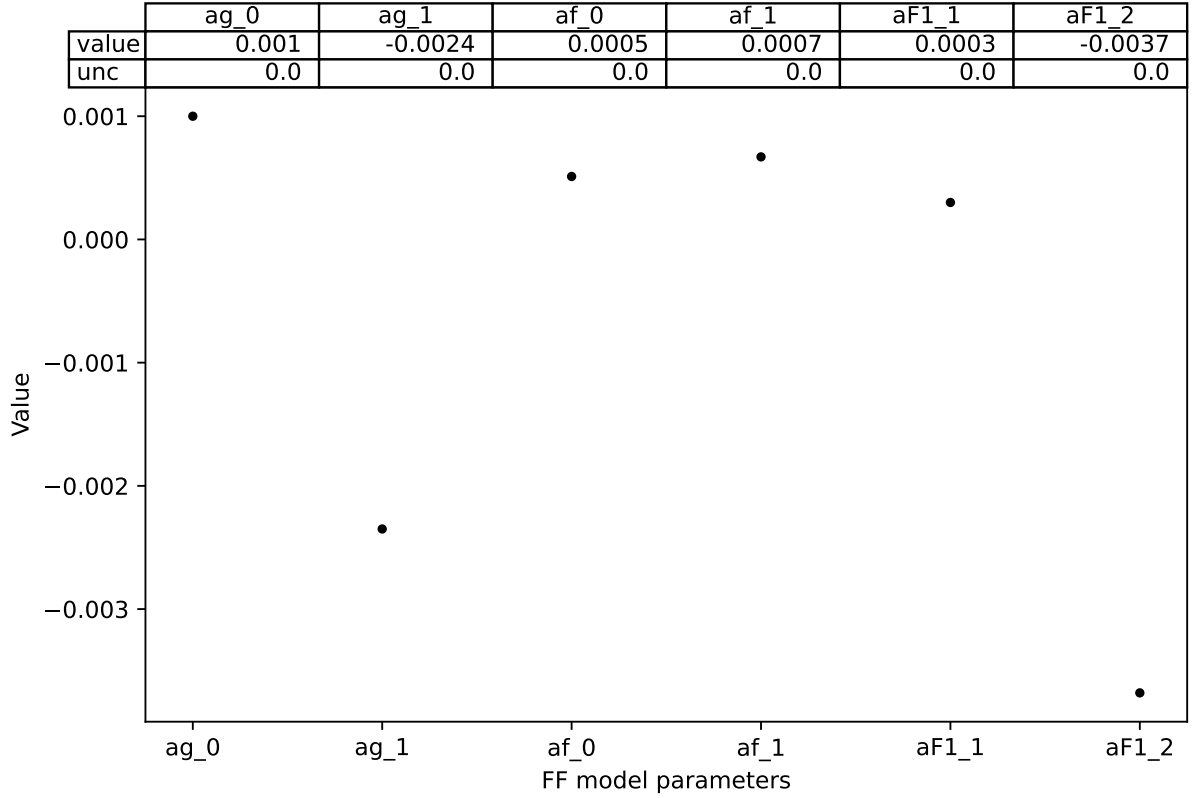


Figure 112: Parameters used in the BGL parameterization for the generation of $B \rightarrow D^* \ell \nu$ decays in the Belle II MC

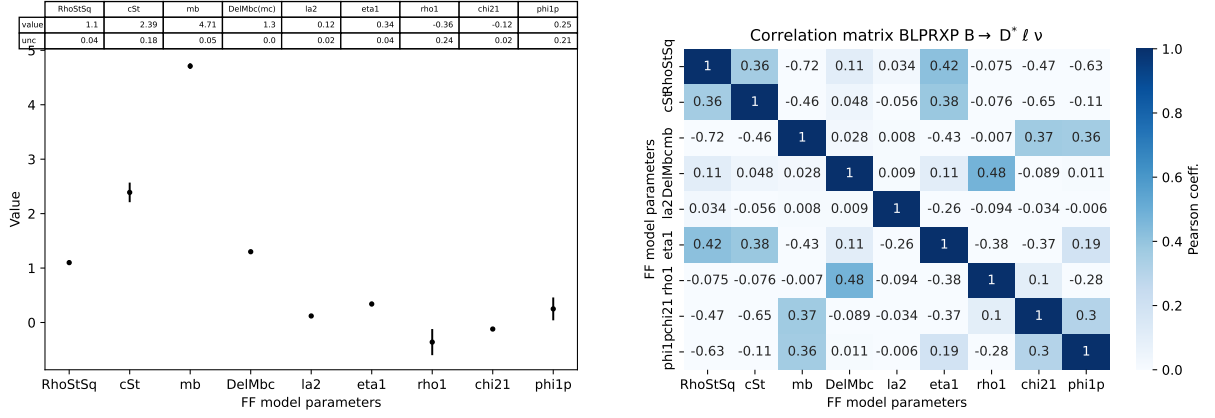


Figure 113: Target parameters used in the BLPRXP parameterization for the reweighting of $B \rightarrow D^* \ell \nu$ with HAMMER

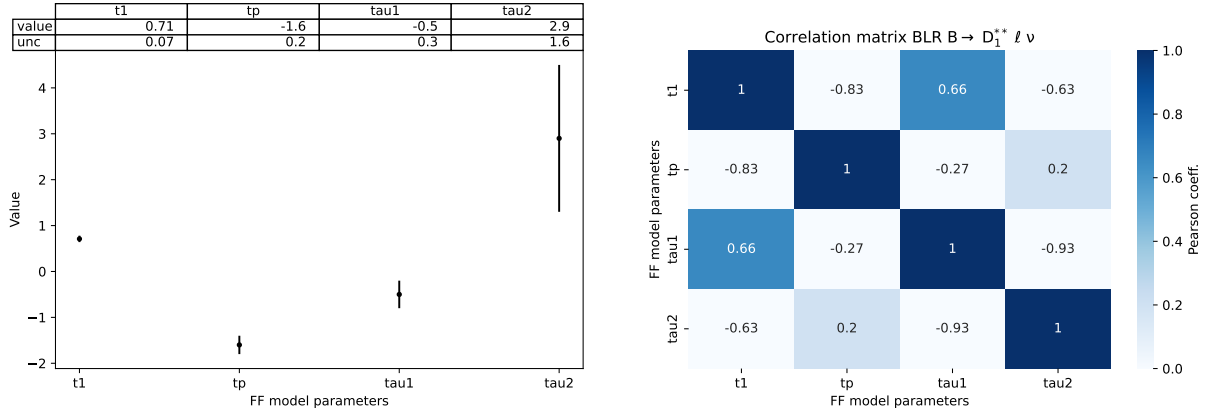


Figure 114: Parameters used in the BLR parameterization for the generation of $B \rightarrow D_1^{**} \ell \nu$ decays in the Belle II MC

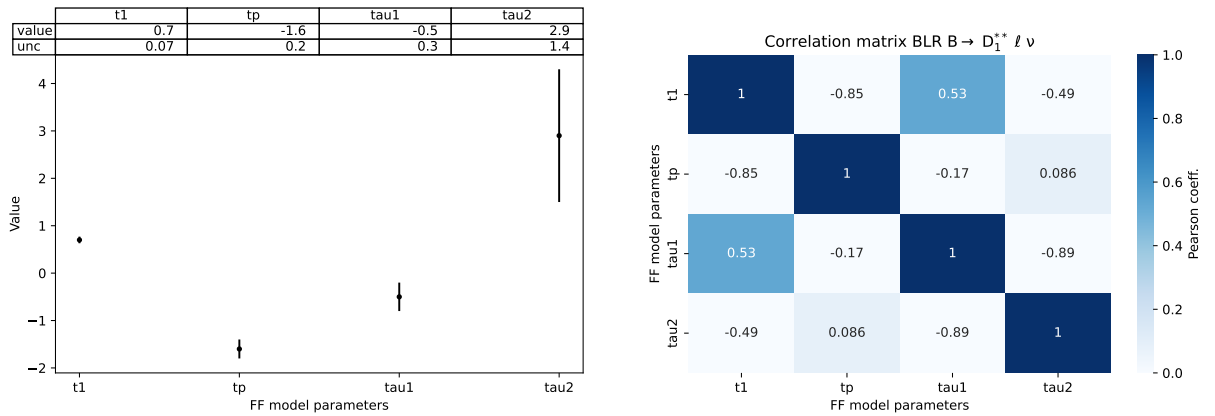


Figure 115: Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_1^{**} \ell \nu$ with HAMMER

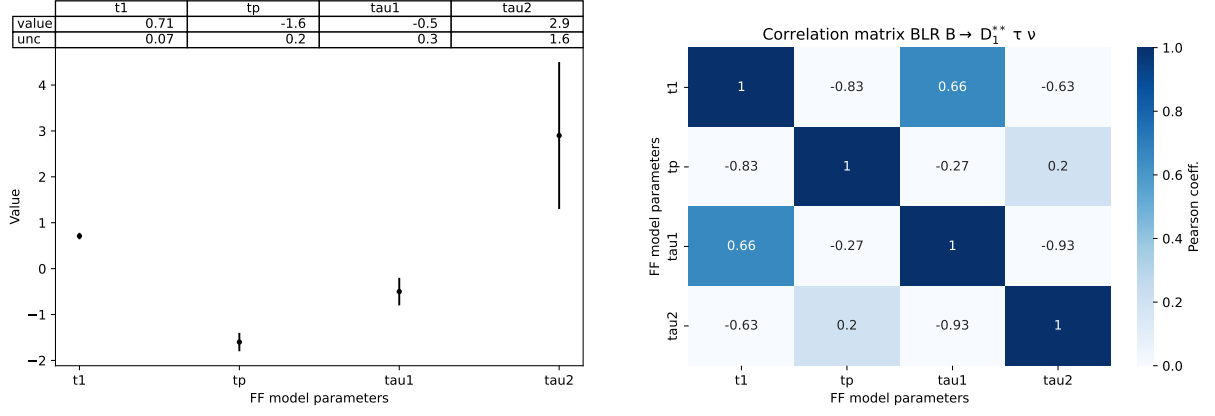


Figure 116: Parameters used in the BLR parameterization for the generation of $B \rightarrow D_1^{**} \tau \nu$ decays in the Belle II MC

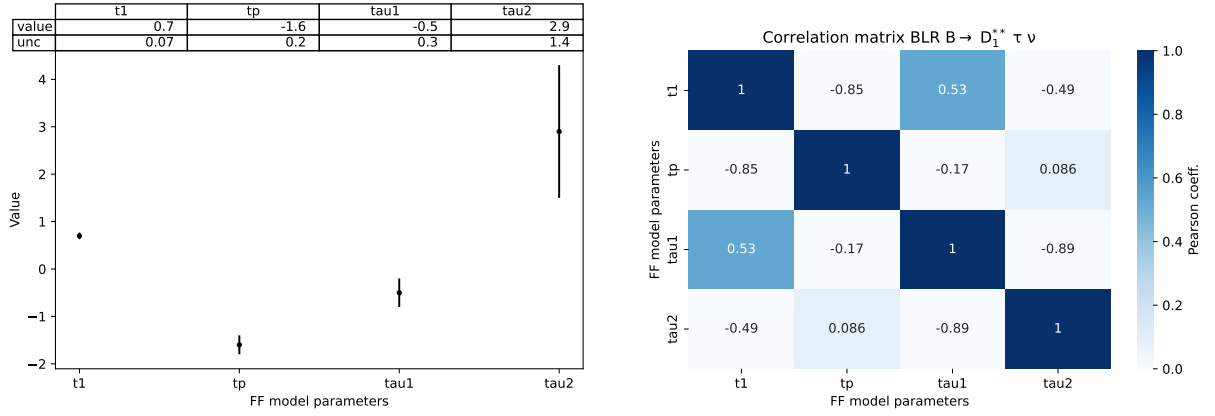


Figure 117: Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_1^{**} \tau \nu$ with HAMMER

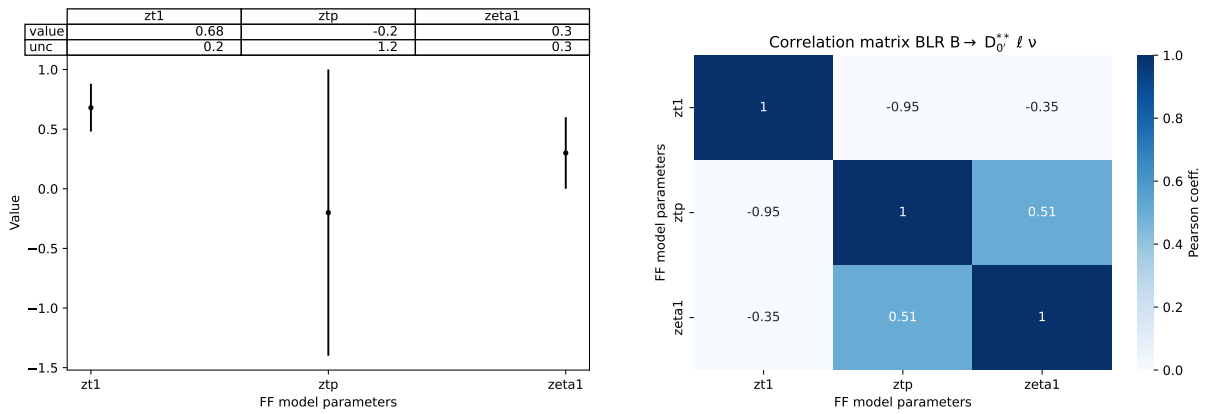


Figure 118: Parameters used in the BLR parameterization for the generation of $B \rightarrow D_0^{**} \ell \nu$ decays in the Belle II MC

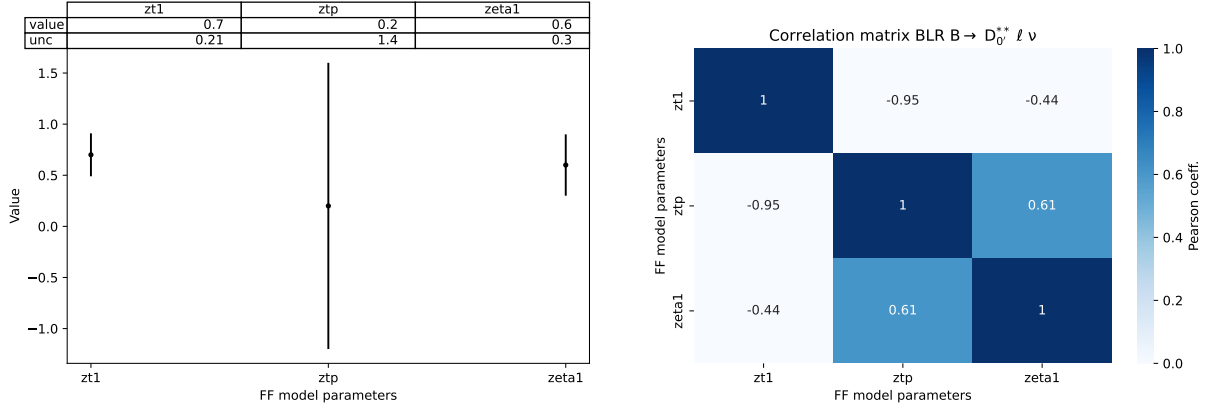


Figure 119: Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_0^{**} \ell \nu$ with HAMMER

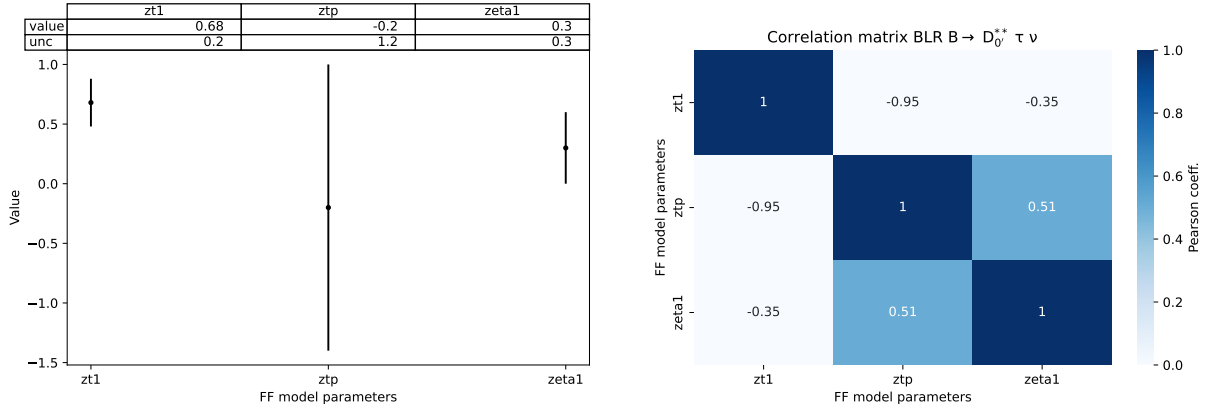


Figure 120: Parameters used in the BLR parameterization for the generation of $B \rightarrow D_0^{**} \tau \nu$ decays in the Belle II MC

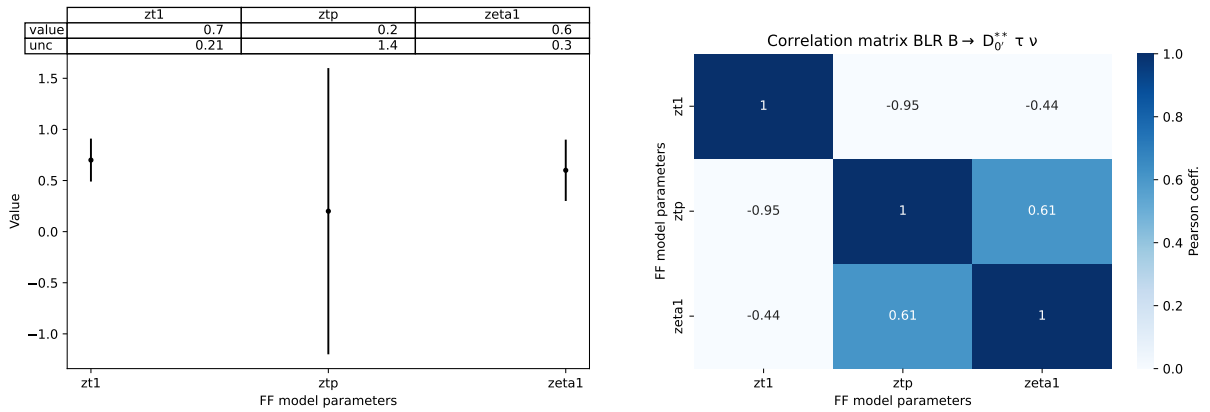


Figure 121: Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_0^{**} \tau \nu$ with HAMMER

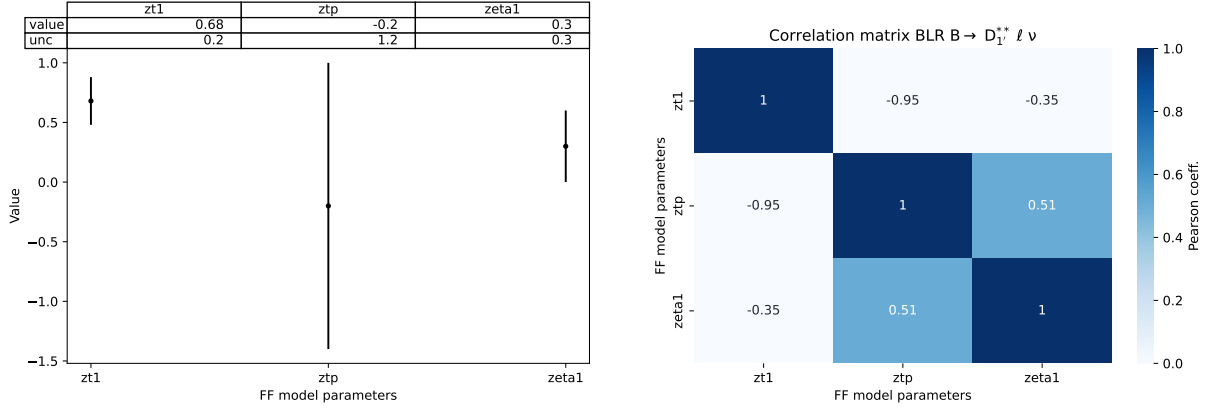


Figure 122: Parameters used in the BLR parameterization for the generation of $B \rightarrow D_1^{**} \ell \nu$ decays in the Belle II MC

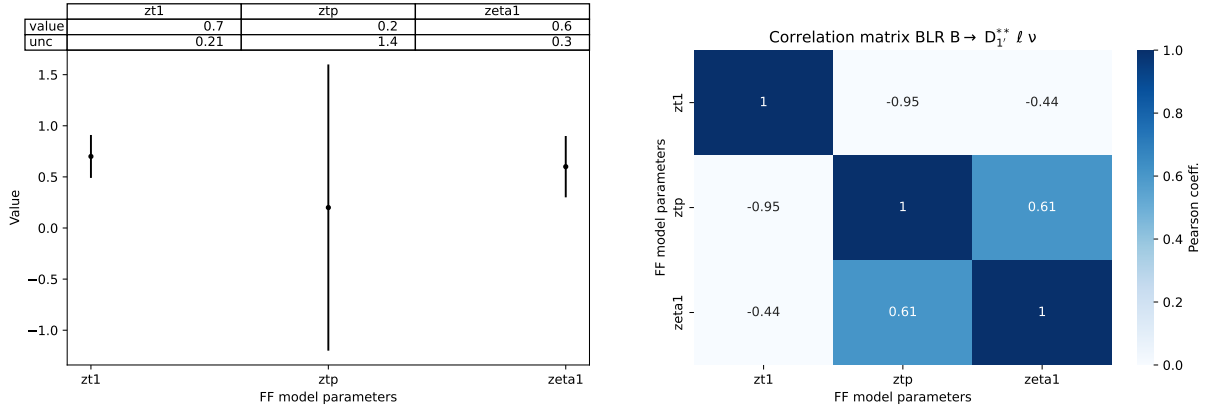


Figure 123: Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_1^{**} \ell \nu$ with HAMMER

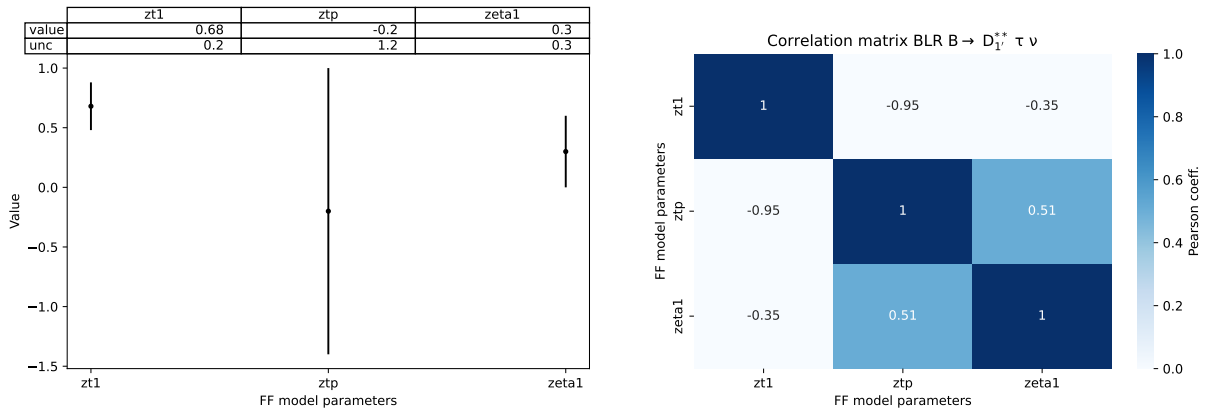


Figure 124: Parameters used in the BLR parameterization for the generation of $B \rightarrow D_1^{**} \tau \nu$ decays in the Belle II MC

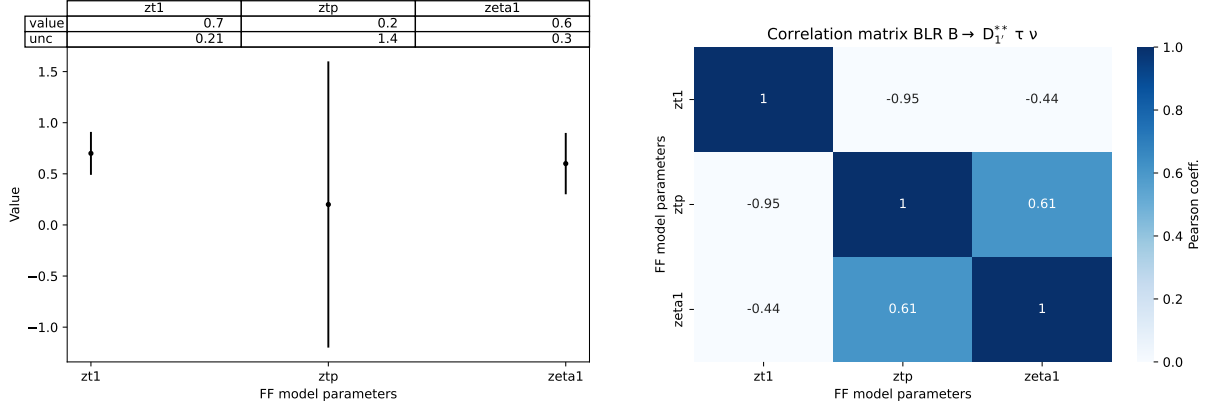


Figure 125: Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_{1'}^{**} \tau \nu$ with HAMMER

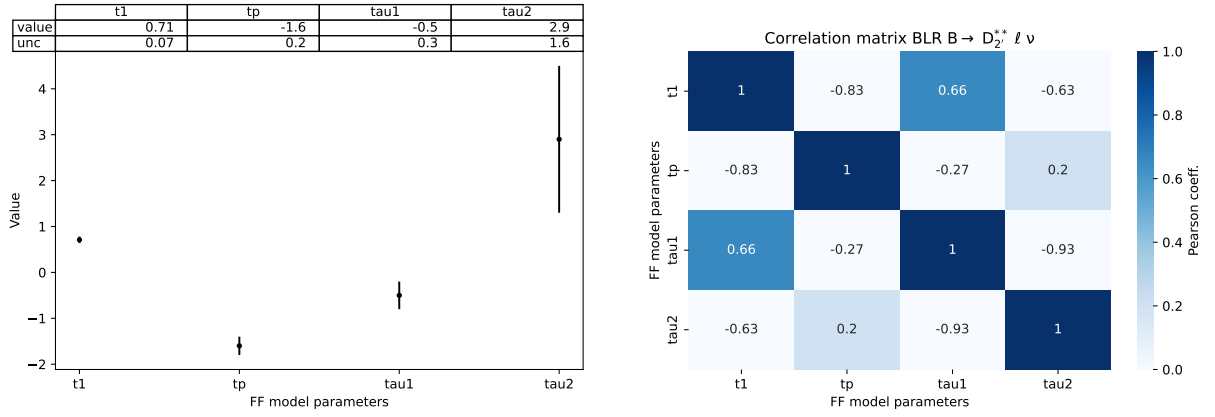


Figure 126: Parameters used in the BLR parameterization for the generation of $B \rightarrow D_2^{**} \ell \nu$ decays in the Belle II MC

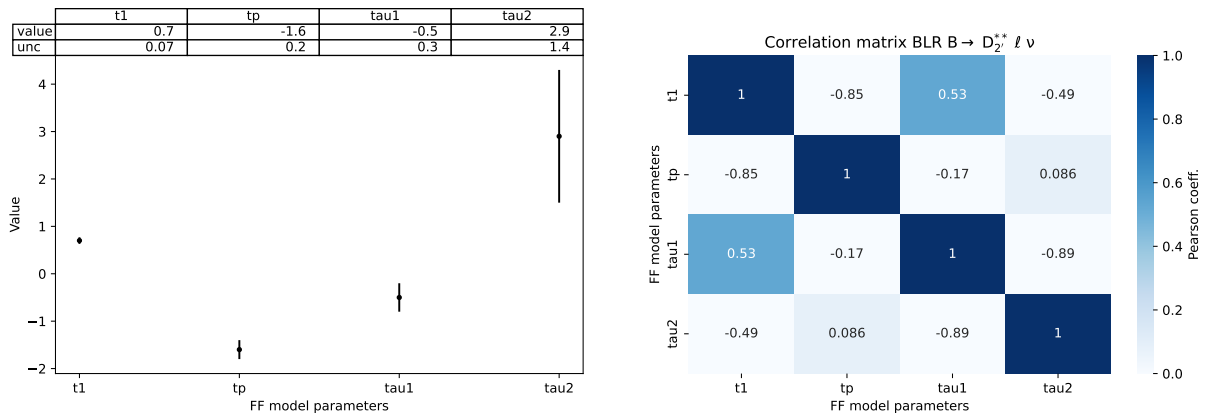


Figure 127: Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_2^{**} \ell \nu$ with HAMMER

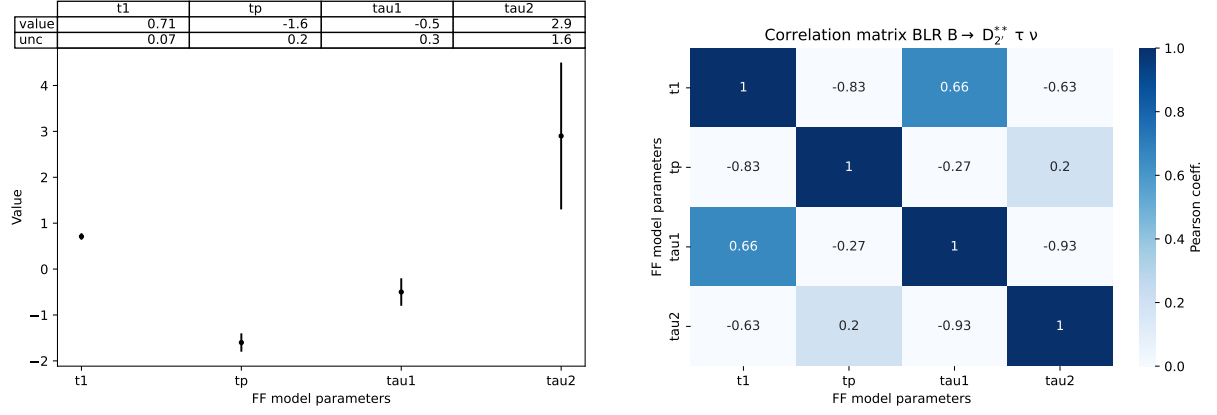


Figure 128: Parameters used in the BLR parameterization for the generation of $B \rightarrow D_2^{*+} \tau \nu$ decays in the Belle II MC

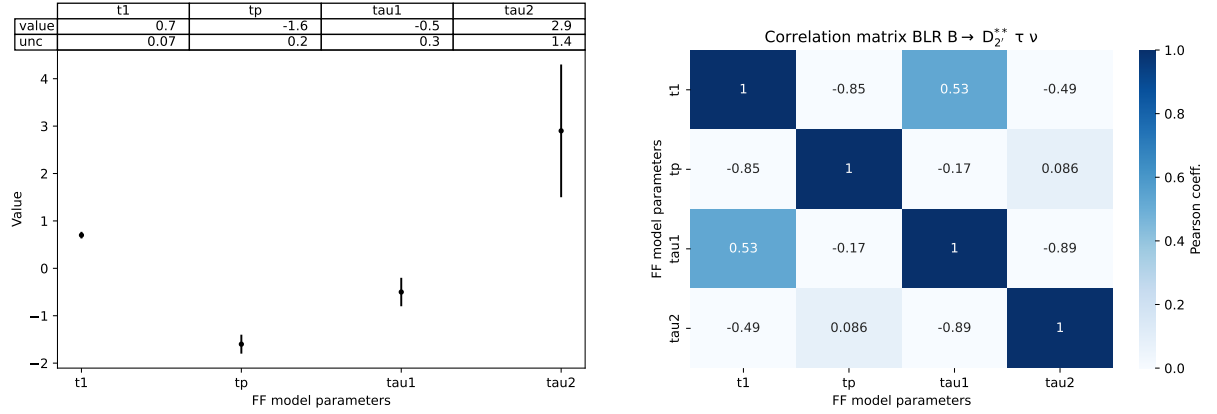


Figure 129: Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_2^{*+} \tau \nu$ with HAMMER

F FEI calibration fit validation

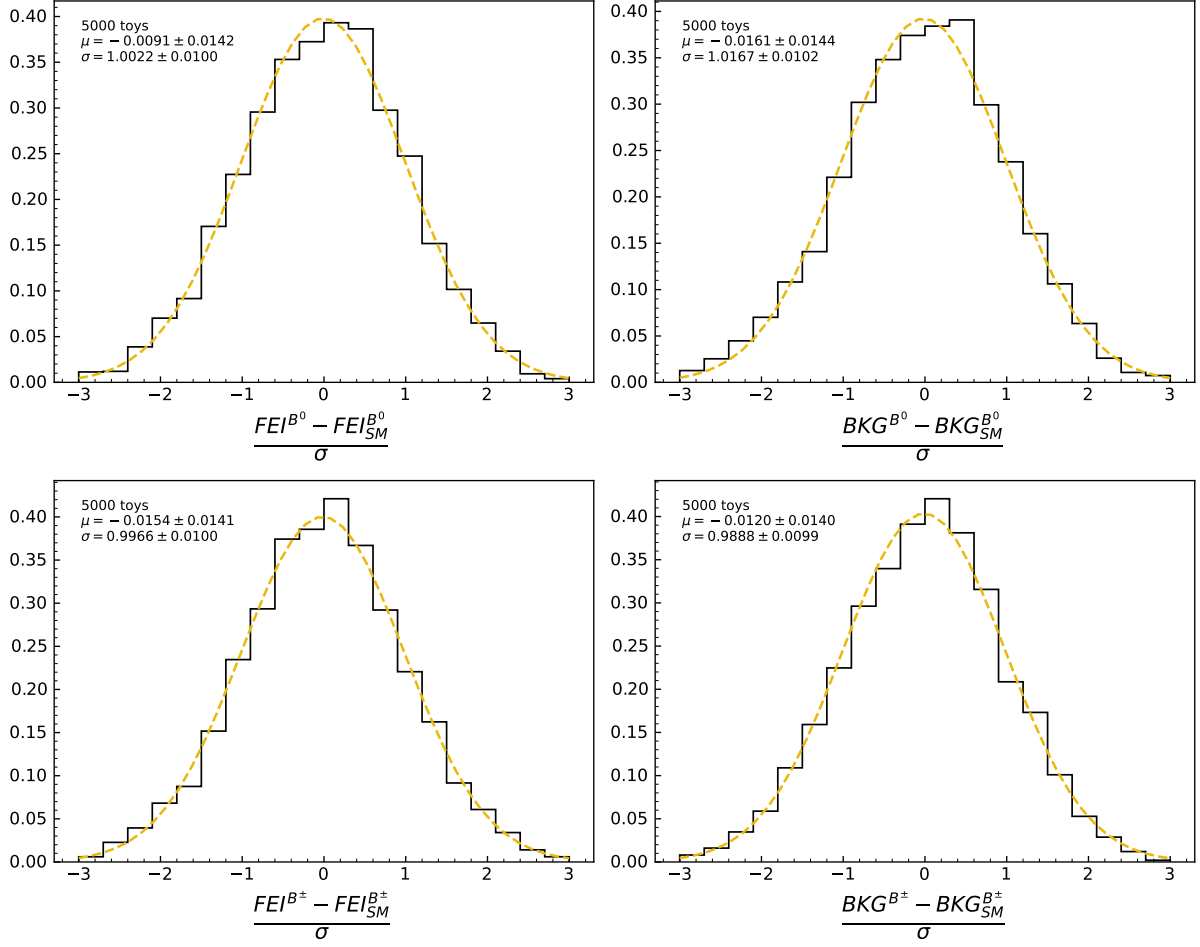


Figure 130: Bias test for FEI calibration estimation. We throw 5000 toys of toy datasets and refit the nominal model to them. All free parameters in the fit are not biased and the uncertainties well estimated based on the mean and standard deviation of the standard normal fit of the pull distribution.

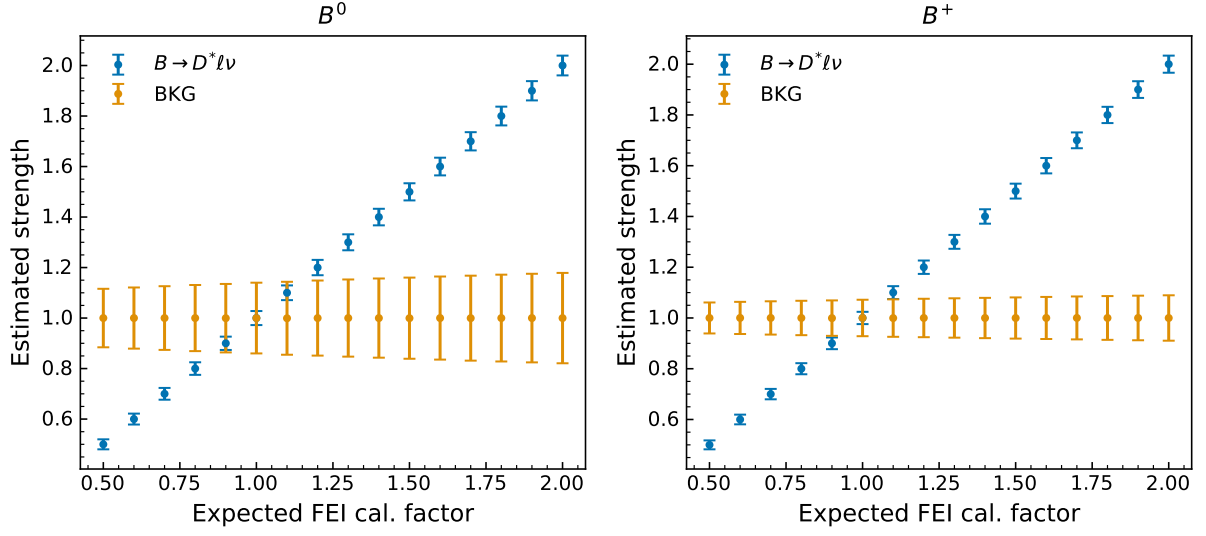


Figure 131: Linearity test for FEI calibration estimation. We bias an asimov dataset by artificially injecting $B \rightarrow D^* \ell \nu$ decays from the nominal histograms. Then we fit this biased asimov dataset to the nominal model. The extracted FEI calibration factors scale as expected while the background strengths remain unaffected.

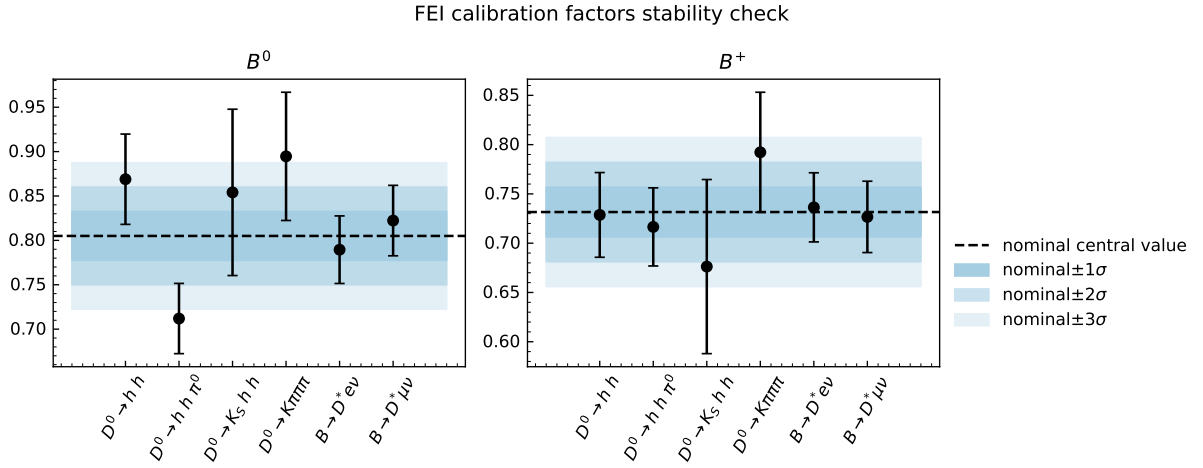


Figure 132: Stability test for FEI calibration. We extract the FEI calibration factors considering different multiplicities by deriving the factors for different D^0 decay modes. We also derive them again for electrons and muons separately. All are consistent within 2σ .

G Statistical tests

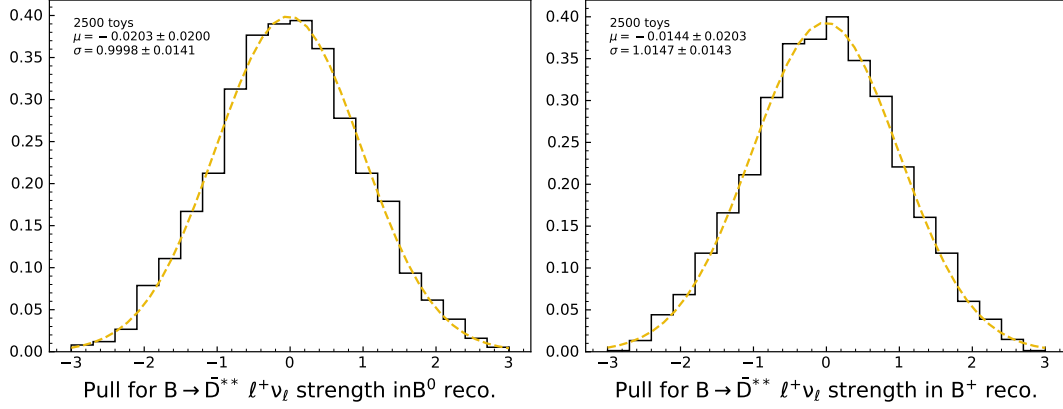


Figure 133: Pull plots for toys $B \rightarrow D^{**} \ell \nu$ yields for B^0 and (left) and B^+ (right) reconstruction channels for the $R(D^*)$ fit. The parameters of the gaussian fit are compatible with a standard normal.

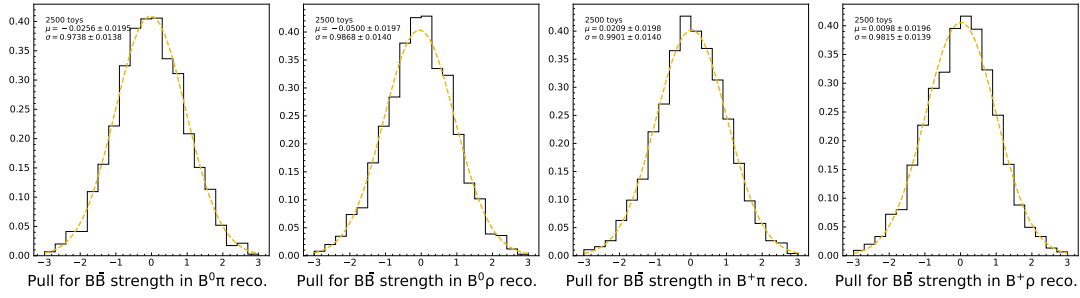


Figure 134: Pull plots for toys $B \bar{B}$ yields for B^0 in the $\tau \rightarrow \pi \nu_\tau$ and $\tau \rightarrow \rho \nu_\tau$ reconstruction channels and B^+ in the $\tau \rightarrow \pi \nu_\tau$ and $\tau \rightarrow \rho \nu_\tau$ reconstruction channels from left to right for the $R(D^*)$ fit. The parameters of the gaussian fit are compatible with a standard normal.

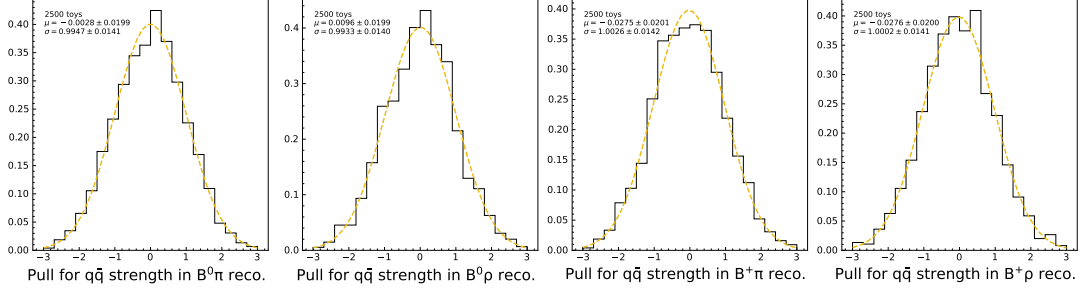


Figure 135: Pull plots for toys $q\bar{q}$ yields for B^0 in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels and B^+ in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels from left to right for the $R(D^*)$ fit. The parameters of the gaussian fit are compatible with a standard normal.

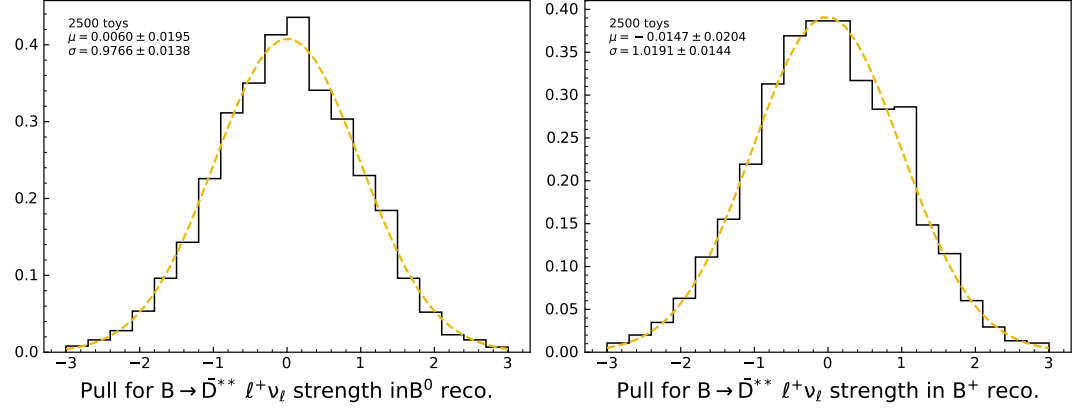


Figure 136: Pull plots for toys $B \rightarrow D^{**}\ell\nu$ yields for B^0 and (left) and B^+ (right) reconstruction channels for the P_τ fit. The parameters of the gaussian fit are compatible with a standard normal.

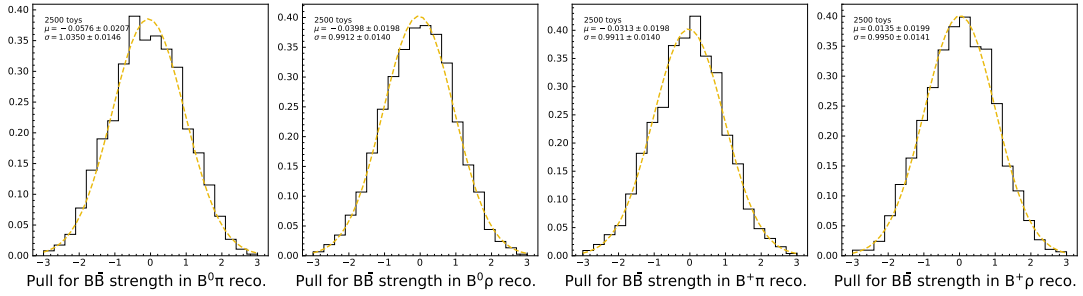


Figure 137: Pull plots for toys $B\bar{B}$ yields for B^0 in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels and B^+ in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels from left to right for the P_τ fit. The parameters of the gaussian fit are compatible with a standard normal.

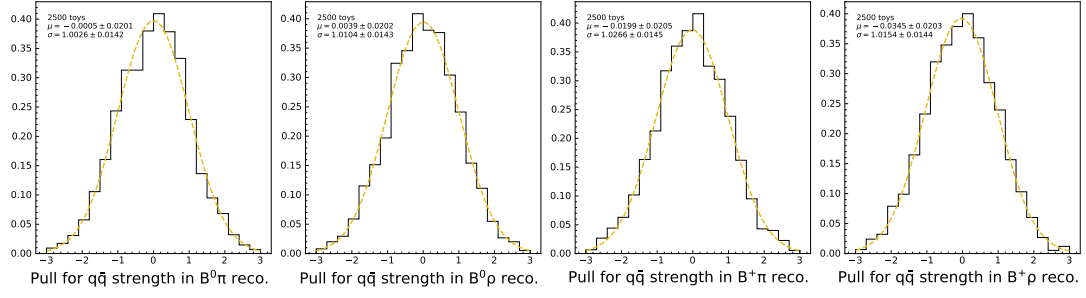


Figure 138: Pull plots for toys $q\bar{q}$ yields for B^0 in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels and B^+ in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels from left to right for the P_τ fit. The parameters of the gaussian fit are compatible with a standard normal.

H Helicity angle window optimization

Even though the physical region of the cosine of the τ helicity angle is between -1 and 1 we expect that resolution effects will make signal events be found in unphysical regions. We perform the following optimization to define the best window per reconstruction channel. We run the simultaneous fit of $R(D^*)$ and P_τ for a set of different $\frac{-\gamma\beta E_d + \gamma p_d \cos\theta_{\tau_d}}{p_d^{\tau_{rest}}}$ windows. For every fit we include only the relevant reconstruction channel. The CR regions are not included in the optimization fit due to technical complications. All backgrounds are allowed to float freely. Figure 139 presents the uncertainty of P_τ for every reconstruction channel as determined by the asimov fit. Based on these results we determine the different cosine helicity angle windows as outlined in Table 20.

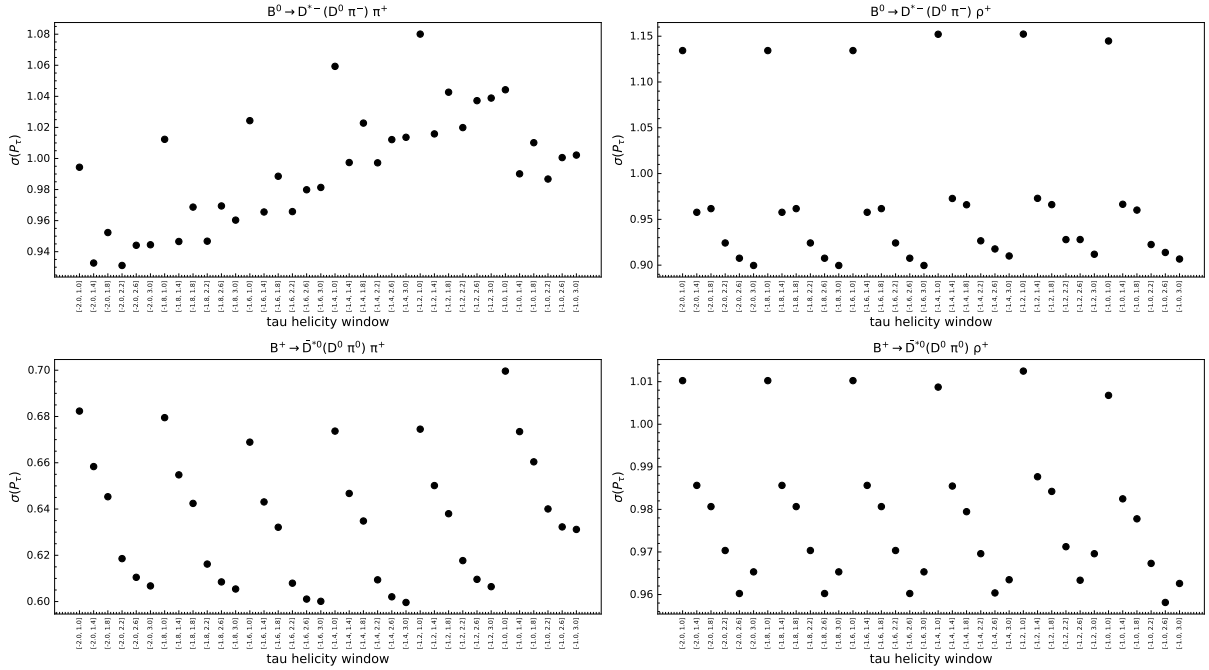


Figure 139: Channel dependant optimization of the helicity angle window

I Selection of D^0 modes

Since the beginning of the analysis we have tried several different reconstruction modes for the decay of D^0 . In the last version of the analysis before the WG review we reconstructed D^0 mesons in eight distinct modes. A final test on the impact of every mode in the sensitivity of the analysis made us keep only six modes.

In order to evaluate each D^0 mode's sensitivity in the analysis I run the Asimov fit once including all the D^0 decay modes and once by excluding one mode at a time. The results can be seen in Figure 140.

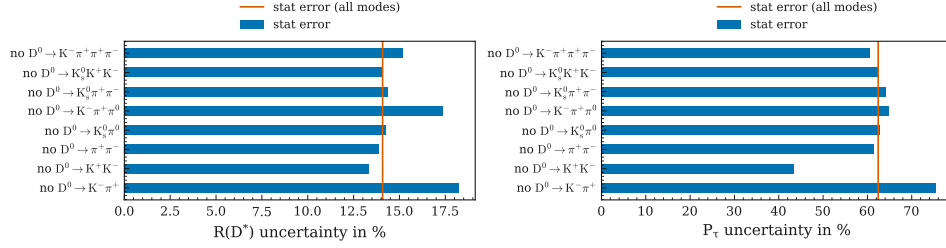


Figure 140: Impact of every D^0 mode on the sensitivity of the parameters of interest. The uncertainties are taken from an Asimov fit. We confirm that we observe closure in the fit (the central values are minimized to the expected ones)

It is very clear that the $D^0 \rightarrow K^- K^+$ mode has the worst sensitivity and it worsens our sensitivity in both parameters of interest. We completely exclude this mode and the repeat the test. The results are presented in Figure 141.

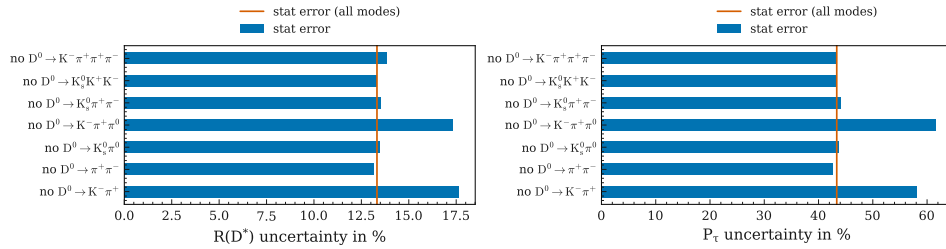


Figure 141: Impact of every D^0 mode on the sensitivity of the parameters of interest. We completely exclude the $D^0 \rightarrow K^- K^+$ mode from this test. The uncertainties are taken from an Asimov fit. We confirm that we observe closure in the fit (the central values are minimized to the expected ones)

We observe that the $D^0 \rightarrow \pi^- \pi^+$ mode still deteriorates the performance of our fit. We completely exclude this mode and then repeat the test. The results are presented in Figure 142.

Since no particular mode is making the performance worse, we decide to move one the remaining six D^0 modes. $D^0 \rightarrow K^- K^+$ and $D^0 \rightarrow \pi^- \pi^+$ will be excluded from the analysis from now on.

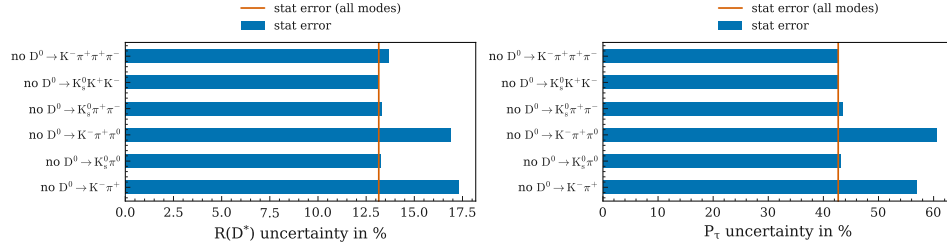


Figure 142: Impact of every D^0 mode on the sensitivity of the parameters of interest. We completely exclude the $D^0 \rightarrow K^- K^+$ and $D^0 \rightarrow \pi^- \pi^+$ modes from this test. The uncertainties are taken from an Asimov fit. We confirm that we observe closure in the fit (the central values are minimized to the expected ones)

J Background composition checks

In order to check that the background composition of the prompt hadronic and double charm decays between the SR and the control regions that are being used are compatible, we're performing the following check. We select the Prompt Hadronic and Double Charm component for every reconstruction channel in the SR and every sideband as described in Table 42. The reconstruction channel together with the helicity bin can be seen in the tile of every figure. We split the components to positive and negative reconstructed helicity as these are treated separately in the control fit. We use the total Data/MC weight (without the luminosity one to increase the statistics) to count the occurrences of every individual decay that makes it into the BBbar template. We renormalize the overall number of events of the most abundant region (SR or sideband) to the least abundant one, in order to not increase our statistics artificially. The regions that has been used for the renormalization can be seen at the title of every subfigure. We only plot decays where we find at least 5 events in one of the two regions as the number of different decays that are reconstructed in these categories is very large. We assume that at least 5 events could have an impact in our determined scaling factors which arguably is a very conservative assumption. In Figures 143 - 158 we present our findings. Event though the composition is not exactly the same, the statistics of every decay mode are very low to claim that they could significantly alter the overall scaling factors that we derive. All these decays contribute only a handful of events out of the hundred or thousand ones that are reconstructed in the BBbar template.

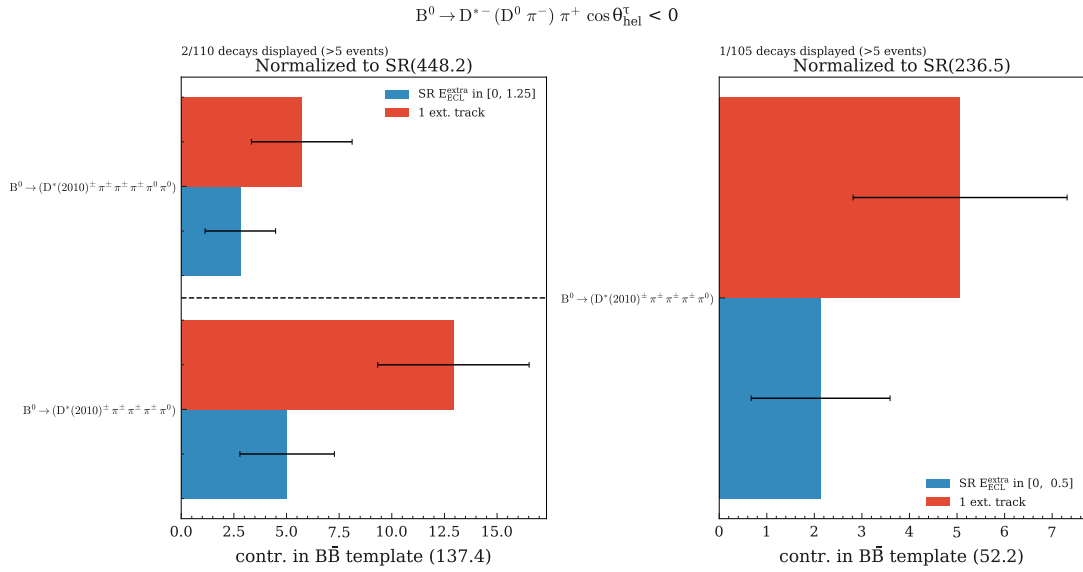


Figure 143: Background composition in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ channel for the Prompt Hadronic component in the negative helicity bin.

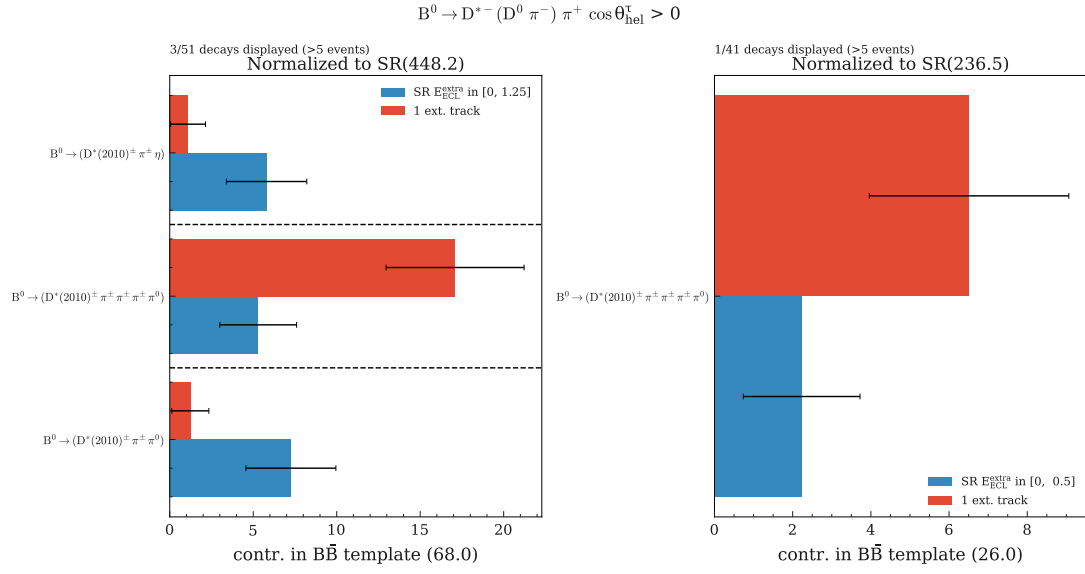


Figure 144: Background composition in the $B^0 \rightarrow D^{*-} (D^0 \pi^-) \pi^+$ channel for the Prompt Hadronic component in the positive helicity bin.

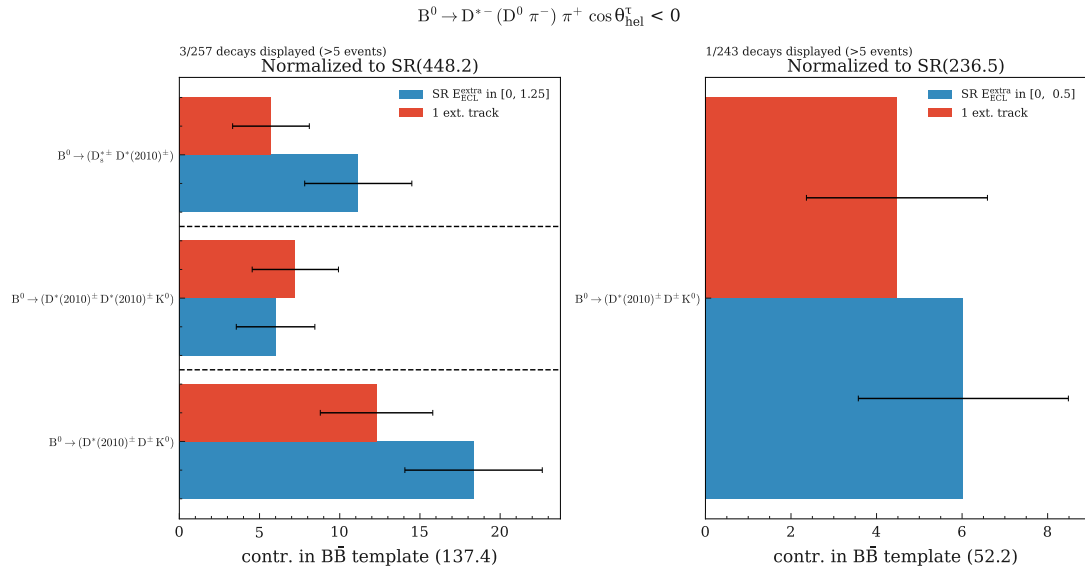


Figure 145: Background composition in the $B^0 \rightarrow D^{*-} (D^0 \pi^-) \pi^+$ channel for the Double Charm component in the negative helicity bin.

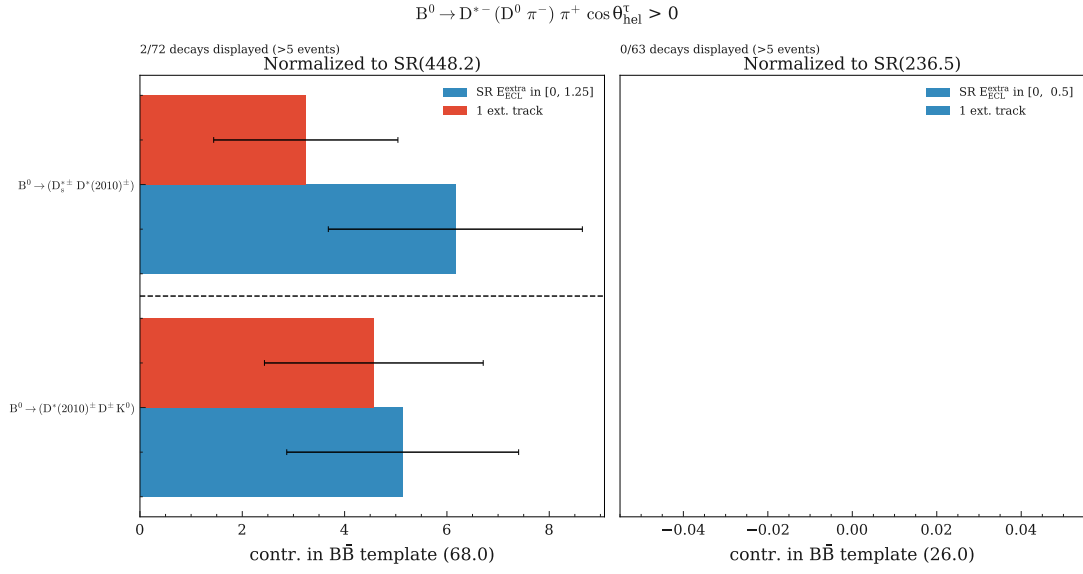


Figure 146: Background composition in the $B^0 \rightarrow D^{*-} (D^0 \pi^-) \pi^+$ channel for the Double Charm component in the positive helicity bin.

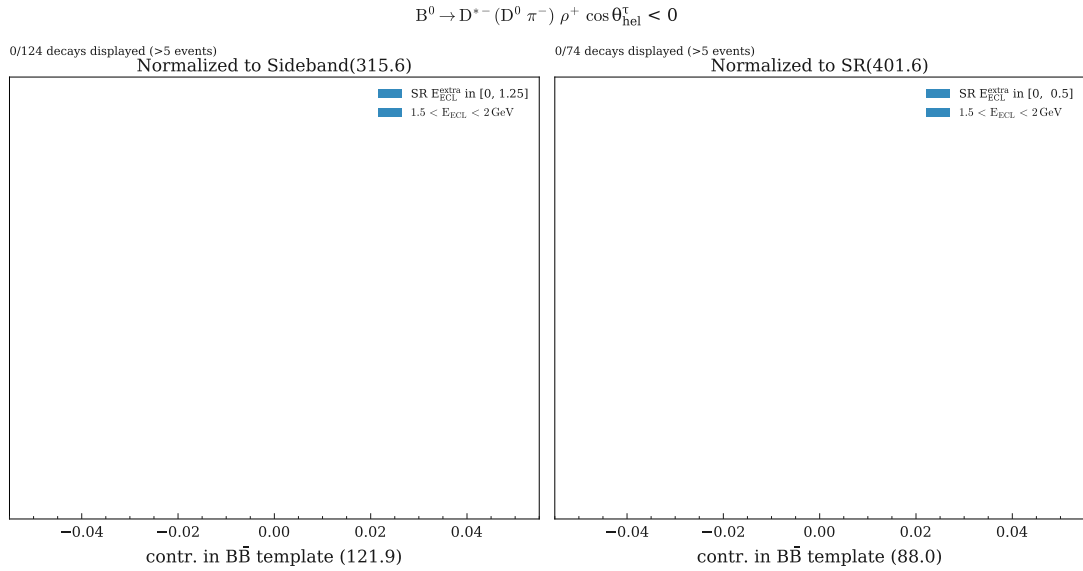


Figure 147: Background composition in the $B^0 \rightarrow D^{*-} (D^0 \pi^-) \rho^+$ channel for the Prompt Hadronic component in the negative helicity bin.

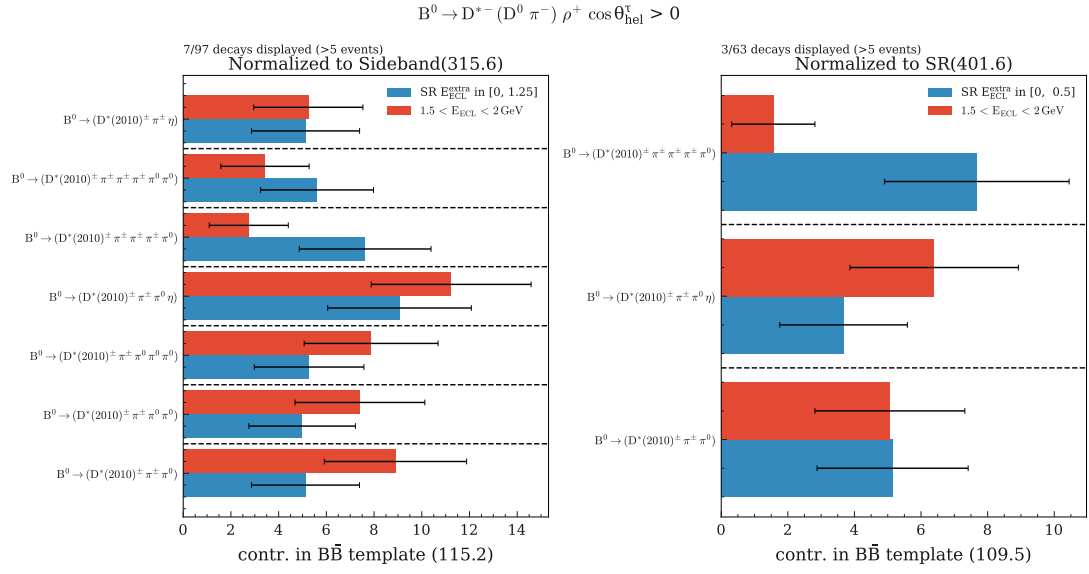


Figure 148: Background composition in the $B^0 \rightarrow D^{*-} (D^0 \pi^-) \rho^+$ channel for the Prompt Hadronic component in the positive helicity bin.

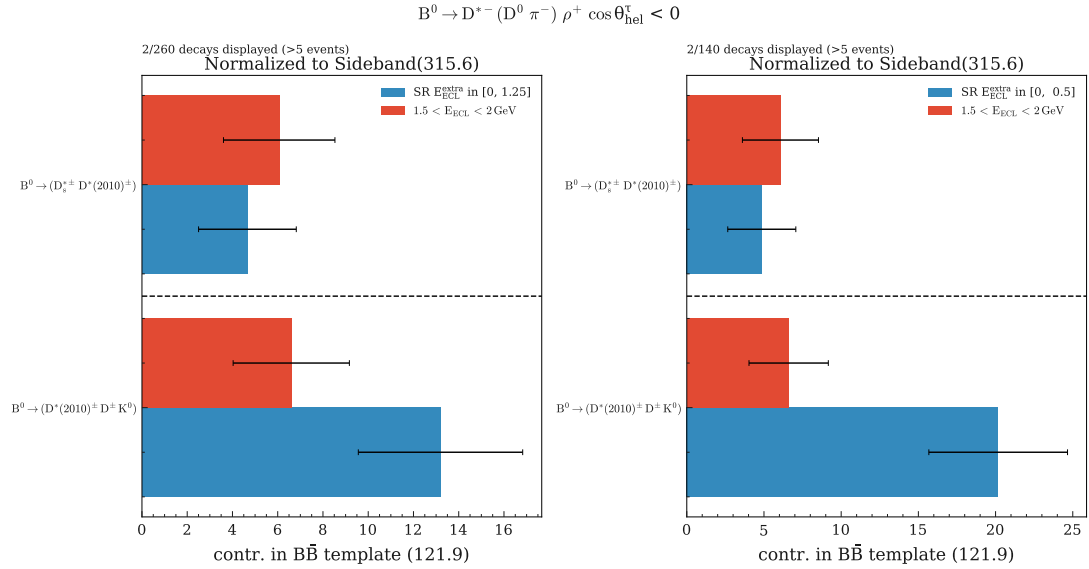


Figure 149: Background composition in the $B^0 \rightarrow D^{*-} (D^0 \pi^-) \rho^+$ channel for the Double Charm component in the negative helicity bin.

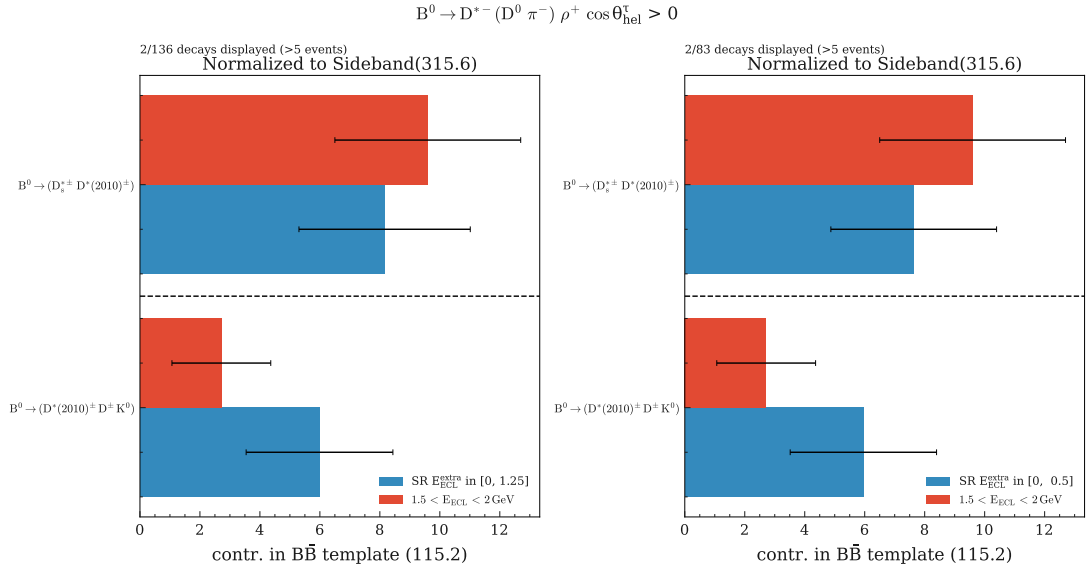


Figure 150: Background composition in the $B^0 \rightarrow D^{*-} (D^0 \pi^-) \rho^+$ channel for the Double Charm component in the positive helicity bin.

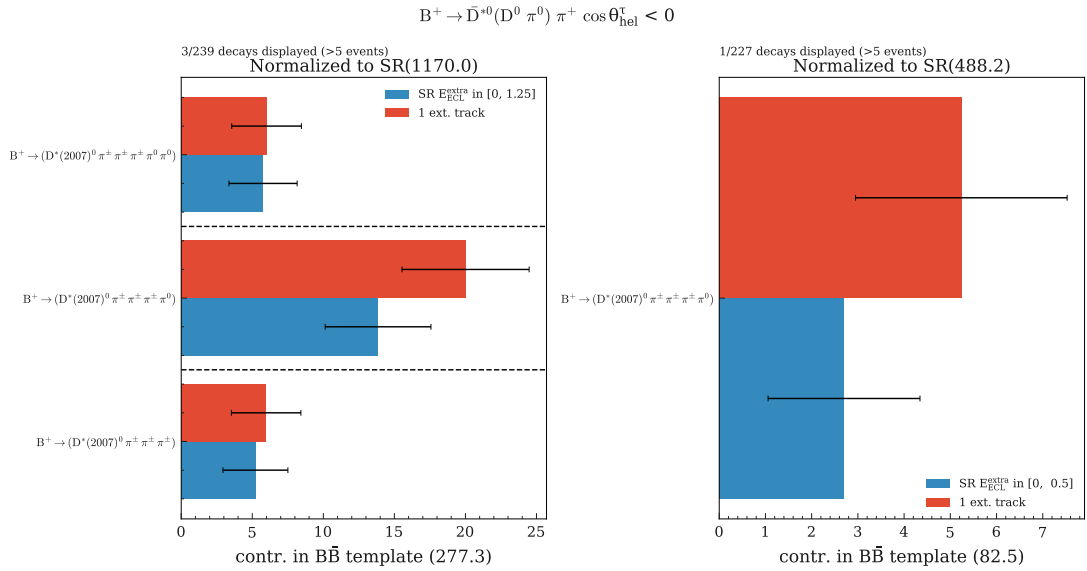


Figure 151: Background composition in the $B^+ \rightarrow D^{*0} (D^0 \pi^0) \pi^+$ channel for the Prompt Hadronic component in the negative helicity bin.

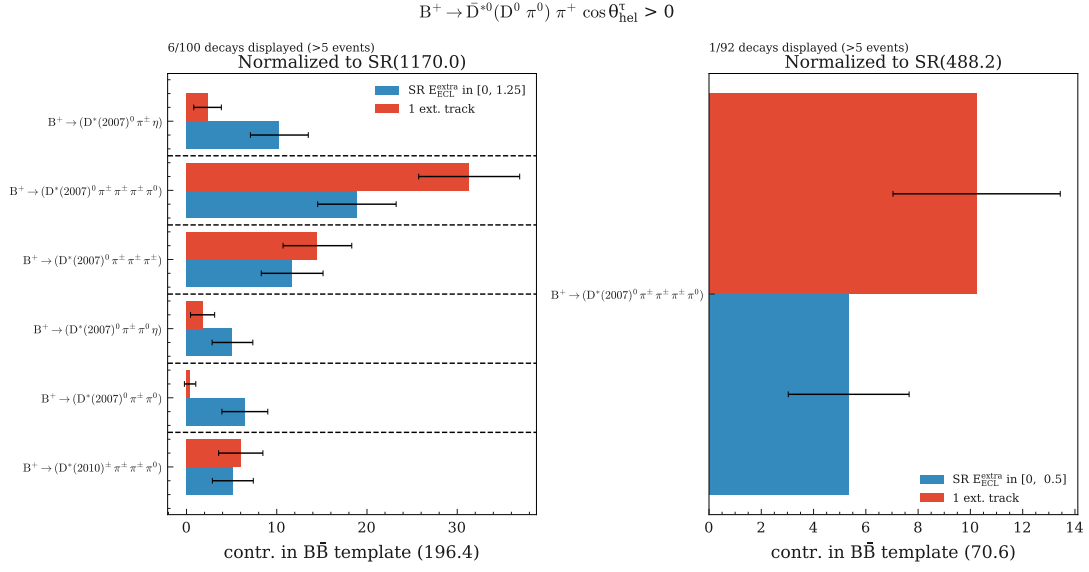


Figure 152: Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ channel for the Prompt Hadronic component in the positive helicity bin.

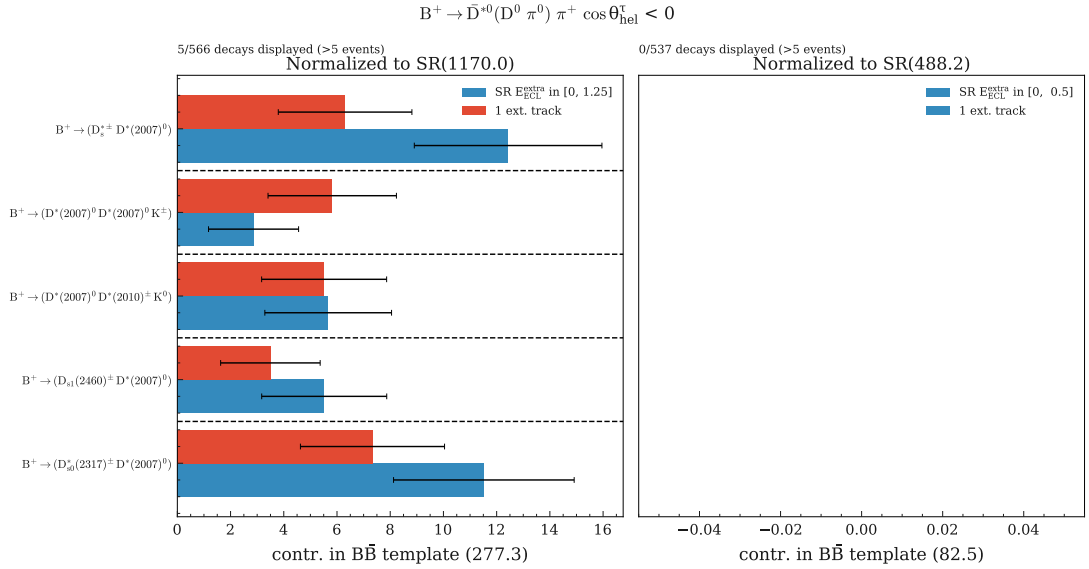


Figure 153: Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ channel for the Double Charm component in the negative helicity bin.

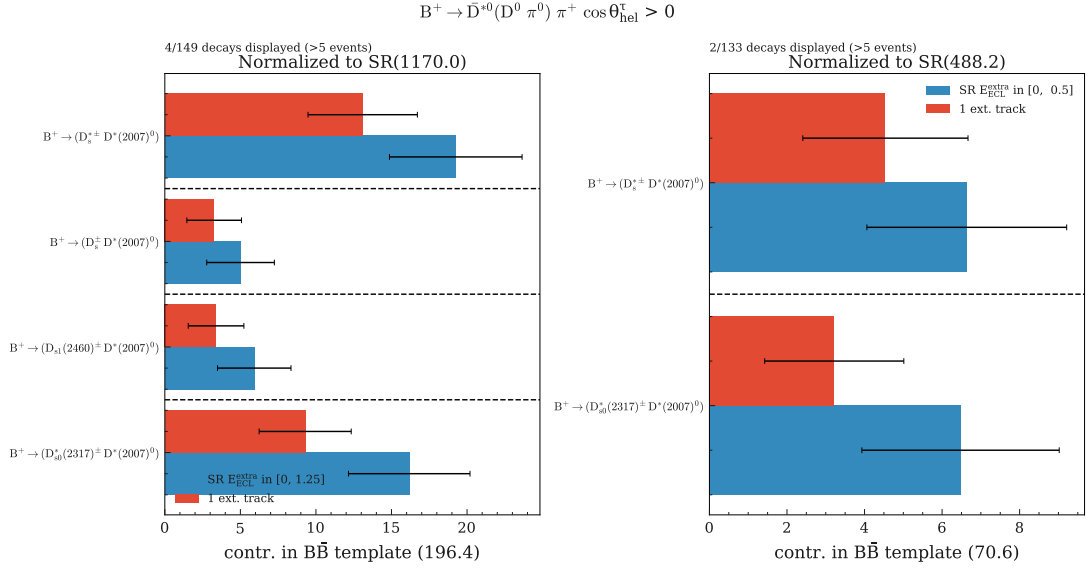


Figure 154: Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ channel for the Double Charm component in the positive helicity bin.

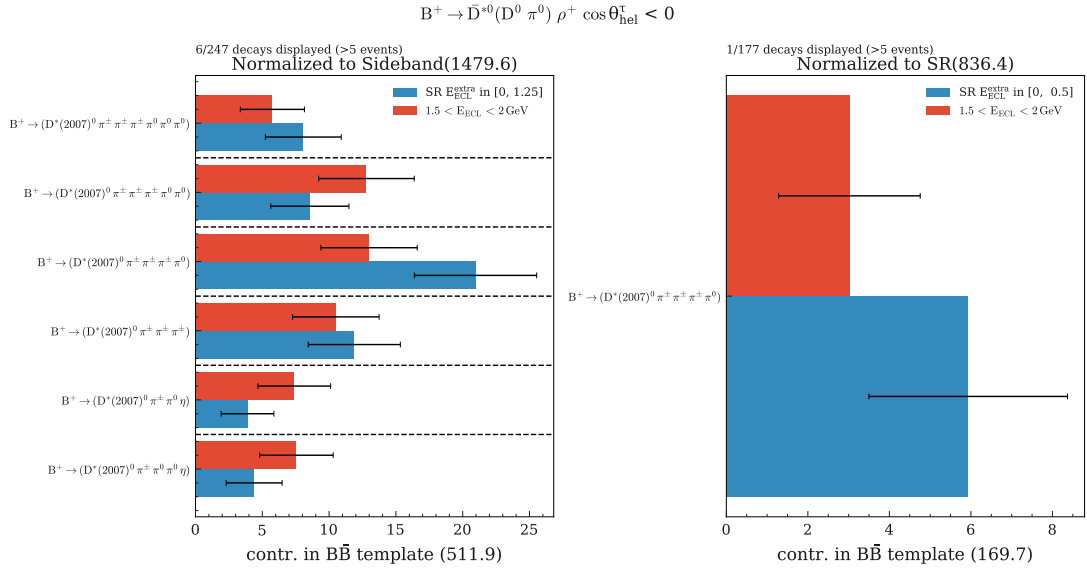


Figure 155: Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ channel for the Prompt Hadronic component in the negative helicity bin.

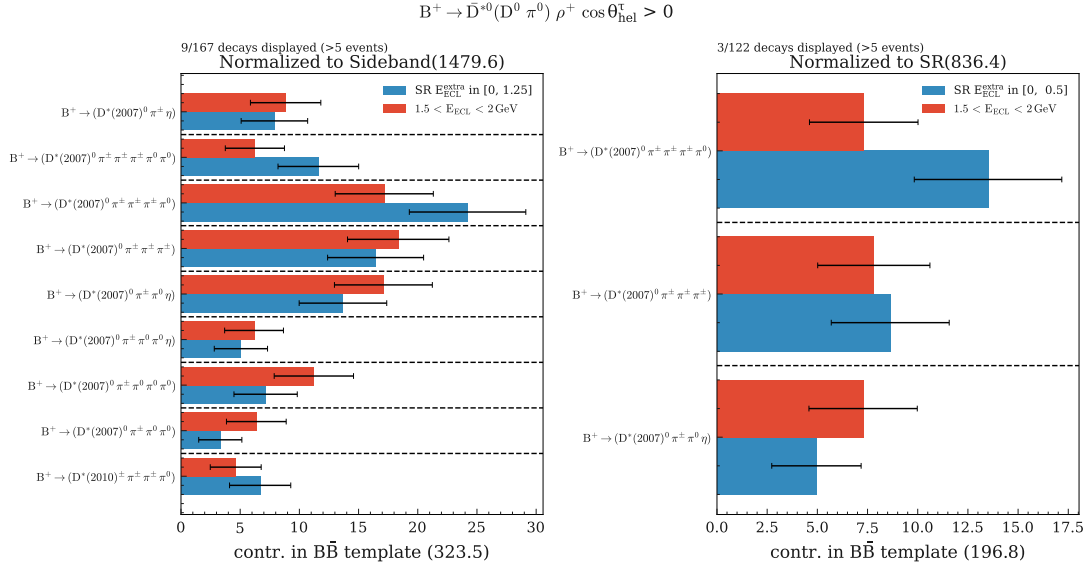


Figure 156: Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ channel for the Prompt Hadronic component in the positive helicity bin.

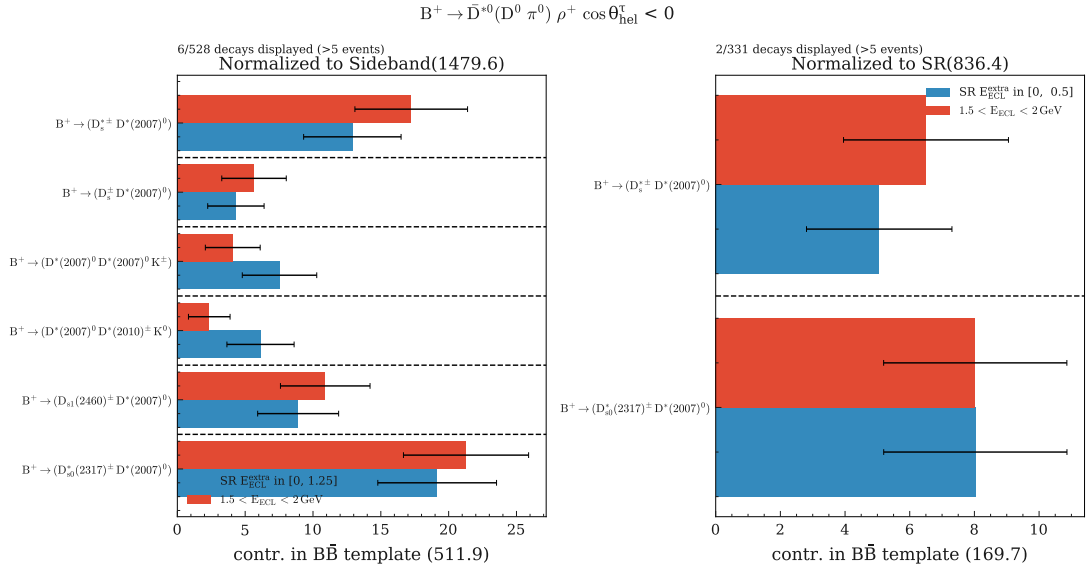


Figure 157: Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ channel for the Double Charm component in the negative helicity bin.

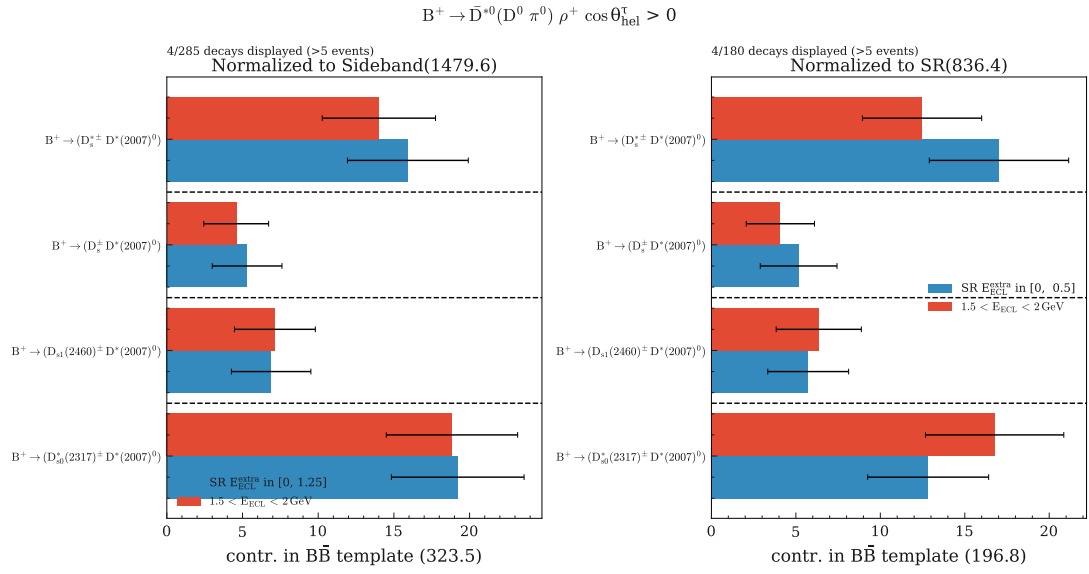


Figure 158: Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ channel for the Double Charm component in the positive helicity bin.

Derivation of Off-Resonance Cut Shifts

Let E_{on} and E_{off} be the center-of-mass (CM) energies for on-resonance and off-resonance data, respectively.

1. Shifting ΔE Cuts

The definition of ΔE for on-resonance data:

$$\Delta E^{\text{on}} = E_B^{\text{on}} - \frac{E_{\text{on}}}{2}$$

For off-resonance data:

$$\Delta E^{\text{off}} = E_B^{\text{off}} - \frac{E_{\text{off}}}{2}$$

To compare off-resonance data with on-resonance cuts, we rescale E_B^{off} :

$$\Delta E^{\text{shifted}} = \frac{E_{\text{on}}}{E_{\text{off}}} E_B^{\text{off}} - \frac{E_{\text{on}}}{2}$$

Suppose we want to apply the on-resonance cut $a < \Delta E^{\text{on}} < b$ to off-resonance data. We solve for the equivalent range in ΔE^{off} .

Set $\Delta E^{\text{shifted}} = x$ and solve for E_B^{off} :

$$x = \frac{E_{\text{on}}}{E_{\text{off}}} E_B^{\text{off}} - \frac{E_{\text{on}}}{2} \implies \frac{E_{\text{on}}}{E_{\text{off}}} E_B^{\text{off}} = x + \frac{E_{\text{on}}}{2} \implies E_B^{\text{off}} = \frac{x + \frac{E_{\text{on}}}{2}}{\frac{E_{\text{on}}}{E_{\text{off}}}}$$

But $E_B^{\text{off}} = \Delta E^{\text{off}} + \frac{E_{\text{off}}}{2}$, so:

$$\Delta E^{\text{off}} + \frac{E_{\text{off}}}{2} = \frac{x + \frac{E_{\text{on}}}{2}}{\frac{E_{\text{on}}}{E_{\text{off}}}} \implies \Delta E^{\text{off}} = \frac{x + \frac{E_{\text{on}}}{2}}{\frac{E_{\text{on}}}{E_{\text{off}}}} - \frac{E_{\text{off}}}{2}$$

Apply this to both lower and upper bounds a and b :

$$\boxed{\Delta E_{\text{low}}^{\text{off}} = \frac{a + \frac{E_{\text{on}}}{2}}{\frac{E_{\text{on}}}{E_{\text{off}}}} - \frac{E_{\text{off}}}{2}} \quad \boxed{\Delta E_{\text{high}}^{\text{off}} = \frac{b + \frac{E_{\text{on}}}{2}}{\frac{E_{\text{on}}}{E_{\text{off}}}} - \frac{E_{\text{off}}}{2}}$$

2. Shifting M_{bc} Cuts

The definition of the beam-constrained mass:

$$M_{\text{bc}} = \sqrt{\left(\frac{E_{\text{beam}}}{2}\right)^2 - |\vec{p}_B|^2}$$

For off-resonance data, if we want to shift $|\vec{p}_B|$ to match on-resonance conditions, we use:

$$M_{\text{bc}}^{\text{shifted}} = \sqrt{\left(\frac{E_{\text{on}}}{2}\right)^2 - \left(\frac{E_{\text{on}}}{E_{\text{off}}}\right)^2 |\vec{p}_B^{\text{off}}|^2}$$

Suppose we want to apply the on-resonance cut $M_{bc}^{\text{shifted}} > M_{\text{cut}}$, but on unshifted off-resonance variables.

Set $M_{bc}^{\text{shifted}} = M_{\text{cut}}$ and solve for $|\vec{p}_B^{\text{off}}|^2$:

$$\begin{aligned} M_{bc}^{\text{shifted}} &= \sqrt{\frac{E_{\text{on}}^2}{4} - \left(\frac{E_{\text{on}}}{E_{\text{off}}}\right)^2 |\vec{p}_B^{\text{off}}|^2} \\ \Rightarrow M_{bc}^{\text{shifted}}{}^2 &= \frac{E_{\text{on}}^2}{4} - \left(\frac{E_{\text{on}}}{E_{\text{off}}}\right)^2 |\vec{p}_B^{\text{off}}|^2 \\ \Rightarrow \left(\frac{E_{\text{on}}}{E_{\text{off}}}\right)^2 |\vec{p}_B^{\text{off}}|^2 &= \frac{E_{\text{on}}^2}{4} - M_{\text{cut}}^2 \\ \Rightarrow |\vec{p}_B^{\text{off}}|^2 &= \frac{E_{\text{off}}^2}{E_{\text{on}}^2} \left(\frac{E_{\text{on}}^2}{4} - M_{\text{cut}}^2 \right) \end{aligned}$$

Now, the unshifted off-resonance M_{bc} is:

$$M_{bc}^{\text{off}} = \sqrt{\frac{E_{\text{off}}^2}{4} - |\vec{p}_B^{\text{off}}|^2}$$

Plug in the value for $|\vec{p}_B^{\text{off}}|^2$:

$$\begin{aligned} M_{bc \text{ cut}}^{\text{off}} &= \sqrt{\frac{E_{\text{off}}^2}{4} - \frac{E_{\text{off}}^2}{E_{\text{on}}^2} \left(\frac{E_{\text{on}}^2}{4} - M_{\text{cut}}^2 \right)} \\ &= \sqrt{\frac{E_{\text{off}}^2}{4} - \frac{E_{\text{off}}^2}{E_{\text{on}}^2} \cdot \frac{E_{\text{on}}^2}{4} + \frac{E_{\text{off}}^2}{E_{\text{on}}^2} M_{\text{cut}}^2} \\ &= \sqrt{\frac{E_{\text{off}}^2}{4} - \frac{E_{\text{off}}^2}{4} + \frac{E_{\text{off}}^2}{E_{\text{on}}^2} M_{\text{cut}}^2} \\ &= \sqrt{\frac{E_{\text{off}}^2}{E_{\text{on}}^2} M_{\text{cut}}^2} \\ &= \frac{E_{\text{off}}}{E_{\text{on}}} M_{\text{cut}} \end{aligned}$$

Wait! This last step suggests a more direct result. Let's verify:

From above:

$$M_{bc \text{ cut}}^{\text{off}} = \sqrt{\frac{E_{\text{off}}^2}{4} - \frac{E_{\text{off}}^2}{E_{\text{on}}^2} \left(\frac{E_{\text{on}}^2}{4} - M_{\text{cut}}^2 \right)}$$

But:

$$\frac{E_{\text{off}}^2}{4} - \frac{E_{\text{off}}^2}{E_{\text{on}}^2} \frac{E_{\text{on}}^2}{4} = \frac{E_{\text{off}}^2}{4} - \frac{E_{\text{off}}^2}{4} = 0$$

So:

$$M_{bc \text{ cut}}^{\text{off}} = \sqrt{\frac{E_{\text{off}}^2}{E_{\text{on}}^2} M_{\text{cut}}^2} = \frac{E_{\text{off}}}{E_{\text{on}}} M_{\text{cut}}$$

3. Final Off-Resonance Cut Equations

For ΔE :

$$\Delta E^{\text{off}} = \frac{\Delta E^{\text{on}} + \frac{E_{\text{on}}}{2}}{\frac{E_{\text{on}}}{E_{\text{off}}}} - \frac{E_{\text{off}}}{2}$$

For M_{bc} :

$$M_{\text{bc}}^{\text{off}} = \frac{E_{\text{off}}}{E_{\text{on}}} M_{\text{bc}}^{\text{on}}$$

4. Numerical Example

Given $E_{\text{on}} = 10.58$ GeV, $E_{\text{off}} = 10.52$ GeV, $M_{\text{bc}}^{\text{on}} = 5.272$ GeV, and $-0.150 < \Delta E^{\text{on}} < 0.100$ GeV:

$$\begin{aligned} \frac{E_{\text{off}}}{E_{\text{on}}} &= \frac{10.52}{10.58} \approx 0.9943 \\ M_{\text{bc}}^{\text{off}} &> 0.9943 \times 5.272 \approx 5.240 \end{aligned}$$

For ΔE :

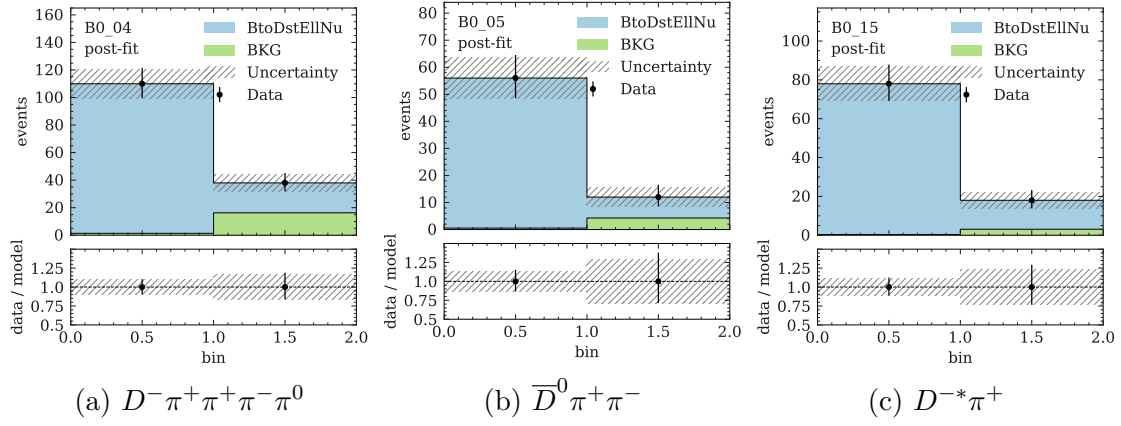
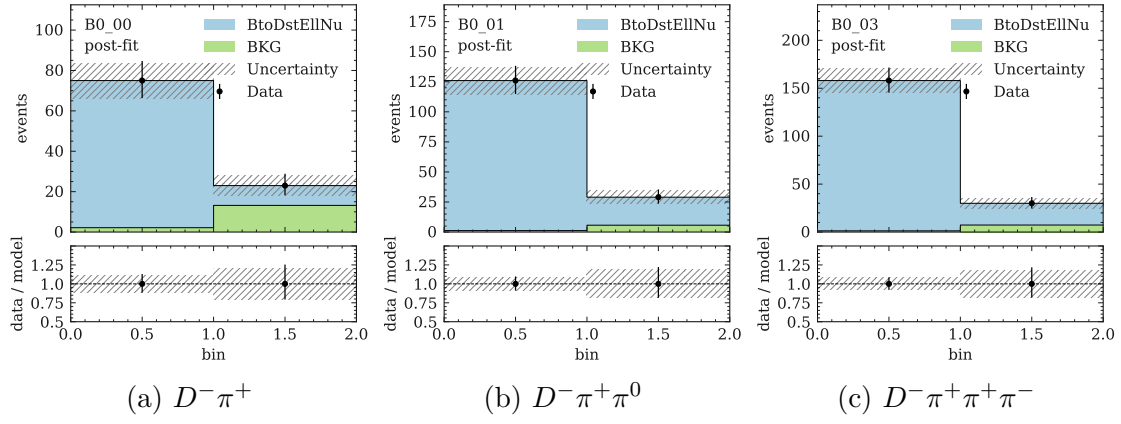
$$\begin{aligned} \Delta E_{\text{low}}^{\text{off}} &= \frac{-0.150 + 5.29}{1.0057} - 5.26 = \frac{5.14}{1.0057} - 5.26 \approx 5.110 - 5.26 = -0.150 \\ \Delta E_{\text{high}}^{\text{off}} &= \frac{0.100 + 5.29}{1.0057} - 5.26 = \frac{5.39}{1.0057} - 5.26 \approx 5.358 - 5.26 = 0.098 \end{aligned}$$

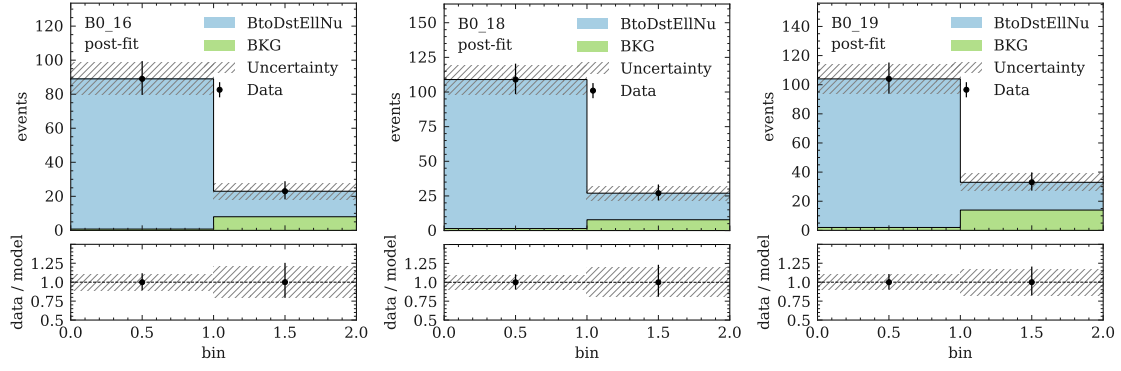
Therefore, the equivalent off-resonance cuts are:

$$\begin{aligned} M_{\text{bc}}^{\text{off}} &> 5.240 \\ -0.150 &< \Delta E^{\text{off}} < 0.098 \end{aligned}$$

K FEI calibration postfit plots

Post fit distributions of the FEI calibration

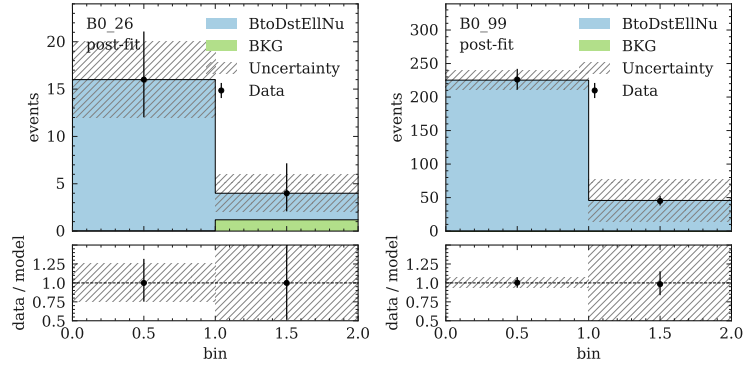




(a) $D^{-*}\pi^+\pi^0$

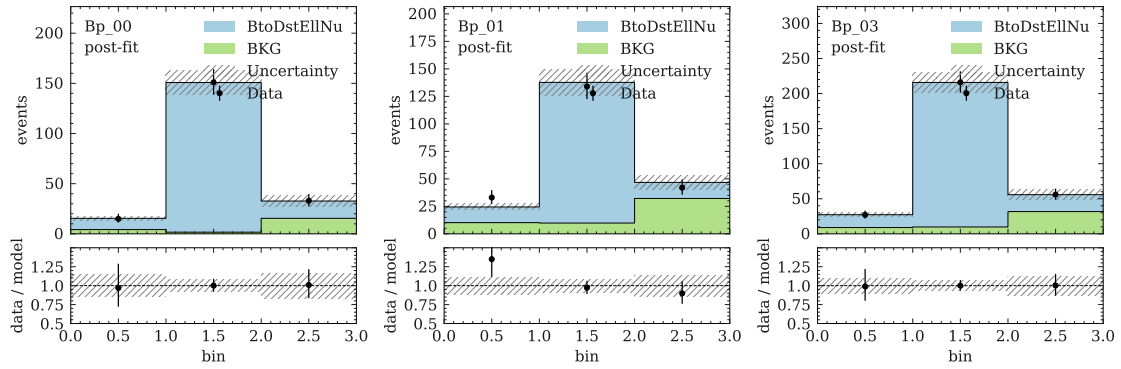
(b) $D^{-*}\pi^+\pi^+\pi^-$

(c) $D^{-*}\pi^+\pi^+\pi^-\pi^0$



(a) $\Lambda_c^- p \pi^+ \pi^-$

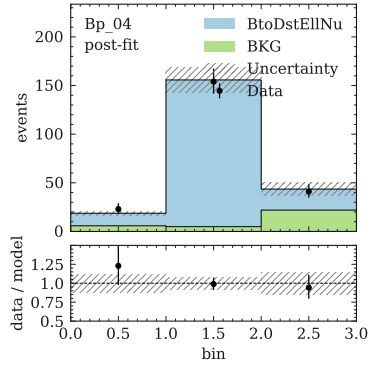
(b) Rest



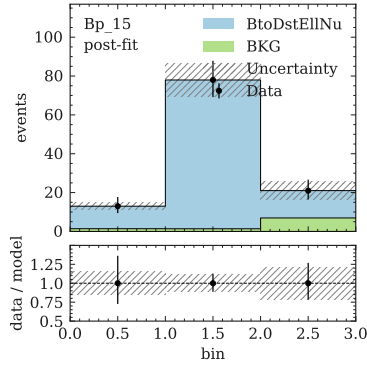
(a) $\bar{D}^0 \pi^+$

(b) $\bar{D}^0 \pi^+ \pi^0$

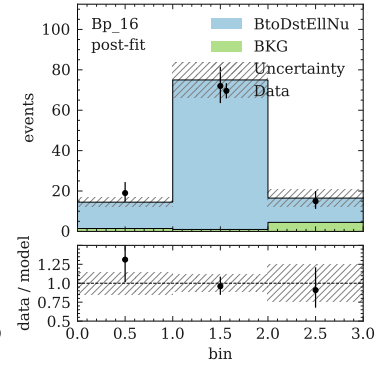
(c) $\bar{D}^0 \pi^+ \pi^+ \pi^-$



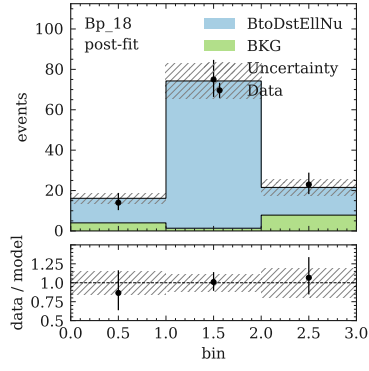
(a) $\bar{D}^0 \pi^+ \pi^+ \pi^- \pi^0$



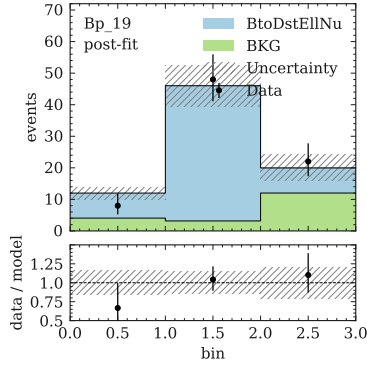
(b) $\bar{D}^{0*} \pi^+$



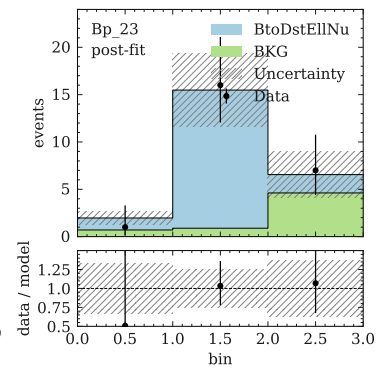
(c) $\bar{D}^{0*} \pi^+ \pi^0$



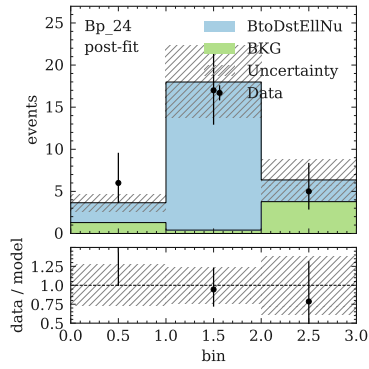
(a) $\bar{D}^{0*} \pi^+ \pi^+ \pi^-$



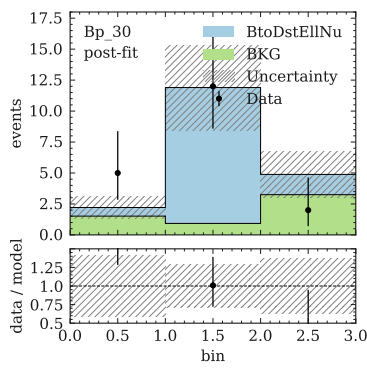
(b) $\bar{D}^{0*} \pi^+ \pi^+ \pi^- \pi^0$



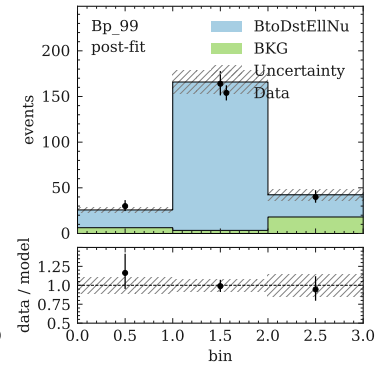
(c) $D^- \pi^+ \pi^+$



(a) $D^- \pi^+ \pi^+ \pi^0$



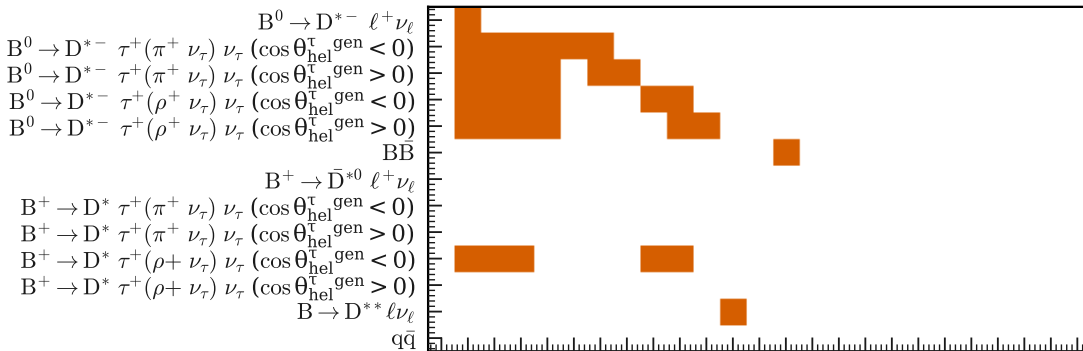
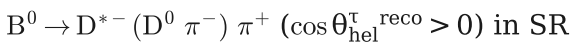
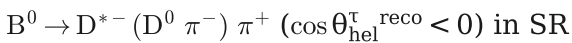
(b) $\Lambda_c^- p \pi^+ \pi^- \pi^+$



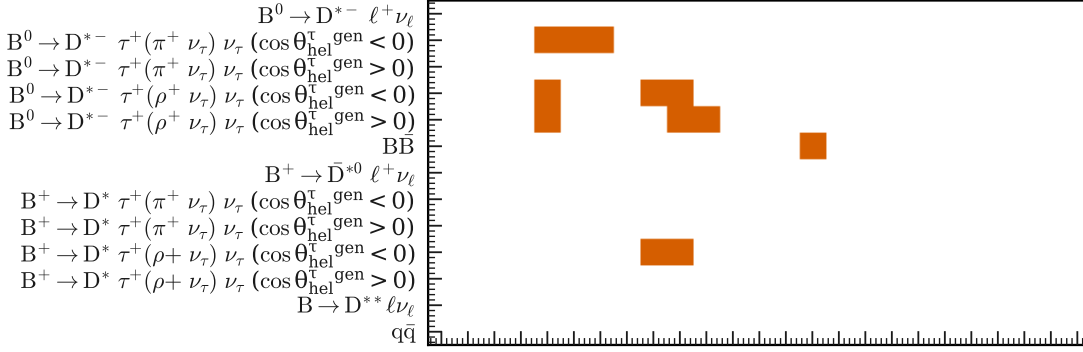
(c) Rest

L Free parameters of the fit

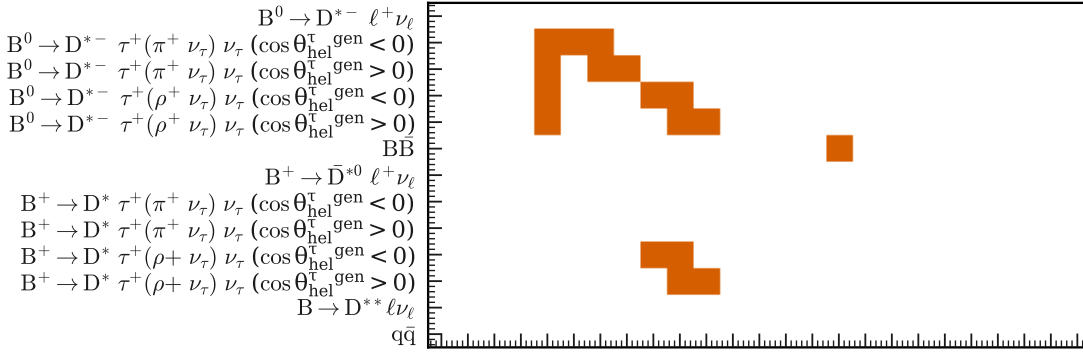
In this section we provide a grid plot of all the free and fixed parameters of the fit. The titles of all the subfigures show the reconstruction channel that the parameters are affecting. The y axis ticks show the template that the parameters is affecting. The x axis ticks show the name of the parameters.



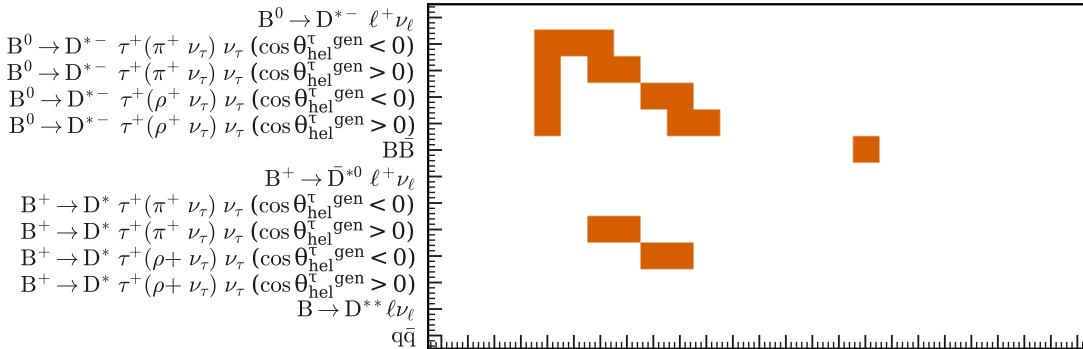
$B^0 \rightarrow D^{*-} (D^0 \pi^-) \rho^+ (\cos \theta_{\text{hel}}^{\tau, \text{reco}} < 0)$ and $-1 < m_{\text{miss}}^2 < 1 \text{ GeV}^2$ in SR



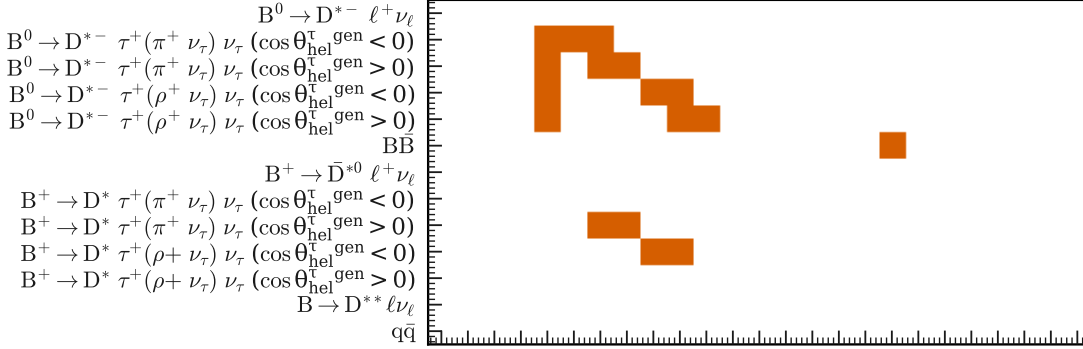
$B^0 \rightarrow D^{*-} (D^0 \pi^-) \rho^+ (\cos \theta_{\text{hel}}^{\tau, \text{reco}} < 0)$ and $1 < m_{\text{miss}}^2 < 7 \text{ GeV}^2$ in SR



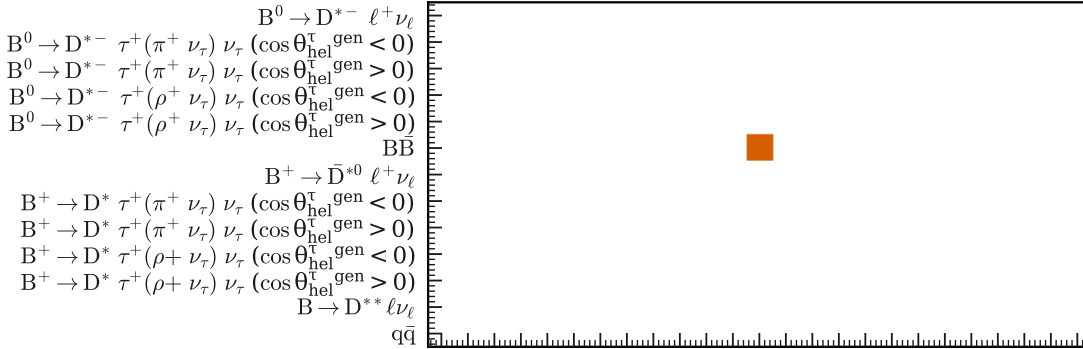
$B^0 \rightarrow D^{*-} (D^0 \pi^-) \rho^+ (\cos \theta_{\text{hel}}^{\tau, \text{reco}} > 0)$ and $-1 < m_{\text{miss}}^2 < 1 \text{ GeV}^2$ in SR



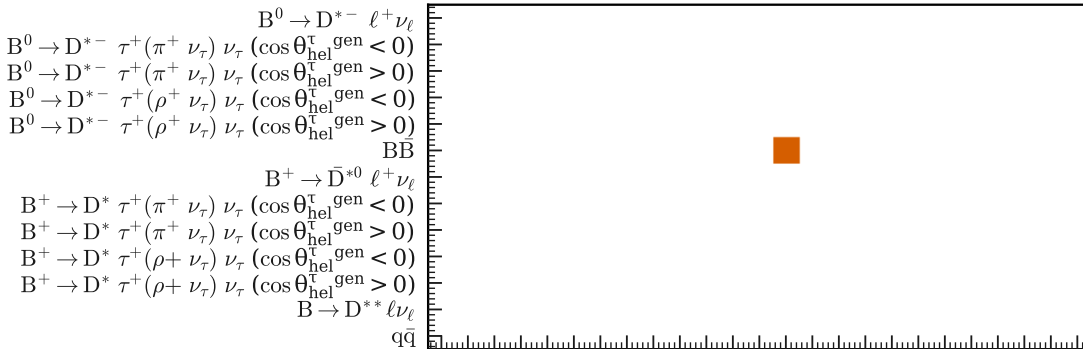
$B^0 \rightarrow D^{*-} (D^0 \pi^-) \rho^+ (\cos \theta_{\text{hel}}^{\tau, \text{reco}} > 0)$ and $1 < m_{\text{miss}}^2 < 7 \text{ GeV}^2$ in SR



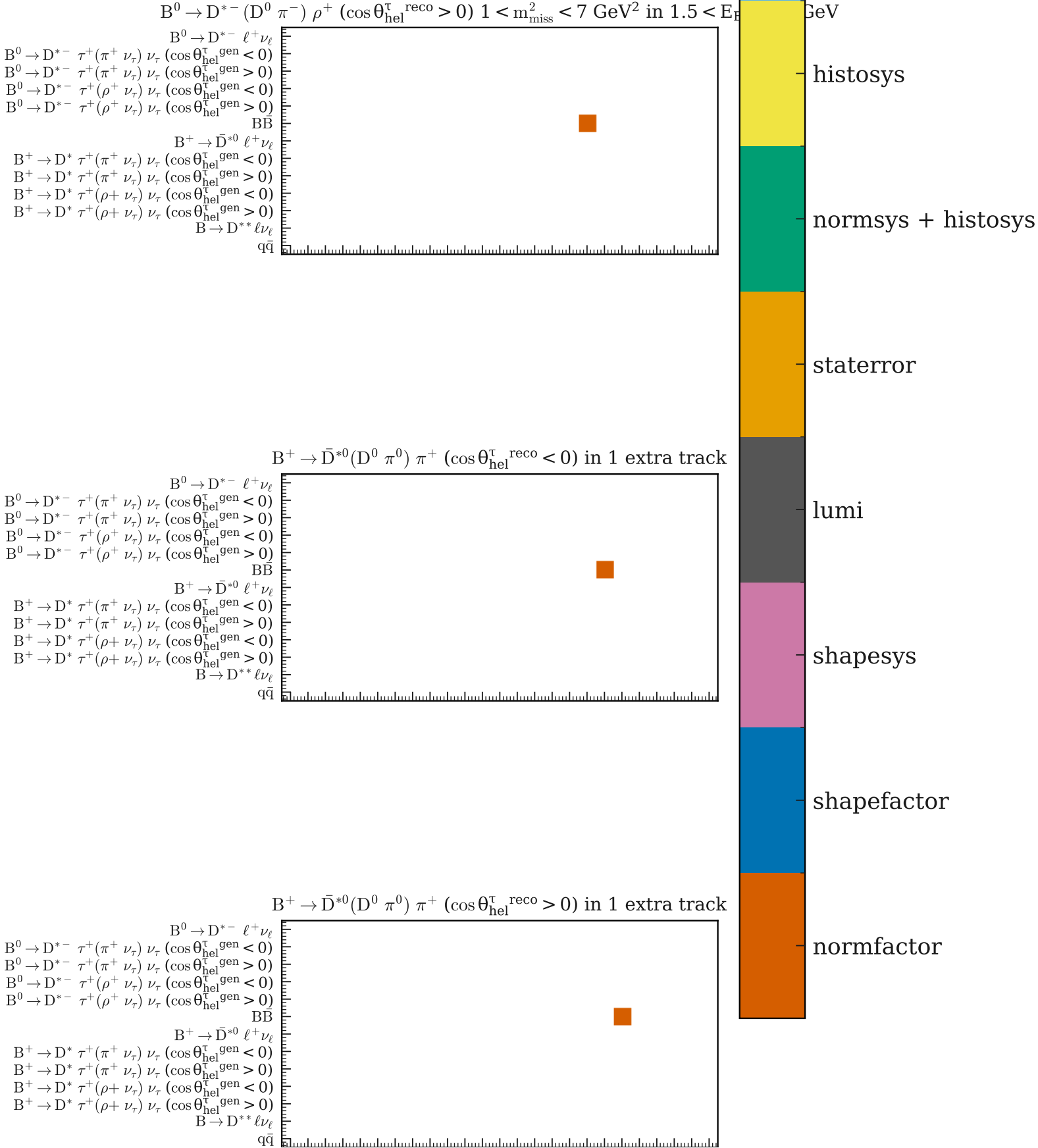
$B^0 \rightarrow D^{*-} (D^0 \pi^-) \pi^+ (\cos \theta_{\text{hel}}^{\tau, \text{reco}} < 0)$ in 1 extra track



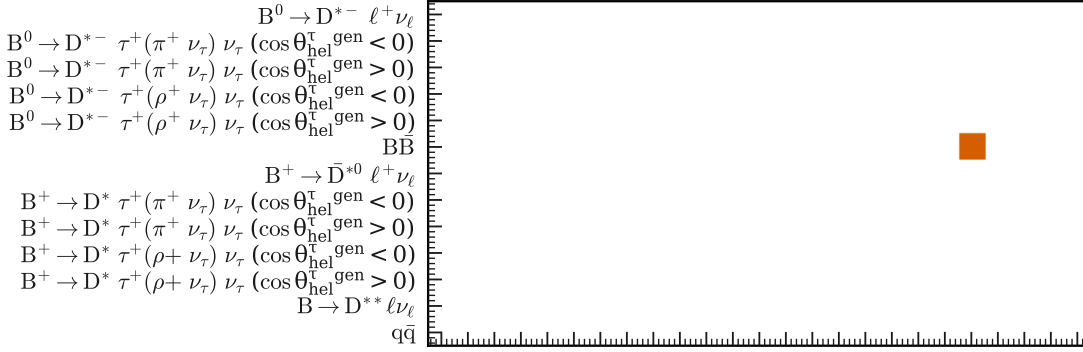
$B^0 \rightarrow D^{*-} (D^0 \pi^-) \pi^+ (\cos \theta_{\text{hel}}^{\tau, \text{reco}} > 0)$ in 1 extra track



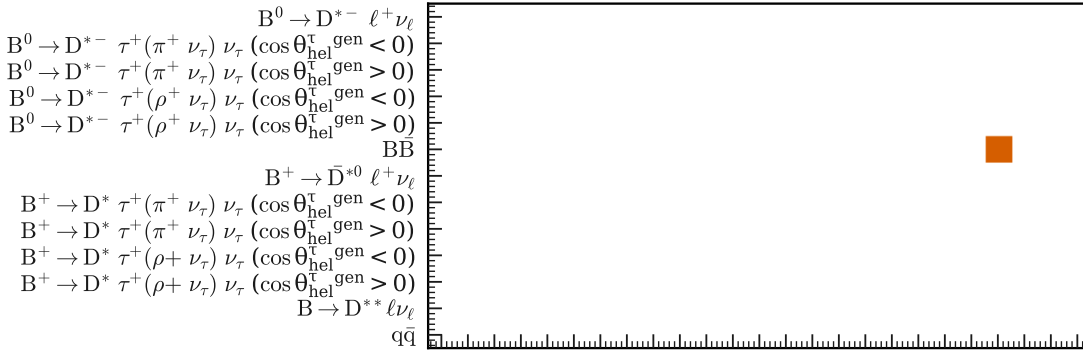
— none



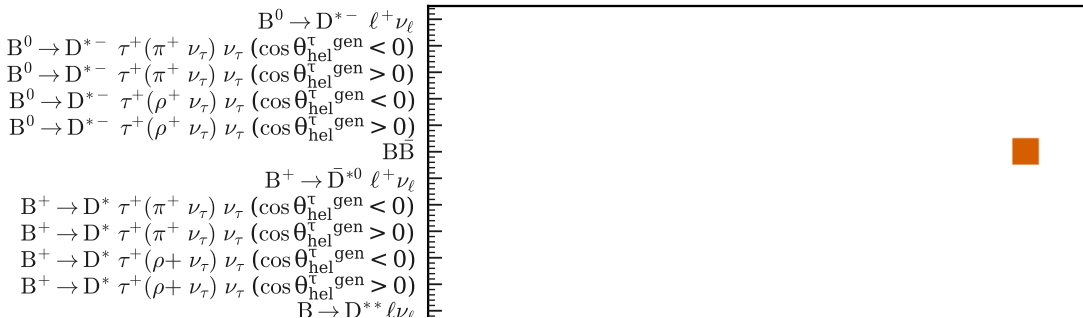
$B^+ \rightarrow \bar{D}^{*0}(D^0 \pi^0) \rho^+ (\cos \theta_{\text{hel}}^{\tau \text{ reco}} < 0)$ in $1.5 < E_{\text{ECL}} < 2$ GeV and $-1 < m_{\text{miss}}^2 < 1$ GeV²



$B^+ \rightarrow \bar{D}^{*0}(D^0 \pi^0) \rho^+ (\cos \theta_{\text{hel}}^{\tau \text{ reco}} < 0)$ in $1.5 < E_{\text{ECL}} < 2$ GeV and $1 < m_{\text{miss}}^2 < 7$ GeV²



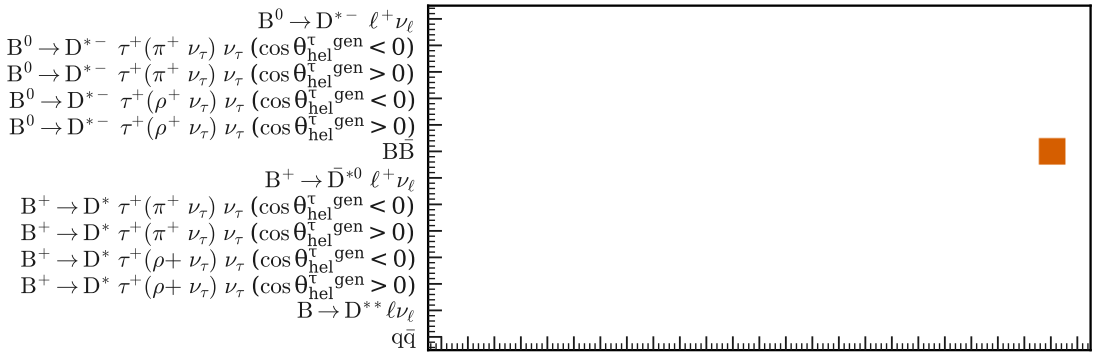
$B^+ \rightarrow \bar{D}^{*0}(D^0 \pi^0) \rho^+ (\cos \theta_{\text{hel}}^{\tau \text{ reco}} > 0)$ in $1.5 < E_{\text{ECL}} < 2$ GeV and $-1 < m_{\text{miss}}^2 < 1$ GeV²



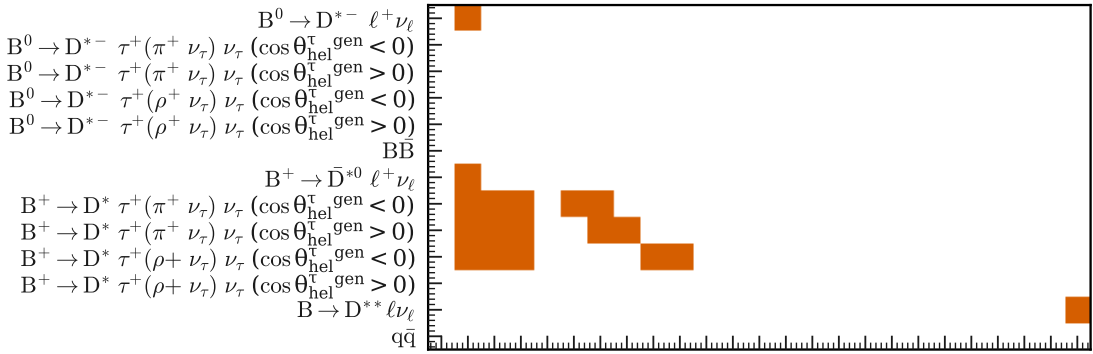


1819

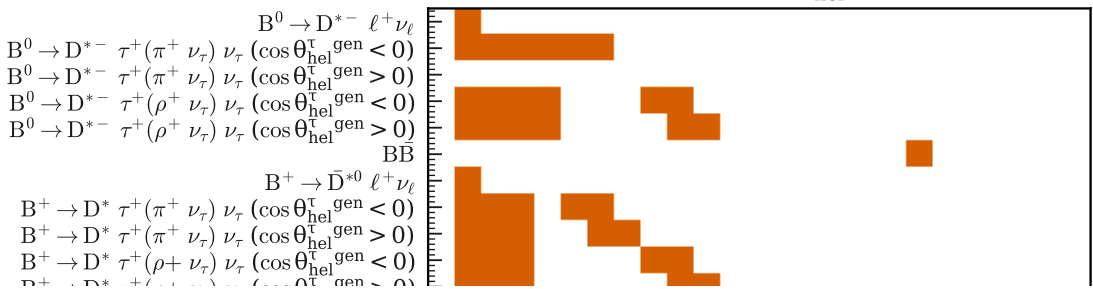
$B^+ \rightarrow \bar{D}^{*0}(D^0 \pi^0) \rho^+ (\cos \theta_{\text{hel}}^{\tau \text{ reco}} > 0)$ in $1.5 < E_{\text{ECL}} < 2 \text{ GeV}$ and $-1 < m_{\text{miss}}^2 < 1 \text{ GeV}^2$

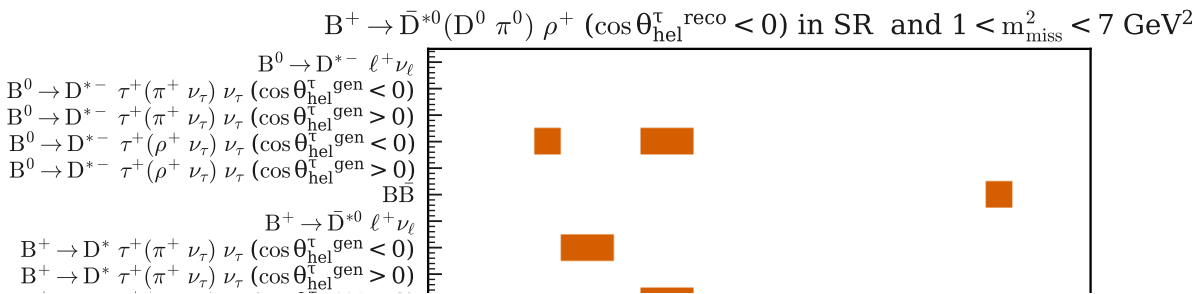
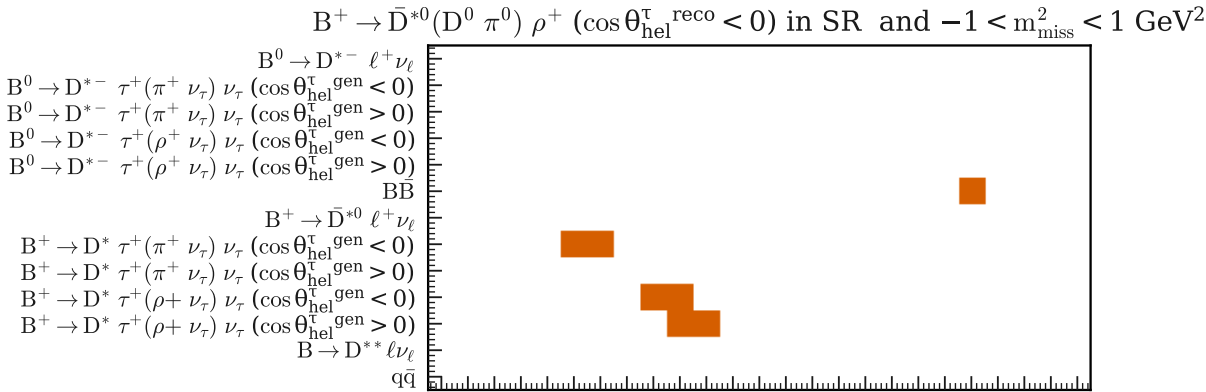
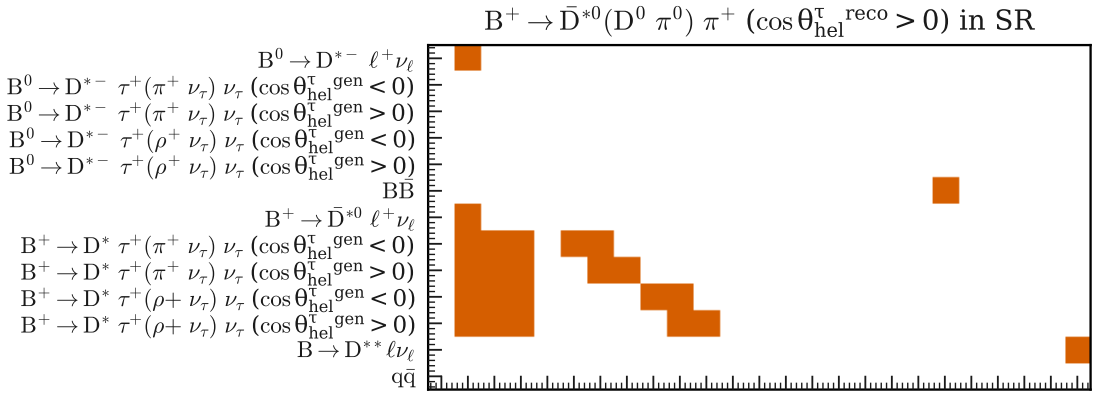
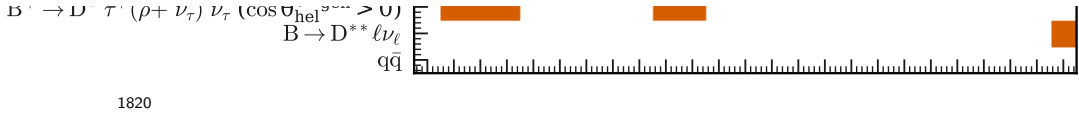


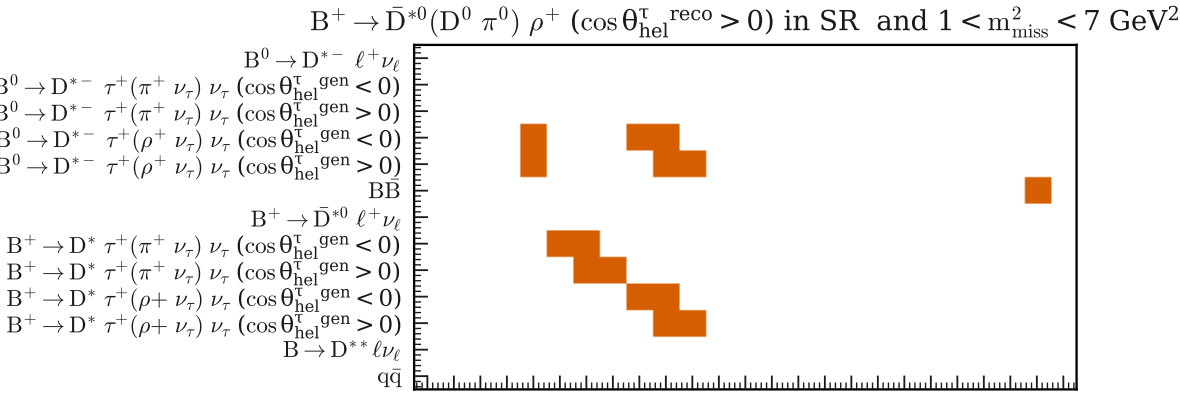
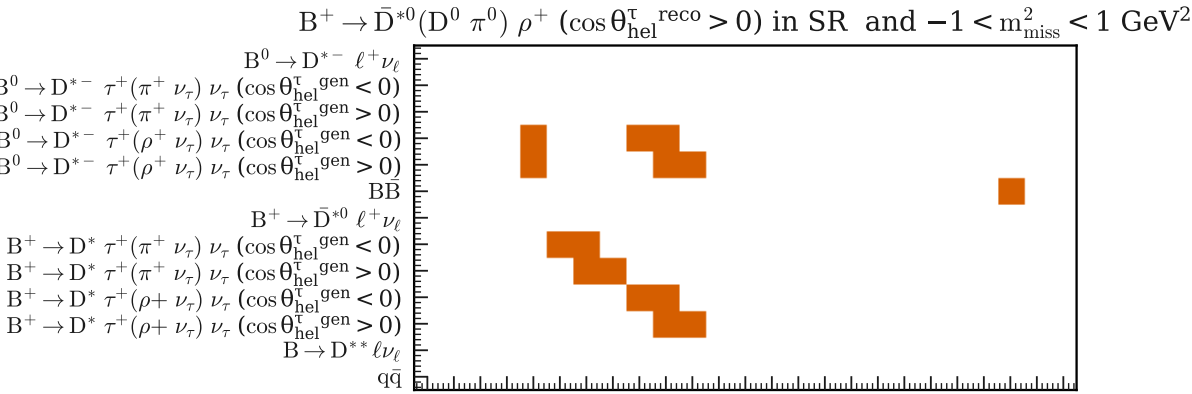
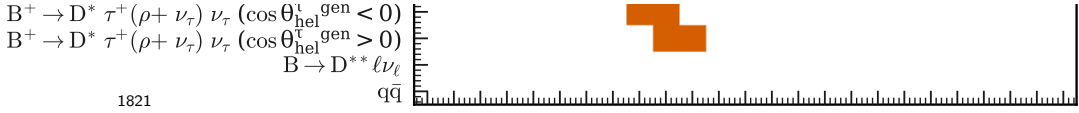
$B^+ \rightarrow \bar{D}^{*0}(D^0 \pi^0) \ell^+$ in NR



$B^+ \rightarrow \bar{D}^{*0}(D^0 \pi^0) \pi^+ (\cos \theta_{\text{hel}}^{\tau \text{ reco}} < 0)$ in SR







P_τ
 $B^+ \rightarrow \bar{D}^{*0} \ell^+ \nu_\ell$ yield
 $R(D^*)^{-1}_{\text{SM}}$ (fixed)
 $R(D^*)^{-1}_{\text{SM}} \ell^{\text{MC}}$ (fixed)
 $(1 - \frac{1}{2} p_{\tau}^{\text{SM}})^{-1}$ (fixed)
 $B(\tau \rightarrow \pi \nu)$ (fixed)
 $(1 + \frac{1}{2} p_{\tau}^{\text{SM}})^{-1}$ (fixed)
 $(1 - \frac{0.45}{2} p_{\tau}^{\text{SM}})^{-1}$ (fixed)
 $B(\tau \rightarrow \rho \nu)$ (fixed)
 $(1 + \frac{0.45}{2} p_{\tau}^{\text{SM}})^{-1}$ (fixed)
 $B \rightarrow D^{**} \ell \nu_\ell$ yield in B^0 reco
 $B\bar{B}$ yield in $B^0 \pi$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} < 0$)
 $B\bar{B}$ yield in $B^0 \pi$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} > 0$)
 $B\bar{B}$ yield in $B^0 \rho$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} < 0$) and $-1 < m_{\text{miss}}^2 < 1 \text{ GeV}^2$
 $B\bar{B}$ yield in $B^0 \rho$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} < 0$) and $1 < m_{\text{miss}}^2 < 7 \text{ GeV}^2$
 $B\bar{B}$ yield in $B^0 \rho$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} > 0$) and $-1 < m_{\text{miss}}^2 < 1 \text{ GeV}^2$
 $B\bar{B}$ yield in $B^0 \rho$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} > 0$) and $1 < m_{\text{miss}}^2 < 7 \text{ GeV}^2$
 $B\bar{B}$ yield in $B^+ \pi$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} < 0$)
 $B\bar{B}$ yield in $B^+ \pi$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} > 0$)
 $B\bar{B}$ yield in $B^+ \rho$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} < 0$) and $-1 < m_{\text{miss}}^2 < 1 \text{ GeV}^2$
 $B\bar{B}$ yield in $B^+ \rho$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} < 0$) and $1 < m_{\text{miss}}^2 < 7 \text{ GeV}^2$
 $B\bar{B}$ yield in $B^+ \rho$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} > 0$) and $-1 < m_{\text{miss}}^2 < 1 \text{ GeV}^2$
 $B\bar{B}$ yield in $B^+ \rho$ ($\cos \theta_{\text{hel}}^{\tau \text{ reco}} > 0$) and $1 < m_{\text{miss}}^2 < 7 \text{ GeV}^2$
 $B \rightarrow D^{**} \ell \nu_\ell$ yield in B^0 reco

Acronyms

basf2 Belle II Analysis Software Framework

BCS Best Candidate Selection

BDT Boosted Decision Tree

BF Branching Fraction

CDF Cumulative Distribution Function

CLT Central Limit Theorem

CoM Center-Of-Mass

CR Control Region

DP Data Production

FEI Full Event Interpretation

FF Form Factor

FS Final State

FSP Final State Particle

GoF Goodness of Fit

HAMMER Helicity Amplitude Module for Matrix Element Reweighting

HEP High Energy Physics

HFLAV Heavy Flavor Averaging Group

HID Hadron IDentification

LFU Lepton Flavour Universality

LHC Large Hadron Collider

LID Lepton IDentification

LS1 Long Shutdown 1

MC Monte Carlo

NR Normalization enhanced Region

PCA Principal Component Analysis

PDF Probability Density Function

- 1849 **PDG** Particle Data Group
- 1850 **POI** Parameter Of Interest
- 1851 **ROE** Rest of Event
- 1852 **SL** SemiLeptonic
- 1853 **SM** Standard Model of Particle Physics
- 1854 **SR** Signal enhanced Region

List of Figures

1	Feynman diagram of a $b \rightarrow c$ transition at the quark level.	9
2	Recent experimental results and theoretical predictions from HFLAV [23]	11
3	Unique experimental result of the P_τ in B semitauonic decays by Belle. The result is presented in contour with the measured value of $R(D^*)$ in the same event topology and with the same dataset. [4]	12
4	Overview of the FEI hierarchical reconstruction approach. At first step clusters and tracks are used to form final state particle candidates. These are given as input to subsequent BDTs in order to build composite particle candidates until a valid B-meson candidate is formed.	18
5	Illustration of event reconstruction for signal.	24
6	Overview of τ lepton BF's	25
7	Total number of events that survive after every offline selection for the most important templates in the signal reconstruction channels.	38
8	Efficiency after every offline selection for the most important templates in the signal reconstruction channels.	38
9	Purity after every offline selection for the most important templates in the signal reconstruction channels.	39
10	Total number of events that survive after every offline selection for the most important templates in the normalization reconstruction channels.	39
11	Efficiency after every offline selection for the most important templates in the normalization reconstruction channels.	40
12	Purity after every offline selection for the most important templates in the normalization reconstruction channels.	40
13	eID efficiency tables.	44
14	μ ID efficiency tables.	45
15	eID fake rate tables where a kaon is faking an electron.	45
16	eID fake rate tables where a pion is faking an electron.	45
17	μ ID fake rate tables where a kaon is faking a muon.	45
18	μ ID fake rate tables where a pion is faking a muon.	46
19	kaonID efficiency tables	46
20	kaonID fake rate tables where a pion is faking a kaon	46
21	π ID efficiency tables	47
22	π ID fake rate tables where a kaon is faking a pion.	47
23	π ID fake rate tables where a muon is faking a pion.	48
24	Change of total MC expectation for B^0 . The largest differences are observed in the low momentum region of the charged hadron or lepton as expected.	51
25	Change of total MC expectation for B^+ . The largest differences are observed in the low momentum region of the charged hadron or lepton as expected.	52
26	Parameters used in the CLN parametrization for the generation of signal decays in the Belle II MC	53

1898	27	Target parameters used in the BLPRX parametrization for the reweighting with HAMMER	53
1899			
1900	28	Available statistics used for the off-resonance calibration of the continuum.	55
1901	29	Prefit distributions for the FEI calibration in the q^2 sideband for B^0 (left) and $B^\pm$ (right).	56
1902			
1903	30	Shape comparison between simulated data and experimental data for the FEI calibration in the q^2 sideband.	57
1904			
1905	31	Normalized shape distribution of m_{miss}^2 which defines the correctly reconstructed and misreconstructed enhanced regions where the photon multiplicity reweighting is derived for B^0 (left) and $B^\pm$ (right).	60
1906			
1907			
1908	32	Normalized shape distribution of the ROE photon multiplicity in the high $\Delta M(D^*)$ where the corrections of Table 39 are derived from for B^0 (left) and $B^\pm$ (right)	61
1909			
1910			
1911	33	m_{miss}^2 distribution in the NR. $B \rightarrow D^* \ell \nu$ decays peak near zero while all other contributions are spread in the tails of the distribution.	62
1912			
1913	34	m_{miss}^2 distribution in the NR using a log scale on the y axis for better visibility of the background distributions.	62
1914			
1915	35	$E_{\text{ECL}}^{\text{extra}}$ distribution in the SR. $B \rightarrow D^* \tau \nu$ decays peak near zero. All other background sources peak toward larger values of $E_{\text{ECL}}^{\text{extra}}$ except $B \rightarrow D^* \ell \nu$ decays, making it hard to distinguish them from signal processes.	63
1916			
1917			
1918	36	Final fit distributions used for the $R(D^*)$ fit.	64
1919	37	Final fit distributions used for the $R(D^*)$ and P_τ fit.	69
1920	38	Confusion matrix between the true generated cosinus of the helicity angle and the reconstructed one.	70
1921			
1922	39	FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ in the negative helicity bin.	71
1923			
1924	40	FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ in the positive helicity bin.	72
1925			
1926	41	FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ in the negative helicity bin and low m_{miss}^2	72
1927			
1928	42	FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ in the negative helicity bin and high m_{miss}^2	73
1929			
1930	43	FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ in the positive helicity bin and low m_{miss}^2	73
1931			
1932	44	FEI decay mode composition of $B\bar{B}$ events in $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ in the positive helicity bin and high m_{miss}^2	74
1933			
1934	45	FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ in the negative helicity bin.	74
1935			
1936	46	FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ in the positive helicity bin.	75
1937			
1938	47	FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ in the negative helicity bin and low m_{miss}^2	75
1939			
1940	48	FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ in the negative helicity bin and high m_{miss}^2	76
1941			

1942	49	FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ in the	
1943		positive helicity bin and low m_{miss}^2 .	76
1944	50	FEI decay mode composition of $B\bar{B}$ events in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ in the	
1945		positive helicity bin and high m_{miss}^2 .	77
1946	51	Statistical only correlation matrix of the P_τ and $R(D^*)$ fit.	79
1947	52	Postfit distributions on Asimov in the SR and control regions for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$	80
1948	53	Postfit distributions on Asimov in the SR and control regions for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	80
1949	54	Postfit distributions on Asimov in the SR and control regions for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$	81
1950	55	Postfit distributions on Asimov in the SR and control regions for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	81
1951	56	Pull plots for toys for $R(D^*)$. P_τ and the yield of $B \rightarrow D^* \ell \nu$ decays.	
1952		The MINOS error has been used to calculate the pull of P_τ for every toy.	
1953		A negligible amount of toys (11/2500) did not converge, therefore we	
1954		excluded them from the pull distribution. The parameters of the gaussian	
1955		fit are compatible with a standard normal.	82
1956	57	The response of the estimator scales as expected when we bias the Asimov	
1957		dataset.	83
1958	58	Linearity test for the three POIs in the simultaneous fit. P_τ scales as	
1959		expected while the other POIs are minimized to their nominal values. We	
1960		conclude that the estimator's response scales as expected.	83
1961	59	Fitting variables in continuum processes on off-resonance data in SR of	
1962		$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$	88
1963	60	Fitting variables in continuum processes on off-resonance data in SR of	
1964		$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	88
1965	61	Fitting variables in continuum processes on off-resonance data in SR of	
1966		$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$	88
1967	62	Fitting variables in continuum processes on off-resonance data in SR of	
1968		$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	89
1969	63	Fitting variables in continuum processes on off-resonance data in SR of	
1970		$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$	89
1971	64	Fitting variables in continuum processes on off-resonance data in SR of	
1972		$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	89
1973	65	Fitting variables in continuum processes on off-resonance data in SR of	
1974		$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$	90
1975	66	Fitting variables in continuum processes on off-resonance data in SR of	
1976		$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	90
1977	67	Post fit distributions of the control fit for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ that are	
1978		constraining $B\bar{B}$ decays.	90
1979	68	Post fit distributions of the control fit for $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ that are	
1980		constraining $B\bar{B}$ decays.	91
1981	69	Post fit distributions of the control fit for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ that are	
1982		constraining $B\bar{B}$ decays.	92
1983	70	Post fit distributions of the control fit for $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ that are	
1984		constraining $B\bar{B}$ decays.	92

1985	71	M_{bc} (top left) and $M(D^0)$ (bottom left), lepton momentum (top right)	
1986		and π_s momentum (bottom right) for the $B^0 \rightarrow D^{*-}(D^0 \pi^-) e^+$ in the q^2	
1987		sideband.	93
1988	72	M_{bc} (top left) and $M(D^0)$ (bottom left), lepton momentum (top right)	
1989		and π_s momentum (bottom right) for the $B^+ \rightarrow D^{*0}(D^0 \pi^0) e^+$ in the q^2	
1990		sideband.	93
1991	73	m_{miss}^2 distribution for B^0 (left) and $B^\pm$ (right) mesons in the q^2 sideband	94
1992	74	E_{ECL}^{extra} distribution for B^0 (left) and $B^\pm$ (right) mesons in the NR sideband	94
1993	75	E_{ECL}^{extra} distribution for B^0 (left) and $B^\pm$ (right) mesons in the NR sideband	
1994		after applying the photon multiplicity correction.	95
1995	76	M_{bc} (top left) and $M(D^0)$ (bottom left), lepton momentum (top right) and	
1996		π_s momentum (bottom right) for the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$ in the NR. . . .	95
1997	77	M_{bc} (top left) and $M(D^0)$ (bottom left), lepton momentum (top right) and	
1998		π_s momentum (bottom right) for the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$ in NR.	96
1999	78	Data/MC agreement of the fitting variables in the control fit region for	
2000		$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$	96
2001	79	Data/MC agreement of the fitting variables in the control fit region for	
2002		$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	96
2003	80	Data/MC agreement of the fitting variables in the control fit region for	
2004		$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$	97
2005	81	Data/MC agreement of the fitting variables in the control fit region for	
2006		$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	97
2007	82	Data/MC agreement of the fitting variables in the validation region for	
2008		$B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$	98
2009	83	Data/MC agreement of the fitting variables in the validation region for	
2010		$B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$	98
2011	84	Data/MC agreement of the fitting variables in the validation region for	
2012		$B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$	98
2013	85	Data/MC agreement of the fitting variables in the validation region for	
2014		$B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$	99
2015	86	Module overview of the SysVar software	102
2016	87	Pseudovariables for signal extraction	103
2017	88	Pseudovariable for correction bins	103
2018	89	Different corrections and error sources. In this example three sources of	
2019		uncertainty are considered for the correction. Two statistical in nature	
2020		and one systematic	104
2021	90		104
2022	91		104
2023	92		105
2024	93	A summary of the variations of the correction weights.	105
2025	94		106
2026	95	The correlation matrix arising from all the varied templates. It is clear	
2027		that the correlation structure in this space becomes highly not trivial to	
2028		model. Therefore the eigendecomposition of it, provides a framework that	
2029		simplifies the treatment of these correlations in a easier way	107

2030	96	The normalized covariance matrix differences as a function of the eigendirections considered. The target precision at 0.5% is set arbitrarily by the user.	108
2031			
2032			
2033	97	Multinomial error scaling with number of total occurrences and rates . . .	110
2034	98	Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K^-\pi^+$, $D^0 \rightarrow \pi^-\pi^+$ and $D^0 \rightarrow K^-K^+$ on MC.	111
2035			
2036	99	Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K_s^0\pi^0$ and $D^0 \rightarrow K^-\pi^+\pi^0$ on MC.	111
2037			
2038	100	Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K_s^0\pi^+\pi^-$ and $D^0 \rightarrow K_s^0K^+K^-$ on MC.	112
2039			
2040	101	Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K^-\pi^+\pi^-\pi^+$ on MC.	112
2041			
2042	102	Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K^-\pi^+$, $D^0 \rightarrow \pi^-\pi^+$ and $D^0 \rightarrow K^-K^+$ on Data.	112
2043			
2044	103	Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K_s^0\pi^0$ and $D^0 \rightarrow K^-\pi^+\pi^0$ on Data.	112
2045			
2046	104	Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K_s^0\pi^+\pi^-$ and $D^0 \rightarrow K_s^0K^+K^-$ on Data.	113
2047			
2048	105	Signal shape, post fit distribution and goodness of fit for $D^0 \rightarrow K^-\pi^+\pi^-\pi^+$ on Data.	113
2049			
2050	106	Parameters used in the CLN parameterization for the generation of $B \rightarrow D\tau\nu$ decays in the Belle II MC	114
2051			
2052	107	Target parameters used in the BLPRXP parameterization for the reweighting of $B \rightarrow D\tau\nu$ with HAMMER	114
2053			
2054	108	Parameters used in the CLN parameterization for the generation of $B \rightarrow D^*\tau\nu$ decays in the Belle II MC	115
2055			
2056	109	Target parameters used in the BLPRXP parameterization for the reweighting of $B \rightarrow D^*\tau\nu$ with HAMMER	115
2057			
2058	110	Parameters used in the BGL parameterization for the generation of $B \rightarrow D\ell\nu$ decays in the Belle II MC	115
2059			
2060	111	Target parameters used in the BLPRXP parameterization for the reweighting of $B \rightarrow D\ell\nu$ with HAMMER	116
2061			
2062	112	Parameters used in the BGL parameterization for the generation of $B \rightarrow D^*\ell\nu$ decays in the Belle II MC	116
2063			
2064	113	Target parameters used in the BLPRXP parameterization for the reweighting of $B \rightarrow D^*\ell\nu$ with HAMMER	117
2065			
2066	114	Parameters used in the BLR parameterization for the generation of $B \rightarrow D_1^{**}\ell\nu$ decays in the Belle II MC	117
2067			
2068	115	Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_1^{**}\ell\nu$ with HAMMER	117
2069			
2070	116	Parameters used in the BLR parameterization for the generation of $B \rightarrow D_1^{**}\tau\nu$ decays in the Belle II MC	118
2071			
2072	117	Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_1^{**}\tau\nu$ with HAMMER	118
2073			

2074	118	Parameters used in the BLR parameterization for the generation of $B \rightarrow D_0^{**}\ell\nu$ decays in the Belle II MC	118
2075			
2076	119	Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_0^{**}\ell\nu$ with HAMMER	119
2077			
2078	120	Parameters used in the BLR parameterization for the generation of $B \rightarrow D_0^{**}\tau\nu$ decays in the Belle II MC	119
2079			
2080	121	Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_0^{**}\tau\nu$ with HAMMER	119
2081			
2082	122	Parameters used in the BLR parameterization for the generation of $B \rightarrow D_{1'}^{**}\ell\nu$ decays in the Belle II MC	120
2083			
2084	123	Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_{1'}^{**}\ell\nu$ with HAMMER	120
2085			
2086	124	Parameters used in the BLR parameterization for the generation of $B \rightarrow D_{1'}^{**}\tau\nu$ decays in the Belle II MC	120
2087			
2088	125	Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_{1'}^{**}\tau\nu$ with HAMMER	121
2089			
2090	126	Parameters used in the BLR parameterization for the generation of $B \rightarrow D_2^{**}\ell\nu$ decays in the Belle II MC	121
2091			
2092	127	Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_2^{**}\ell\nu$ with HAMMER	121
2093			
2094	128	Parameters used in the BLR parameterization for the generation of $B \rightarrow D_2^{**}\tau\nu$ decays in the Belle II MC	122
2095			
2096	129	Target parameters used in the BLR parameterization for the reweighting of $B \rightarrow D_2^{**}\tau\nu$ with HAMMER	122
2097			
2098	130	Bias test for FEI calibration estimation. We throw 5000 toys of toy datasets and refit the nominal model to them. All free parameters in the fit are not biased and the uncertainties well estimated based on the mean and standard deviation of the standard normal fit of the pull distribution.	123
2099			
2100			
2101			
2102	131	Linearity test for FEI calibration estimation. We bias an asimov dataset by artificially injecting $B \rightarrow D^{**}\ell\nu$ decays from the nominal histograms Then we fit this biased asimov dataset to the nominal model. The extracted FEI calibration factors scale as expected while the background strengths remains unaffected.	124
2103			
2104			
2105			
2106			
2107	132	Stability test for FEI calibration. We extract the FEI calibraation factors considering different multiplicities by deriving the factors for different D^0 decays modes. We also derive them again for electrons and muons separately. All are consistent within 2σ	124
2108			
2109			
2110			
2111	133	Pull plots for toys $B \rightarrow D^{**}\ell\nu$ yields for B^0 and (left) and B^+ (right) reconstruction channels for the $R(D^*)$ fit. The parameters of the gaussian fit are compatible with a standard normal.	125
2112			
2113			
2114	134	Pull plots for toys $B\bar{B}$ yields for B^0 in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels and B^+ in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels from left to right for the $R(D^*)$ fit. The parameters of the gaussian fit are compatible with a standard normal.	125
2115			
2116			
2117			

2118	135	Pull plots for toys $q\bar{q}$ yields for B^0 in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels and B^+ in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels from left to right for the $R(D^*)$ fit. The parameters of the gaussian fit are compatible with a standard normal.	126
2119			
2120			
2121			
2122	136	Pull plots for toys $B \rightarrow D^{**}\ell\nu$ yields for B^0 and (left) and B^+ (right) reconstruction channels for the P_τ fit. The parameters of the gaussian fit are compatible with a standard normal.	126
2123			
2124			
2125	137	Pull plots for toys $B\bar{B}$ yields for B^0 in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels and B^+ in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels from left to right for the P_τ fit. The parameters of the gaussian fit are compatible with a standard normal.	126
2126			
2127			
2128			
2129	138	Pull plots for toys $q\bar{q}$ yields for B^0 in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels and B^+ in the $\tau \rightarrow \pi\nu_\tau$ and $\tau \rightarrow \rho\nu_\tau$ reconstruction channels from left to right for the P_τ fit. The parameters of the gaussian fit are compatible with a standard normal.	127
2130			
2131			
2132			
2133	139	Channel dependant optimization of the helicity angle window	128
2134	140	Impact of every D^0 mode on the sensitivity of the parameters of interest. The uncertainties are taken from an Asimov fit. We confirm that we observe closure in the fit (the central values are minimized to the expected ones)	129
2135			
2136			
2137	141	Impact of every D^0 mode on the sensitivity of the parameters of interest. We completely exclude the $D^0 \rightarrow K^-K^+$ mode from this test. The uncertainties are taken from an Asimov fit. We confirm that we observe closure in the fit (the central values are minimized to the expected ones)	129
2138			
2139			
2140			
2141	142	Impact of every D^0 mode on the sensitivity of the parameters of interest. We completely exclude the $D^0 \rightarrow K^-K^+$ and $D^0 \rightarrow \pi^-\pi^+$ modes from this test. The uncertainties are taken from an Asimov fit. We confirm that we observe closure in the fit (the central values are minimized to the expected ones)	130
2142			
2143			
2144			
2145			
2146	143	Background composition in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ channel for the Prompt Hadronic component in the negative helicity bin.	131
2147			
2148	144	Background composition in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ channel for the Prompt Hadronic component in the positive helicity bin.	132
2149			
2150	145	Background composition in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ channel for the Double Charm component in the negative helicity bin.	132
2151			
2152	146	Background composition in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \pi^+$ channel for the Double Charm component in the positive helicity bin.	133
2153			
2154	147	Background composition in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ channel for the Prompt Hadronic component in the negative helicity bin.	133
2155			
2156	148	Background composition in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ channel for the Prompt Hadronic component in the positive helicity bin.	134
2157			
2158	149	Background composition in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ channel for the Double Charm component in the negative helicity bin.	134
2159			
2160	150	Background composition in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \rho^+$ channel for the Double Charm component in the positive helicity bin.	135
2161			

2162	151	Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ channel for the Prompt	
2163		Hadronic component in the negative helicity bin.	135
2164	152	Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ channel for the Prompt	
2165		Hadronic component in the positive helicity bin.	136
2166	153	Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ channel for the Dou-	
2167		ble Charm component in the negative helicity bin.	136
2168	154	Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \pi^+$ channel for the Dou-	
2169		ble Charm component in the positive helicity bin.	137
2170	155	Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ channel for the Prompt	
2171		Hadronic component in the negative helicity bin.	137
2172	156	Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ channel for the Prompt	
2173		Hadronic component in the positive helicity bin.	138
2174	157	Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ channel for the Dou-	
2175		ble Charm component in the negative helicity bin.	138
2176	158	Background composition in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ channel for the Dou-	
2177		ble Charm component in the positive helicity bin.	139
2178	167	HID recommendations from charged hadrons as per the conference readi-	
2179		ness page.	171
2180	168	minC2TDist on the photons of the ROE in all three detector. Most of the	
2181		hadronic split-offs are discarded without any Data/MC disagreement being	
2182		introduced.	173
2183	169	PDG codes of the daughters of the τ lepton in $B^0 \rightarrow D^* \tau(\rightarrow \rho \nu) \nu$ and	
2184		$B^+ \rightarrow D^* \tau(\rightarrow \rho \nu) \nu$ events	173
2185	170	Demonstration of proportions of fake and true leptons in the retained sam-	
2186		ples of $B \rightarrow D^*(D^0 \pi) \ell^+$	174
2187	171	Demonstration of proportions of fake and true leptons in the retained sam-	
2188		ples of $B \rightarrow D^*(D^0 \pi) \ell^+$	175
2189	172	Difference between the unconstrained factors in the control fits without and	
2190		with the background reweighting.	176
2191	173	Correlation matrix for unconstrained parameters of the fit if the control	
2192		regions have not been added.	178
2193	174	Impact of nCDCHits cut on fitting variable	181
2194	175	Data/MC comparison of the flight of distance of K_s^0 in the low q^2 sideband	
2195		in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$ channels (left) and $B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$ channels	
2196		(right). The p-values indicate that the modelling of this observable is ac-	
2197		ceptable with our current statistical precision.	181
2198	176	Invariant mass of the ρ meson	182
2199	177	Generated and reconstructed mass of the ρ^+ meson in signal MC.	182
2200	178	ρ^+ invariant mass cut at the Belle analysis.	182
2201	179	FOM for the ρ mass window. The best window is shown with a red square.	183
2202	180	Invariant mass of the ρ meson with the optimized cut	183
2203	181	Invariant mass of D meson for different modes.	184
2204	182	Impact of every D^0 mode on the sensitivity of the parameters of interest.	
2205		The uncertainties are taken from an Asimov fit. We confirm that we observe	
2206		closure in the fit (the central values are minimized to the expected ones)	184

2207	183	Impact of every D^0 mode on the sensitivity of the parameters of interest.	
2208		We completely exclude the $D^0 \rightarrow K^- K^+$ mode from this test. The uncer-	
2209		tainties are taken from an Asimov fit. We confirm that we observe closure	
2210		in the fit (the central values are minimized to the expected ones)	185
2211	184	Impact of every D^0 mode on the sensitivity of the parameters of interest.	
2212		We completely exclude the $D^0 \rightarrow K^- K^+$ and $D^0 \rightarrow \pi^- \pi^+$ modes from	
2213		this test. The uncertainties are taken from an Asimov fit. We confirm	
2214		that we observe closure in the fit (the central values are minimized to the	
2215		expected ones)	185
2216	185	Cluster energy distributions of photons in the ROE in the three detector	
2217		regions.	186
2218	186	Reconstructed invariant mass of slow π^0 . A clear shift from the theoretical	
2219		value can be seen.	187
2220	187	Original finer binning in m_{miss}^2 to extract the $B \rightarrow D^* \ell \nu$ yield.	190
2221	188	Updated coarse binning in m_{miss}^2 to extract the $B \rightarrow D^* \ell \nu$ yield.	190
2222	189	Events that would be discarded by a tighter momentum cut at 0.7 GeV.	
2223		The impact on the signal template is larger than the normalization template	
2224		in our fitting variable.	191
2225	190	Pre and post fit impact of gap mode BF uncertainties in [30]	195
2226	191	Results by running toys with nuisance parameters on.	197
2227	192	Overview of different crossfeed modes of semitauonic B meson decays. . .	198

List of Tables

2228			
2229	1	Overview of experimental measurements of $R(D^*)$ [23]	11
2230	2	List of <code>pyhf</code> modifier used in this analysis. All the above modifier are	
2231		controlled by a single nuisance parameter. The subscripts stand for: s	
2232		for sample, c for channel, b for bin. f and g are functions determined by	
2233		interpolating the nominal template between $+1$ and -1 sigma. a is an	
2234		auxiliary measurement and delta is the relative uncertainty of a template	
2235		in a particular bin with respect to the total uncertainty.	15
2236	3	Pre-selection cuts of the FEI skim.	19
2237	4	Luminosity for different samples in the Generic MC for MC15rd	19
2238	5	Event types and corresponding decays with number of events and luminos-	
2239		ity for gap modes including light leptons. $N_{leptons}$ has been set 2 in Eq 5 to	
2240		account for the fact that the light lepton gap modes are not produced sep-	
2241		arately. The calculated luminosity is used to scale the events that survive	
2242		the full set of event selections to the luminosity of experimental data. . . .	20
2243	6	Latest values of BF's for the light lepton $B \rightarrow H_c \ell \nu$ decays and $R(H_c)$ ratios.	
2244		The BF of the semitauonic modes is calculated using $R(H_c) = \frac{B \rightarrow H_c \tau \nu}{B \rightarrow H_c \ell \nu}$	21
2245	7	Event types and corresponding decays with number of events and luminos-	
2246		ity for gap semitauonic modes. The calculated luminosity is used to scale	
2247		the events that survive the full set of event selections to the luminosity of	
2248		experimental data.	22
2249	8	Experiment Offline Luminosity for 4S	22
2250	9	Experiment Offline Luminosity for 4S_offres	23
2251	10	Global tags used in the steering file. The names of the global tags are used	
2252		as input to the <code>basf2</code> helper function <code>b2.conditions.prepend_globaltag</code>	25
2253	11	Selections for tracks and cluster considered in the B_{ROE}^{tag}	26
2254	12	LID selection criteria	27
2255	13	Charmed meson reconstruction modes	29
2256	14	Overview of online cuts for building B meson signal candidates	30
2257	15	Selections for tracks and cluster considered in the ROE against the $\Upsilon(4S)$	
2258		candidate	30
2259	16	$\Upsilon(4S)$ reconstructions modes and decays of interest	31
2260	17	Main decay truth matching categories. For D^* both charged and neutral	
2261		mesons are considered. For D^* and D , τ , ℓ , ν , charge conjugation is taken	
2262		into account. The definitions of the PDG codes that are used for D^{**} , H_c ,	
2263		H_c^{gap} , $X_{u/s}^{gap}$, X_s , and X_u are given in Appendix B.	33
2264	18	τ decay truth matching categories. The order of the truth matched daugh-	
2265		ters is not important as all possible permutation are considered.	34
2266	19	Final truth matching categories. The entries of the column B^{sig} mcID are	
2267		described in Table 17. The entries of the column τ mcID are described in	
2268		Table 18.	35
2269	20	List of final tight cuts applied in the sample.	37
2270	21	BCS, number of candidates and event efficiencies for the signal reconstruc-	
2271		tion channels. Results are not scaled to data luminosity.	41

2272	22	BCS, number of candidates and event efficiencies the normalization reconstruction channels. Results are not scaled to data luminosity.	42
2273			
2274	23	Luminosity and scaling factors for different samples simulated data. The assumed $\mathcal{L}$ for experimental data for the LS1 dataset is $365.29 \pm 1.70 \text{ fb}^{-1}$	43
2275			
2276	24	Correction tables for $\pi_s^\pm$ efficiency corrections derived by the Tracking group.	44
2277			
2278	25	Correction tables for π_s^0 efficiency corrections derived by the Neutrals group.	44
2279	26	BF corrections for charmed meson decays	49
2280	27	BF corrections for τ lepton decays	49
2281	28	BF corrections for double charm decays	50
2282	29	BF corrections for prompt hadronic decays	50
2283	30	Percentage increase in MC expectation for different templates in the B^0 reconstruction	51
2284			
2285	31	Percentage increase in MC expectation for different templates in the B^+ reconstruction	52
2286			
2287	32	FF models and parameters used for the HAMMER reweighting	52
2288	33	Smearing factors determined for every different D^0 decay mode reconstruction.	54
2289			
2290	34	Continuum calibration factors as derived by off-resonance data.	55
2291	35	Signal strengths for $B \rightarrow D^* \ell \nu$ which are interpreted as the FEI calibration factors fit on an Asimov dataset in the B^0 channels. The first uncertainty is statistical while the second is the total systematic uncertainty.	58
2292			
2293			
2294	36	Signal strengths for $B \rightarrow D^* \ell \nu$ which are interpreted as the FEI calibration factors fit on an Asimov dataset in the B^+ channels. The first uncertainty is statistical while the second is the total systematic uncertainty.	58
2295			
2296			
2297	37	Signal strengths for $B \rightarrow D^* \ell \nu$ and all other background processes in the FEI calibration fit on experimental data in the B^0 channels. The first uncertainty is statistical while the second is the total systematic uncertainty. We get a GoF p-value of 10.19%.	59
2298			
2299			
2300			
2301	38	Signal strengths for $B \rightarrow D^* \ell \nu$ and all other background processes in the FEI calibration fit on experimental data in the B^+ channels. The first uncertainty is statistical while the second is the total systematic uncertainty. We get a GoF p-value of 10.19%.	59
2302			
2303			
2304			
2305	39	Photon multiplicity weights for B^0 and B^+ for correctly and misreconstructed events.	60
2306			
2307	40	Final templates for the fit	64
2308	41	Final templates for the $R(D^*)$ and P_τ fit	70
2309	42	Overview of fit regions	77
2310	43	Asimov fit uncertainties of all free unconstrained parameters for the simultaneous $R(D^*)$ and P_τ fit. The uncertainties are presented in terms of percentages in case the free parameter's nominal value is not 1. The nominal values of all parameters is presented in parenthesis in the first column. We confirm that we observe full closure in all our Asimov fits, i.e. all free parameters are minimized to their nominal values.	78
2311			
2312			
2313			
2314			
2315			
2316	44	Summary of Systematic Uncertainties	86

2317	45	Sideband checks summary	87
2318	46	Results for control fits	91
2319	47	PDG codes of categories for the first layer of truth matching. Charge	
2320		conjugation is used for all PDG codes.	109

Working group review questions and answers

- Comments received on 27-05-2025 by WG reader Michele Mantovano

1. *The FEI training with the MC15rd is already available (https://docs.belle2.org/pub_data/documents/25/), so why are you still using the one for the MC15ri?*

I believe that the document that you sent refers to the MC15rd calibration of the FEI. Even if the inference and the calibration of the efficiency on MC happens on the rd sample, to the best of my knowledge the training weights that are used, still rely on the training on 200 fb^{-1} of MC15ri. Please let me know if this is wrong. Nonetheless I removed the table with the MC15ri luminosities to avoid confusion. The original plan to cross check the analysis by default on MC15ri was too optimistic and will now pursue this only if requested. Let me stress out that the official FEI calibration is not used throughout this analysis. The argument in Section 3.2.1 was only meant to explain why we do not use the full amount of generic MC15ri available (potentially for RC readers outside of the S(L) WG were not familiar with the ins and outs of the FEI).

2. *Could you clarify that you're setting $N_{\text{lepton}}=2$ in the calculation of the gap modes with light leptons?*

Yes we do. I explicitly now mention this in the caption of Table 5.

3. *Tab.6: Could you clarify why you set $f=0.5$ in the luminosity calculation for $D_{\pi\pi}$ and $D^*\pi\pi$?*

In the MC cocktail that is recommended to use, the non resonant gap modes are replaced by the same final state, through a resonant D^{**} decay. In some cases two different resonances are chosen to model the same final state. To the best of my knowledge this is just a matter of choice and a by-product of our ignorance on how the gap should be filled. If we examine the example of B^0 for the $D\pi\pi$ final state we see that two decay modes are being used, mode 1196700001 and mode 1196700003. In the first one the $D\pi\pi$ final state goes through a D_1^* resonance and in the second through a D_0^* . Therefore half of the $D\pi\pi$ events will be found in a D_1^* and the other half in D_0^* events, even though they are meant to replace the same non-resonant final states. In order to account for that we set the factor f to 0.5 for such cases. The full list of the available modes can be found in [this xwiki page](#).

4. *Tab.6: The calculated luminosity values seem too large to me, especially for the resonant $D\pi\pi$ modes — could you double-check them? I think the values for the NBB are wrong; they should be $5.4e5$ and $5.1e5$.*

You are right! Thank you! I changed the values for the number of $B\bar{B}$ events. Now Tables 5 and 23 should have the correct values for the luminosities and the scaling factors.

5. *Can you explain why the nominal values and uncertainties in Tab.7 of the gap modes are different from those reported in Tab.6?*

Apologies. There was an update in the values and I forgot to also update Table 6. Now both tables have the correct values.

- 2364 6. *Tab. 7, for the D^*0 are you using only the measurement from Belle?*
 2365 We're using as input the latest values that Markus Prim has advertised in the
 2366 working group. I cannot give you definite answer on what measurements made
 2367 it into the HFLAV average calculation but by looking at table 37 of HFLAV
 2368 report [3] and comparing them to the values of Table 6 in this document, I
 2369 believe that both the Belle and BABAR measurements must have been used by
 2370 the HFLAV group. Please also see Question 32.
- 2371 7. *I think you should also include 'chargedpidmva_rel6_v5' as a global tag since*
 2372 *you're using the BDT electron ID.*
 2373 You're absolutely right! I included this global tag in the last processing but
 2374 forgot to update Table 10. Now Table 10 is updated with all the global tags.
 2375 In the current and previous version of the note for all results the chargedpid-
 2376 mva_rel6_v5 globaltag has been prepended. This oversight has affected only
 2377 Table 10.
- 2378 8. *Are you sure that the cuts $-dr- < 2$ cm and $-dz- < 4$ cm are the ones*
 2379 *recommended by the LID group?*
 2380 I cannot find any official recommendation on the selections on dr and dz from
 2381 the LID group on their [xwiki page](#). However I did find that the `stdCharged`
 2382 module (which is used for the reconstruction of leptons in this analysis as
 2383 mentioned in Section 4.5.2 is applying ipCuts as `ipCut = 'dr < 0.5 and`
 2384 `abs(dz) < 2'` ([link](#) to documentation). These are tighter than the ones we
 2385 apply on all tracks, therefore our manually applied cuts on all all charged
 2386 tracks (including the leptons) should actually have zero impact as they are
 2387 covered by the cuts applied by the `stdCharged` module.
- 2388 9. *Are you sure that the cut on $CDChits > 20$ is still necessary to calculate the*
 2389 *hadron ID corrections? I remember that with the SF you can compute the cor-*
 2390 *rections even without this cut. Is it still recommended to use this cut?*
 2391 Please see below in Figure 167 the latest recommendations from HID group
 taken from [Conference readiness page on xwiki](#) It is recommended to use the

Recommendations

- There are no requirements for using hadron ID variables
- The following recommended track selection may improve hadron ID performance, depending on your analysis
 - `[ abs(dr) < 2 ] and [ abs(dz) < 4 ]`
 - `nCDCHits > 20`
- **Attention:** All ntuple files of hadron samples of the Systematics Framework are produced without the cut on `nCDCHits > 20`, but the **calculations within the Systematics Framework apply this cut per default as a constraint to all hadron samples**
 - If you do not use this cut or if you use a different cut in your analysis, please see this [tutorial on Custom PID weight production](#) on how to **adjust the constraints to your analysis**

Figure 167: HID recommendations from charged hadrons as per the conference readiness page.

nCDCHits > 20 cut on charged hadrons. This cut has also been applied when determining corrections with the systematics framework. As far as I know from attending the HID meetings in the past this cut is used to ensure that the charged tracks have reached the TOP detector which ensures good kaon/pion separation. If a track has less than 20 hits in the CDC then it's a very slow one

which will never reach the TOP detector. This means that the PID likelihood will rely mainly on information from CDC which does not provide great discrimination power between kaons and pions. We believe that this cuts should remain as high HID performance is very important in this analysis (especially for the pion coming from the τ lepton in the $B \rightarrow D^*(D^0 \pi) h^+$ reconstruction channels.

FOLLOW-UP

Please see my follow up in Question 36

10. *For the Ks selection, you only require the flight distance and the significance of the distance to be positive. Aren't there any recommended cuts for the Ks? I remember that a cut of significance of distance larger than 10 was imposed for the Ks.*

To the best of my knowledge there are no official recommendations for the selections applied to K_S^0 . I couldn't find any official recommendations in the conference readiness or the tracking group page on xwiki. Furthermore I do not see any cuts applied on the flight distance or the significance of distance from the `stdKshorts` module in `basf2` (which is the one that is used in this analysis. The only selection seems to be `conf_level = 0.0`, imposed by `treeFit`.

11. *Can you justify why you're using $\text{minC2TDist} > 50 \text{ cm}$? I remember that this cut is no longer among the recommended ones.*

I believe you're referring to the cut on the photons of the ROE. As far as I know there are not official recommendations on what cuts to apply on the objects that make it into the ROE. The `minC2TDist` variable is one of the well-model ones as per the recommendations of the Neutrals group (see https://indico.belle2.org/event/13722/contributions/84645/attachments/31512/46646/2024_11_17_ferber_ecl.pdf). During the work-shopping of the analysis I have tried using at least eight different ROE masks (and their counterparts by applying the photon killing procedure) that have been used in other analyses from WG1 and WG2 or have been recommended to me privately in the past. Through these studies I have found that this mask is the one with the best Data/MC agreement. Furthermore I have made extensive Data/MC agreement studies on different observables for tracks and clusters that are used to build the ROE (these are however not included in this note to avoid making this document unnecessarily even longer). Regarding the cut in question I provide Figure 168 that shows that a cut at 50 cm on the `minC2TDist` on the photons of the ROE in all three detector regions discards most of the hadronic split-offs without any Data/MC disagreement being introduced (the overall number of photons on MC has been normalized to the one on data. Therefore these plots are evaluating the modeling of the shape of `minC2TDist`)

12. *Could you explain more clearly why the truth-matching doesn't works for $\tau \rightarrow \rho \nu$? Are you running TauolaBelle for the truth-matching?*

I looked closer into the `DECAY.dec` file and apparently the decay $\tau \rightarrow \rho \nu_\tau$ is commented out in favor of

```
0.255100000 pi- pi0 nu_tau TAUHADNU -0.108 0.775 0.149 1.364 0.400
```

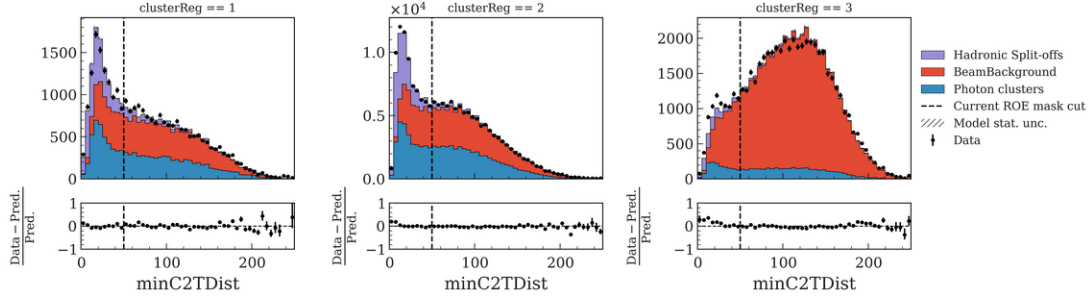


Figure 168: minC2TDist on the photons of the ROE in all three detector. Most of the hadronic split-offs are discarded without any Data/MC disagreement being introduced.

0.001124109 nu_tau gamma pi- pi0 PYTHIA 21;

I just checked again and apparently the misorderings of the radiative photons are not there anymore. I had developed this truth matching code still on MC14 when the problem was way more prominent. Perhaps some things have changed since then, so I think that the algorithm is still robust against any weird orderings just in case. For completion I paste below in Figure 169 the PDG codes of the τ daughters of the truth matched $B^0 \rightarrow D^*\tau(\rightarrow \rho\nu)\nu$ and $B^+ \rightarrow D^*\tau(\rightarrow \rho\nu)\nu$ events. I have added a respective comment in Section 5.1.4 in line 622. Regarding the TaubaBelle tool, as far as I understand this is a tool meant to be used in τ -pair analyses and not supposed to be used to truth match τ decays in $B\bar{B}$ events. Please correct me if I'm wrong.

```
cols = ["B_g_d10_mcPDG", "B_g_d11_mcPDG", "B_g_d12_mcPDG", "B_g_d13_mcPDG", "B_g_d14_mcPDG"]
result = (
    mc.query("isDstRhoNu == 1")
    .assign(**{col: lambda x, c=col: x[c].abs() for col in cols})
    .groupby(cols)
    .size()
)
result
```

B_g_d10_mcPDG	B_g_d11_mcPDG	B_g_d12_mcPDG	B_g_d13_mcPDG	B_g_d14_mcPDG	
16.0	22.0	211.0	111.0	9999.0	8
211.0	111.0	16.0	22.0	9999.0	125
			9999.0	9999.0	1511

dtype: int64

Figure 169: PDG codes of the daughters of the τ lepton in $B^0 \rightarrow D^*\tau(\rightarrow \rho\nu)\nu$ and $B^+ \rightarrow D^*\tau(\rightarrow \rho\nu)\nu$ events

13. *In Table 20, could you clarify why you are defining a signal region (SR) that also includes the normalization region (NR) in M_{miss}^2 ? Isn't the SR defined for $M_{\text{miss}}^2 > 2 \text{ GeV}^2$? Also because, in Fig. 38, I don't see any signal (blue components) in the NR region.*

The NR is defined only for the $B \rightarrow D^*(D^0 \pi) \ell^+$ reconstruction channels, while the SR is defined only for the $B \rightarrow D^*(D^0 \pi) h^+$ reconstruction channels. If we include the $m_{\text{miss}}^2 > 2 \text{ GeV}^2$ for channels $B \rightarrow D^*(D^0 \pi) \ell^+$ we are effectively repeating the Hadronic FEI-leptonic tau measurement which we want to avoid as this would complicate the setup significantly. For channels $B \rightarrow D^*(D^0 \pi) h^+$ we

still observe some events in the low m_{miss}^2 region (better visible in Figure 36 and this is why we do not discard this region. However since these are different reconstruction channels (lepton vs hadron) there's no overlap between them. Any potential small overlap is already taken care of from the BCS. I updated Table 20 so that it is clear for which reconstruction channels every m_{miss}^2 cut is referring to.

14. *Why do you choose the pi_slow candidate with the highest momentum when there are multiple candidates?*

This is what the Belle analysis did and we replicated their strategy on that one. This criterion is relevant only for the charged slow pions which anyways do not yield a lot of duplicates. In Table 21 you can see that choosing the slow pion with the highest momentum changes the candidate multiplicity from 1.737 to 1.724 (only 55 duplicate candidates were discarded) therefore the effect of this is expected to be minimal. If you have strong arguments against doing that we can switch to a random selection of the charged slow pion.

15. *Tab. 23 most likely has the same error in the calculated luminosity.*

Answered above at 4.

16. *Can you clarify how you apply the photon efficiency to the photons in the ROE? Do you need to save all the photons in the ROE to apply the weight?*

I doubled checked and we have the photon efficiency weights in the tuples for all γ (daughters of π^0 and photons in the ROE) but we consider none of them into the final MC weight. I'm sorry for the oversight! I removed the misinformation part from 5.5.5. It would be challenging to apply the photon efficiency weight from the photons of the ROE to the $\Upsilon(4S)$ tuples, but we can look into it if you think that this is necessary.

17. *For the efficiency tables for lepton ID, the recommendation should be to use v1 tables (/group/belle2/users2022/unok/leptonid/combinat/perf/PID/methods/moriond_20 instead of v0.*

Thanks for the heads up! The LID xwiki page recommends to use `v0_coarse`. When I check on kekcc `v0_coarse` is just a symbolic link to `v1`. So I am indeed using `v1`, event if this was not very transparent... This is all shown in Figure 170. I updated the versioning information of section 5.5.8 with the above details.

```

Path to csv tables (**)
Normal theta binning: /group/belle2/users2022/unok/leptonid/combinat/perf/
PID/methods/moriond_2024/proc13prompt/MC15rd/v0_coarse

/group/belle2/users2022/unok/leptonid/combinat/perf/PID/methods/moriond_2024/proc13prompt/MC15rd
-bash-5.1$ ls -l
total 2
lrwxrwxrwx 1 unok b2_belle2 2 Dec 13 2023 v0_coarse -> v1
drwxr-xr-x 2 unok b2_belle2 4096 Dec 13 2023 v0_dense
drwxr-xr-x 3 unok b2_belle2 4096 Dec 2 2023 v1

```

Figure 170: Demonstration of proportions of fake and true leptons in the retained samples of $B \rightarrow D^*(D^0 \pi) \ell^+$.

18. *For the fake rate corrections (e.g., Figs. 15-18), how did you choose the binning? There are some bins with large uncertainties on the correction; have you*

tried optimizing the binning?

The initial choice of the binning has been aligned with the one that is used for the official efficiency correction tables. Of course this is the reason that the uncertainties are much larger for the fake rate corrections as the control samples there are limited. We could consider optimizing this binning however as you can see in Figure 171 the amount of fake leptons is very small in this analysis. Moreover since for signal decays we use the hadronic reconstruction, this correction is affecting $R(D^*)$ only partially through the yield of $B \rightarrow D^* \ell \nu$ decays in the denominator (not relevant for the nominator where we reconstruct a charged hadron). However the uncertainty of this yield is expected to be already very small as you can see in Table 43 with our Asimov results. This is also reflected in the systematic budget in Table ?? where the Lepton fakes systematic impact on $R(D^*)$ is already negligible, even with these statistically inflated uncertainties for the fake rate corrections. We believe that changing the binning of the fake rate corrections will not have any impact on the final result of this analysis but let us know if you think differently.

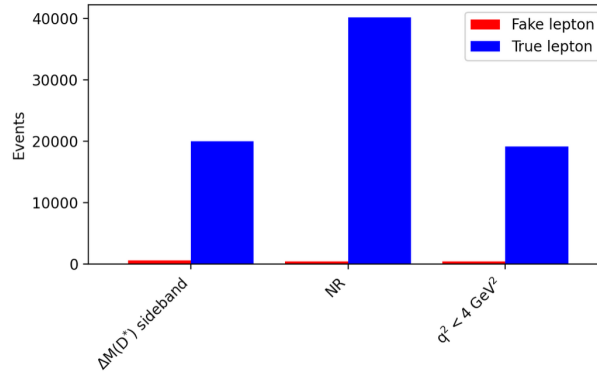


Figure 171: Demonstration of proportions of fake and true leptons in the retained samples of $B \rightarrow D^*(D^0 \pi) \ell^+$.

19. *Are you sure that the $BR(\text{tau to } 3\pi)$ in the DECFILE is wrong by a factor of 10 (Tab.27)?*

Sorry this is a typo. I confirmed that the values used in the code are the correct ones. This typo affects only the documentation. I corrected Table 27.

20. *In Fig. 29 (and 44, 46, 50, 58, etc.) there actually seems to be a difference in shapes.*

Yes you're right, some of the Figures indeed demonstrate small differences in the shape of some variables. Actually since the beginning of the WG review we had some internal discussions about the adequacy of our background reweighting. We have good reasons to believe that it's not necessary in the end to reweight the backgrounds the way we do. To test this hypothesis we ran the control fits as described in Section 8.2 once with and once without including the background reweighting. In Figure 172 we show the difference between all the unconstrained factors of the control fit with and without the background reweighting. Since in the two versions of the fit we fit the same data we expect

the two versions of the factors we extract to be highly correlated. We do not expect the correlation to be exactly 1 since the weights are different and some of them differ quite a lot from one and have sizeable uncertainties. Since we do not know exact correlation between the two versions of the scaling factors we decide to use a 95% correlation between them when we subtract them. Almost all of the differences are consistent with zero within the uncertainties. There's one that demonstrates a $\simeq 2\sigma$ difference and one of them a $\simeq 1.3\sigma$ tensions which we consider acceptable. We therefore conclude that since the impact of the background reweighing on the central values that we derive is minimal, it's better if we completely drop it from the analysis. The main advantages of such decision is that the analysis becomes much simpler and we get rid of sizable extra systematics that we have already assigned as could be seen in Table 46 of v3 of this note. We modify the text in Section 6.3 and we exclude the description of the background reweighing from the documentation. We also modify the systematic budget in Section ??.

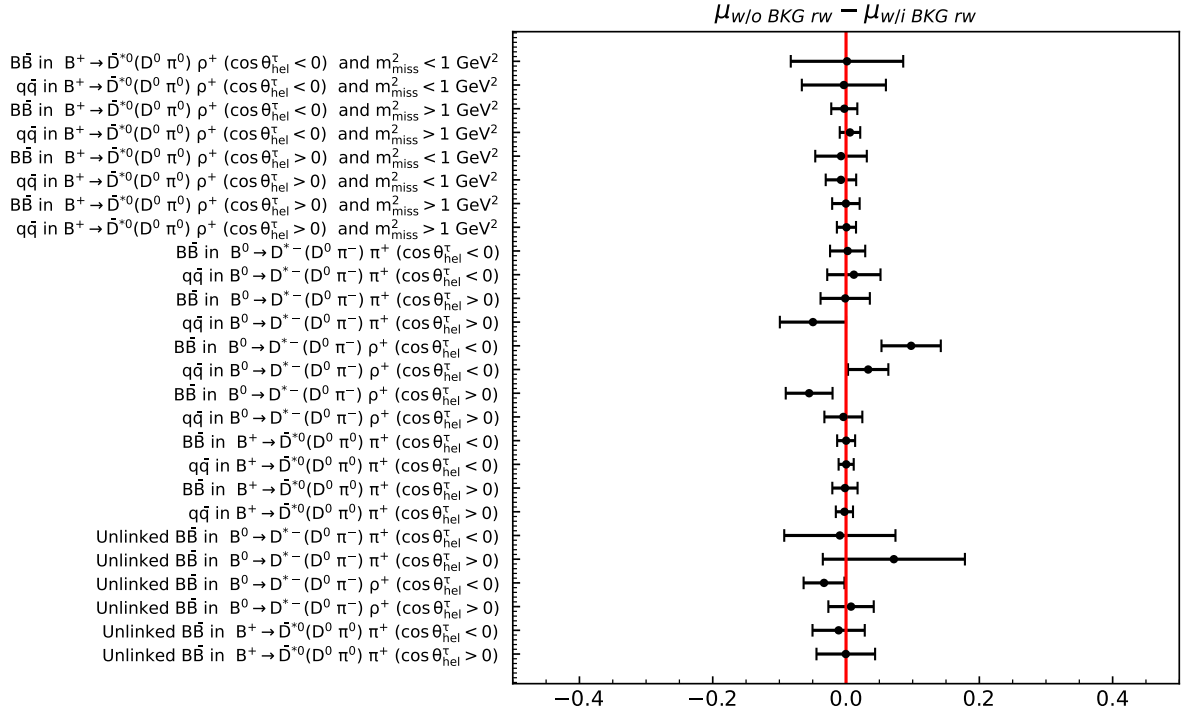


Figure 172: Difference between the unconstrained factors in the control fits without and with the background reweighing.

21. *Fig. 42: Could you explain why for $B^0 \rightarrow D\pi$ you use 'one extra charged track' as the sideband to constrain the $B\bar{B}$, while for $B^0 \rightarrow D\rho$ you use the region $1.5 < ECL < 2$? Couldn't you use the same regions for both?*

One of the main effects that we're trying to correct with our control regions is the FEI efficiency of these background events. The criterion that we have

defined to ensure that the FEI performance between the SR and the control regions is the similarity between the FEI decay mode distributions in the two regions. Figures ?? shows that the FEI decay mode distributions in the one extra charged track sideband is compatible with the one in the SR for the $\tau \rightarrow \pi\nu_\tau$ channel. This is not the case for the $\tau \rightarrow \rho\nu_\tau$ channel. As can be seen in Figure ?? the 95% confidence interval that we set for the fraction of the residuals that are within -1 and 1 does not cover the 68.3% for the comparison between the SR and the one extra charged track sideband. Therefore we conclude that the residuals are not normally distributed and therefore the two orthogonal regions are not compatible. This is not the case for the $1.5 < E_{\text{ECL}}^{\text{extra}} < 2$ sideband where based on the statistical power of our test we cannot exclude the possibility that the residuals are normally distributed. Therefore we choose the high $E_{\text{ECL}}^{\text{extra}}$ sideband to constrain $B\bar{B}$ decays in the $B^0 \rightarrow D^*\tau(\rightarrow \rho\nu)\nu$ channel.

22. *Line 1594: You say to normalize everything to the ssbar component, but in Fig. 44 you say you normalize everything to the cobar component. Could you clarify this point?*

Sorry for this typo! The title of plot contains the correct information (normalized to ssbar) not the captions. I updated all the captions.

23. *Fig. 76: Why do you observe a high correlation for some yields off the diagonal as well?*

The observed strong anti-correlation in the correlation matrix is coming from the $B\bar{B}$ and $q\bar{q}$ templates. These have very similar shapes in the the observables used for the extraction of $R(D^*)$ and P_τ . This can be even better seen in Figure 173 which presents the correlation matrix of the unconstrained parameters in a fit where the control regions have not been included (only the SR is being fitted). We observe a strong anti-correlation between the $B\bar{B}$ and the $q\bar{q}$ yields $< -96\%$. Including the control regions helps in reducing the anti-correlation between these yields (at least 20% decrease). Ideally if our control fits were perfect this anti-correlation would completely vanish, but since all our control regions are contaminated with both $B\bar{B}$ and $q\bar{q}$ decays, this is the best we can do right now. The impact of including the control regions in the fit in terms of the uncertainties of the parameters of interest can be seen in Table 43.

24. ** Fig. 77: You have a bias in the mean of pT of about 3 sigma. Do you take this into account as an additional systematic uncertainty?*

Thank you for this question! In Figure 77 from v3 of the note, the HESSE error has been used to calculate the pull distribution. Since P_τ due to it's definition (we measure it as an asymmetry) can have also asymmetric errors, especially when throwing toys, we decided to use the MINOS error for the pulls of P_τ . We now updated Figure 56 with the MINOS uncertainty. When the nominator of the pull is larger than zero, we use the lower MINOS uncertainty. When the nominator of the pull is smaller than zero we use the upper MINOS uncertainty. With this modification one can see that the bias in the fit vanishes, therefore no extra systematic uncertainty is required.

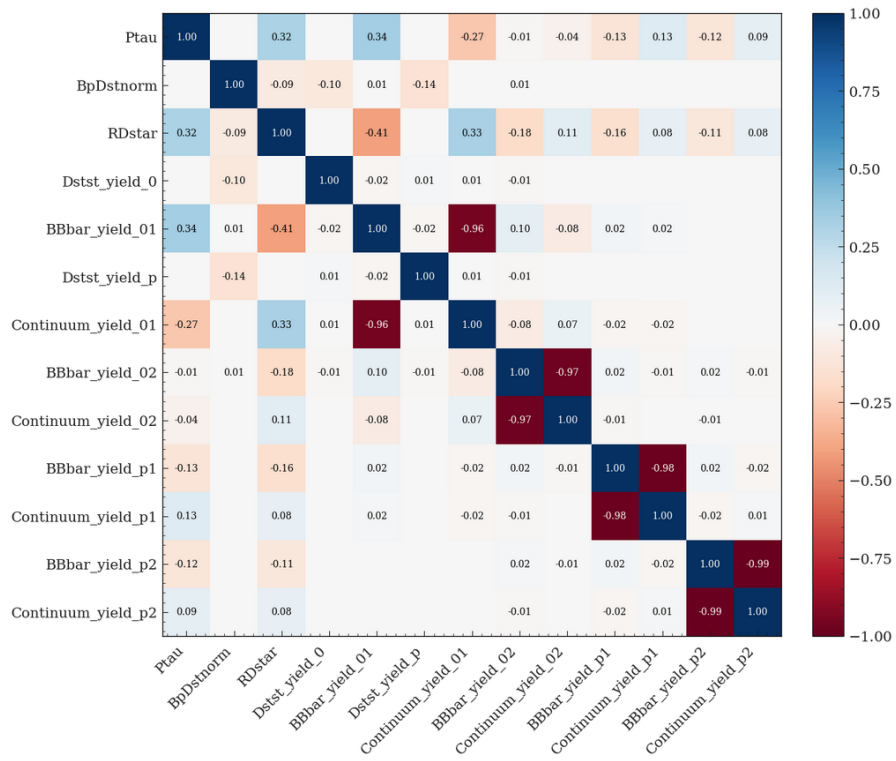


Figure 173: Correlation matrix for unconstrained parameters of the fit if the control regions have not been added.

- Comments received on 01-06-2025 by WG reader Taichiro Koga

25. *line 528 FEM -> FEI ?*

Corrected the typo. Thanks!

26. *-Figure1 Could you add c in the diagram ?*

Thank you! I updated the diagram in Figure 1.

27. *-Table1 Could you add Belle II semileptonic result ? You should add a caption at the top of Table (not only this table but also all tables)*

I updated Table 1 with our 2025 SL tagged result. Thank you for remarking upon the proper position of the caption of the tables. I updated all tables in the document and now the caption is on the top.

28. *-Section3.1 I think the explanation is too detailed and part of them can be moved to Appendix.*

Thank you for your suggestion regarding Section 3.1. I intentionally chose to provide a more detailed overview of the software tools early in the document as part of my narrative style. I believe this level of detail will be particularly valuable for newcomers or future researchers who may wish to repeat the measurements or build upon this work. From personal experience, I've learned that clear documentation of tools and their purposes can save considerable time and confusion—especially in the early stages of an analysis. With these five pages, I've tried to be the kind of culture change I hope to see in future documentation. That said, if you believe it's truly essential to reduce the length of this section, I'm open to condensing it and moving the more technical parts to the Appendix, while keeping the core narrative intact.

29. *-line 820 3 → Table 3*

Corrected.

30. *-Table 4 Is this MCri ? Could you write so ?*

This Table is now removed as part of Michele's questions. For more please refer to Question 1.

31. *-Table 6,7 D(star)etaellnu branching ratio is different between Table 6 and Table 7. Which is correct ?*

You're absolutely right. Apologies for the oversight. Michele had the same question. This is now corrected, please refer to Question 5.

32. *-Table 7 Branching ratio of D_0^* is different from https://indico.belle2.org/event/14125/contributions/89012/attachments/32814/48431/followup_180225.pdf Could you confirm with Michele ?*

Yes you're right. I got the latest values by Markus for the reweighting, but forgot to update Table 6. Now the Table is updated

33. *-line 519, 889, Table 23 integrated luminosity is different in each other. Which value is correct ?*

Actually none of them. Sorry for the oversight. Now I aligned all the LS1 luminosity references across the document using a custom command. The official luminosity of the LS1 dataset on the $\Upsilon(4S)$ resonance as per the [luminosity page](#) of the Data Production group should be $365.29 \pm 1.70 \text{ fb}^{-1}$.

- 2635 34. *-line 933 Does “three neutral clusters” include clusters from charged particles*
 2636 *? If so, could you modify the text ? If it is not, the selection looks too tight.*
 2637 Apologies for the confusion. With the text in Section 4.3 I was aiming to refer
 2638 again to the pre-cuts that the FEI skim is applying to the events. I do not ex-
 2639 plicitly apply these event cuts as they would be redundant, since the FEI skim
 2640 is already applying them (see `fei_precuts` method in [https://software.](https://software.belle2.org/light-2503-ceres/sphinx/_modules/skim/WGs/fei.html#BaseFEISkim)
 2641 [belle2.org/light-2503-ceres/sphinx/_modules/skim/WGs/fei.html#BaseFEISkim](https://software.belle2.org/light-2503-ceres/sphinx/_modules/skim/WGs/fei.html#BaseFEISkim).
 2642 The only event cuts that I apply using the `applyEventCuts` module are the
 2643 ones mentioned in Line 460, and refer to discarding events where no FEI can-
 2644 didates were found. I modified the text in Section 4.3 by just referring back
 2645 to Table 3 to avoid the confusion of the reader thinking that I explicitly apply
 2646 these cuts in my steering file.
- 2647 35. *-line 977, table 13 Could you apply a lepton momentum cut ? At least $p > 0.2$ is*
 2648 *needed, because $p < 0.2$ is not available in Figure 13. If you prefer conservative*
 2649 *way, $p > 0.7\text{GeV}$ is needed for muon PID by reaching KLM. (at $p < 0.7\text{GeV}$,*
 2650 *muon and pion can not be identified and both are included.)*
 2651 Thank you for this comment! I can apply a pre-cut online at 0.2 on the leptons
 2652 as per your suggestion. However later offline we apply a cut at 0.5 GeV for e ,
 2653 μ and π^+ and ρ^+ coming from the τ decay. This is documented in Table 20.
- 2654 36. *-line 984 If you require $n\text{CDCHits} > 20$, signal efficiency will reduce significantly.*
 2655 *Could you optimize the number of CDCHits threshold ?*
 2656

2657 Michele had the same question. Please see my answer to Question 9. Nonethe-
 2658 less optimizing this cut would require reprocessing online which would take
 2659 some extra time, so I will follow up in the next iteration.

2660 FOLLOW-UP

2661 I removed the $n\text{CDCHits}$ online and in Figure 174 I’m showing the distribution
 2662 of events in the fitting variable that would be added in the analysis if we indeed
 2663 dropped the $n\text{CDCHits}$ cut from all our hadrons. It is clear that only a handful
 2664 of signal events can be found in this range of $n\text{CDCHits}$. On the contrary we
 2665 see a lot of $B \rightarrow D^* \ell \nu$ events being included, confirming hypothesis as de-
 2666 scribe in Question 9 that not requiring $n\text{CDCHits} > 20$ deteriorates hadronID
 2667 performance (these $B \rightarrow D^* \ell \nu$ event are events where we tried to reconstruct
 2668 a pion but actually picked up a lepton). Based on this plot we consider that
 2669 the removal of this cut will worsen our result, therefore we decide to keep it as
 2670 is i.e. apply $n\text{CDCHits} > 20$ to all pions and kaons that we reconstruct.

- 2671 37. *-line 1000 I think K s efficiency correction needs to be done in each analysis, as*
 2672 *a function of flight distance. Could you show me data/MC comparison of the*
 2673 *flight distance ?*

2674 Due to a very naive bug (a simple typo) in our reconstruction code this variable
 2675 is not saved. We have to reprocess the tuples to give you this information. I
 2676 will follow up on that at a next iteration.

2677 FOLLOW-UP

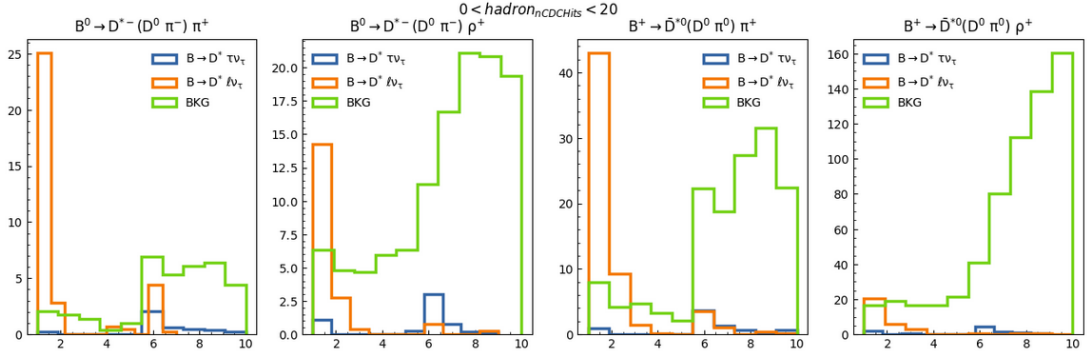


Figure 174: Impact of nCDCHits cut on fitting variable

Please see in Figure 175 the Data/MC agreement for the flight of distance of reconstructed K_S^0 candidates in the low q^2 sideband. All Data/MC weights and the FEI calibration has been applied. The p-values indicate that the modelling of this observable is acceptable with our current statistical precision, therefore we argue that no efficiency corrections is required for this analysis.

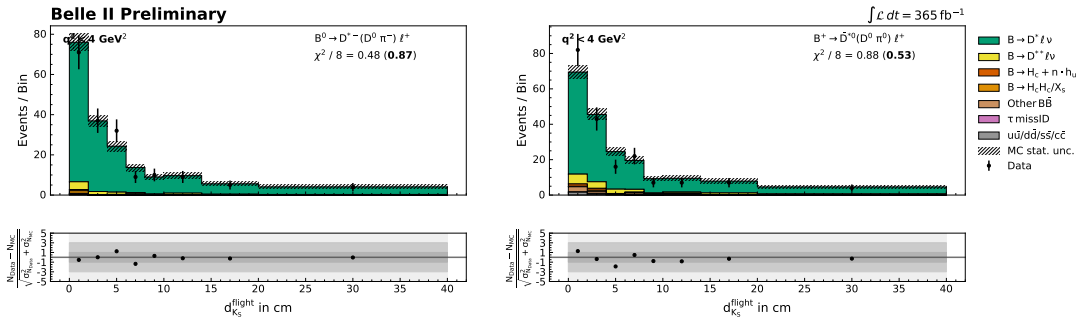


Figure 175: Data/MC comparison of the flight of distance of K_S^0 in the low q^2 sideband in the $B^0 \rightarrow D^{*-}(D^0 \pi^-) \ell^+$ channels (left) and $B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$ channels (right). The p-values indicate that the modelling of this observable is acceptable with our current statistical precision.

38. *-line 1009 Could you add an invariant mass distribution of rho ? How do you optimize the 0.66 and 0.96 GeV ?*

We adopted this cut from the Belle analysis. As can be seen in Figure 176 the window is indeed a bit narrow as it cuts off the tail of true ρ^+ mesons (especially on the lower end). Unfortunately this cut is applied online.

We generated some signal MC to study the properties of the ρ^+ invariant mass. This can be seen in Figure 177. The tails of the distribution of the generated mass range from $\simeq 0.35$ to $\simeq 1.4$ GeV. Of course we cannot broaden the cut by that much as this would significantly increase the background levels.

I couldn't find reason that the Belle analysis chose exactly this cut, but I found the distributions of true and fake ρ^+ , shown in Figure 178, from where it is clear that the backgrounds steeply increase in the lower end of the distribution. I

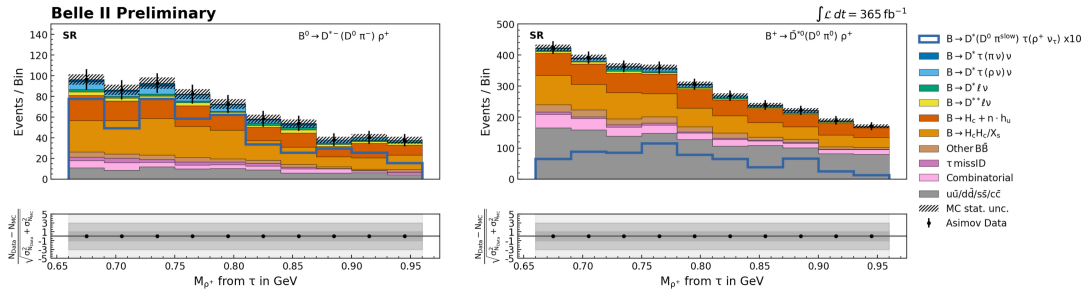


Figure 176: Invariant mass of the ρ meson

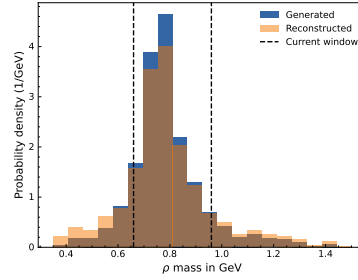


Figure 177: Generated and reconstructed mass of the ρ^+ meson in signal MC.

could broaden the online cut at a future processing if required and study offline the effect of this tighter mass window in terms of sensitivity and Data/MC impact. Please let me know about your thoughts on this.

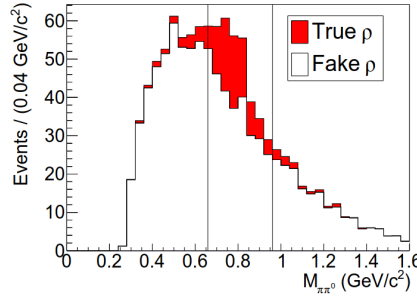


Figure 178: ρ^+ invariant mass cut at the Belle analysis.

FOLLOW-UP

We broaden the online cut of the reconstructed ρ invariant mass to $[0.560, 1.060]$. We tried broadening it even more however the combinatorics exploded due to reconstruction of a ρ meson along side with a D^* meson. The steering file was taking way too long to go through and we were reaching 10,000 candidates per event very often (when `basf2` stops combining more candidates). We found this range to be the best compromise between broadening the online cut and keeping our online reconstruction feasible. We then optimized the mass window offline by maximizing the figure of merit $\frac{S}{\sqrt{S+B}}$. Our results are shown in

Figure 179. We find that the best mass window is $[0.590, 1.050]$ GeV, inside the online mass window that we chose. We report that the FOM improves from 5.28 to 6.19 when we switch from the old $[0.660, 0.960]$ GeV window to the optimized $[0.590, 1.050]$ GeV. We will be using this cut for the rest of the analysis. For completion we also provide the rho mass distribution with the optimized cut in Figure 180.

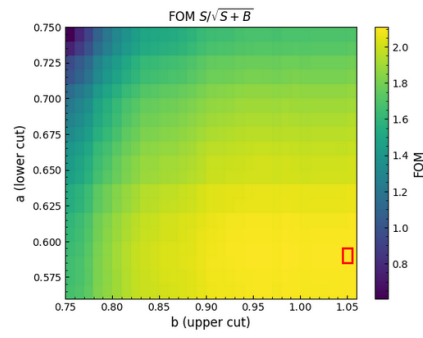


Figure 179: FOM for the ρ mass window. The best window is shown with a red square.

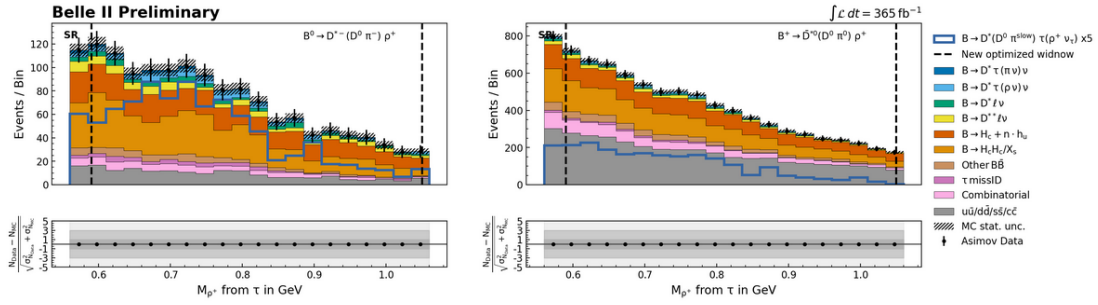


Figure 180: Invariant mass of the ρ meson with the optimized cut

39. -Table 14 Could you add $K_S \pi \pi \pi$? BF is about 5%. Is there any reason to exclude it ?

We had including this mode initially in the analysis due to this BF argument. However our studies in the past have shown that this mode has very low $S/\sqrt{S+B}$. This can be see in Figure 181 where we show the reconstructed invariant mass of D^0 in a broad window (without any offline cuts applied). $D^0 \rightarrow K_S^0 \pi^+ \pi^- \pi^0$ demonstrates a very low figure of merit compared to the other modes as can be seen in the bottom right plot. We believe that we cannot use this mode contrary to other semileptonic analyses due to the fact that we reconstruct a fully hadronic final state with missing energy for the signal, therefore it becomes very challenging to properly reconstruct high multiplicity final states with many neutrals. In the worst case in the $B^+ \rightarrow D^{*0}(D^0 \pi^0) \rho^+$ reconstruction we would reconstruct four neutral particles on the signal side (slow π^0 , π^0 from ρ^+ , K_S^0 and π^0 from D^0) which really smears out the resolution and significantly increases the combinatorics. We

believe that is best if we do not include this mode. Please let us know if you think differently.

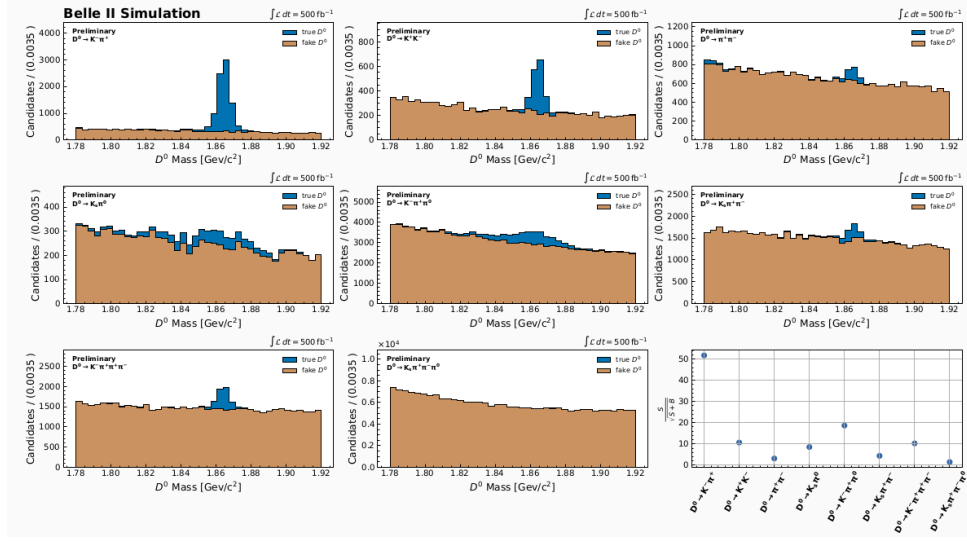


Figure 181: Invariant mass of D meson for different modes.

40. *-Table 14 BF of $D \rightarrow KK$ and $D \rightarrow p\pi\pi$ are opposite ?*

You're right, sorry for the typo. Corrected.

41. *-Table 14 Could you show me that the figure of merit (or Asimov fit sensitivity) is improved by adding each channel, after applying final tight selection ? I worry some of the channels are dirty (or low signal statistics) and not useful to improve the analysis sensitivity.*

In order to evaluate each D^0 mode's sensitivity in the analysis I run the Asimov fit once including all the D^0 decay modes and once by excluding one mode at a time. The results can be seen in Figure 182.

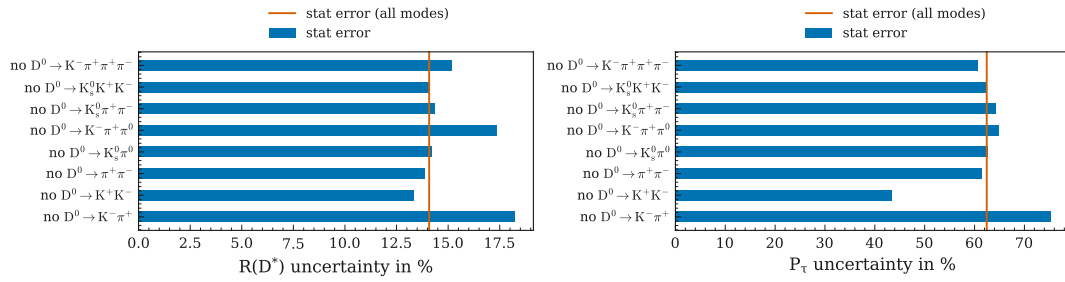


Figure 182: Impact of every D^0 mode on the sensitivity of the parameters of interest. The uncertainties are taken from an Asimov fit. We confirm that we observe closure in the fit (the central values are minimized to the expected ones)

It is very clear that the $D^0 \rightarrow K^-K^+$ mode has the worst sensitivity and it worsens our sensitivity in both parameters of interest. We completely exclude this mode and the repeat the test. The results are presented in Figure 183.

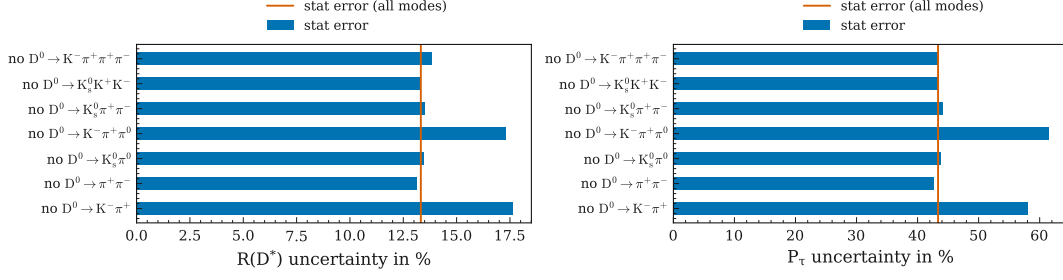


Figure 183: Impact of every D^0 mode on the sensitivity of the parameters of interest. We completely exclude the $D^0 \rightarrow K^- K^+$ mode from this test. The uncertainties are taken from an Asimov fit. We confirm that we observe closure in the fit (the central values are minimized to the expected ones)

We observe that the $D^0 \rightarrow \pi^- \pi^+$ mode still deteriorates the performance of our fit. We completely exclude this mode and then repeat the test. The results are presented in Figure 184.

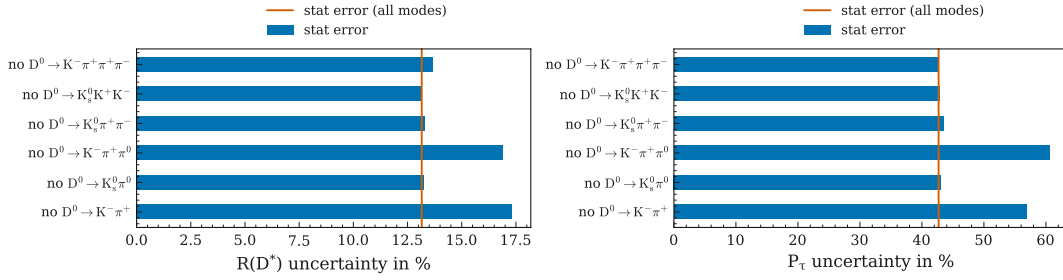


Figure 184: Impact of every D^0 mode on the sensitivity of the parameters of interest. We completely exclude the $D^0 \rightarrow K^- K^+$ and $D^0 \rightarrow \pi^- \pi^+$ modes from this test. The uncertainties are taken from an Asimov fit. We confirm that we observe closure in the fit (the central values are minimized to the expected ones)

Since no particular mode is making the performance worse, we decide to move on with the remaining six D^0 modes. $D^0 \rightarrow K^- K^+$ and $D^0 \rightarrow \pi^- \pi^+$ will be excluded from the analysis from now on.

I added Appendix I with this information and updated Section 4.6.4

42. -Table 15 How do you optimize the energy threshold of ECL clusters ?

This is somehow linked to Question 11 by Michele. During the development of the analysis we have tried several well motivated masks that have been used also in other analyses at Belle II and this is the only one that give good Data/MC agreement in the $E_{\text{ECL}}^{\text{extra}}$ distribution. Furthermore in Figure 185 I provide a Data/MC check on the cluster energies of the photons in the ROE. The dashed line shows the chosen cut. The MC has been normalized to data, so here we're comparing the shapes. Since our cuts are more or less on the switch of the sign of the disagreement we are not introducing any extra normalization

or shape disagreement. This is also justified by our last checks on $E_{\text{ECL}}^{\text{extra}}$ 75, Please let us know if you think we should further optimize this.

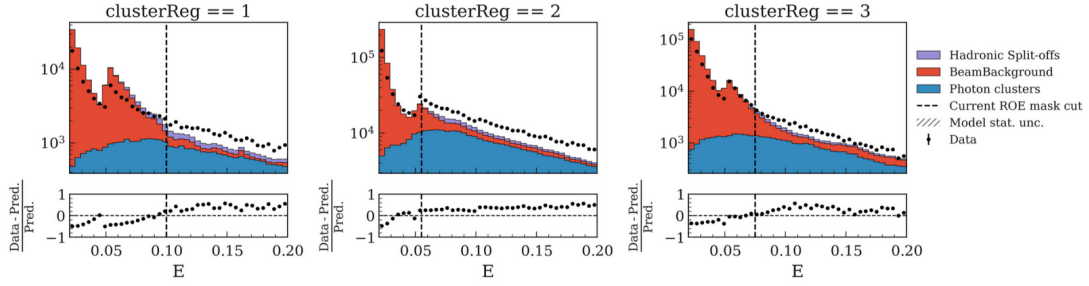


Figure 185: Cluster energy distributions of photons in the ROE in the three detector regions.

43. *-Table 15 Did you try to improve ROE photon selection by using fakePhoton-Suppression and beambackgroundPhotonSuppression ? If not, could you check ?*

The Neutrals group has made public the list of ECL variables that are recommended for use. (see https://indico.belle2.org/event/13722/contributions/84645/attachments/31512/46646/2024_11_17_ferber_ecl.pdf). The `clusterZernikeM` has been used for the training which belongs to the non-recommended ones. Nevertheless the Neutrals group recommends to *Use the $E_{\text{ECL}}^{\text{extra}}$ distribution to check for the data-MC agreement* <https://xwiki.desy.de/xwiki/bin/view/BI/Belle%20II%20Internal/Physics%20Performance%20Webhome/Neutrals%20Performance/>. We can say that all masks that we have tried and included the MVAs did not demonstrate good Data/MC agreement, therefore we have decided to not use them.

44. *-Table 20 Could you improve the format of the table ? It is difficult to see. (I can not understand which left column corresponds to which right column.)*

Apologies, I added horizontal lines between each cut, I hope it is more visible now. Let me know if not.

45. *-line1218, 1220 Reconstructed mass peak of π^0 can be different from the theoretical one, due to the response of ECL. Could you show me the reconstructed π^0 mass distribution ?*

You are right! I provide in Figure 186 the reconstructed mass of the slow π^0 in $B^+ \rightarrow D^{*0}(D^0 \pi^0) \ell^+$ channel in the low q^2 sideband after applying the FEI calibration factor. This is a reconstruction that is affected by the BCS as described in 5.4. Luckily the shift is common for MC and Data. The two means are compatible with zero within the uncertainty. Since there's no bias introduced and we still keep all signal candidates as can be seen in Table 21 we believe it's okay to keep the BCS as it is. Please let us know if you disagree.

46. *-line 1223 I think it is better to use purity than the BF in general. From figure 9, it looks like the purity of the π channel is better than the ρ channel. How about prioritizing the π channel rather than the ρ channel ?*

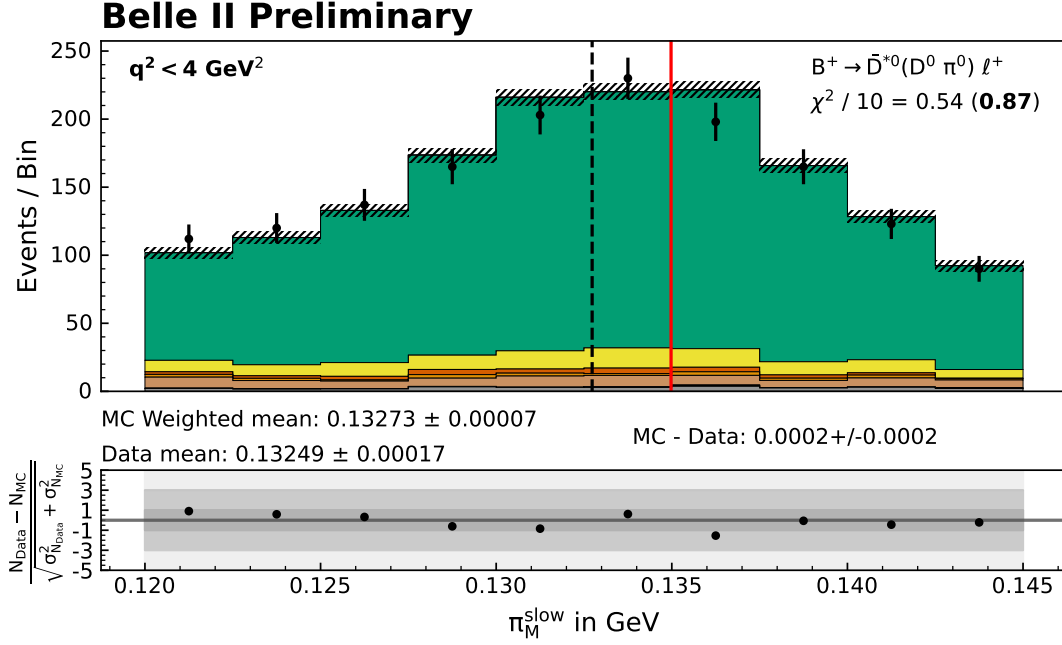


Figure 186: Reconstructed invariant mass of slow π^0 . A clear shift from the theoretical value can be seen.

Thank you for this question! I tried both approaches and I found that with our current strategy (prioritizing BF) the Asimov result fits are $R(D^*) = 0.258 \pm 0.0335$ and $P_\tau = -0.4970 \pm 0.2076$. When we prioritize the $\tau \rightarrow \pi \nu_\tau$ mode (prioritizing purity) the results are $R(D^*) = 0.258 \pm 0.0363$ and $P_\tau = -0.4970 \pm 0.3045$, worse than before, especially for P_τ . The main reason seems to be that a lot of the duplicates are also found in the $B\bar{B}$ template. That means that when we prioritize the $\tau \rightarrow \pi \nu_\tau$ channel we're contaminating it with more background events which messes up its relatively high purity observed with the current strategy. Based on these results we decide to not change this step of your BCS, but keep it as described in 5.4.

47. *-line 1233 In the end if we find multiple candidates having been reconstructed in two types of reconstruction channels e.g. signal and normalization, then we keep one candidate at random. How frequently such a case happens ?*

This happens in 314 out of 231,123 total events on MC (all reconstruction channels and all sideband together) which represents a $0.136\% \pm 0.008\%$ of the total events. On Data this happens in 27 out of 38,310 total events (all reconstruction channels and all sideband together) which represents $0.071\% \pm 0.014\%$ of the total events. The errors on these ratios have been calculated using the binomial approximation. Even if this happens at a different frequency on Data and MC, we believe that choosing a random candidate does not introduce any bias in our sample.

48. *-line 1413 we also set a 95% Confidence Interval (CI) for the fraction of residuals that are ± 1 sigma for every comparison. Sorry, what do you mean ? The*

1sigma does not equal 95% CI.

We are calculating the residuals for the fraction of events that are reconstructed in every FEI decay mode. These residuals are expected to be normally distributed around 0 if the compositions of the different templates are compatible with each other. In other words we expect 68.3% of the residuals to be within $[-1, 1]$ the composition of the modes that are reconstructed by the FEI is compatible across the templates. To test this we are calculating the 95% CI of the fraction of the total number of the residuals (e.g. in Figure ?? we have 15 residuals coming from 15 FEI decay modes), that are within $\pm 1 \sigma$. If 68.3% is covered by the CI then we argue that we have no statistical evidence that the residuals are not normally distributed, since if we had 100 replicas of our MC and we repeated the test 100 times, in 95 of them we would find the fraction of residuals which are between $\pm 1 \sigma$ within the 95% CI which contains the 68.3%. I hope this clarifies our method. Please let me know if it is still unclear. However this might be irrelevant once I follow up to Question 53.

49. *-line 1414 If 68.3% is covered by the CI then we conclude that with the current dataset we have no statistical evidence that the tag-side decay mode distribution is different between the three modes of interest. Sorry, I can not understand this statement. For example, in $D^*pipipipi0$ mode, the fraction of green line ($B \rightarrow D^*taunu$ in SR) is significantly (3σ) smaller than the others. So, we have statistical evidence of the difference locally.*

Yes you're right, in Figure ?? there is one residual that is demonstrating a slightly larger tension than 3σ . However we believe that only one mode is not enough to make us argue that the FEI composition overall is not compatible between two templates in two different regions. That is the reason that we have developed the test that I tried to clarify further in my answer to Question 48. If we wanted to take into account local differences and outliers we would then need to come up with a reweighting strategy, which could be possible, but needs to be discussed. In general I believe that quantifying such a test about the FEI composition is an ill defined problem as many modes have very low counts and the aggregation of several modes into the "Rest" category, heavily depends on the number of events that have reconstructed e.g. for the signal template in SR we have in general way less events than in the normalization templates in NR and the q^2 sideband. So even if the test can be defined for more modes for the normalization template, there's certain modes that need to be excluded when testing for the signal template, due to the lower expected yield. However this might be irrelevant once I follow up to Question 53.

50. *-Figure 29-32 Could you add error bars to the histograms ?*
See Question 53

51. *-Figure30 Is the p-value calculated correctly ? For example, in the left figure, four points are deviated more than 3σ but p-value is high. I think there is something wrong.*
See Question 53

52. *-Figure 32 In the left plot, The pull distribution has a clear tendency (at low*

Btag_deltaE value, the blue line is smaller than the red line. at high value, the blue line is larger.). In this kind of case, even if the p-value is large, we should conclude that there is a significant shape difference. (because the chi2 and p-value do not take into account the sign of each bin.) Similar things happen to Figure 30(left).

See Question 53

53. *-line1467, table 35 I think you are estimating the FEI efficiency correction factor by integrating all Btag decay mode. However, the correction factor is significantly different in each Btag decay mode. Is there any reason to estimate the correction with integrated mode ? In order to check possible impact to analysis, could you compare the PDF shape of fit variables in each Btag decay mode ? If there is no difference, you do not need to estimate the correction factor in each Btag decay mode. If there is a difference, you need to estimate the correction factor in each Btag decay mode. (or assign systematic error.)*

The reason for not deriving one calculation factor per FEI decay mode is of course the complexity of the fit and the much lower statistics that would increase the uncertainty of the per mode, calibration factors. Nonetheless since we have all the technical tools available we will pursue a mode-by-mode calibration and follow up on this in the next iteration of the note.

54. *-line 1471 I think it is not enough to check $B \rightarrow D^* \ell \nu$ channel only for photon multiplicity reweighting, because one of main reason of the data/MC inconsistency is fake photon from hadronic particles. Depending on the final state of the decay, the correction can be different. Because you are using tau hadronic decay, you must check other sideband in the π and ρ channels.*

I totally agree with you! Figures ?? - ?? and ??-?? present the $E_{\text{ECL}}^{\text{extra}}$ distribution among other observables and for all of those plots the photon multiplicity reweighting has been applied on top of the weights derived from the control regions to constrain the backgrounds. In Section 8.6 I had already mention that in Line 1553. I updated the text in both Sections 8.5 and 8.6 to make it more clear.

55. *-Table 36 Could you show me the data/MC comparison plot of the photon multiplicity ?*

I added Figure 32 in Section 5.12.

- Comments received on 03-06-2025 by WG reader Taichiro Koga

56. *-Figure40 Even in higher missing mass region, there are still $B \rightarrow D^* \ell \nu$ events in $\tau \rightarrow \pi \nu_\tau$ sample. I think this is coming from the higher tail of $M_{\text{miss}2}$ as below plot. How do you validate the shape of the tail is correctly modeled by MC ?(For example, in our analysis, we measured the resolution of $M_{\text{miss}2}$ on data and MC then applied correction.)*

As shown in the main text we initially used a fine binning in the normalization enhanced region to extract the $B \rightarrow D^* \ell \nu$ yield. The binning used so far can be see in Figure 187.

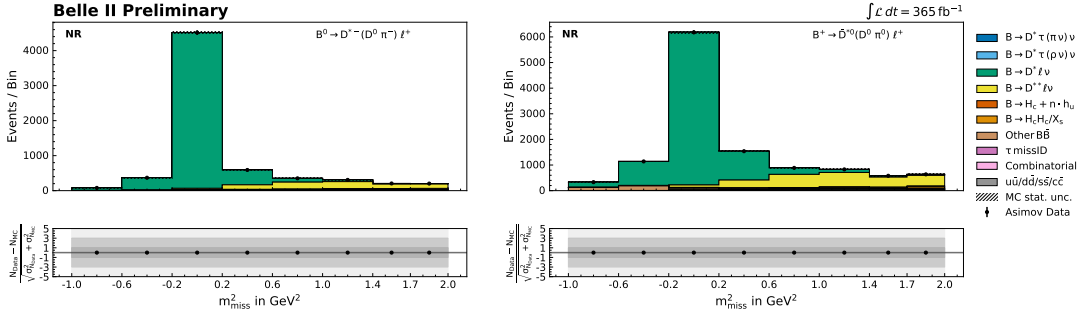


Figure 187: Original finer binning in m_{miss}^2 to extract the $B \rightarrow D^* \ell \nu$ yield.

In order to avoid further complicating the analysis we tried to bin this region much more coarsely as can be seen in Figure 188. Such a binning is not sensitive to the missmodelling of the peak of the tail of the m_{miss}^2 . In order to evaluate what is the impact on our extracted parameters we ran the Asimov fit using both binning scenarios. We found that the coarser binning worsens the sensitivity by < 0.0002 for P_τ (nominal value -0.497), < 0.0001 for $R(D^*)$ (nominal value 0.258) and < 0.0004 for the strength of $B \rightarrow D^* \ell \nu$ decays (nominal value 1.0). Therefore we conclude we were not gaining much by this finer binning which was causing the mismodelling concerns in your question. We decide to keep this coarse binning for the rest of the analysis.

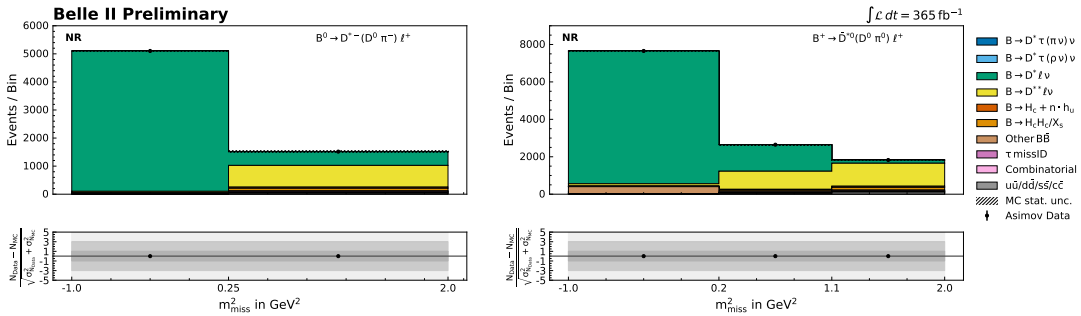


Figure 188: Updated coarse binning in m_{miss}^2 to extract the $B \rightarrow D^* \ell \nu$ yield.

57. *-Figure40 Even in higher missing mass region, there are still $B \rightarrow D^* \ell \nu$ events in $\tau \rightarrow \pi \nu_\tau$ sample. If you increase muon momentum threshold up to 0.7GeV,*

how much the contaminated $B \rightarrow D^ \ell \nu$ is reduced ?*

Thank you for this question. I believe you wanted to write pion momentum as in the $\tau \rightarrow \pi \nu_\tau$ channel we're not reconstructing a muon but a pion. I provide Figure 189 where I show the reconstructed events in the range $[0.5, 0.7]$ GeV for three categories, signal, normalization and other background in the fitting variable. If we discarded these events by tightening the pion momentum cut we would be losing way more events from the signal template rather than from the misreconstructed normalization events. We therefore think that such a cut would worsen the sensitivity of our measurement, due to the large number of discarded signal events. We do not worry that these $B \rightarrow D^* \ell \nu$ events would impact our $R(D^*)$ measurement by a lot as their yield is linked to the correctly reconstructed $B \rightarrow D^* \ell \nu$ events from $B \rightarrow D^*(D^0 \pi) \ell^+$ whose uncertainty is very small. Therefore these misreconstructed events will be scaled properly by the NR.

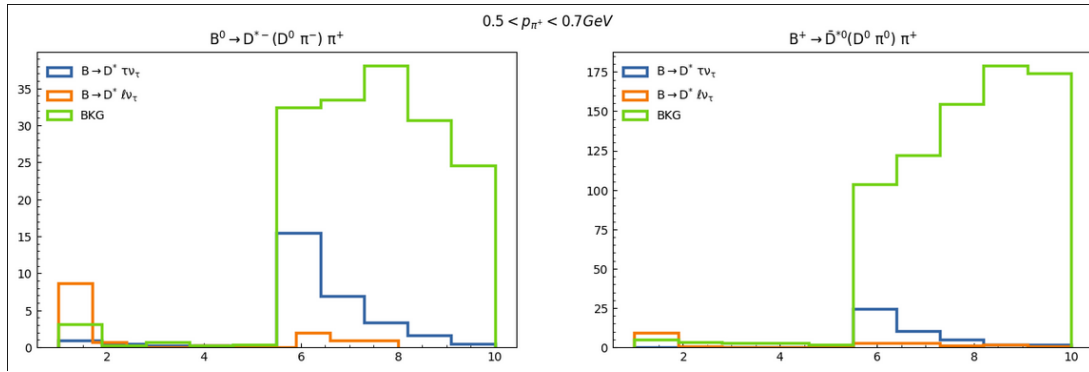


Figure 189: Events that would be discarded by a tighter momentum cut at 0.7 GeV. The impact on the signal template is larger than the normalization template in our fitting variable.

58. *-Figure42 Could you prepare another table to list the name and definition of selection criteria of all sideband ? (Figure42 is not enough to explain it, and Table 42 is too late to appear)*

Perhaps you missed Table 42? I added a dedicated reference in the text.

59. *-line1575 What is the definition of Region01 and Region02*

The definition of all the control regions is given in Table 42.

60. *-Figure 45 $\tau \rightarrow \pi \nu_\tau$ channel $\rightarrow \tau \rightarrow \rho \nu_\tau$ channel ?*

You're right! Sorry for that. These figures have been removed as part of Michele's Question 20.

61. *-Figure 45, 47 Could you unify the name (label in the figure) of each sideband ?*

Same as answer just above.

62. *-line1654 I think there are many kind of decay modes in "Prompt hadronic" and "Double charm". I am not sure if the kind of decay mode in SR is similar to that in BR01, BR02. Could you check the MC generated decay mode ID (by*

mcGentool or Topoana or whatever) in each SR, BR01 and BR02 ? Please check and compare it too with that at the lowest bin of E_{ECL}^{extra} in the SR (signal enhanced region).

In order to answer to this questions I had to produce a lot of plots which I put in an Appendix. Please refer to Appendix J.

63. *-line1736 Sorry, I can not understand the meaning of ζ . Could you add more explanation ?*

When we apply a Data/MC correction e.g. charged slow pion efficiency correction of LID efficiency correction, then these corrections can alter both the normalization (overall number of events) and the shape (no change in the overall number of events). These two effects are disentangled in our setup and treated separately. ζ controls the impact of the normalization part on the yield of every template. Since this effect can alter the measured yield of every template it has to be multiplied to the expected number of events and the free parameters that are modifying the yield of every template. I'm added some more text in Line 1106 to make this more clear.

64. *-line1752 I think it is better to move Table2 and related explanations here.*
Please see my answer to Question 28.

65. *-line1764 What is $N_s B^0$? I firstly thought it is the number of reconstructed signals as mentioned in line1760, but according to line 1732, N is defined as the number of true generated events. Could you check and clarify ?*

These were supposed to be the reconstructed events that are used to create the templates, but my notation was somewhat unfortunate. I modified the notation a bit to make it more clear. Now there events are denoted as $N_{sB^0}^{reco}$.

66. *line1782 What is $N_{nB\pm}$ in bold ? Is that $\sum_i N_n B\pm^i$*

Fixed by Question 65.

67. *-line1764 What is $B(D^* \rightarrow)$? Could you add a daughter state ?*

Apologies for the ambiguous notation. I intended to denote the sum of exclusive BF's of the D^* or D^0 mesons that have been used in the reconstruction, but this was not a great choice. I updated all the equations from $(D^* \rightarrow)$ to $(D^{(*)} \rightarrow \sum_i f_i)$ which I believe it more clear. I explain that in text now in Line 1142. Let me know if you still find the notation problematic.

68. *-line1779 In order to connect this equation to Eq.(39), could you write down not only v but also v_{cb} ?*

Good idea! Thank you! I added this and explain what the subscript cb is in text. Also simplified a bit the notation for the superscript e.g. $s_{B^0} \rightarrow s^0$. I also explain the new notation in text.

69. *Figure 69-71 There are no explanations in the main text ?*

Fixed.

70. *-Table43 Before showing this table, could you put all distributions used for the fit with Asimov data and post fit distribution ? Please show the pull too. (In Asimov fit, pull should stay 0 in all bins. We need to confirm it.) Similar*

comment to table45

I added Figures 52 - 55.

71. *-Table43 I can not understand the meaning of the variable and values listed in Table43. Before showing this table, could you list and explain all parameters in your fit, and which parameters are fixed (f00, efficiency, tau branching ratio,...) with their value, and which parameters are floated with their limit and constraint ? Similar comment to table45.*

I added Appendix L which present a modifier grid plot of all the free and fixed parameters of the fit as provided by `cabinetry`. I also reference this explicitly in Line 1306

72. *-Table43 I guess you are showing fit error only, but you should show the central value of the fit. (you need to show the output of the fit is consistent with the expected Asimov input.) Similar comment to table45.*

Sorry this should have been clearer in these Tables. I confirm that in all of our Asimov fits we get full closure. All free parameters in the fit are minimized to their nominal values. The nominal values were presented in parentheses in the first column of both Tables. I now confirm in the captions that we get closure in the central values of the free parameters.

73. *-Section 6.3.1, 6.3.2 Is this study done by separating positive and negative helicity angle bins ? If so, please move these study after line1910 (at the beginning of section 6.4.1.) If not, please repeat the same study by separating them. (I assume final result of paper will be given by the separated bin.)*

This study is not carried out in the two helicity bins. This study serves only as a proof of concept to show that fitting $R(D^*)$ directly is possible. For the final result the next study with the simultaneous fit of $R(D^*)$ and P_τ will be used which by design is performed in two bins of the helicity angle. Therefore I think it is not necessary to perform the $R(D^*)$ fit only in two helicity bins as this fit will not actually be used for our result but is described in the note for demonstration purposes. The reason that I did not show the simultaneous fit straight away is that it is a way more complicated fit, with an extra observable and way more templates and free parameters. I thought it might be more instructive to first demonstrate the method with a normal $R(D^*)$ setup, which is more intuitive to more people and then demonstrate the method to the more complicated P_τ fit.

74. *-Figure74 Sorry, I thought the sideband was fitted too simultaneously. Am I correct ? If so, please add the distribution of sideband too. (even if it is shown in previous pages, I think you need to show it again here. probably, Figure 69-71 should be moved here ?)*

Indeed the sidebands are fitted simultaneously to the SR and NR. I'm adding a reference in the text that the previously shown distributions are also fitted simultaneously together with these distributions. All the regions that are fitted simultaneously can now be seen in Figures 52 - 55.

75. *-Figure 75 caption is wrong*

Apologies! The caption of Figure 38 is now corrected.

- 3027 76. *-Figure 75 What is the reason for the large (>10%) fraction of the non-diagonal*
3028 *term ?*
3029 We believe that the migrations to the opposite sign of the cosinus of the helicity
3030 angle arises because of poor resolution in the reconstruction of the B^{tag} and D^*
3031 kinematics. Furthermore the migrations are more prominent for the $\tau \rightarrow \rho\nu_\tau$
3032 channels which is not surprizing since in these modes we reconstruct one extra
3033 π^0 .
- 3034 77. *-Figure 75, line1901-1909 Sorry, I can not understand how you are performing*
3035 *the fitting with true helicity angle. I can not understand how Figure75 is used.*
3036 *Could you explain the method (with equation if needed) ?*
3037 We use the true helicity angle in order to build the templates as they are
3038 outlined in Table 41. e.g. $B^0 \rightarrow D^*\tau(\pi\nu)\nu$ and $\cos\theta_{hel}^\tau > 0$ means that
3039 we have a signal decay where the τ lepton decays as $\tau \rightarrow \rho\nu_\tau$ and the true
3040 helicity angle (not the reconstucted one) is positive. We still use two bins of
3041 the reconstructed helicity angle for our MC and Data in the fit as illustrated
3042 in Figure 37 but these two bins can have signal events with either positive or
3043 negative generated helicity. Our assumption is that our fit is sensitive enough
3044 to properly scale events e.g with true positive helicity reconstructed in the
3045 positive or negative bin since the template is common. The confusion matrix
3046 in Figure 38 is only presented for illustration purposes. Since we use the true
3047 helicity angle to build the templates we do not need to do any unfolding and
3048 this matrix is not futher used in the fit.
- 3049 78. *-page101 table is out of page and there is no caption*
3050 I removed this big table now.
- 3051 79. *-Figure77, middle By eye, the distribution is not modeled by Gaussian well.*
3052 *By eye, the mean looks negative. Could you add chi2/ndf and error bar to*
3053 *histogram ? If chi2 is bad, could you test asymmetric Gaussian ?*
3054 You're absolutely right! This was adressed in Michele's Question 24. Please
3055 let me know if you still see a problem with Figure 56.
- 3056 80. *-line 1928 Explanation of Section 3.1.4 should be moved here.*
3057 Please refer to my answer in Question 28. I would prefer to keep the methods
3058 and tools at the beginning of the document and refer back to them when they
3059 are being used.
- 3060 81. *-line 1933 It seems you are focusing on the technical explanation of the tool*
3061 *itself, but I feel the explanation of the analysis method is not enough. Could*
3062 *you modify the text ? For example, could you add how the nuisance parameters*
3063 *are implemented to yourlikelihood in Eq.39 ? (Could you write down the c_x*
3064 *with the nuisance parameter?)*
3065 Table 2 outlines the different kind of parameters that are being used in the pyhf
3066 framework to build the likelihood. The column *Constraint term* shows how the
3067 nuisance parameter is being implemented mathematically and I added a short
3068 description in the last column where I explain the role of every parameter in
3069 text. I also added Appendix L which present a modifier grid plot of all the free
3070 and fixed parameters of the fit as provided by `cabinetry`.

82. *-line 1952 This is general feature in any analysis tool (not only sysvar but also all physics analysis has this kind of feature in order to estimate systematic error), so probably you can reduce to emphasis it.*

Excuse my enthousiasm, I spent quite some time developing SysVar and I'm quite proud for some parts of it including how flexible and intuitive it is (and of course not so proud of some other parts) I removed this sentence.

83. *-line1988-1991 I surprised that explanation of systematics errors are not enough. Could you explain how you evaluate each systematics one by one, by adding each subsection?*

We have a unified approach to all systematic uncertainties where we include and exclude the NPs and the subtract the uncertainties we get in quadrature. This is now described starting from Line 1435. Please let me know what additional information you would require as I don't see any other obvious information to include besides Tables ?? and ??.

84. *I think error of branching ratio of $D(^*)\eta_{\text{aellnu}}$ need to be assigned $\pm 100\%$ (errors on Table 6 should not be used). How are you assigning errors for now ? Does sysvar support uniform function ?*

I agree that introducing a 100% BF error is a valid and conservative approach. However, in practice, this introduces significant technical challenges in the fit.

From previous experience with the semileptonic tagged $R(D^*)$ measurement (see [30]), we observed that such a large prior tends to be overly conservative. Below I show a part of Figure 8 from [30]. Even when a 100% prior was assigned to gap modes, the post-fit impact on $R(D^*)$ was substantially smaller. This demonstrates that the fit can constrain these contributions in a data-driven manner. Given that the 100% prior is somewhat arbitrary—reflecting our current lack of precise knowledge about these branching fractions—and that it may not materially affect the final result, we propose starting with a less conservative prior (e.g., 50%). If the post-fit impact appears underestimated, we can iteratively broaden the prior as needed.

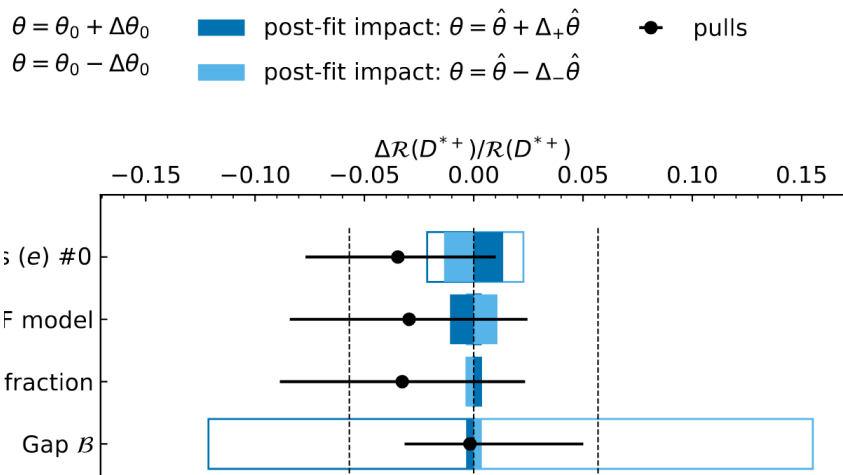


Figure 190: Pre and post fit impact of gap mode BF uncertainties in [30]

Regarding the use of a uniform distribution: while this is valid for toy sampling to estimate impact, it is not consistent with the Gaussian constraints used by pyhf for NPs. Therefore, we prefer to stay within a Gaussian treatment for consistency between toys and the statistical model.

The errors I'm assigning now are the ones assumed from the HFLAV average as outlined in Table 6, but since you say that they should not be used I can move to a more conservative approach as described above.

85. *For major error, please*

I think you didn't finish this question.

86. *-Table 47 What is the meaning of "Multiplicative" "Additive" "Combined $\rho = 0$ " "Combined $\rho = 1$ " ?*

The meaning of the terms Multiplicative and Additive are given in the paragraph starting from Line 1400. Multiplicative is the part of the systematic uncertainty associated with changes in the overall yield of every template. Additive is the part of the systematic uncertainty associated with changes in the shape of the template that do not affect the overall yield. The reason that we refer to them as Multiplicative and Additive becomes clear if one looks closer at Equation 38 since the normalization related term multiply the nominal rate, while the shape related terms are being added to it. Combined $\rho = 0$ means that we treat the multiplicative NP and the additive NP of a given principal component as fully uncorrelated, while $\rho = 1$ means that these two are treated as fully correlated. For completion we presented both approaches. We will follow up on our final decision on how to treat these two effect which will be done case by case.

87. *-Table 47 Could you add total systematics ?*

Added in Table 44.

88. *-Table 47 Which is the error from fitter bias, mentioned in line 1922*

Since we argue that there is not bias in our measurement as per Question 24, we are not modifying the Table. But you're right, back then I should have included the error. Sorry for that.

89. *-Section 6 Are the Asimov and toy studies performed with the systematic error explained in Section 7 ? If so, please explain it clearly. If not, please repeat the same study with systematics*

In Figure 191 we present our results from the toy studies with the NPs turned on. The top row shows the mean of the pull distribution for every parameter of the fit extracted by performing a single gaussian fit to the pull distribution we acquire from the toys. The middle plot shows the withd extracted in the same way. The bottom row shows the pull distribution for every parameter of the fit. We choose to show those distributions in this way to avoid presenting hundrends of pull distributions for every single parameter of the fit. We first sample nuisance parameters based on a standard normal distribution. Then we evaluate the likelihood on those nuisance parameters and the nominal free uncronstrained parameters. Once we evaluate the likelihood we then sample from a poisson distribution for every bin.

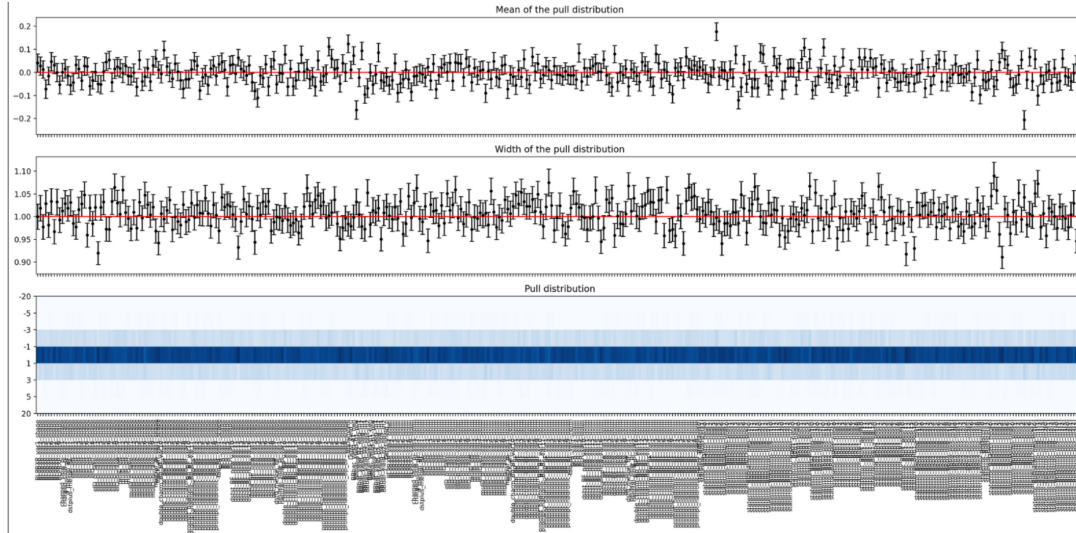


Figure 191: Results by running toys with nuisance parameters on.

The plots shown above have small caveat, we have excluded some regions with very small bin counts, since the gaussian sampling of the nuisance parameters was leading to negative yields for some bins in the toys. Since P_τ is the only parameter in the fit that can take both positive and negative values, these negative bin counts were always flipping the sign of P_τ to create negative yields, introducing biases in the toy results. Of course such a scenario is unrealistic as we would never observe negative overall yields on the experimental dataset. After removing these pathogenic regions, the toy fit results are more sensible.

90. *-Section 6.4 You should perform linearity test (like Fig.73) of tau polarization too*

I added now Figure 58 which show the results of the linearity study as described in Section 6.5.

91. *Section 7 I think you should perform Asimov and toy study to check behavior of the nuisance parameters. Could you check the pre fit, post fit values and pulls of all nuisance parameters by Asimov and the toy ? I worry the fit behavior can be out of our understanding, because the number of nuisance parameters are huge. What is the format of nuisance parameters ? Are those physics variables, which behavior/meaning can be understandable by human ?*

We performed a study with systematics on as described in questions 89

- Comments received on 30-10-2025 during the Full Status Report

92. *Quote your absolute systematic uncertainties for P_{τ} , not the percentage ones.*
Updated Table 44.

93. *What is the tau-missID component that has a similar distribution as your signal? How do you treat it ?*

In Figure 192 we present the overview of the different τ modes that contribute to this component. These are B meson semitauonic events where the τ lepton has decayed in a different mode than the ones that we tried to reconstruct. For comparison we always include the number of signal events (correctly reconstructed τ mode) with the blue color. We present this information for all eight different reconstruction modes. We see that for the $\tau \rightarrow \pi \nu_{\tau}$ modes a muon that has faked a pion is the dominant crossfeed component. For the $\tau \rightarrow \rho \nu_{\tau}$ modes it's mostly events where τ decayed into one charged and two neutral pions where one of the two neutral pions has been missed. For the leptonic reconstruction we see that we have a handful of semitauonic events that survived our m_{miss}^2 cut (we expect these semitauonic events to have higher values of m_{miss}^2). However these events are very few compared to the overall semileptonic yield (for electrons and muons).

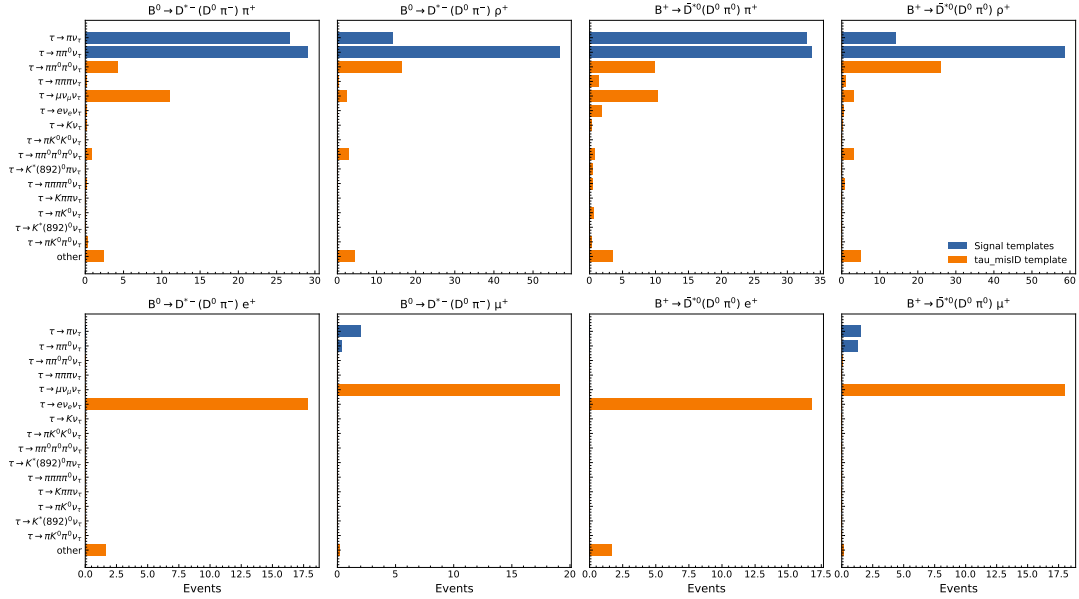


Figure 192: Overview of different crossfeed modes of semitauonic B meson decays.

This component was mistakenly merged into the signal templates so far. This is not a problem for $R(D^*)$ however it would affect the estimation of P_{τ} , since the helicity angle is not well defined for these decays. In order to circumvent this problem, we decide to create a separate template for this component in the fit. This component is fully linked via $R(D^*)$ but does not affect P_{τ} .

Currently we do not have access to corrections for leptons faking pions for the relevant momentum range within the systematics framework. However we

believe that this component is small enough to not really affect our extraction of $R(D^*)$. Nevertheless if the RC believes that we should treat this component differently, we can assign a conservative 20% uncertainty to these $\tau \rightarrow \mu\nu_\tau\nu_\mu$ events and evaluate the uncertainty on $R(D^*)$.

We update Tables 40 and 41 with the overview of the templates and all respective results in the note by adding this extra template.

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3390

Additional Material

3391

This sections contains all figures and numbers that are requested to be approved for public presentation.

3392